The Moduli Schemes of Elliptic Curves

P. Deligne and M. Rapoport

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THE MODULI SCHEMES OF ELLIPTIC CURVES

by P. Deligne(1) and M. Rapoport(2)

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(2) The second author considers the political activities of NATO to be incompatible with the peaceful activity of the mathematician and regrets, for this reason, that this congress was partially financed by that organization.

LES SCHÉMAS DE MODULES DE COURBES ELLIPTIQUES

par P. Deligne(1) et M. Rapoport(2)

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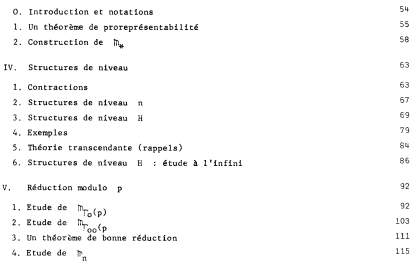

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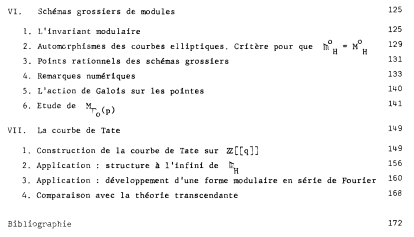

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Introduction

1. Let X be the Poincaré half-plane

$$X = \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\}.$$

The group $SL(2, \mathbb{R})$ acts on X by homographic transformations

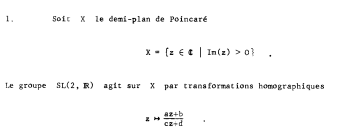
$$z \mapsto \frac{az+b}{cz+d}.$$

If Γ is a subgroup of $SL(2, \mathbb{Z})$ defined by congruence conditions, the Riemann surface X/Γ is the complement of a finite set of points (”at infinity”) in a compact Riemann surface. It is therefore an algebraic curve. Its points are in bijective correspondence with the isomorphism classes of elliptic curves equipped with a ”level structure” of a suitable kind. It is known from this interpretation that it admits as a field of definition a subfield of a cyclotomic field. In this article, we study the structure at infinity and the reduction modulo p of X/Γ.

Let us concentrate on the case in which Γ is the subgroup of SL(2, ℤ) formed by the matrices congruent modulo n to the identity matrix. Suppose that n ≥ 3, in which case Γ acts on X without a fixed point.

For z ∈ X, let E_z denote the quotient of ℂ by the lattice ℤ + ℤz. The addition law on ℂ descends to the quotient and equips E_z with the structure of an elliptic curve (= an abelian variety of dimension one). The elliptic curves E_z and E_z′ are isomorphic if and only if z and z′ are conjugate under SL(2, ℤ). The kernel E(z)_n of multiplication by n on E(z) is the image of (1/n)(ℤ + ℤz); we shall denote by α(z) the isomorphism

E(z)_n ≃ (ℤ/nℤ)^2 : (a+bz)/n ↦ (a,b).


1. Soit X le demi-plan de Poincaré

$$X = \{z \in \mathbb{C} \mid \mathrm{Im}(z) > 0\} .$$

Le groupe SL(2, ℝ) agit sur X par transformations homographiques

$$z \mapsto \frac{az+b}{cz+d} .$$

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For the pairs (E(z),α(z)) and (E(z′),α(z′)) to be isomorphic, it is necessary and sufficient that z and z′ be conjugate under Γ.

For every elliptic curve E over an algebraically closed field k of characteristic p not dividing n, the group E_n is equipped with a nondegenerate alternating form e_n with values in the nth roots of unity in k. A level-n structure on E is an isomorphism α:E_n ≃ (ℤ/n)^2. If α is a level-n structure, we shall denote by ζ(α) the primitive nth root of unity

ζ(α) = e_n(α^−1((1,0)), α^−1((0,1))).

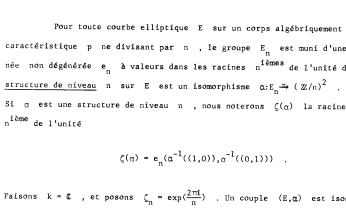
Put k = ℂ, and set ζ_n = exp(2πi/n). A pair (E,α) is isomorphic to a pair (E(z),α(z)) (z ∈ X) if and only if

ζ(α) = ζ_n

(or ζ_n^−1, according to the sign convention used in the definition of e_n).

2. The foregoing generalizes to families of elliptic curves parametrized by a scheme S. One finds that X/Γ represents the following functor F_n on schemes over ℂ: F_n(T) is the set of isomorphism classes of algebraic families, parametrized by T, of elliptic curves E equipped with a level-n structure such that ζ(α) = ζ_n.

The definition of F_n(T) remains meaningful when T is merely assumed to be a scheme over ℤ[ζ_n, 1/n]. Igusa proved in [11] that the resulting functor F_n is representable by a scheme M_n^∘[1/n] over ℤ[ζ_n,1/n]. Igusa's method is as follows.


Pour toute courbe elliptique E sur un corps algébriquement clos k de caractéristique p ne divisant par n, le groupe E_n est muni d'une forme alternée non dégénérée e_n à valeurs dans les racines $n^{\text{ièmes}}$ de l'unité de k. Une structure de niveau n sur E est un isomorphisme $\alpha : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$. Si α est une structure de niveau n, nous noterons ζ(α) la racine primitive $n^{\text{ième}}$ de l'unité

$$\zeta(\alpha) = e_n(\alpha^{-1}((1,0)),\alpha^{-1}((0,1))) .$$

Faisons $k = \mathbb{C}$, et posons $\zeta_n = \exp(\frac{2\pi i}{n})$. Un couple (E,α) est isomorphe à un

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a) If M_n^∘[1/n] exists, then M_nm^∘[1/nm] exists. If E is an elliptic curve over an algebraically closed field, multiplication by m induces an isomorphism of E_nm/nE_nm with E_n; a level-nm structure α:E_nm ≃ (ℤ/nm)^2 thereby defines, by reduction modulo n, a level-n structure. This carries over to an arbitrary base scheme and defines a morphism of functors F_nm → F_n. This morphism is relatively representable. It makes M_nm^∘[1/nm] an unramified Galois covering, with group Ker(SL(2, ℤ/nm) → SL(2, ℤ/n)), of

M_n^∘[1/nm] ⊗_{ℤ[ζ_n]} ℤ[ζ_nm]

(⊗ denotes an extension of scalars).

b) If M_nm^∘[1/nm] exists, M_n^∘[1/nm] exists. One obtains M_n^∘[1/nm] by passing to the quotient.

c) M_3^∘[1/3] and M_4^∘[1/4] exist. Explicitly, one has

M_3^∘[1/3] = Spec(ℤ[ζ_3,1/3,μ,(μ^3−1)^−1]),

with universal elliptic curve the plane cubic with equation

X^3 + Y^3 + Z^3 = 3μ XYZ.

The zero section is (1,−1,0), and the basis of the group of points of order 3 is ((−1,0,1), (−1,ζ_3,0)).

At level 4, one has

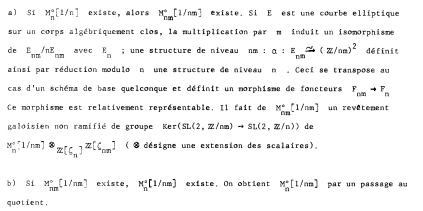
M_4^∘[1/4] = Spec(ℤ[i,1/2,σ,(σ(σ^4−1))^−1]),

where i = ζ_4, with universal elliptic curve the plane cubic whose nonhomogeneous equation is

y^2 = x(x−1)(x−λ), where

λ = (1/4)(σ + 1/σ)^2.

The zero section is the point at infinity, and the basis (r,s) of the group of points


a) Si $M_n^\circ[1/n]$ existe, alors $M_{nm}^\circ[1/nm]$ existe. Si E est une courbe elliptique sur un corps algébriquement clos, la multiplication par m induit un isomorphisme de E_{nm}/nE_{nm} avec E_n ; une structure de niveau nm : $\alpha : E_{nm} \xrightarrow{\sim} (\mathbb{Z}/nm)^2$ définit ainsi par réduction modulo n une structure de niveau n . Ceci se transpose au cas d'un schéma de base quelconque et définit un morphisme de foncteurs $F_{nm} \to F_n$ Ce morphisme est relativement représentable. Il fait de $M_{nm}^\circ[1/nm]$ un revêtement galoisien non ramifié de groupe $\mathrm{Ker}(SL(2,\mathbb{Z}/nm) \to SL(2,\mathbb{Z}/n))$ de $M_n^\circ[1/nm] \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{Z}[\zeta_{nm}]$ (⊗ désigne une extension des scalaires).

b) Si $M_{nm}^\circ[1/nm]$ existe, $M_n^\circ[1/nm]$ existe. On obtient $M_n^\circ[1/nm]$ par un passage au quotient.

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of order 4 is given by

r = (1/2(σ + 1/σ), i(σ²+1)(σ−1)²/(4·σ²))

s = (−i/2(σ − 1/σ) + 1, −(σ²−1)(σ+1)²/(4·σ²)).

The formulas above for M_4^∘[1/4] are copied from Shioda [31]. Igusa used not M_4^∘[1/4], but the moduli scheme of elliptic curves equipped with a structure intermediate between a level-4 structure and a level-2 structure.

d) M_n^∘[1/n] is obtained by gluing M_n^∘[1/3n] and M_n^∘[1/4n].

3. The Riemann surface X/Γ is noncompact. Geometrically, this fact translates as follows: if E_η is an elliptic curve equipped with a level-n structure over $\mathbb{C}((T))$, it may happen that the minimal model of E_η over $\mathbb{C}[[T]]$ has bad reduction. In that case, the special fiber E'_o of the Néron model E' of E_η over $\mathbb{C}[[T]]$ is isomorphic to $\mathbb{C}^*\times\mathbb{Z}/kn$ for a suitable k. Let E_o be the subgroup of E'_o that is the union of the components of E'_o whose order in π_o(E'_o) divides n. This subgroup is isomorphic to $\mathbb{C}^*\times\mathbb{Z}/n$. It has the following advantage over E'_o: after extending scalars from $\mathbb{C}((T))$ to $\mathbb{C}((T^{\wedge}\{1/\ell\}))$, the number of connected components of E'_o is multiplied by ℓ, while E_o remains unchanged.

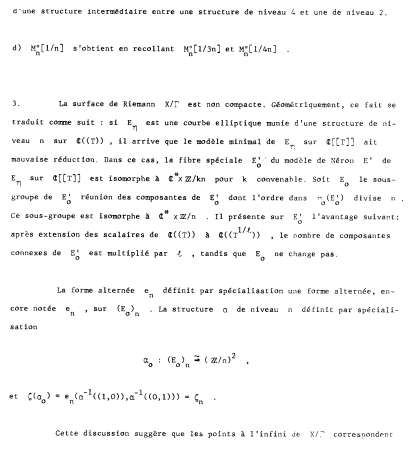
By specialization, the alternating form e_n defines an alternating form, again denoted e_n, on (E_o)_n. By specialization, the level-n structure α defines

α_o : (E_o)_n ≃ $(\mathbb{Z}/n)^2$,

and

ζ(α_o) = e_n(α^−1((1,0)), α^−1((0,1))) = ζ_n.

This discussion suggests that the points at infinity of X/Γ correspond


d'une structure intermédiaire entre une structure de niveau 4 et une de niveau 2.

d) $M_n^*[1/n]$ s'obtient en recollant $M_n^*[1/3n]$ et $M_n^*[1/4n]$.

3. La surface de Riemann X/Γ est non compacte. Géométriquement, ce fait se traduit comme suit : si E_η est une courbe elliptique munie d'une structure de niveau n sur $\mathbb{C}((T))$, il arrive que le modèle minimal de E_η sur $\mathbb{C}[[T]]$ ait mauvaise réduction. Dans ce cas, la fibre spéciale E'_o du modèle de Néron E' de E_η sur $\mathbb{C}[[T]]$ est isomorphe à $\mathbb{C}^*\times\mathbb{Z}/kn$ pour k convenable. Soit E_o le sous-groupe de E'_o réunion des composantes de E'_o dont l'ordre dans $\pi_o(E'_o)$ divise n . Ce sous-groupe est isomorphe à $\mathbb{C}^*\times\mathbb{Z}/n$. Il présente sur E'_o l'avantage suivant: après extension des scalaires de $\mathbb{C}((T))$ à $\mathbb{C}((T^{1/\ell}))$, le nombre de composantes connexes de E'_o est multiplié par ℓ , tandis que E_o ne change pas.

La forme alternée e_n définit par spécialisation une forme alternée, encore notée e_n , sur $(E_o)_n$. La structure α de niveau n définit par spécialisation

$$\alpha_o : (E_o)_n \xrightarrow{\sim} (\mathbb{Z}/n)^2 ,$$

et $\zeta(\alpha_o) = e_n(\alpha^{-1}((1,0)),\alpha^{-1}((0,1))) = \zeta_n$.

Cette discussion suggère que les points à l'infini de X/Γ correspondent

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to the isomorphism classes of systems (E_o,e_n,α_n) as above. This is indeed the case.

In the text, we refine this modular interpretation of the set of points at infinity of X/Γ into a modular interpretation of the projective curve X/Γ, the compactification of X/Γ. Here is the principle.

For E_η over $\mathbb{C}((T))$ as above, the special fiber Ē'_o of the minimal model Ē' of E_η over $\mathbb{C}[[T]]$ is a cycle of kn projective lines:

[cycle of kn projective lines: see authority asset P0008-A01]

Let Ē be obtained from Ē' by contracting to a point the irreducible components of Ē'_o whose order does not divide n. The special fiber Ē_o of Ē is a cycle of n projective lines meeting transversely, and is a compactification of E_o (E_o identifies with the smooth locus Ē_o^reg of Ē_o). By specialization, + and α define a group law on the smooth locus Ē_o^reg of Ē_o, an action of Ē_o^reg on Ē_o, and an isomorphism

α_o : (Ē_o^reg)_n ≃ $(\mathbb{Z}/n)^2$.

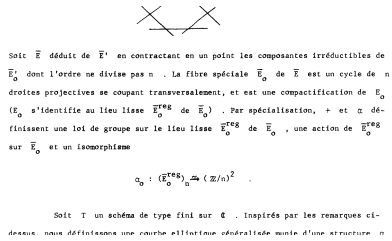
Let T be a scheme of finite type over $\mathbb{C}$. Inspired by the remarks above, we define a

generalized elliptic curve over T, equipped with a level-n structure α such that ζ(α) = ζ_n, as consisting of:

a) a proper flat morphism p : E → T;

b) a commutative group-scheme structure on the open subset E^reg of E where p is smooth; addition is assumed to extend to an action

+ : E^reg ×_T E → E;



Soit $\bar{E}$ déduit de $\bar{E}'$ en contractant en un point les composantes irréductibles de $\bar{E}'_o$ dont l'ordre ne divise pas n. La fibre spéciale $\bar{E}_o$ de $\bar{E}$ est un cycle de n droites projectives se coupant transversalement, et est une compactification de E_o (E_o s'identifie au lieu lisse $\bar{E}_o^{reg}$ de $\bar{E}_o$). Par spécialisation, + et α définissent une loi de groupe sur le lieu lisse $\bar{E}_o^{reg}$ de $\bar{E}_o$, une action de $\bar{E}_o^{reg}$ sur $\bar{E}_o$ et un isomorphisme

$$\alpha_o : (\bar{E}_o^{reg})_n \xrightarrow{\sim} (\mathbb{Z}/n)^2 .$$

Soit T un schéma de type fini sur $\mathbb{C}$. Inspirés par les remarques ci-dessus, nous définissons une courbe elliptique généralisée munie d'une structure n

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c) an isomorphism of group schemes

(E^reg)_n ≃ (ℤ/n)²_T.

The geometric fibers of p are assumed either to be elliptic curves equipped with a level-n structure such that ζ(α) = ζ_n, or to be isomorphic to one of the systems (Ē_o,α_o) constructed above.

We prove that the curve X/Γ represents the following functor F̄_n on schemes of finite type over ℂ. F̄_n(T) is the set of isomorphism classes of generalized elliptic curves E over T, equipped with a level-n structure α such that ζ(α) = ζ_n.

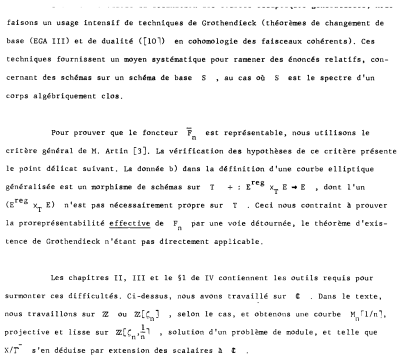
To make the definition of generalized elliptic curves viable, we make intensive use of Grothendieck techniques (base-change theorems (EGA III) and duality ([10]) in coherent-sheaf cohomology). These techniques provide a systematic means of reducing relative statements concerning schemes over a base scheme S to the case in which S is the spectrum of an algebraically closed field.

To prove that the functor F̄_n is representable, we use the general criterion of M. Artin [3]. Verifying the hypotheses of this criterion presents the following delicate point. Datum b) in the definition of a generalized elliptic curve is a morphism of schemes over T

+ : E^reg ×_T E → E,

whose source E^reg ×_T E is not necessarily proper over T. This forces us to prove the effective prorepresentability of F_n by an indirect route, since Grothendieck's existence theorem is not directly applicable.

Chapters II and III and §1 of IV contain the tools required to overcome these difficulties. Above, we worked over ℂ. In the text, we work over ℤ or ℤ[ζ_n], as appropriate, and obtain a curve M_n[1/n], projective and smooth over ℤ[ζ_n,1/n], solving a moduli problem, and such that X/Γ is obtained from it by extension of scalars to ℂ.



faisons un usage intensif de techniques de Grothendieck (théorèmes de changement de base (EGA III) et de dualité ([10]) en cohomologie des faisceaux cohérents). Ces techniques fournissent un moyen systématique pour ramener des énoncés relatifs, concernant des schémas sur un schéma de base S, au cas où S est le spectre d'un corps algébriquement clos.

Pour prouver que le foncteur $\overline{F}_n$ est représentable, nous utilisons le critère général de M. Artin [3]. La vérification des hypothèses de ce critère présente le point délicat suivant. La donnée b) dans la définition d'une courbe elliptique généralisée est un morphisme de schémas sur T $+ : E^{reg} \times_T E \to E$, dont l'un $(E^{reg} \times_T E)$ n'est pas nécessairement propre sur T. Ceci nous contraint à prouver la proreprésentabilité effective de F_n par une voie détournée, le théorème d'existence de Grothendieck n'étant pas directement applicable.

Les chapitres II, III et le §1 de IV contiennent les outils requis pour surmonter ces difficultés. Ci-dessus, nous avons travaillé sur $\mathbb{C}$. Dans le texte, nous travaillons sur $\mathbb{Z}$ ou $\mathbb{Z}[\zeta_n]$, selon le cas, et obtenons une courbe $M_n[1/n]$, projective et lisse sur $\mathbb{Z}[\zeta_n, \frac{1}{n}]$, solution d'un problème de module, et telle que X/Γ s'en déduise par extension des scalaires à $\mathbb{C}$.

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4. To speak of the ”reduction mod p of X/Γ” presupposes the choice of a model M_n of X/Γ over $\mathbb{Z}[\zeta_n]$. The definition of M_n given in the text is equivalent to the following: M_n is the normalization of the j-line (the projective line over $\mathbb{Z}[\zeta_n]$, to which M_n maps by the ”modular invariant of the universal elliptic curve” map) in the function field of M_n^∘[1/n]. Also let M_n^∘ be the affine curve whose coordinate ring is the normalization of $\mathbb{Z}[\zeta_n][j]$ in the coordinate ring of M_n^∘. It is shown in the text that

M_n[1/n] = M_n ⊗_{$\mathbb{Z}[\zeta_n]$} $\mathbb{Z}[\zeta_n,1/n]$ and

M_n^∘[1/n] = M_n^∘ ⊗_{$\mathbb{Z}[\zeta_n]$} $\mathbb{Z}[\zeta_n,1/n]$.

We prove the following results:

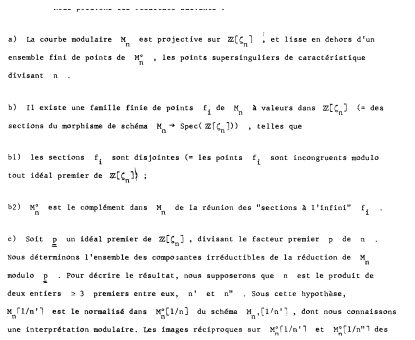
a) The modular curve M_n is projective over $\mathbb{Z}[\zeta_n]$, and smooth away from a finite set of points of M_n^∘, namely the supersingular points whose characteristic divides n.

b) There exists a finite family of points f_i of M_n with values in $\mathbb{Z}[\zeta_n]$ (= sections of the scheme morphism M_n → Spec($\mathbb{Z}[\zeta_n]$)), such that

b1) the sections f_i are disjoint (= the points f_i are incongruent modulo every prime ideal of $\mathbb{Z}[\zeta_n]$);

b2) M_n^∘ is the complement in M_n of the union of the ”sections at infinity” f_i.

c) Let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\zeta_n]$ dividing the prime factor p of n. We determine the set of irreducible components of the reduction of M_n modulo $\mathfrak{p}$. To describe the result, we suppose that n is the product of two relatively prime integers ≥ 3, n’ and n”. Under this hypothesis, M_n[1/n’] is the normalization in M_n^∘[1/n] of the scheme M_{n’}[1/n’], for which we know a modular interpretation. The pullbacks to M_{n’}^∘[1/n’] and M_{n”}^∘[1/n”] of the



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universal elliptic curves over M_{n'}^∘[1/n'] and M_{n''}^∘[1/n''] glue together and furnish a universal elliptic curve E over M_n^∘. The level structure

α : E_n ≃ (ℤ/n)²

over M_n^∘[1/n] extends to a morphism of group schemes over M_n^∘,

α^−1 : (ℤ/n)² → E_n.

The p-primary component of α^−1 is a morphism

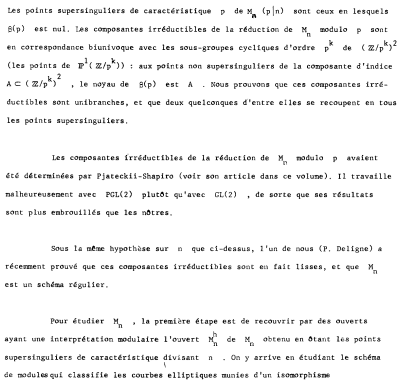
β(p) : (ℤ/p^k)² → E_{p^k}.

The supersingular points of characteristic p of M_𝔭 (p|n) are those at which β(p) is zero. The irreducible components of the reduction of M_n modulo 𝔭 are in bijective correspondence with the cyclic subgroups of order p^k of (ℤ/p^k)² (the points of ℙ¹(ℤ/p^k)): at the nonsupersingular points of the component indexed by A ⊂ (ℤ/p^k)², the kernel of β(p) is A. We prove that these irreducible components are unibranch, and that any two of them meet at every supersingular point.

The irreducible components of the reduction of M_n modulo p had been determined by Pjateckii-Shapiro (see his article in this volume). Unfortunately, he works with PGL(2) rather than GL(2), so that his results are more tangled than ours.

Under the same hypothesis on n as above, one of us (P. Deligne) recently proved that these irreducible components are in fact smooth, and that M_n is a regular scheme.

To study M_n, the first step is to cover by open subsets having a modular interpretation the open subset M_n^h of M_n obtained by removing the supersingular points whose characteristic divides n. This is achieved by studying the moduli scheme classifying elliptic curves equipped with an isomorphism



Les points supersinguliers de caractéristique p de $M_n(p|n)$ sont ceux en lesquels $\beta(p)$ est nul. Les composantes irréductibles de la réduction de M_n modulo p sont en correspondance biunivoque avec les sous-groupes cycliques d'ordre p^k de $(\mathbb{Z}/p^k)^2$ (les points de $\mathbb{P}^1(\mathbb{Z}/p^k)$) : aux points non supersinguliers de la composante d'indice $A \subset (\mathbb{Z}/p^k)^2$, le noyau de $\beta(p)$ est A. Nous prouvons que ces composantes irréductibles sont unibranches, et que deux quelconques d'entre elles se recoupent en tous les points supersinguliers.

Les composantes irréductibles de la réduction de M_n modulo p avaient été déterminées par Pjateckii-Shapiro (voir son article dans ce volume). Il travaille malheureusement avec $PGL(2)$ plutôt qu'avec $GL(2)$, de sorte que ses résultats sont plus embrouillés que les nôtres.

Sous la même hypothèse sur n que ci-dessus, l'un de nous (P. Deligne) a récemment prouvé que ces composantes irréductibles sont en fait lisses, et que M_n est un schéma régulier.

Pour étudier M_n, la première étape est de recouvrir par des ouverts M_n^h ayant une interprétation modulaire l'ouvert M_n^h de M_n obtenu en ôtant les points supersinguliers de caractéristique divisant n. On y arrive en étudiant le schéma de modules M_n^t qui classifie les courbes elliptiques munies d'un isomorphisme

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α : E_n ≃ μ_n × ℤ/n.

A modular interpretation makes it possible

a) to determine the set of points of M_n over an algebraically closed field;

b) to prove the smoothness of M_n^h over ℤ[ζ_n]. It remains, and this is the most delicate point, to study the supersingular points.

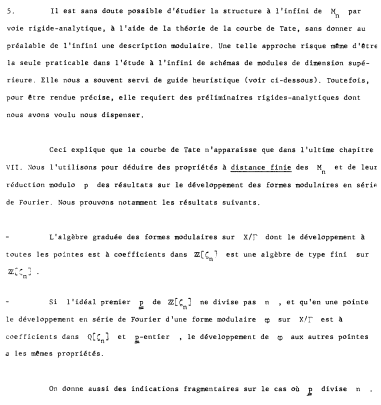
5. It is no doubt possible to study the structure of M_n at infinity by rigid- analytic methods, using the theory of the Tate curve, without first giving the points at infinity a modular description. Such an approach may even be the only practicable one for studying the boundary of higher-dimensional moduli schemes. It has often served us as a heuristic guide (see below). However, to make it precise, it requires rigid-analytic preliminaries that we wished to avoid.

This explains why the Tate curve appears only in the final chapter, Chapter VII. We use it to deduce properties at finite distance of the M_n, and of their reduction modulo p, from results on the Fourier-series expansions of modular forms. In particular, we prove the following results.

− The graded algebra of modular forms on X/Γ whose expansions at all cusps have coefficients in ℤ[ζ_n] is a finitely generated algebra over ℤ[ζ_n].

− If the prime ideal 𝔭 of ℤ[ζ_n] does not divide n, and if at one cusp the Fourier-series expansion of a modular form φ on X/Γ has coefficients in ℚ[ζ_n] and is 𝔭-integral, then the expansion of φ at the other cusps has the same properties.

Fragmentary indications are also given for the case where 𝔭 divides n.



5. Il est sans doute possible d'étudier la structure à l'infini de M_n par voie rigide-analytique, à l'aide de la théorie de la courbe de Tate, sans donner au préalable de l'infini une description modulaire. Une telle approche risque même d'être la seule praticable dans l'étude à l'infini de schémas de modules de dimension supérieure. Elle nous a souvent servi de guide heuristique (voir ci-dessous). Toutefois, pour être rendue précise, elle requiert des préliminaires rigides-analytiques dont nous avons voulu nous dispenser.

Ceci explique que la courbe de Tate n'apparaisse que dans l'ultime chapitre VII. Nous l'utilisons pour déduire des propriétés à distance finie des M_n et de leur réduction modulo p des résultats sur le développement des formes modulaires en série de Fourier. Nous prouvons notamment les résultats suivants.

- L'algèbre graduée des formes modulaires sur X/Γ dont le développement à toutes les pointes est à coefficients dans $\mathbb{Z}[\zeta_n]$ est une algèbre de type fini sur $\mathbb{Z}[\zeta_n]$.

- Si l'idéal premier $\mathfrak{p}$ de $\mathbb{Z}[\zeta_n]$ ne divise pas n, et qu'en une pointe le développement en série de Fourier d'une forme modulaire φ sur X/Γ est à coefficients dans $\mathbb{Q}[\zeta_n]$ et $\mathfrak{p}$-entier, le développement de φ aux autres pointes a les mêmes propriétés.

On donne aussi des indications fragmentaires sur le cas où $\mathfrak{p}$ divise n.

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Authority page 13; collected-paper page 506; original folio 155.

6. Our heuristic use of the Tate curve rests on the following principles.

a) With the aid of the Tate curve, one can guess or prove what happens in a neighborhood of infinity.

b) Outside the supersingular points, the situation is everywhere the same, and in particular the same as in a neighborhood of infinity.

To these principles is added the following one.

c) Geometrically, i.e. after extension of scalars (or of the residue field) to $\mathbb{F}_p$, the situation is the same at the various supersingular points of characteristic p.

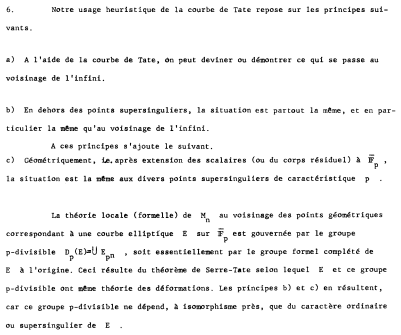
The local (formal) theory of M_n in a neighborhood of the geometric points corresponding to an elliptic curve E over $\mathbb{F}_p$ is governed by the p-divisible group

$$D_p(E) = \bigcup E_{p^n},$$

that is, essentially, by the completed formal group of E at the origin. This follows from the Serre–Tate theorem, according to which E and this p-divisible group have the same deformation theory. Principles b) and c) follow from it, since this p-divisible group depends, up to isomorphism, only on whether E is ordinary or supersingular.

7. Let H be an indefinite quaternion algebra, of discriminant D, over $\mathbb{Q}$, and let Δ be a congruence subgroup of the group of units of H. The group Δ is a discrete subgroup of $SL(2,\mathbb{R})$ with compact quotient. The complete curve X/Δ parametrizes abelian varieties A of dimension 2, equipped with a homomorphism from an order of H into End(A) ("false elliptic curves," in Serre's terminology). Our methods apply to the study of X/Δ and of its reduction modulo p, provided that p is prime to D.

The following remarks even make it possible to deduce, from theorems for M_n, theorems on the local structure of X/Δ.



6. Notre usage heuristique de la courbe de Tate repose sur les principes suivants.

a) A l'aide de la courbe de Tate, on peut deviner ou démontrer ce qui se passe au voisinage de l'infini.

b) En dehors des points supersinguliers, la situation est partout la même, et en particulier la même qu'au voisinage de l'infini.

A ces principes s'ajoute le suivant.

c) Géométriquement, i.e. après extension des scalaires (ou du corps résiduel) à $\bar{F}_p$, la situation est la même aux divers points supersinguliers de caractéristique p.

La théorie locale (formelle) de M_n au voisinage des points géométriques correspondant à une courbe elliptique E sur $\bar{F}_p$ est gouvernée par le groupe p-divisible $D_p(E)=\bigcup E_{p^n}$, soit essentiellement par le groupe formel complété de E à l'origine. Ceci résulte du théorème de Serre-Tate selon lequel E et ce groupe p-divisible ont même théorie des déformations. Les principes b) et c) en résultent, car ce groupe p-divisible ne dépend, à isomorphisme près, que du caractère ordinaire ou supersingulier de E.

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d) Let $\mathcal{O}$ be an order of H, with $\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Z}_p \simeq GL(2,\mathbb{Z}_p)$, and let e be a nontrivial idempotent of $\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Z}_p$. If A is an abelian scheme of dimension 2 over an algebraically closed field k of characteristic p, equipped with $\mu : \mathcal{O} \to \mathrm{End}(A)$, the completion $\mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Z}_p$ acts on the p-divisible group $D_p(A) = \bigcup A_{p^n}$, and

$D_p A) = e \cdot D_p(A) \oplus (1-e) D_p(A)$.

The abelian scheme A, equipped with μ, and the p-divisible group $eD_p(A)$, have the same deformation theory.

e) the p-divisible group $eD_p(A)$ is isomorphic to $D_p(E)$ for an elliptic curve E that is either ordinary or supersingular.

The reduction modulo p of X/Δ was studied by Morita. He also obtains results in the much more difficult case where $p|D$.

8. In the literature on moduli schemes, the language of Weil's Foundations is sometimes used. According as one uses this language or Grothendieck's, emphasis is placed on one of the following two views of what a scheme X over a number field K is.

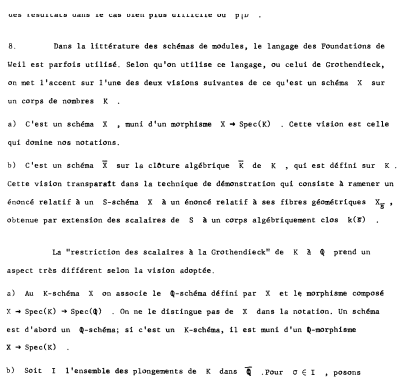
a) It is a scheme X, equipped with a morphism $X \to \mathrm{Spec}(K)$. This is the view that predominates in our notation.

b) It is a scheme $\bar{X}$ over the algebraic closure $\bar{K}$ of K, which is defined over K. This view shows through in the proof technique that consists in reducing a statement concerning an S-scheme X to a statement concerning its geometric fibers $X_{\bar{s}}$, obtained by extension of scalars from S to an algebraically closed field $k(\bar{s})$.

The "restriction of scalars à la Grothendieck" from K to $\mathbb{Q}$ takes on a very different appearance according to the view adopted.

a) To the K-scheme X one associates the $\mathbb{Q}$-scheme defined by X and the composite morphism $X \to \mathrm{Spec}(K) \to \mathrm{Spec}(\mathbb{Q})$. It is not distinguished from X in the notation. A scheme is first of all a $\mathbb{Q}$-scheme; if it is a K-scheme, it is equipped with a $\mathbb{Q}$-morphism $X \to \mathrm{Spec}(K)$.

b) Let I be the set of embeddings of K into $\bar{\mathbb{Q}}$. For $\sigma \in I$, set



des résultats dans le cas bien plus difficile où p|D .

8. Dans la littérature des schémas de modules, le langage des Foundations de Weil est parfois utilisé. Selon qu'on utilise ce langage, ou celui de Grothendieck, on met l'accent sur l'une des deux visions suivantes de ce qu'est un schéma X sur un corps de nombres K .

a) C'est un schéma X , muni d'un morphisme X → Spec(K) . Cette vision est celle qui domine nos notations.

b) C'est un schéma $\overline{X}$ sur la clôture algébrique $\overline{K}$ de K , qui est défini sur K . Cette vision transparaît dans la technique de démonstration qui consiste à ramener un énoncé relatif à un S-schéma X à un énoncé relatif à ses fibres géométriques $X_{\overline{s}}$, obtenue par extension des scalaires de S à un corps algébriquement clos k(s̄) .

La "restriction des scalaires à la Grothendieck" de K à ℚ prend un aspect très différent selon la vision adoptée.

a) Au K-schéma X on associe le ℚ-schéma défini par X et le morphisme composé X → Spec(K) → Spec(ℚ) . On ne le distingue pas de X dans la notation. Un schéma est d'abord un ℚ-schéma; si c'est un K-schéma, il est muni d'un ℚ-morphisme X → Spec(K) .

b) Soit I l'ensemble des plongements de K dans $\overline{\mathbb{Q}}$ .Pour σ ∈ I , posons

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$\bar{X}_\sigma = X \otimes_{K,\sigma} \bar{\mathbb{Q}}$. We have $X \otimes_{\mathbb{Q}} \bar{\mathbb{Q}} = \sum_{\sigma\in I} \bar{X}_\sigma$. The ℚ-scheme deduced, as in a), from a K-scheme appears as the disjoint sum of the conjugates $\bar{X}_\sigma$ of $\bar{X}$, this sum being moreover defined over ℚ. The ℚ-morphism X → Spec(K) is interpreted as the map from this sum to the index set I = Spec(K) $\otimes_{\mathbb{Q}} \bar{\mathbb{Q}}$.

For $M_n^\circ[1/n]$, the situation is as follows: the scheme $M_n^\circ[1/n]$ represents the functor of isomorphism classes of elliptic curves E/T, equipped with an isomorphism

$\alpha : E_n \simeq (\mathbb{Z}/n)^2$.

Such a pair (E,α) defines (by the theory of the alternating form e_n) a primitive n-th root of unity ζ(α) over T, hence a morphism of schemes

$(E,\alpha) \mapsto \zeta(\alpha) : M_n^\circ[1/n] \to \mathrm{Spec}(\mathbb{Z}[\zeta_n,1/n])$.

Viewed as a $\mathbb{Z}[\zeta_n,1/n]$-scheme, $M_n^\circ[1/n]$ classifies the (E,α)/T, with T a $\mathbb{Z}[\zeta_n,1/n]$-scheme, such that ζ(α) = ζ_n.

We have

$M_n^\circ[1/n] \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C} = X/\Gamma$;

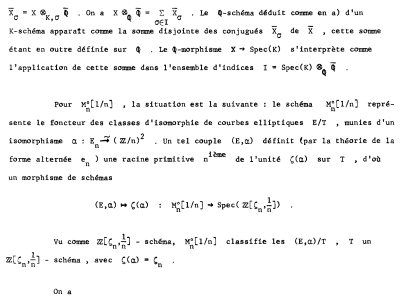
whereas, if one puts $X^\pm = \mathbb{C} - \mathbb{R}$,

$M_n^\circ[1/n] \otimes_{\mathbb{Z}} \mathbb{C} = X^\pm \times GL(2,\mathbb{Z}/n) / GL(2,\mathbb{Z})$

is the sum of φ(n) copies of X/Γ.

9. An elliptic curve over an algebraically closed field k has one or more nontrivial automorphisms. This rules out representability of the functor that assigns to a scheme T the set of isomorphism classes of elliptic curves E over T: it is not even a sheaf. This can be remedied in two ways.

a) Systematically impose additional level structures that eliminate these automorphisms. This is what we have done in this introduction.



$\overline{X}_\sigma = X \otimes_{K,\sigma} \overline{\mathbb{Q}}$. On a $X \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} = \coprod_{\sigma \in I} \overline{X}_\sigma$. Le $\mathbb{Q}$-schéma déduit comme en a) d'un K-schéma apparaît comme la somme disjointe des conjugués $\overline{X}_\sigma$ de $\overline{X}$, cette somme étant en outre définie sur $\mathbb{Q}$. Le $\mathbb{Q}$-morphisme $X \to \mathrm{Spec}(K)$ s'interprète comme l'application de cette somme dans l'ensemble d'indices $I = \mathrm{Spec}(K) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$.

Pour $M_n^\alpha[1/n]$, la situation est la suivante : le schéma $M_n^\alpha[1/n]$ représente le foncteur des classes d'isomorphie de courbes elliptiques E/T , munies d'un isomorphisme $\alpha : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$. Un tel couple (E,α) définit (par la théorie de la forme alternée e_n) une racine primitive $n^{\text{ième}}$ de l'unité $\zeta(\alpha)$ sur T , d'où un morphisme de schémas

$$(E,\alpha) \mapsto \zeta(\alpha) \quad : \quad M_n^\alpha[1/n] \to \mathrm{Spec}\left(\mathbb{Z}\left[\zeta_n, \tfrac{1}{n}\right]\right) \quad .$$

Vu comme $\mathbb{Z}[\zeta_n, \frac{1}{n}]$ - schéma, $M_n^\alpha[1/n]$ classifie les $(E,\alpha)/T$, T un $\mathbb{Z}[\zeta_n, \frac{1}{n}]$ - schéma , avec $\zeta(\alpha) = \zeta_n$.

On a

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b) Use the language of algebraic stacks, for which we refer to [8] and [17]. This is what we do in the text, for the following reasons.

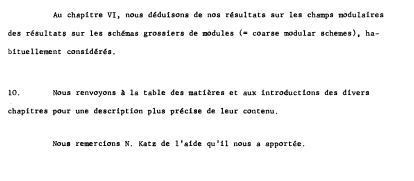
1. Imposing an additional level structure modifies the behavior at infinity of the objects studied (M_{ñm} ramifies over M_n along infinity).

2. The calculations and proofs become simpler when one does not have to concern oneself with troublesome additional structures.

In Chapter VI, from our results on modular stacks we deduce results on the coarse moduli schemes (= coarse modular schemes) usually considered.

10. We refer to the table of contents and to the introductions of the various chapters for a more precise description of their contents.

We thank N. Katz for the help he has given us.



Au chapitre VI, nous déduisons de nos résultats sur les champs modulaires des résultats sur les schémas grossiers de modules (= coarse modular schemes), habituellement considérés.

10. Nous renvoyons à la table des matières et aux introductions des divers chapitres pour une description plus précise de leur contenu.

Nous remercions N. Katz de l'aide qu'il nous a apportée.

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Notations

The work is divided into chapters; each chapter is divided into sections. Internal references within a single chapter do not mention the number of that chapter.

We shall freely use the definitions and notation of the theory of algebraic stacks as set out in [8] §4. However, contrary to the terminology of loc. cit., a morphism of categories fibered in groupoids

$\varphi : \mathfrak{M}_1 \to \mathfrak{M}_2$

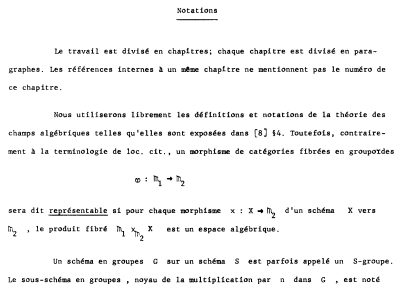
will be called representable if, for every morphism $x : X \to \mathfrak{M}_2$ from a scheme X to $\mathfrak{M}_2$, the fiber product $\mathfrak{M}_1 \times_{\{\mathfrak{M}_2\}} X$ is an algebraic space.

A group scheme G over a scheme S is sometimes called an S-group. The subgroup scheme that is the kernel of multiplication by n in G is denoted by G_n.

Let X be an S-scheme. If T is an S-scheme, we denote by X_T the fiber product X ×_S T. If S = Spec(A) and T = Spec(B), X_B, or X ⊗_A B, have the same meaning. For n ∈ ℕ, we set X[1/n] = X ⊗_ℤ ℤ[1/n].

We set ζ_n = exp(2πi/n).

For the principles governing our notation for modular stacks, we refer to §0 of Chapter III.



Notations

Le travail est divisé en chapitres; chaque chapitre est divisé en paragraphes. Les références internes à un même chapitre ne mentionnent pas le numéro de ce chapitre.

Nous utiliserons librement les définitions et notations de la théorie des champs algébriques telles qu'elles sont exposées dans [8] §4. Toutefois, contrairement à la terminologie de loc. cit., un morphisme de catégories fibrées en groupoïdes

$$\varpi : \mathcal{H}_1 \to \mathcal{H}_2$$

sera dit représentable si pour chaque morphisme $x : X \to \mathcal{H}_2$ d'un schéma X vers $\mathcal{H}_2$, le produit fibré $\mathcal{H}_1 \times_{\mathcal{H}_2} X$ est un espace algébrique.

Un schéma en groupes G sur un schéma S est parfois appelé un S-groupe. Le sous-schéma en groupes $\hat{}$, noyau de la multiplication par n dans G, est noté

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I Preliminaries

This chapter gathers well-known results of which we shall have occasion to make use. The reader is invited to consult it only as needed.

1. Curve schemes.

1.0. In this section, a curve scheme over a scheme S means a proper and flat morphism of finite presentation, of relative dimension at most 1.

1.1. If C/S is a curve scheme over S, the Euler–Poincaré characteristic $\chi(\mathcal{O}_{C_s})$ of the fibers is locally constant (EGA III, 7.9.). The arithmetic genus g_s of C_s is defined by

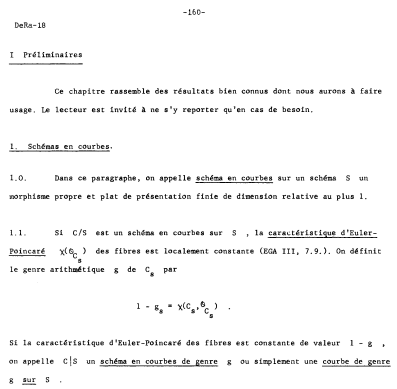
$1 - g_s = \chi(C_s, \mathcal{O}_{C_s})$.

If the Euler–Poincaré characteristic of the fibers is constant with value $1 - g$, C|S is called a curve scheme of genus g, or simply a curve of genus g over S.

1.2. A morphism $p : X \to S$ is said to be Cohen–Macaulay if it is flat and of finite presentation and its (geometric) fibers are Cohen–Macaulay schemes.

A Cohen–Macaulay curve scheme over S is a curve scheme $p : C \to S$ that is Cohen–Macaulay and whose geometric fibers are purely of dimension one. Likewise, when we speak of smooth, reduced, or integral curves, ”purely of dimension one” will be understood.

For S = Spec(A), we shall speak interchangeably of curves over S and curves over A.



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I Préliminaires

Ce chapitre rassemble des résultats bien connus dont nous aurons à faire usage. Le lecteur est invité à ne s'y reporter qu'en cas de besoin.

1. Schémas en courbes.

1.0. Dans ce paragraphe, on appelle schéma en courbes sur un schéma S un morphisme propre et plat de présentation finie de dimension relative au plus 1.

1.1. Si C/S est un schéma en courbes sur S, la caractéristique d'Euler-Poincaré $\chi(\mathcal{O}_{C_s})$ des fibres est localement constante (EGA III, 7.9.). On définit le genre arithmétique g de C_s par

$$1 - g_s = \chi(C_s, \mathcal{O}_{C_s}) .$$

Si la caractéristique d'Euler-Poincaré des fibres est constante de valeur $1 - g$, on appelle $C|S$ un schéma en courbes de genre g ou simplement une courbe de genre g sur S.

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2. Duality in the cohomology of coherent sheaves.

2.1. We propose to recall what Grothendieck duality theory provides for a morphism p : X → S when

a) p is Cohen–Macaulay, purely of relative dimension d; and

b) S is Noetherian.

In the applications we have in view, we could always reduce to the case in which p is moreover quasi-projective.

The relative dualizing complex Rp^!$\mathcal{O}$_S [10] has only one nonzero cohomology sheaf, that of degree −d; it is the sheaf of regular differentials

(2.1.1.) ω_{X/S} = $\mathcal{H}$^{−d}(Rp^!$\mathcal{O}$_S)

([10] V. 9.7.). This sheaf is flat over S. Its formation commutes with every base change S’ → S and with étale localization on X (see [10] or [34]).

2.2. If p is moreover proper, one defines a ”trace” morphism

(2.2.1) Tr : R^d p_*(ω_{X/S}) → $\mathcal{O}$_S

whose formation is compatible with every base change. The duality theorem asserts that, for every bounded complex of coherent sheaves K, the arrow deduced from Tr,

(2.2.2.) Rp_* R Hom_{$\mathcal{O}$_X}(K,ω_{X/S})[−d] ≃ R Hom_{$\mathcal{O}$_S}(Rp_*K,$\mathcal{O}$_S),

is an isomorphism.

If K is reduced to a locally free sheaf F placed in degree 0 and the sheaves R^i p_*(F) are locally free, (2.2.2.) becomes

(2.1.1.) $\omega_{X/S} = \underline{H}^{-d}(Rp^! \mathcal{O}_S)$

([10] V. 9.7.). Ce faisceau est plat sur S. Sa formation commute à tout changement de base $S' \to S$ et à la localisation étale sur X (voir [10] ou [34]).

2.2. Si p est de plus propre, on définit un morphisme "trace"

(2.2.1) $\mathrm{Tr} : R^d p_*(\omega_{X/S}) \to \mathcal{O}_S$

de formation compatible à tout changement de base. Le théorème de dualité affirme que, pour tout complexe borné de faisceaux cohérents K, la flèche déduite de Tr

(2.2.2.) $Rp_* R \underset{X}{\mathcal{H}om}(K, \omega_{X/S})[-d] \to R \underset{\mathcal{O}_S}{\mathcal{H}om}(Rp_* K, \mathcal{O}_S)$

est un isomorphisme.

Si K est réduit à un faisceau localement libre F placé en degré 0 et que les faisceaux $R^i p_*(F)$ sont localement libres, (2.2.2.) devient

(2.2.3.) R^{d−j}p_*(F^∨ ⊗ ω_{X/S}) ≃ R^j p_*(F)^∨

(where ∨ denotes the dual of an $\mathcal{O}$_X-module, respectively of an $\mathcal{O}$_S-module).

ω_{X/S} is an invertible sheaf if and only if p is Gorenstein [10].

For S not necessarily Noetherian, one defines ω_{X/S} and Tr by passage to the limit; for K a bounded complex of locally free sheaves, (2.2.2.) remains valid and is proved by passage to the limit.

When S is the spectrum of a field k, ω_{X/S} will also be denoted ω_{X/k}, or simply ω_X.

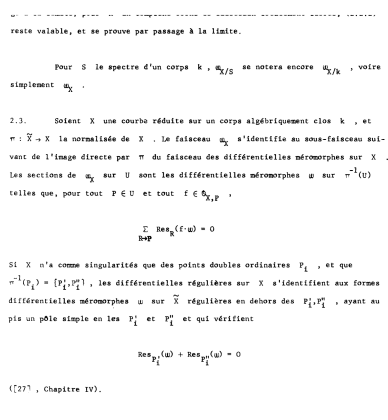
2.3. Let X be a reduced curve over an algebraically closed field k, and let π : $\tilde{X}$ → X be the normalization of X. The sheaf ω_X is identified with the following subsheaf of the direct image under π of the sheaf of meromorphic differentials on X. The sections of ω_X over U are the meromorphic differentials ω on π^{−1}(U) such that, for every P ∈ U and every f ∈ $\mathcal{O}$_{X,P},

∑_{R↦P} Res_R(f·ω) = 0.

If the only singularities of X are ordinary double points P_i, and π^{−1}(P_i) = {P'_i,P"_i}, the regular differentials on X are identified with the meromorphic differential forms ω on $\tilde{X}$ that are regular away from P'_i,P"_i, have at worst a simple pole at P'_i and P"_i, and satisfy

Res_{P'_i}(ω) + Res_{P"_i}(ω) = 0

([27], Chapter IV).


reste valable, et se prouve par passage à la limite.

Pour S le spectre d'un corps k, $\omega_{X/S}$ se notera encore $\omega_{X/k}$, voire simplement ω_X.

2.3. Soient X une courbe réduite sur un corps algébriquement clos k, et $\pi : \tilde{X} \to X$ la normalisée de X. Le faisceau ω_X s'identifie au sous-faisceau suivant de l'image directe par π du faisceau des différentielles méromorphes sur $\tilde{X}$. Les sections de ω_X sur U sont les différentielles méromorphes ω sur $\pi^{-1}(U)$ telles que, pour tout P ∈ U et tout $f \in \mathcal{O}_{X,P}$,

$$\sum_{R \mapsto P} \mathrm{Res}_R(f\cdot\omega) = 0$$

Si X n'a comme singularités que des points doubles ordinaires P_i, et que $\pi^{-1}(P_i) = \{P_i', P_i''\}$, les différentielles régulières sur X s'identifient aux formes différentielles méromorphes ω sur $\tilde{X}$ régulières en dehors des P_i', P_i'', ayant au pis un pôle simple en les P_i' et P_i'' et qui vérifient

$$\mathrm{Res}_{P_i'}(\omega) + \mathrm{Res}_{P_i''}(\omega) = 0$$

([27], Chapitre IV).

3. Review of the Picard functor.

3.1. Let p : X → S be a proper flat morphism of finite presentation. The relative Picard functor of X over S, denoted Pic_{X/S}, is the fppf sheaf associated with the presheaf

(3.1.1) S' ↦ Pic(X ×_S S') = the group of isomorphism classes of invertible sheaves on X ×_S S'.

The canonical morphism Pic(X)/Pic(S) → Pic_{X/S}(S) is injective if p_*$\mathcal{O}$_X = $\mathcal{O}$_S. It is bijective if, in addition, p has a section (TDTE V.2.4.).

3.2. One says that p is cohomologically flat in dimension 0 (EGA III, 7.), or simply cohomologically flat, if the formation of p_*$\mathcal{O}$_X commutes with every base change S' → S. A fundamental theorem of M. Artin asserts that, if p is proper, flat of finite presentation, and cohomologically flat, then Pic_{X/S} is representable by an algebraic space locally of finite presentation over S ([2] Thm. 7.3). If, in addition, p : X → S is a curve scheme, Pic_{X/S} is smooth over S.

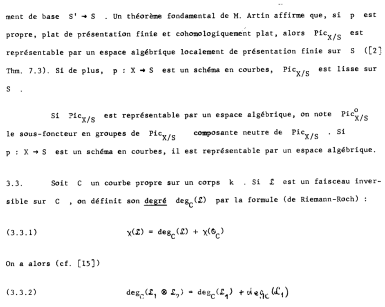
If Pic_{X/S} is representable by an algebraic space, Pic^o_{X/S} denotes the subfunctor in groups of Pic_{X/S} that is the neutral component of Pic_{X/S}. If p : X → S is a curve scheme, it is representable by an algebraic space.

3.3. Let C be a proper curve over a field k. If $\mathcal{L}$ is an invertible sheaf on C, its degree deg_C($\mathcal{L}$) is defined by the Riemann–Roch formula

(3.3.1) χ($\mathcal{L}$) = deg_C($\mathcal{L}$) + χ($\mathcal{O}$_C).

Then (cf. [15])

(3.3.2) deg_C($\mathcal{L}$_1 ⊗ $\mathcal{L}$_2) = deg_C($\mathcal{L}$_1) + deg_C($\mathcal{L}$_1).


ment de base $S' \to S$. Un théorème fondamental de M. Artin affirme que, si p est propre, plat de présentation finie et cohomologiquement plat, alors $\mathrm{Pic}_{X/S}$ est représentable par un espace algébrique localement de présentation finie sur S ([2] Thm. 7.3). Si de plus, $p : X \to S$ est un schéma en courbes, $\mathrm{Pic}_{X/S}$ est lisse sur S.

Si $\mathrm{Pic}_{X/S}$ est représentable par un espace algébrique, on note $\mathrm{Pic}^{0}_{X/S}$ le sous-foncteur en groupes de $\mathrm{Pic}_{X/S}$ composante neutre de $\mathrm{Pic}_{X/S}$. Si $p : X \to S$ est un schéma en courbes, il est représentable par un espace algébrique.

3.3. Soit C un courbe propre sur un corps k. Si $\mathcal{L}$ est un faisceau inversible sur C, on définit son degré $\deg_C(\mathcal{L})$ par la formule (de Riemann-Roch) :

(3.3.1) $\chi(\mathcal{L}) = \deg_C(\mathcal{L}) + \chi(\mathcal{O}_C)$

On a alors (cf. [15])

(3.3.2) $\deg_C(\mathcal{L}_1 \otimes \mathcal{L}_2) = \deg_C(\mathcal{L}_1) + \mathrm{rk}_{\mathcal{O}_C}(\mathcal{L}_1)$

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Authority page 22; collected-paper page 515; original folio 164.

(3.3.3) If the reduced irreducible components of C are the subschemes D_i, and if D_i has multiplicity n_i (= the length of the local ring of C at the generic point of D_i), then

deg_C($\mathcal{L}$) = ∑ n_i deg_{D_i}($\mathcal{L}$), where one has set

(3.3.4) deg_{D_i}($\mathcal{L}$) = deg_{D_i}($\mathcal{L}$ ⊗ $\mathcal{O}$_{D_i}).

3.4. Let p : C → S be a curve over S and let $\mathcal{L}$ be an invertible sheaf on C. The degree of $\mathcal{L}$ on the fibers C_s (s ∈ S) of C is locally constant and defines a morphism of abelian sheaves (for the fppf topology):

(3.4.1) deg : Pic_{C/S} → $\mathbb{Z}$.

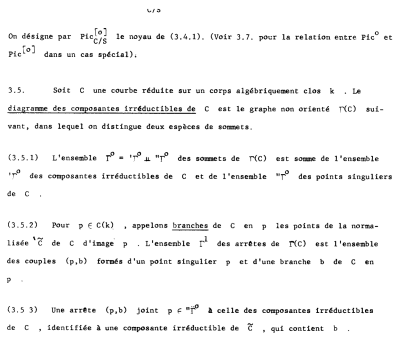
The kernel of (3.4.1) is denoted Pic^{[o]}_{C/S}. (See 3.7 for the relation between Pic^o and Pic^{[o]} in a special case.)

3.5. Let C be a reduced curve over an algebraically closed field k. The diagram of the irreducible components of C is the following unoriented graph Γ(C), in which two kinds of vertices are distinguished.

(3.5.1) The set Γ^o = 'Γ^o ⊔ "Γ^o of vertices of Γ(C) is the sum of the set 'Γ^o of irreducible components of C and the set "Γ^o of singular points of C.

(3.5.2) For p ∈ C(k), call the points of the normalization C̃ of C mapping to p the branches of C at p. The set Γ^1 of edges of Γ(C) is the set of pairs (p,b) formed by a singular point p and a branch b of C at p.

(3.5 3) An edge (p,b) joins p ∈ ”Γ^o to the irreducible component of C, identified with an irreducible component of C̃, that contains b.



On désigne par $\mathrm{Pic}^{[o]}_{C/S}$ le noyau de (3.4.1). (Voir 3.7. pour la relation entre Pic° et $\mathrm{Pic}^{[o]}$ dans un cas spécial).

3.5. Soit C une courbe réduite sur un corps algébriquement clos k. Le diagramme des composantes irréductibles de C est le graphe non orienté Γ(C) suivant, dans lequel on distingue deux espèces de sommets.

(3.5.1) L'ensemble $\Gamma^0 = {}'\Gamma^0 \amalg {}''\Gamma^0$ des sommets de Γ(C) est somme de l'ensemble ${}'\Gamma^0$ des composantes irréductibles de C et de l'ensemble ${}''\Gamma^0$ des points singuliers de C.

(3.5.2) Pour $p \in C(k)$, appelons branches de C en p les points de la normalisée ${}^{\nu}C$ de C d'image p. L'ensemble Γ^1 des arrêtes de Γ(C) est l'ensemble des couples (p,b) formés d'un point singulier p et d'une branche b de C en p.

(3.5 3) Une arrête (p,b) joint $p \in {}''\Gamma^0$ à celle des composantes irréductibles de C, identifiée à une composante irréductible de ${}^{\nu}C$, qui contient b.

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Authority page 23; collected-paper page 516; original folio 165.

Examples 3.5.4. Denote the points of ’Γ^o by ○ and those of ”Γ^o by ⊗. The following curves have the following diagrams:

curve diagram

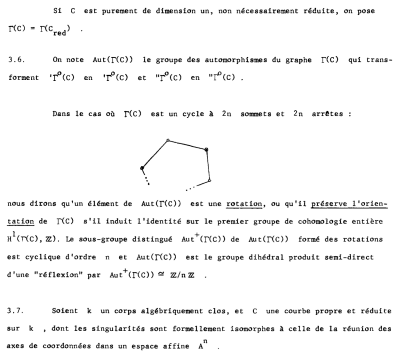
If C is purely of dimension one, but not necessarily reduced, set Γ(C) = Γ(C_red).

3.6. Let Aut(Γ(C)) denote the group of automorphisms of the graph Γ(C) that carry ’Γ^o(C) into ’Γ^o(C) and ”Γ^o(C) into ”Γ^o(C).

In the case where Γ(C) is a cycle with 2n vertices and 2n edges:

we shall say that an element of Aut(Γ(C)) is a rotation, or that it preserves the orientation of Γ(C), if it induces the identity on the first integral cohomology group H^1(Γ(C),ℤ). The distinguished subgroup Aut^+(Γ(C)) of Aut(Γ(C)) formed by the rotations is cyclic of order n, and Aut(Γ(C)) is the dihedral group that is the semidirect product of a ”reflection” by Aut^+(Γ(C)) ≃ ℤ/nℤ.

3.7. Let k be an algebraically closed field, and let C be a proper reduced curve over k whose singularities are formally isomorphic to that of the union of the coordinate axes in an affine space 𝔸^n.



Si C est purement de dimension un, non nécessairement réduite, on pose $\Gamma(C) = \Gamma(C_{red})$.

3.6. On note Aut(Γ(C)) le groupe des automorphismes du graphe Γ(C) qui transforment ${}'\Gamma^0(C)$ en ${}'\Gamma^0(C)$ et ${}''\Gamma^0(C)$ en ${}''\Gamma^0(C)$.

Dans le cas où Γ(C) est un cycle à 2n sommets et 2n arrêtes :

nous dirons qu'un élément de Aut(Γ(C)) est une rotation, ou qu'il préserve l'orientation de Γ(C) s'il induit l'identité sur le premier groupe de cohomologie entière $H^1(\Gamma(C), \mathbb{Z})$. Le sous-groupe distingué $\mathrm{Aut}^+(\Gamma(C))$ de Aut(Γ(C)) formé des rotations est cyclique d'ordre n et Aut(Γ(C)) est le groupe dihédral produit semi-direct d'une "réflexion" par $\mathrm{Aut}^+(\Gamma(C)) \cong \mathbb{Z}/n\mathbb{Z}$.

3.7. Soient k un corps algébriquement clos, et C une courbe propre et réduite sur k, dont les singularités sont formellement isomorphes à celle de la réunion des axes de coordonnées dans un espace affine A^n.

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The Picard scheme Pic_{C/k} decomposes as follows. Let (D_i)_{i∈I} be the irreducible components of C, let $\tilde{D}$_i be the normalization of D_i, and let $\tilde{C} = \amalg \tilde{D}$_i be the normalization of C.

a) The ”total degree” morphism

deg_I = (deg_{D_i})_{i∈I} : Pic_{C/k} → $\mathbb{Z}$^I

is surjective, with kernel Pic^o_{C/k}.

b) The morphism

Pic^o_{C/k} → Pic^o_{$\tilde{C}$/k} = ∏ Pic^o_{$\tilde{D}$_i/k}

is surjective, with image the largest quotient of Pic^o_{C/k} that is an abelian scheme. Its kernel is a torus, canonically isomorphic to

H^1(Γ(C),$\mathbb{Z}$) ⊗ $\mathbb{G}$_m.

If D is proper and purely of dimension one over k, there exists a unique curve C as above, equipped with a radicial morphism C → D; the map

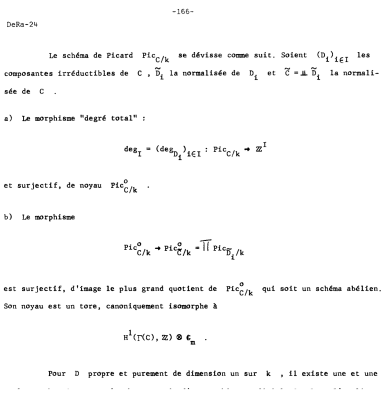
Pic_{D/k} → Pic_{C/k}

is surjective with connected unipotent kernel.

4. Non-smoothness subscheme.

Let f : X → S be a flat morphism of finite presentation, purely of relative dimension d.

It follows from the Jacobian criterion for smoothness that the subscheme X^sing of X defined by the Jacobian ideal, i.e. by the d-th determinantal ideal of Ω^1_{X/S}



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Le schéma de Picard $\mathrm{Pic}_{C/k}$ se dévisse comme suit. Soient $(D_i)_{i\in I}$ les composantes irréductibles de C, $\tilde{D}_i$ la normalisée de D_i et $\tilde{C} = \coprod \tilde{D}_i$ la normalisée de C.

a) Le morphisme "degré total" :

$$\deg_I = (\deg_{D_i})_{i\in I} : \mathrm{Pic}_{C/k} \to \mathbb{Z}^I$$

et surjectif, de noyau $\mathrm{Pic}^0_{C/k}$.

b) Le morphisme

$$\mathrm{Pic}^0_{C/k} \to \mathrm{Pic}^0_{\tilde{C}/k} = \overline{\prod} \mathrm{Pic}^0_{\tilde{D}_i/k}$$

est surjectif, d'image le plus grand quotient de $\mathrm{Pic}^0_{C/k}$ qui soit un schéma abélien. Son noyau est un tore, canoniquement isomorphe à

$$H^1(\Gamma(C), \mathbb{Z}) \otimes \mathbb{G}_m .$$

Pour D propre et purement de dimension un sur k, il existe une et une

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(Bourbaki; Commutative Algebra, Chapter 7, §4, Exercise 10) has as its underlying topological space the set of points of X at which f is not smooth. We shall call it the non-smoothness subscheme of X [SGA 7, VI]. The locus on which f is smooth is denoted X^reg

5. Review of deformation theory.

5.1. Let Λ be a complete Noetherian local ring with residue field k. Two particularly useful cases are the following.

a) Λ = k.

b) k is perfect of characteristic p > 0 and Λ is the ring of Witt vectors W(k) (the unique complete discrete valuation ring with residue field k and maximal ideal (p)).

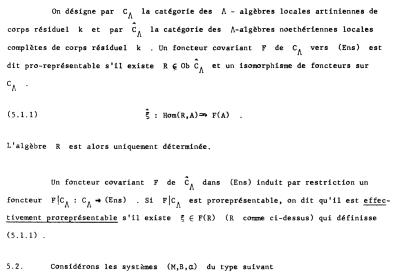
Let C_Λ denote the category of local Artinian Λ-algebras with residue field k, and let Ĉ_Λ denote the category of complete Noetherian local Λ-algebras with residue field k. A covariant functor F from C_Λ to (Sets) is said to be pro-representable if there exist R ∈ Ob Ĉ_Λ and an isomorphism of functors on C_Λ

(5.1.1) $\hat{\xi}$: Hom(R,A) ⟶̃ F(A).

The algebra R is then uniquely determined.

A covariant functor F from Ĉ_Λ to (Sets) induces by restriction a functor F|C_Λ : C_Λ → (Sets). If F|C_Λ is prorepresentable, it is said to be effectively prorepresentable if there exists ξ ∈ F(R) (with R as above) that defines (5.1.1).

5.2. Consider systems (M,B,α) of the following type



On désigne par C_Λ la catégorie des Λ - algèbres locales artiniennes de corps résiduel k et par $\hat{C}_\Lambda$ la catégorie des Λ-algèbres noethériennes locales complètes de corps résiduel k . Un foncteur covariant F de C_Λ vers (Ens) est dit pro-représentable s'il existe $R \in Ob\, \hat{C}_\Lambda$ et un isomorphisme de foncteurs sur C_Λ .

(5.1.1) $\hat{\xi} : Hom(R,A) \xrightarrow{\sim} F(A)$.

L'algèbre R est alors uniquement déterminée.

Un foncteur covariant F de $\hat{C}_\Lambda$ dans (Ens) induit par restriction un foncteur $F|C_\Lambda : C_\Lambda \to (Ens)$. Si $F|C_\Lambda$ est proreprésentable, on dit qu'il est effectivement proreprésentable s'il existe $\xi \in F(R)$ (R comme ci-dessus) qui définisse (5.1.1) .

5.2. Considérons les systèmes (M,B,α) du type suivant

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a) M is a complete Noetherian local Λ-algebra with residue field k;

b) B is a complete flat Noetherian local M-algebra, with residue field k, equipped with an isomorphism

α : B ⊗_M k ≃ k[[X,Y]]/(XY).

Such a system (Λ[[t]], A, α_{o/0}) is provided by the Λ[[t]]-algebra

A = Λ[[t,X,Y]]/(XY−t).

It is proved in [8] §1 that this last system is versal. For every (B,M,α), there exists a homomorphism Λ[[t]] → M such that (B,α) is deduced from (A,α_{o/0}) by extension of scalars:

B = A ⊗_{Λ[[t]]} M.

One easily deduces the following statement:

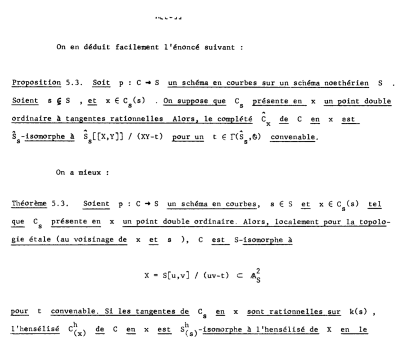
Proposition 5.3. Let p : C → S be a scheme of curves over a Noetherian scheme S. Let s ∈ S and x ∈ C_s(s). Suppose that C_s has at x an ordinary double point with rational tangent directions. Then the completion Ĉ_x of C at x is Ŝ_s-isomorphic to Ŝ_s[[X,Y]]/(XY−t) for a suitable t ∈ Γ(Ŝ_s,$\mathcal{O}$).

Better still:

Theorem 5.3. Let p : C → S be a scheme of curves, let s ∈ S, and let x ∈ C_s(s) be such that C_s has an ordinary double point at x. Then, locally for the étale topology (near x and s), C is S-isomorphic to

X = S[u,v]/(uv−t) ⊂ $\mathbb{A}$^2_S

for a suitable t. If the tangent directions of C_s at x are rational over k(s), the henselization C^h_{(x)} of C at x is S^h_{(s)}-isomorphic to the henselization of X at the



On en déduit facilement l'énoncé suivant :

Proposition 5.3. Soit p : C → S un schéma en courbes sur un schéma noethérien S. Soient s ∈ S, et x ∈ C_s(s). On suppose que C_s présente en x un point double ordinaire à tangentes rationnelles. Alors, le complété $\hat{C}_x$ de C en x est $\hat{\mathcal{O}}_s$-isomorphe à $\hat{\mathcal{O}}_s[[X,Y]]/(XY-t)$ pour un $t \in \Gamma(\hat{\mathcal{O}}_s, \mathcal{O})$ convenable.

On a mieux :

Théorème 5.3. Soient p : C → S un schéma en courbes, s ∈ S et x ∈ C_s(s) tel que C_s présente en x un point double ordinaire. Alors, localement pour la topologie étale (au voisinage de x et s), C est S-isomorphe à

$$X = S[u,v]/(uv-t) \subset \mathbb{A}^2_S$$

pour t convenable. Si les tangentes de C_s en x sont rationnelles sur k(s), l'hensélisé $C^h_{(x)}$ de C en x est $S^h_{(s)}$-isomorphe à l'hensélisé de X en le

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point (0,0) lying over s, for a suitable t ∈ Γ(S^h_{(s)},$\mathcal{O}$).

This theorem follows at once from the special case in which S is assumed to be of finite type over $\mathbb{Z}$. It follows from 5.2 and R. Elkik's theory of algebraization of modules of isolated singularities.

6. Division points of elliptic curves.

Let E be an elliptic curve (an abelian variety of dimension one) over an algebraically closed field k.

For every integer n ≥ 1, the group scheme E_n, the kernel of multiplication by n, is known to be the spectrum of a Hopf algebra of dimension n^2 over k. If n is invertible in k, then E_n ≃ ($\mathbb{Z}$/n)^2.

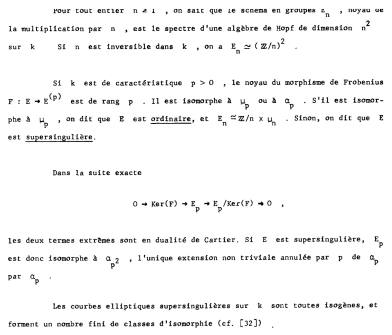
If k has characteristic p > 0, the kernel of the Frobenius morphism F : E → E^(p) has rank p. It is isomorphic to μ_p or to α_p. If it is isomorphic to μ_p, E is said to be ordinary, and E_n ≃ $\mathbb{Z}$/n × μ_n. Otherwise E is said to be supersingular.

In the exact sequence

0 → Ker(F) → E_p → E_p/Ker(F) → 0,

the two extreme terms are Cartier dual. If E is supersingular, E_p is therefore isomorphic to α_{p^2}, the unique nontrivial extension killed by p of α_p by α_p.

The supersingular elliptic curves over k are all isogenous and form a finite number of isomorphism classes (cf. [32]).



Pour tout entier $n \neq 1$, on sait que le schema en groupes α_n, noyau de la multiplication par n, est le spectre d'une algèbre de Hopf de dimension n^2 sur k. Si n est inversible dans k, on a $E_n \simeq (\mathbb{Z}/n)^2$.

Si k est de caractéristique $p > 0$, le noyau du morphisme de Frobenius $F : E \to E^{(p)}$ est de rang p. Il est isomorphe à μ_p ou à α_p. S'il est isomorphe à μ_p, on dit que E est ordinaire, et $E_n \simeq \mathbb{Z}/n \times \mu_n$. Sinon, on dit que E est supersingulière.

Dans la suite exacte

$$0 \to \mathrm{Ker}(F) \to E_p \to E_p/\mathrm{Ker}(F) \to 0 ,$$

les deux termes extrêmes sont en dualité de Cartier. Si E est supersingulière, E_p est donc isomorphe à α_{p^2}, l'unique extension non triviale annulée par p de α_p par α_p.

Les courbes elliptiques supersingulières sur k sont toutes isogènes, et forment un nombre fini de classes d'isomorphie (cf. [32]).

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7. Review of commutative algebra.

We shall often use the following results from commutative algebra; for their proofs we refer to EGA.

(7.1) Let $f : X \to Y$ be a morphism of finite type between Noetherian schemes. If Y is Cohen–Macaulay and f is flat, then X is Cohen–Macaulay if and only if the fibers of f are. For example, if Y is Cohen–Macaulay and f is quasi-finite and flat, then X is Cohen–Macaulay.

If Y is regular, X is Cohen–Macaulay, and at the point $x \in X$ with image $y \in Y$ one has

(7.1 1) $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} k(y))$,

then f is flat at x. Condition (7.1.1) is satisfied if f is quasi-finite and dominant and X is irreducible.

EGA IV 6.3.5 and EGA IV 6.1.5.

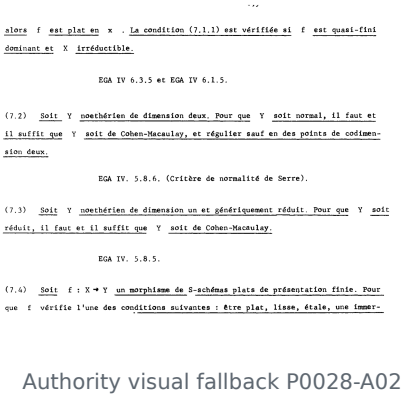
(7.2) Let Y be Noetherian of dimension two. For Y to be normal, it is necessary and sufficient that Y be Cohen–Macaulay and regular except at points of codimension two.

EGA IV. 5.8.6. (Serre's normality criterion).

(7.3) Let Y be Noetherian of dimension one and generically reduced. For Y to be reduced, it is necessary and sufficient that Y be Cohen–Macaulay.

EGA IV. 5.8.5.

(7.4) Let $f : X \to Y$ be a morphism of flat S-schemes of finite presentation. For f to satisfy one of the following conditions: to be flat, smooth, étale, an open immersion



alors f est plat en x. La condition (7.1.1) est vérifiée si f est quasi-fini dominant et X irréductible.

EGA IV 6.3.5 et EGA IV 6.1.5.

(7.2) Soit Y noethérien de dimension deux. Pour que Y soit normal, il faut et il suffit que Y soit de Cohen-Macaulay, et régulier sauf en des points de codimension deux.

EGA IV. 5.8.6. (Critère de normalité de Serre).

(7.3) Soit Y noethérien de dimension un et génériquement réduit. Pour que Y soit réduit, il faut et il suffit que Y soit de Cohen-Macaulay.

EGA IV. 5.8.5.

(7.4) Soit $f : X \to Y$ un morphisme de S-schémas plats de présentation finie. Pour que f vérifie l'une des conditions suivantes : être plat, lisse, étale, une imme[illegible]

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an isomorphism, or a flat relative complete intersection, it is necessary and sufficient that for every geometric point $\bar{s}$ of S, $f_{\bar{s}} : X_{\bar{s}} \to Y_{\bar{s}}$ satisfy the stated condition.

For ”flat,” this is EGA IV 11.3.10. The properties ”smooth,” ”étale,” and ”flat relative complete intersection” are equivalent to ”flat” plus a property of the fibers. Finally, ”open immersion” is equivalent to ”étale and radicial” (EGA IV.17.9.1), where ”radicial” is a property of the fibers, and ”isomorphism” is equivalent to ”open immersion and surjective.”

8. Coarse moduli schemes

Let $\mathfrak{M}$ be an algebraic stack over a base scheme S.

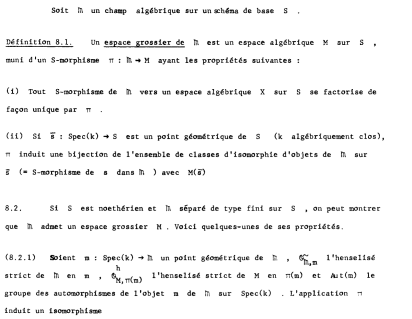
Definition 8.1. A coarse space of $\mathfrak{M}$ is an algebraic space M over S, equipped with an S-morphism $\pi : \mathfrak{M} \to M$ having the following properties:

(i) Every S-morphism from $\mathfrak{M}$ to an algebraic space X over S factors uniquely through π.

(ii) If $\bar{s} : \mathrm{Spec}(k) \to S$ is a geometric point of S (with k algebraically closed), π induces a bijection between the set of isomorphism classes of objects of $\mathfrak{M}$ over $\bar{s}$ (= an S-morphism from s to $\mathfrak{M}$) and $M(\bar{s})$.

8.2. If S is Noetherian and $\mathfrak{M}$ is separated and of finite type over S, one can show that $\mathfrak{M}$ admits a coarse space M. Here are some of its properties.

(8.2.1) Let $m : \mathrm{Spec}(k) \to \mathfrak{M}$ be a geometric point of $\mathfrak{M}$, let $\mathcal{O}_{\{\mathfrak{M},m\}}$ be the strict henselization of $\mathfrak{M}$ at m, let $\mathcal{O}^h_{\{M,\pi(m)\}}$ be the strict henselization of M at π(m), and let Aut(m) be the group of automorphisms of the object m of $\mathfrak{M}$ over Spec(k). The map π induces an isomorphism


Soit $\mathfrak{M}$ un champ algébrique sur un schéma de base S.

Définition 8.1. Un espace grossier de $\mathfrak{M}$ est un espace algébrique M sur S, muni d'un S-morphisme $\pi : \mathfrak{M} \to M$ ayant les propriétés suivantes :

(i) Tout S-morphisme de $\mathfrak{M}$ vers un espace algébrique X sur S se factorise de façon unique par π.

(ii) Si $\bar{s} : \mathrm{Spec}(k) \to S$ est un point géométrique de S (k algébriquement clos), π induit une bijection de l'ensemble de classes d'isomorphie d'objets de $\mathfrak{M}$ sur $\bar{s}$ (= S-morphisme de s dans $\mathfrak{M}$) avec $M(\bar{s})$

8.2. Si S est noethérien et $\mathfrak{M}$ séparé de type fini sur S, on peut montrer que $\mathfrak{M}$ admet un espace grossier M. Voici quelques-unes de ses propriétés.

(8.2.1) Soient $m : \mathrm{Spec}(k) \to \mathfrak{M}$ un point géométrique de $\mathfrak{M}$, $\mathcal{O}^h_{\mathfrak{M},m}$ l'hensélisé strict de $\mathfrak{M}$ en m, $\mathcal{O}^h_{M,\pi(m)}$ l'hensélisé strict de M en π(m) et Aut(m) le groupe des automorphismes de l'objet m de $\mathfrak{M}$ sur Spec(k). L'application π induit un isomorphisme

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$$\mathrm{Spec}(\mathcal{O}^h_{\{\mathfrak{M},m\}})/\mathrm{Aut}(m) \simeq \mathrm{Spec}(\mathcal{O}^h_{\{M,\pi(m)\}}).$$

Let $H \subset \mathrm{Aut}(m)$ be the subgroup of automorphisms of the object m of $\mathfrak{M}$ over k that extend to an automorphism of the object $\mathrm{Spec}(\mathcal{O}^h_{\{\mathfrak{M},m\}}) \to \mathfrak{M}$ of $\mathfrak{M}$ over $\mathrm{Spec}(\mathcal{O}^h_{\{\mathfrak{M},m\}})$. The group Aut(m)/H acts effectively on $\mathcal{O}^h_{\{\mathfrak{M},m\}}$.

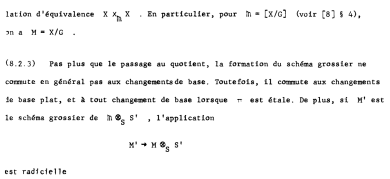
(8.2.2) If $u : X \to \mathfrak{M}$ is étale and surjective, M is the quotient of X by the equivalence relation $X \times_{\mathfrak{M}} X$. In particular, for $\mathfrak{M} = [X/G]$ (see [8] §4), one has $M = X/G$.

(8.2.3) No more than passage to the quotient does formation of the coarse scheme

commute in general with base changes. It does, however, commute with flat base changes, and with every base change when π is étale. Moreover, if M' is the coarse scheme of $\mathfrak{M} \otimes_S S'$, the map

$M' \to M \otimes_S S'$

is radicial


lation d'équivalence $X \times_h X$. En particulier, pour $h = [X/G]$ (voir [8] § 4), on a $M = X/G$.

(8.2.3) Pas plus que le passage au quotient, la formation du schéma grossier ne commute en général pas aux changements de base. Toutefois, il commute aux changements de base plat, et à tout changement de base lorsque π est étale. De plus, si M' est le schéma grossier de $\mathfrak{M} \otimes_S S'$, l'application

$$M' \to M \otimes_S S'$$

est radicielle

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Authority page 31; collected-paper page 524; original folio 173.

II Generalized elliptic curves.

In this chapter, we define generalized elliptic curves over an arbitrary base scheme and prove the theorems that make the adopted definition workable.

1. Néron polygons.

1.1. Let $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$ be the disjoint union of n copies of $\mathbb{P}^1$, indexed by $\mathbb{Z}/n$ ($n \geq 1$). By gluing the i-th copy of $\mathbb{P}^1$ to the (i+1)-st, identifying the section 0 of the i-th copy with the section ∞ of the (i+1)-st, one obtains a genus-one curve C over Spec($\mathbb{Z}$), with normalization $\tilde{C}$.

For every scheme S, the scheme over S obtained from this one by extension of scalars is called the standard Néron polygon with n sides over S, or simply the standard n-gon over S. A Néron polygon (respectively, an n-gon) over an algebraically closed field k is a scheme over k isomorphic to one of the standard polygons (respectively, to the standard n-gon).

Examples.

n = 1

n = 3

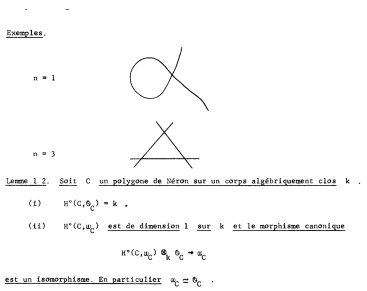
Lemma 1 2. Let C be a Néron polygon over an algebraically closed field k.

(i) $H^0(C,\mathcal{O}_C) = k$.

(ii) $H^0(C,\omega_C)$ has dimension 1 over k, and the canonical morphism

$H^0(C,\omega_C) \otimes_k \mathcal{O}_C \to \omega_C$

is an isomorphism. In particular, $\omega_C \simeq \mathcal{O}_C$.


Exemples.

n = 1

n = 3

Lemme 1.2. Soit C un polygone de Néron sur un corps algébriquement clos k .

(i) $H^0(C,\mathcal{O}_C) = k$.

(ii) $H^0(C,\omega_C)$ est de dimension 1 sur k et le morphisme canonique

$$H^0(C,\omega_C) \otimes_k \mathcal{O}_C \to \omega_C$$

est un isomorphisme. En particulier $\omega_C \simeq \mathcal{O}_C$.

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Proof: (i) follows from the fact that C is reduced and connected.

(ii) We may suppose that C is the standard n-gon.

Let η be the differential form on the normalization $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$ whose restriction to the i-th copy of $\mathbb{P}^1$ is dx/x. It is clear that η has at most simple poles, with residues +1 and −1 at the “0” and “∞,” respectively, of the i-th copy of $\mathbb{P}^1$. The form η therefore defines a global section of ω_{C/S} (cf. I. 2.3.). It generates ω_C at every point of C.

Lemma 1.3. Let C be a reduced, connected genus-one curve over an algebraically closed field, whose only singularities are ordinary double points, and such that $\omega_C \simeq \mathcal{O}$. Then C is smooth or a Néron polygon.

Proof: Let $\pi : \tilde{C} \to C$ be the normalization of C, with irreducible components $(C_i)_{1\le i\le a}$; let $g_i \ge 0$ be the genus of C_i, and let b be the number of double points of C. The long exact cohomology sequence deduced from

$$0 \to \mathcal{O}_C \to \pi_*\mathcal{O}_{\tilde{C}} \to \text{cokernel} \to 0$$

and 1.2(i) gives

$$a = \sum g_i + b \text{ and}$$

$$b \ge a-1 \ge 0.$$

Let $b_i \ge 0$ be the number of points of C_i whose image is a singular point of C. One has

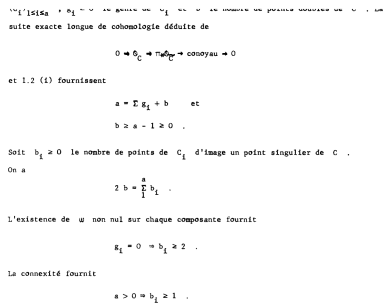
$$2b = \sum_{i=1}^{a} b_i.$$

The existence of a nonzero ω on each component gives

$$g_i = 0 \Rightarrow b_i \ge 2.$$

Connectedness gives

$$a > 0 \Rightarrow b_i \ge 1.$$



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Authority page 33; collected-paper page 526; original folio 175.

These inequalities have the following solutions:

a) a = 1, b = 0, g = 1 (the nonsingular case)

b) a = n, b = n, g_i = 0, b_i = 2 (n ≥ 1): the C_i are ≃ $\mathbb{P}^1$,

and on each C_i two points map to a double point of C. Since C is connected, it is a Néron polygon.

Definition 1.4. A stable genus-one curve C over a scheme S (or an algebraic space S, or an algebraic stack S) is a scheme of curves over S (cf. I.1.0.) every geometric fiber of which is either a proper smooth connected genus-one curve or a Néron polygon.

Proposition 1.5. : Let p : C → S be a proper flat morphism of finite presentation. The set of points s ∈ S such that C_s is a stable genus-one curve is an open subscheme of S.

Proof: The conditions that C_s be reduced, reduced and connected, reduced and purely one-dimensional, of genus one, and have only ordinary quadratic singularities are all open. That the condition ω ≃ $\mathcal{O}$ of 1.3 is open follows from the proof of 1.6 below.

Proposition 1.6. Let p : C → S be a stable genus-one curve.

(i) p_*$\mathcal{O}$_C = $\mathcal{O}$_S universally.

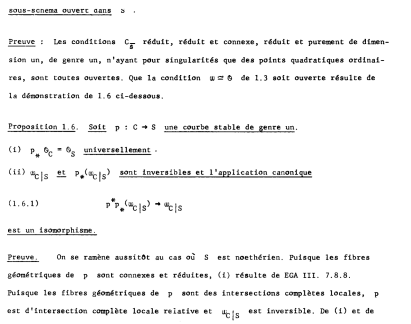
(ii) ω_{C/S} and p_*(ω_{C/S}) are invertible, and the canonical map

(1.6.1) p^*p_*(ω_{C/S}) → ω_{C/S}

is an isomorphism.

Proof. We immediately reduce to the case where S is Noetherian. Since the geometric fibers of p are connected and reduced, (i) follows from EGA III. 7.8.8.

Since the geometric fibers of p are local complete intersections, p is a relative local complete intersection and ω_{C/S} is invertible. From (i) and from



sous-schema ouvert dans S .

Preuve : Les conditions $C_{\overline{s}}$ réduit, réduit et connexe, réduit et purement de dimension un, de genre un, n'ayant pour singularités que des points quadratiques ordinaires, sont toutes ouvertes. Que la condition $\omega \simeq \mathcal{O}$ de 1.3 soit ouverte résulte de la démonstration de 1.6 ci-dessous.

Proposition 1.6. Soit $p : C \to S$ une courbe stable de genre un.

(i) $p_* \mathcal{O}_C = \mathcal{O}_S$ universellement .

(ii) $\omega_{C|S}$ et $p_*(\omega_{C|S})$ sont inversibles et l'application canonique

(1.6.1) $$p^* p_*(\omega_{C|S}) \to \omega_{C|S}$$

est un isomorphisme.

Preuve. On se ramène aussitôt au cas où S est noethérien. Puisque les fibres géométriques de p sont connexes et réduites, (i) résulte de EGA III. 7.8.8. Puisque les fibres géométriques de p sont des intersections complètes locales, p est d'intersection complète locale relative et $\omega_{C|S}$ est inversible. De (i) et de

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the fact that $R^2p_*\mathcal{O} = 0$, it follows that $R^1p_*\mathcal{O}$ is locally free and its formation is compatible with every base change S' → S (EGA III. 7.8.). Coherent-sheaf duality (I.2.) then shows that p_*(ω_{C/S}) is locally free and its formation is compatible with every base change. Since (ω_{C/S})_s = ω_{C_s/s}, it remains to prove (ii) when S is the spectrum of an algebraically closed field. In that case, 1.6.1 is well known when C is smooth, and follows from 1.2 for a Néron polygon.

The following Lemma 1.7 is a special case of I 3.7.

Lemma 1.7. Let C be a Néron polygon over an algebraically closed field k.

(i) Pic^o(C) = $\mathbb{G}_m \otimes H^1(\Gamma(C),\mathbb{Z})$ is isomorphic to $\mathbb{G}_m$.

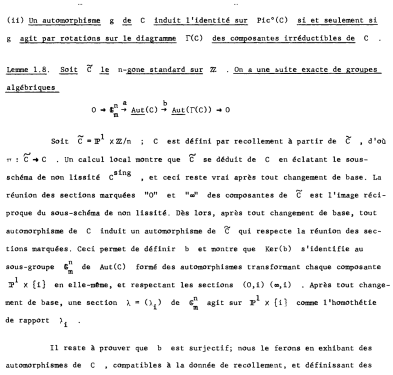
(ii) An automorphism g of C induces the identity on Pic^o(C) if and only if g acts by rotations on the component diagram Γ(C) of C.

Lemma 1.8. Let $\tilde{C}$ be the standard n-gon over $\mathbb{Z}$. There is an exact sequence of algebraic groups

$$0 \to \mathbb{G}_m^n \longrightarrow^a \mathrm{Aut}(C) \longrightarrow^b \mathrm{Aut}(\Gamma(C)) \to 0.$$

Let $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$; C is defined by gluing from $\tilde{C}$, whence $\pi : \tilde{C} \to C$. A local computation shows that $\tilde{C}$ is obtained from C by blowing up the non-smoothness subscheme C^sing, and this remains true after every base change. The union of the marked sections "0" and "∞" of the components of $\tilde{C}$ is the inverse image of the non-smoothness subscheme. Hence, after every base change, every automorphism of C induces an automorphism of $\tilde{C}$ that preserves the union of the marked sections. This defines b and shows that Ker(b) is identified with the subgroup $\mathbb{G}_m^n$ of Aut(C) consisting of the automorphisms that take every component $\mathbb{P}^1 \times \{i\}$ to itself and preserve the sections (0,i) and (∞,i). After every base change, a section λ = (λ_i) of $\mathbb{G}_m^n$ acts on $\mathbb{P}^1 \times \{i\}$ as scaling by the factor λ_i.

It remains to prove that b is surjective; we shall do so by exhibiting automorphisms of C compatible with the gluing data and defining



(ii) Un automorphisme g de C induit l'identité sur $\mathrm{Pic}^o(C)$ si et seulement si g agit par rotations sur le diagramme $\Gamma(C)$ des composantes irréductibles de C .

Lemme 1.8. Soit $\tilde{C}$ le n-gone standard sur $\mathbb{Z}$. On a une suite exacte de groupes algébriques

$$0 \to \mathbb{G}_m^n \xrightarrow{a} \mathrm{Aut}(C) \xrightarrow{b} \mathrm{Aut}(\Gamma(C)) \to 0$$

Soit $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$; C est défini par recollement à partir de $\tilde{C}$, d'où $\pi : \tilde{C} \to C$. Un calcul local montre que $\tilde{C}$ se déduit de C en éclatant le sous-schéma de non lissité C^{sing} , et ceci reste vrai après tout changement de base. La réunion des sections marquées "0" et "∞" des composantes de $\tilde{C}$ est l'image réciproque du sous-schéma de non lissité. Dès lors, après tout changement de base, tout automorphisme de C induit un automorphisme de $\tilde{C}$ qui respecte la réunion des sections marquées. Ceci permet de définir b et montre que Ker(b) s'identifie au sous-groupe $\mathbb{G}_m^n$ de Aut(C) formé des automorphismes transformant chaque composante $\mathbb{P}^1 \times \{i\}$ en elle-même, et respectant les sections $(0,i)$ (∞,i) . Après tout changement de base, une section $\lambda = (\lambda_i)$ de $\mathbb{G}_m^n$ agit sur $\mathbb{P}^1 \times \{i\}$ comme l'homothétie de rapport λ_i .

Il reste à prouver que b est surjectif; nous le ferons en exhibant des automorphismes de C , compatibles à la donnée de recollement, et définissant des

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automorphisms of C whose images under b generate the dihedral group Aut(Γ(C)). They are

τ : (x,i) ↦ (x^{−1},−i)

σ_j : (x,i) ↦ (x,i+j) (j ∈ ℤ/n).

1.9. Let C be the standard n-gon over Spec(ℤ). Identify the normalization $\tilde{C}$ of C with $\mathbb{P}^1 \times \mathbb{Z}/n\mathbb{Z}$ and the inverse image of C^reg in $\tilde{C}$ with $\mathbb{G}_m \times \mathbb{Z}/n\mathbb{Z}$. An action of $\mathbb{G}_m \times \mathbb{Z}/n\mathbb{Z}$ on $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n\mathbb{Z}$ is obtained by setting

(a,i) + (b,j) = (a·b,i+j),

where a·b denotes the natural action of $\mathbb{G}_m$ on $\mathbb{P}^1$.

This action descends to the quotient and defines a morphism

(1.9.1) + : C^reg × C → C.

The restriction of + to C^reg × C^reg defines a commutative group-scheme structure on C^reg, and + is an action of the group C^reg on C.

Automorphisms of (C,+) are defined as follows:

a) the automorphism of $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$

τ : (x,i) ↦ (x^{−1},−i)

descends to the quotient and defines an automorphism τ of (C,+);

b) for every scheme S and every ζ ∈ μ_n(S), the automorphism u(ζ) of C_S

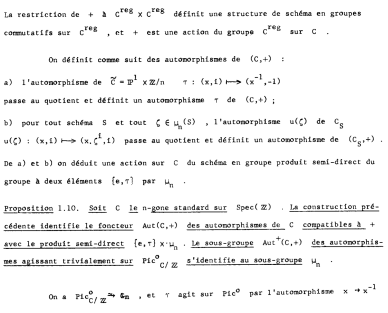
u(ζ) : (x,i) ↦ (x·ζ^i,i)

descends to the quotient and defines an automorphism of (C_S,+).

Parts a) and b) give an action on C by the semidirect-product group scheme of the two-element group {e,τ} by μ_n.

Proposition 1.10. Let C be the standard n-gon over Spec(ℤ). The preceding construction identifies the functor Aut(C,+) of automorphisms of C compatible with + with the semidirect product {e,τ} ×′ μ_n. The subgroup $\mathrm{Aut}^+(C,+)$ of automorphisms acting trivially on Pic^o_{C/ℤ} is identified with the subgroup μ_n.

One has Pic^o_{C/ℤ} ≃ $\mathbb{G}_m$, and τ acts on Pic^o by the automorphism x ↦ x^{−1}



La restriction de + à $C^{reg} \times C^{reg}$ définit une structure de schéma en groupes commutatifs sur C^{reg}, et + est une action du groupe C^{reg} sur C.

On définit comme suit des automorphismes de (C,+) :

a) l'automorphisme de $\tilde{C} = \mathbb{P}^1 \times \mathbb{Z}/n$ $\tau : (x,i) \mapsto (x^{-1},-i)$ passe au quotient et définit un automorphisme τ de (C,+) ;

b) pour tout schéma S et tout $\zeta \in \mu_n(S)$, l'automorphisme $u(\zeta)$ de C_S $u(\zeta) : (x,i) \mapsto (x,\zeta^i,i)$ passe au quotient et définit un automorphisme de $(C_S,+)$.

De a) et b) on déduit une action sur C du schéma en groupe produit semi-direct du groupe à deux éléments $\{e,\tau\}$ par μ_n.

Proposition 1.10. Soit C le n-gone standard sur Spec($\mathbb{Z}$). La construction précédente identifie le foncteur Aut(C,+) des automorphismes de C compatibles à + avec le produit semi-direct $\{e,\tau\} \times \mu_n$. Le sous-groupe $Aut^0(C,+)$ des automorphismes agissant trivialement sur $Pic^0_{C/\mathbb{Z}}$ s'identifie au sous-groupe μ_n.

On a $Pic^0_{C/\mathbb{Z}} \xrightarrow{\sim} \mathbb{G}_m$, et τ agit sur Pic^0 par l'automorphisme $x \mapsto x^{-1}$

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of $\mathbb{G}_m$. Since $Aut(\mathbb{G}_m) \simeq \mathbb{Z}/2$, the diagram

$$0 \to \mu_n \to \{e,\tau\} \times' \mu_n \to \{e,\tau\} \to 0 \downarrow \downarrow \downarrow 0 \to Aut^+(C,+) \to Aut(C,+) \to Aut(\mathbb{G}_m)$$

shows that it is enough to verify that

$$\mu_n \xrightarrow{\sim} Aut^+(C,+).$$

Let $g \in Aut^+(C,+)(S)$. From 1.8 and 1.7(ii), there exists $\lambda = (\lambda_i) \in \mathbb{G}_m^n$ such that g is $a(\lambda)$ (notation of 1.8). The condition that g commute with + then means that $\lambda_i \lambda_j = \lambda_{i+j}$: λ defines a homomorphism from $\mathbb{Z}/n$ to $\mathbb{G}_m$, and g is defined by a section of μ_n.

1.11. Let k be an algebraically closed field, and let (C,+) be obtained from the curve of 1.10 by extension of scalars from $\mathbb{Z}$ to k. One checks that, for $x \in C^{reg}(k)$, translation $y \mapsto x+y$ acts by rotations on the component diagram $\Gamma(C)$.

Definition 1.12. A generalized elliptic curve (over a base S) is a stable genus-one curve

$$p : C \to S,$$

equipped with a morphism

$$(1\ 12.1)\quad + : C^{reg} \times_S C \to C$$

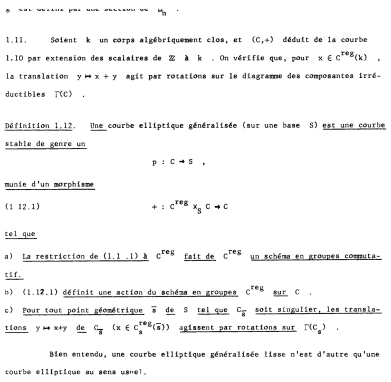
such that

a) The restriction of (1.1 .1) to C^{reg} makes C^{reg} a commutative group scheme.

b) (1.12.1) defines an action of the group scheme C^{reg} on C.

c) For every geometric point $\bar{s}$ of S such that $C_{\bar{s}}$ is singular, the translations $y \mapsto x+y$ of C_s ($x \in C_s^{reg}(\bar{s})$) act by rotations on $\Gamma(C_s)$.

Of course, a smooth generalized elliptic curve is just an elliptic curve in the usual sense.



1.11. Soient k un corps algébriquement clos, et (C,+) déduit de la courbe 1.10 par extension des scalaires de $\mathbb{Z}$ à k . On vérifie que, pour $x \in C^{reg}(k)$, la translation $y \mapsto x + y$ agit par rotations sur le diagramme des composantes irréductibles $\Gamma(C)$.

Définition 1.12. Une courbe elliptique généralisée (sur une base S) est une courbe stable de genre un

$$p : C \to S ,$$

munie d'un morphisme

$$(1\ 12.1) \qquad + : C^{reg} \times_S C \to C$$

tel que

a) La restriction de (1.1 .1) à C^{reg} fait de C^{reg} un schéma en groupes commutatif.

b) (1.12.1) définit une action du schéma en groupes C^{reg} sur C .

c) Pour tout point géométrique $\bar{s}$ de S tel que $C_{\bar{s}}$ soit singulier, les translations $y \mapsto x+y$ de $C_{\bar{s}}$ $(x \in C_{\bar{s}}^{reg}(\bar{s}))$ agissent par rotations sur $\Gamma(C_{\bar{s}})$.

Bien entendu, une courbe elliptique généralisée lisse n'est d'autre qu'une courbe elliptique au sens usuel.

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Proposition 1.13. Let (C,+) satisfy 1.12 a) and b). The set U of s ∈ S such that C_s/s satisfies c) is open and closed in S. Over U, C^reg acts trivially on Pic⁰_{C/S}.

Proof: Let x be a geometric point of C_s̄. Translation y ↦ x+y acts by translation if and only if it acts trivially on Pic⁰_{C_s}. Apply the following lemma (a variation on a theme of SGA 3) to Pic⁰_{C/S} ×_S C^reg over C^reg.

Lemma 1.14. Let p : A → S be a group scheme over S whose fibers are extensions of abelian varieties by tori. Let g and h be two endomorphisms of A. There exists an open and closed part of S such that, for every S-scheme T,

g_T = h_T if and only if T factors through V.

The lemma shows that the U of 1.13 is closed, since its complement is the image of an open (and closed) part C^reg−V of C^reg. It also proves that, if S = U, C^reg acts trivially on Pic⁰_{C/S}. We shall prove that U is open at the same time as 1.15 below.

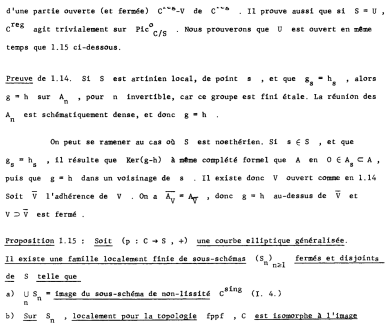
Proof of 1.14. If S is local Artinian, with point s, and g_s = h_s, then g = h on A_n, for n invertible, since this group is finite étale. The union of the A_n is schematically dense, and hence g = h.

We may reduce to the case where S is Noetherian. If s ∈ S and g_s = h_s, it follows that Ker(g−h) has the same formal completion as A at 0 ∈ A_s ⊂ A, and then that g = h in a neighborhood of s. There therefore exists an open V as in 1.14. Let V̄ be the closure of V. One has Ā_V = A_V̄, and hence g = h over V̄, and V ⊃ V̄ is closed.

Proposition 1.15: Let (p : C → S, +) be a generalized elliptic curve. There exists a locally finite family of closed and pairwise disjoint subschemes (S_n)_{n≥1} of S such that

a) ⋃_n S_n = image of the non-smoothness subscheme C^sing (I. 4.)

b) Over S_n, locally for the fppf topology, C is isomorphic to the inverse



d'une partie ouverte (et fermée) C^{sing}-V de C^{sing} . Il prouve aussi que si $S = U$, C^{reg} agit trivialement sur $Pic^{0}_{C/S}$. Nous prouverons que U est ouvert en même temps que 1.15 ci-dessous.

Preuve de 1.14. Si S est artinien local, de point s , et que $g_s = h_s$, alors $g = h$ sur A_n , pour n invertible, car ce groupe est fini étale. La réunion des A_n est schématiquement dense, et donc $g = h$.

On peut se ramener au cas où S est noethérien. Si $s \in S$, et que $g_s = h_s$, il résulte que $Ker(g-h)$ à même complété formel que A en $O \in A_s \subset A$, puis que $g = h$ dans un voisinage de s . Il existe donc V ouvert comme en 1.14 Soit $\overline{V}$ l'adhérence de V . On a $\overline{A_V} = A_{\overline{V}}$, donc $g = h$ au-dessus de $\overline{V}$ et $V \supset \overline{V}$ est fermé .

Proposition 1.15 : Soit $(p : C \to S , +)$ une courbe elliptique généralisée. Il existe une famille localement finie de sous-schémas $(S_n)_{n\geq 1}$ fermés et disjoints de S telle que

a) $\bigcup_n S_n$ = image du sous-schéma de non-lissité C^{sing} (I. 4.)

b) Sur S_n , localement pour la topologie fppf , C est isomorphe à l'image

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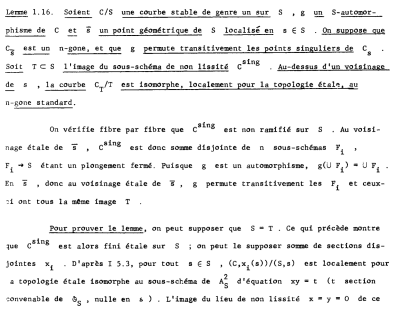
image of the standard generalized elliptic curve with n sides of 1.9.

Proof: The problem is local on S, and trivial in a neighborhood of the points of S where C_s is smooth. Let therefore s̄ be a geometric point of S, localized at s ∈ S, such that the curve C_s̄ is an n-gon. Let α be an isomorphism between C_s̄ and the standard polygon, such that α(e) is the point 1 of the component of index 0 ∈ ℤ/n, and index the components of C_s̄ by ℤ/n using α. Translation by x ∈ C_s̄^reg, in the component of index i, induces the rotation of Γ(C_s̄) taking the component of index 0 to that of index i: addition induces on $\pi_0(C_{s}^{reg}) \simeq \mathbb{Z}/n$ addition in ℤ/n, and, if x lies in the component of index one of C_s̄^reg, translation by x transitively permutes the singular points of C_s̄. In an étale neighborhood of s̄, C_reg/S admits sections x such that x(s̄) lies in the component of index one of C_s̄. The automorphism g = x + of C/S then satisfies the hypothesis of the following lemma.

Lemma 1.16. Let C/S be a stable genus-one curve over S, let g be an S-automorphism of C, and let s̄ be a geometric point of S localized at s ∈ S. Suppose that C_s̄ is an n-gon and that g transitively permutes the singular points of C_s. Let T ⊂ S be the image of the non-smoothness subscheme C^sing. Over a neighborhood of s, the curve C_T/T is isomorphic, locally for the étale topology, to the standard n-gon.

One verifies fiber by fiber that C^sing is unramified over S. In an étale neighborhood of s̄, C^sing is therefore a disjoint sum of n subschemes F_i, each F_i → S being a closed immersion. Since g is an automorphism, g(⋃ F_i) = ⋃ F_i. At s̄, and hence in an étale neighborhood of s̄, g transitively permutes the F_i, and all of these have the same image T.

To prove the lemma, we may suppose that S = T. What precedes shows that C^sing is then finite étale over S; we may suppose it is a sum of disjoint sections x_i. By I 5.3, for every s ∈ S, (C,x_i(s))/(S,s) is locally for the étale topology isomorphic to the subscheme of $\mathbb{A}^2$_S with equation xy = t, where t is a suitable section of $\mathcal{O}$_S, vanishing at s. The image of the non-smooth locus x = y = 0 of this



Lemme 1.16. Soient C/S une courbe stable de genre un sur S, g un S-automorphisme de C et s̄ un point géométrique de S localisé en s ∈ S. On suppose que $C_{\bar{s}}$ est un n-gone, et que g permute transitivement les points singuliers de $C_{\bar{s}}$. Soit T ⊂ S l'image du sous-schéma de non lissité C^{sing}. Au-dessus d'un voisinage de s, la courbe C_T/T est isomorphe, localement pour la topologie étale, au n-gone standard.

On vérifie fibre par fibre que C^{sing} est non ramifié sur S. Au voisinage étale de s̄, C^{sing} est donc somme disjointe de n sous-schémas F_i, $F_i \to S$ étant un plongement fermé. Puisque g est un automorphisme, $g(\cup F_i) = \cup F_i$. En s̄, donc au voisinage étale de s̄, g permute transitivement les F_i et ceux-ci ont tous la même image T.

Pour prouver le lemme, on peut supposer que S = T. Ce qui précède montre que C^{sing} est alors fini étale sur S; on peut le supposer somme de sections disjointes x_i. D'après I 5.3, pour tout s ∈ S, $(C,x_i(s))/(S,s)$ est localement pour la topologie étale isomorphe au sous-schéma de A^2_S d'équation xy = t (t section convenable de $\mathfrak{d}_S$, nulle en s). L'image du lieu de non lissité x = y = 0 de ce

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scheme is the subscheme t = 0 of S; the non-smooth locus can be étale only if t = 0 in a neighborhood of s. Locally for the étale topology, (C,x_i(s))/(S,s) is therefore isomorphic to the subscheme of $\mathbb{A}^2$_S with equation xy = 0.

Blow up C^sing in C to obtain C̃. A local calculation shows that C̃ is smooth over S, with fibers isomorphic to the normalizations of the fibers of C, and that the inverse image of C^sing in C̃ is étale over S.

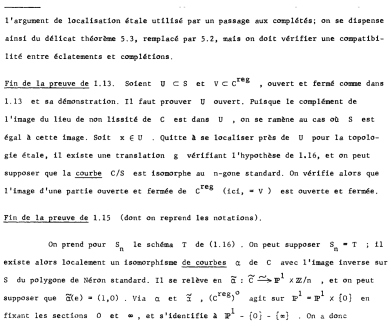
Let π : C̃ → S' be the Stein factorization of C̃. The scheme S' is finite étale over S, and the geometric fibers of π are isomorphic to $\mathbb{P}^1$. Locally on S for the étale topology, C̃ is therefore a sum of n copies of $\mathbb{P}^1$. In an étale neighborhood of s, C is obtained from C̃ just as C_s̄ is obtained from C̃_s̄, and hence is a standard Néron polygon.

Variant: We easily reduce to supposing S Noetherian. We may then replace the étale-localization argument used above by passage to completions; this avoids the delicate theorem 5.3, replaced by 5.2, but one must verify compatibility between blowups and completions.

End of the proof of 1.13. Let U ⊂ S and V ⊂ C^reg be open and closed as in 1.13 and its proof. We must prove U open. Since the complement of the image of the non-smooth locus of C lies in U, we reduce to the case where S equals that image. Let x ∈ U. After localizing near U for the étale topology, there exists a translation g satisfying the hypothesis of 1.16, and we may suppose that the curve C/S is isomorphic to the standard n-gon. One then verifies that the image of an open and closed part of C^reg (here, = V) is open and closed.

End of the proof of 1.15 (whose notation we resume).

Take for S_n the scheme T of (1.16). We may suppose S_n = T; there then exists locally an isomorphism of curves α from C to the inverse image over S of the standard Néron polygon. It lifts to $\tilde{\alpha} : \tilde{C} \xrightarrow{\sim} \mathbb{P}^1 \times \mathbb{Z}/n$, and we may suppose that α̃(e) = (1,0). Via α and α̃, $(C^{reg})^0$ acts on $\mathbb{P}^1 = \mathbb{P}^1 \times \{0\}$ while fixing the sections 0 and ∞, and is identified with $\mathbb{P}^1 - \{0\} - \{\infty\}$. Thus one has



l'argument de localisation étale utilisé par un passage aux complétés; on se dispense ainsi du délicat théorème 5.3, remplacé par 5.2, mais on doit vérifier une compatibilité entre éclatements et complétions.

Fin de la preuve de 1.13. Soient $U \subset S$ et $V \subset C^{reg}$, ouvert et fermé comme dans 1.13 et sa démonstration. Il faut prouver U ouvert. Puisque le complément de l'image du lieu de non lissité de C est dans U, on se ramène au cas où S est égal à cette image. Soit $x \in U$. Quitte à se localiser près de U pour la topologie étale, il existe une translation g vérifiant l'hypothèse de 1.16, et on peut supposer que la courbe C/S est isomorphe au n-gone standard. On vérifie alors que l'image d'une partie ouverte et fermée de C^{reg} (ici, $= V$) est ouverte et fermée.

Fin de la preuve de 1.15 (dont on reprend les notations).

On prend pour S_n le schéma T de (1.16). On peut supposer $S_n = T$; il existe alors localement un isomorphisme de courbes α de C avec l'image inverse sur S du polygone de Néron standard. Il se relève en $\tilde{\alpha} : \tilde{C} \xrightarrow{\sim} \mathbb{P}^1 \times \mathbb{Z}/n$, et on peut supposer que $\tilde{\alpha}(e) = (1,0)$. Via α et $\tilde{\alpha}$, $(C^{reg}/S)^0$ agit sur $\mathbb{P}^1 = \mathbb{P}^1 \times \{0\}$ en fixant les sections 0 et ∞, et s'identifie à $\mathbb{P}^1 - \{0\} - \{\infty\}$. On a donc

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$(C^{reg})^0 \simeq \mathbb{G}_m$ (compatibly with +). In a neighborhood of s, one may normalize α so that $\tilde{\alpha}(x^i) = (1,i)$ $(0 \le i < n)$. If $x^n = 1$, then α is an isomorphism of (C,+) with the curve (1.9). In the general case, $x^n = a \in \mathbb{G}_m(S)$; a admits an n-th root locally for the fppf topology of S (for the étale topology if n is invertible on S), and one replaces x by $x \cdot a^{-1/n}$.

The following proposition will play a key technical role in questions of representability.

Proposition 1.17. Let X/S be a generalized elliptic curve over S, with identity section $e \in X(S)$, and let $u : Y \to X$ be a finite étale cover of X, equipped with $e \in Y(S)$ over e. Suppose that the geometric fibers of Y/S are connected. There is then one and only one generalized-elliptic-curve structure on Y, with identity e, such that the diagram

$$\begin{array}{ccc} Y^{reg} \times_S Y & \longrightarrow & Y \\ \downarrow & & \downarrow u \\ X^{reg} \times_S Y & \xrightarrow{+} & X \end{array} \qquad (1.17.1)$$

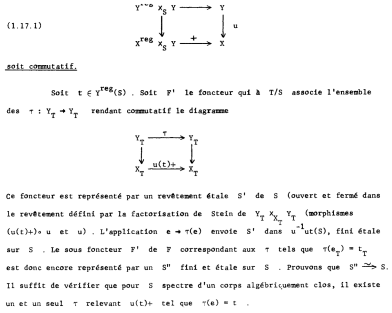
is commutative.

Let $t \in Y^{reg}(S)$. Let F' be the functor associating to T/S the set of $\tau : Y_T \to Y_T$ making the diagram

$$\begin{array}{ccc} \tau\, Y_T & \longrightarrow & Y_T \\ \downarrow & & \downarrow \\ X_T & \xrightarrow{u(t)+} & X_T \end{array}$$

commutative.

This functor is represented by an étale cover S' of S (open and closed in the cover defined by the Stein factorization of $Y_T \times_{\{X_T\}} Y_T$ (for the morphisms $(u(t)+)\circ u$ and u)). The map $e \mapsto \tau(e)$ sends S' into $u^{-1}ut(S)$, finite étale over S. The subfunctor F' of F corresponding to the τ such that $\tau(e_T) = t_T$ is therefore again represented by an S'' finite and étale over S. Let us prove that $S'' \xrightarrow{\sim} S$. It is enough to verify that, when S is the spectrum of an algebraically closed field, there exists one and only one τ lifting $u(t)+$ such that $\tau(e) = t$.



(1.17.1)

soit commutatif.

Soit $t \in Y^{reg}(S)$. Soit F' le foncteur qui à T/S associe l'ensemble des $\tau : Y_T \to Y_T$ rendant commutatif le diagramme

Ce foncteur est représenté par un revêtement étale S' de S (ouvert et fermé dans le revêtement défini par la factorisation de Stein de $Y_T \times_{X_T} Y_T$ (morphismes $(u(t)+)\circ u$ et u). L'application $e \to \tau(e)$ envoie S' dans $u^{-1}ut(S)$, fini étale sur S. Le sous foncteur F' de F correspondant aux τ tels que $\tau(e_T) = t_T$ est donc encore représenté par un S'' fini et étale sur S. Prouvons que $S'' \xrightarrow{\sim} S$. Il suffit de vérifier que pour S spectre d'un corps algébriquement clos, il existe un et un seul τ relevant $u(t)+$ tel que $\tau(e) = t$.

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This amounts to showing that the cover Y of X and its inverse image under u(t)+ : X → X are isomorphic. For smooth X, this follows from the fact that u(t) lies in the neutral component of X^reg. For X a Néron polygon, one observes that X has only one irreducible cover of a given degree.

The unique section of S” over S defines a morphism t+ : Y → Y. Since this construction is compatible with base changes, it defines

+ : Y^reg × Y → Y,

making (1.17.1) commutative, such that t + e = t, and characterized by these properties. It remains to prove that + makes Y a generalized elliptic curve, i.e. that various identities, such as x+y = y+x (x,y ∈ Y^reg(S) are satisfied. Since u(x+y) = u(x) + u(y) = u(y) + u(x) = u(y+x), x+y is a section of u^{−1}(u(y+x)(S)), étale over S, and it is enough to verify that x+y = y+x on the geometric fibers. For proper smooth fibers, a composition law with identity is automatically an abelian group law. For the others, the Néron polygon with n sides, with its standard + law, has as covers only Néron polygons with nk sides, with their standard + law.

The other identities to be verified are proved in the same way.

1.18. Let (C,+) be a generalized elliptic curve over S. One verifies fiber by fiber that the morphism

n : x ↦ nx : C^reg → C^reg

is flat. Its kernel C_n is therefore flat over S.

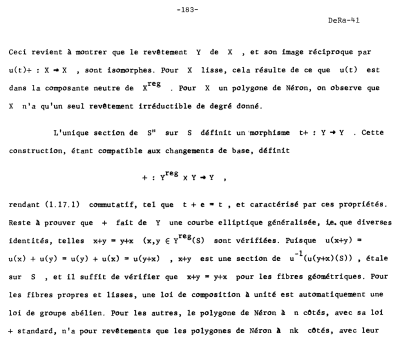
Suppose that S is the spectrum of an algebraically closed field k.

a) if C is smooth, one knows that C_n is finite of rank n^2

b) if C is an m-gon, one has C^reg ≃ $\mathbb{G}$_m × $\mathbb{Z}$/m.

Writing (n,m) for the gcd of n and m, one therefore has

C_n ≃ μ_n × $\mathbb{Z}$/((n,m))



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Ceci revient à montrer que le revêtement Y de X , et son image réciproque par u(t)+ : X → X , sont isomorphes. Pour X lisse, cela résulte de ce que u(t) est dans la composante neutre de X^{reg} . Pour X un polygone de Néron, on observe que X n'a qu'un seul revêtement irréductible de degré donné.

L'unique section de S" sur S définit un morphisme t+ : Y → Y . Cette construction, étant compatible aux changements de base, définit

+ : Y^{reg} x Y → Y ,

rendant (1.17.1) commutatif, tel que t + e = t , et caractérisé par ces propriétés. Reste à prouver que + fait de Y une courbe elliptique généralisée, i.e. que diverses identités, telles x+y = y+x (x,y ∈ Y^{reg}(S) sont vérifiées. Puisque u(x+y) = u(x) + u(y) = u(y) + u(x) = u(y+x) , x+y est une section de u^{-1}(u(y+x)(S)) , étale sur S , et il suffit de vérifier que x+y = y+x pour les fibres géométriques. Pour les fibres propres et lisses, une loi de composition à unité est automatiquement une loi de groupe abélien. Pour les autres, le polygone de Néron à n côtés, avec sa loi + standard, n'a pour revêtements que les polygones de Néron à nk côtés, avec leur

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In particular, C_n is finite of rank n.(n,m).

Recall the following lemma.

Lemma 1.19. Let f : X → S be a quasi-finite, flat, and separated morphism, with S Noetherian. If the rank of the fibers of f is constant, then f is finite.

Proof: Let us prove that f is proper. By the valuative criterion for properness, we may suppose that S is a trait. By the Main theorem of Zariski, X is an open subscheme of a scheme $\bar{X}$ finite over S, which may even be taken flat over S. Comparing the ranks of the special and generic fibers of X and $\bar{X}$, one finds that X = $\bar{X}$, whence 1.19.

Corollary 1.20. Let p : C → S be a generalized elliptic curve over S, and let n be an integer. Suppose that, for every geometric point $\bar{s}$ of S, C_$\bar{s}$ is smooth, or is an m-gon with n|m. Then C_n is finite locally free over S, of rank n^2.

We reduce to supposing S Noetherian, and apply 1.18 and 1.19.

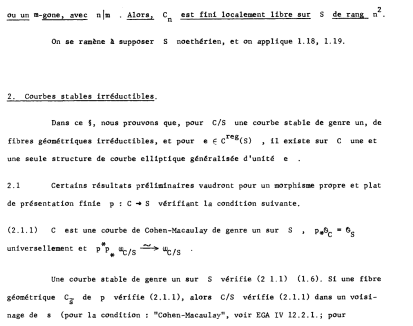
2. Irreducible stable curves.

In this §, we prove that, for C/S a stable genus-one curve with irreducible geometric fibers, and for e ∈ C^reg(S), there exists on C one and only one generalized-elliptic-curve structure with identity e.

2.1 Certain preliminary results will hold for a proper flat morphism of finite presentation p : C → S satisfying the following condition.

(2.1.1) C is a Cohen-Macaulay genus-one curve over S, p_*$\mathcal{O}$_C = $\mathcal{O}$_S universally, and p^*p_*ω_{C/S} ⟶̃ ω_{C/S}.

A stable genus-one curve over S satisfies (2 1.1) (1.6). If a geometric fiber C_$\bar{s}$ of p satisfies (2.1.1), then C/S satisfies (2.1.1) in a neighborhood of s (for the condition: “Cohen-Macaulay,” see EGA IV 12.2.1.; for



ou un m-gone, avec n|m . Alors, C_n est fini localement libre sur S de rang n^2.

On se ramène à supposer S noethérien, et on applique 1.18, 1.19.

2. Courbes stables irréductibles.

Dans ce §, nous prouvons que, pour C/S une courbe stable de genre un, de fibres géométriques irréductibles, et pour e ∈ $C^{reg}(S)$, il existe sur C une et une seule structure de courbe elliptique généralisée d'unité e .

2.1 Certains résultats préliminaires vaudront pour un morphisme propre et plat de présentation finie p : C → S vérifiant la condition suivante.

(2.1.1) C est une courbe de Cohen-Macaulay de genre un sur S , $p_*\mathcal{O}_C = \mathcal{O}_S$ universellement et $p^*p_*\omega_{C/S} \xrightarrow{\sim} \omega_{C/S}$.

Une courbe stable de genre un sur S vérifie (2 1.1) (1.6). Si une fibre géométrique $C_{\bar{s}}$ de p vérifie (2.1.1), alors C/S vérifie (2.1.1) dans un voisinage de s (pour la condition : "Cohen-Macaulay", voir EGA IV 12.2.1.; pour

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p_* $\mathcal{O}$_C = $\mathcal{O}$_S, see EGA III. 7.8.8.; if p_* $\mathcal{O}$_C = $\mathcal{O}$_S universally, p_* ω_{C/S} is locally free with formation compatible with every base change, and we conclude as in 1.6). The special fibers of the Néron models of elliptic curves satisfy (2.1.1)(SGA 7 X 1.15).In particular, curves p : C → S of genus one with geometrically integral fibers (= elliptic curves, nodal plane cubics, or cuspidal plane cubics) satisfy (2 1.1).

For every section t of C/S, write m(t) for the ideal sheaf defining the subscheme t(S).

Proposition 2.2. Let S be a noetherian scheme, $\bar{s}$ a geometric point of S localized at s ∈ S, p : C → S satisfying (2.1.1), and $\mathcal{J}$ a coherent sheaf on C flat over S. Suppose that $\mathcal{J}$_{$\bar{s}$} on C_{$\bar{s}$} is isomorphic to a sheaf m(t_{$\bar{s}$}), for t_{$\bar{s}$} ∈ C_{$\bar{s}$}($\bar{s}$). Then, over

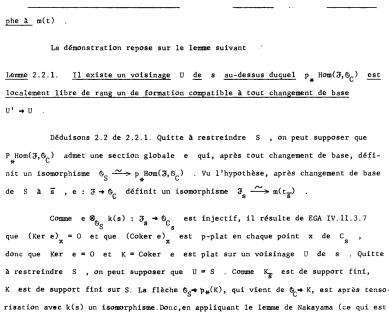
a neighborhood U of s in S, there exists a unique section t ∈ C(U) such that $\mathcal{J}$|_{p^{-1}(U)} is isomor- phic to m(t).

The proof rests on the following lemma.

Lemma 2.2.1. There is a neighborhood U of s over which p_* Hom($\mathcal{J}$,$\mathcal{O}$_C) is locally free of rank one, with formation compatible with every base change U' → U.

Let us deduce Proposition 2.2 from 2.2.1. After shrinking S, suppose that p_* Hom($\mathcal{J}$,$\mathcal{O}$_C) has a global section e which, after every base change, defines an isomorphism $\mathcal{O}$_S ≃ p_* Hom($\mathcal{J}$,$\mathcal{O}$_C). By the hypothesis, after base change from S to s̄, e : $\mathcal{J}$ → $\mathcal{O}$_C defines an isomorphism $\mathcal{J}$_{s̄} ≃ m(t_{s̄}).

Since e ⊗_{$\mathcal{O}$_S} k(s) : $\mathcal{J}$_s → $\mathcal{O}$_{C_s} is injective, EGA IV.11.3.7 gives (Ker e)_x = 0 and says that (Coker e)_x is p-flat at every x of C_s, hence Ker e = 0 and K = Coker e is flat near s. After shrinking S, suppose U = S. Since K_{s̄} has finite support, K has finite support over S. The map $\mathcal{O}$_S → p_*(K), induced by $\mathcal{O}$_C → K, becomes an isomorphism after tensoring with k(s). Thus, by Nakayama's lemma (which is



phe à m(t) .

La démonstration repose sur le lemme suivant .

Lemme 2.2.1. Il existe un voisinage U de s au-dessus duquel $P_*Hom(\mathcal{J},\mathcal{O}_C)$ est localement libre de rang un de formation compatible à tout changement de base U' → U .

Déduisons 2.2 de 2.2.1. Quitte à restreindre S , on peut supposer que $P_*Hom(\mathcal{J},\mathcal{O}_C)$ admet une section globale e qui, après tout changement de base, définit un isomorphisme $\mathcal{O}_S \xrightarrow{\sim} p_*Hom(\mathcal{J},\mathcal{O}_C)$. Vu l'hypothèse, après changement de base de S à $\bar{s}$, e : $\mathcal{J} \to \mathcal{O}_C$ définit un isomorphisme $\mathcal{J}_{\bar{s}} \xrightarrow{\sim} m(t_{\bar{s}})$.

Comme $e \otimes_{\mathcal{O}_S} k(s) : \mathcal{J}_s \to \mathcal{O}_{C_s}$ est injectif, il résulte de EGA IV.11.3.7 que $(\mathrm{Ker}\ e)_x = 0$ et que $(\mathrm{Coker}\ e)_x$ est p-plat en chaque point x de C_s , donc que Ker e = 0 et K = Coker e est plat sur un voisinage U de s . Quitte à restreindre S , on peut supposer que U = S . Comme $K_{\bar{s}}$ est de support fini, K est de support fini sur S. La flèche $\mathcal{O}_S \to p_*(K)$, qui vient de $\mathcal{O}_S \to K$, est après tensorisation avec k(s) un isomorphisme.Donc,en appliquant le lemme de Nakayama (ce qui est

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permissible, since the support of K is finite over S) this morphism is surjective and, because K is flat over S, it is also injective.

After shrinking S, we may therefore suppose that e is injective and that $\mathcal{O}$_S ≃ p_*(K). The subscheme of C defined by e($\mathcal{J}$) is consequently isomorphic to S and defines the unique section t such that $\mathcal{J}$ ≃ m(t).

Let us prove 2.2.1. The long exact cohomology sequence associated with

$0 \to m(t_{\bar{s}}) \to \mathcal{O}_{C_{\bar{s}}} \to \mathcal{O}(t_{\bar{s}}) \to 0$

and (2 1.1) give

(2.2.2) $H^0(C_{\bar{s}},m(t_{\bar{s}})) = 0$

(2.2.3) $\dim_{k(\bar{s})} H^1(C_{\bar{s}},m(t_{\bar{s}})) = 1$.

It is therefore enough to prove the following lemma.

Lemma 2.3. With the notation of 2.2, suppose $H^0(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 0$ and $\dim_{k(\bar{s})} H^1(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 1$. There is then a neighborhood U of s in S over which p_* Hom($\mathcal{J}$,$\mathcal{O}$_C) is locally free of rank one, with formation compatible with every base change U' → U.

Proof. By hypothesis $H^i(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 0$ for i ≠ 1, while $H^1(C_{\bar{s}},\mathcal{J}_{\bar{s}})$ has rank one. It follows that near s, $R^i p_*(\mathcal{J})$ vanishes for i ≠ 1, remains so after every base

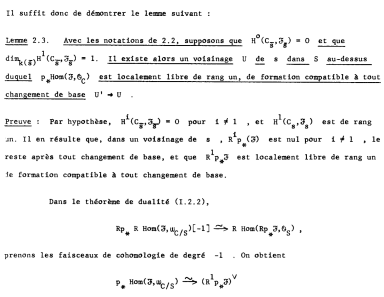
change, and $R^1p_*\mathcal{J}$ is locally free of rank one with formation compatible with every base change.

In the duality theorem (1.2.2),

Rp_* R Hom($\mathcal{J}$,ω_{C/S})[-1] ≃ R Hom(Rp_*$\mathcal{J}$,$\mathcal{O}$_S),

take the cohomology sheaves of degree −1. We obtain

p_* Hom($\mathcal{J}$,ω_{C/S}) ≃ ($R^1p_*\mathcal{J}$)^∨



Il suffit donc de démontrer le lemme suivant :

Lemme 2.3. Avec les notations de 2.2, supposons que $H^0(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 0$ et que $\dim_{k(\bar{s})} H^1(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 1$. Il existe alors un voisinage U de s dans S au-dessus duquel $p_*\mathrm{Hom}(\mathcal{J},\mathcal{O}_C)$ est localement libre de rang un, de formation compatible à tout changement de base U' → U .

Preuve : Par hypothèse, $R^i(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = 0$ pour $i \neq 1$, et $H^1(C_{\bar{s}},\mathcal{J}_{\bar{s}})$ est de rang un. Il en résulte que, dans un voisinage de s, $R^ip_*(\mathcal{J})$ est nul pour $i \neq 1$, le reste après tout changement de base, et que $R^1p_*\mathcal{J}$ est localement libre de rang un de formation compatible à tout changement de base.

Dans le théorème de dualité (I.2.2),

$$Rp_* R\,\mathrm{Hom}(\mathcal{J},\omega_{C/S})[-1] \xrightarrow{\sim} R\,\mathrm{Hom}(Rp_*\mathcal{J},\mathcal{O}_S),$$

prenons les faisceaux de cohomologie de degré −1. On obtient

$$p_*\,\mathrm{Hom}(\mathcal{J},\omega_{C/S}) \xrightarrow{\sim} (R^1p_*\mathcal{J})^\vee$$

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(where ^∨ denotes the dual $\mathcal{O}$_S-module). In particular p_* Hom($\mathcal{J}$,ω_{C/S}) is locally free of rank one, compatible with every base change. It follows from (2.1.1) that p_*ω_{C/S} is locally free of rank one with formation compatible with every base change. Thus (2.1.1) also implies that, locally on S, ω_{C/S} is isomorphic to $\mathcal{O}$_C, and 2.3 follows.

Corollary 2.4. : Let p : C → S be a genus-one curve over S satisfying (2.1.1). Let $\mathcal{L}$ be an invertible sheaf on C defining an element of Pic⁰_{C/S}(S) (cf. I 3.2). For every section t of C over S, there is a unique section t’ such that, locally on S, m(t’) is isomorphic to m(t) ⊗ $\mathcal{L}$.

Proof. We reduce to the case where S is noetherian. By 2.2 we may suppose that S is the spectrum of an algebraically closed field. We treat a “universal” case.

Make the base change to T = C ×_S Pic⁰_{C/S} → S, take t to be the universal section and $\mathcal{L}$ the universal invertible sheaf. For every point of the form (x,1_{Pic}) (x ∈ C) of T the assertion is trivially true. Hence by 2.2 it holds on a neighborhood of C × 1_{Pic} ⊂ T. Since C is proper, there is an open neighborhood U of 1_{Pic} in Pic⁰_{C/S} over which the assertion holds above C × U ⊂ T. The assertion therefore holds on the product of C with the group generated by U, which is all of Pic⁰_{C/S}, and hence on T = C ×_S Pic⁰_{C/S}.

2.5. We write + for the morphism

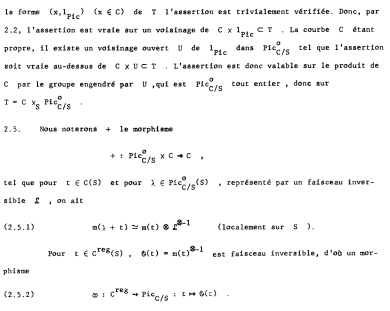
+ : Pic⁰_{C/S} × C → C,

such that, for t ∈ C(S) and λ ∈ Pic⁰_{C/S}(S), represented by an invertible sheaf $\mathcal{L}$,

(2.5.1) m(λ + t) ≃ m(t) ⊗ $\mathcal{L}$^{⊗−1} (locally on S).

For t ∈ C^reg(S), $\mathcal{O}$(t) = m(t)^{⊗−1} is invertible, giving a morphism

(2.5.2) φ : C^reg → Pic_{C/S} : t ↦ $\mathcal{O}$(t).



la forme $(x, 1_{Pic})$ $(x \in C)$ de T l'assertion est trivialement vérifiée. Donc, par 2.2, l'assertion est vraie sur un voisinage de $C \times 1_{Pic} \subset T$. La courbe C étant propre, il existe un voisinage ouvert U de 1_{Pic} dans $Pic^0_{C/S}$ tel que l'assertion soit vraie au-dessus de $C \times U \subset T$. L'assertion est donc valable sur le produit de C par le groupe engendré par U, qui est $Pic^0_{C/S}$ tout entier, donc sur $T = C \times_S Pic^0_{C/S}$.

2.5. Nous noterons $+$ le morphisme

$$+ : Pic^0_{C/S} \times C \to C,$$

tel que pour $t \in C(S)$ et pour $\lambda \in Pic^0_{C/S}(S)$, représenté par un faisceau inversible $\mathcal{L}$, on ait

(2.5.1) $\quad m(\lambda + t) \simeq m(t) \otimes \mathcal{L}^{\otimes -1}$ (localement sur S).

Pour $t \in C^{reg}(S)$, $\Phi(t) = m(t)^{\otimes -1}$ est faisceau inversible, d'où un morphisme

(2.5.2) $\quad \varphi : C^{reg} \to Pic_{C/S} : t \mapsto \Phi(t)$.

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We have

(2.5.3) φ(λ + t) = λ + φ(t).

Theorem 2.6. Let p : C → S be a genus-one curve satisfying (2.1.1).

(i) φ is an open immersion.

(ii) If p has geometrically integral fibers, the image of φ is a torsor under Pic⁰_{C/S}.

(iii) If p has geometrically reduced fibers, the action + of Pic⁰_{C/S} on C induces the trivial action of Pic⁰_{C/S} on Pic^[0]_{C/S}.

Proof. It is enough to prove (i) and (ii) when S is the spectrum of an algebraically closed field k (EGA IV. 17.8.2.). The images under φ of the various connected components of C^reg are then disjoint. Let D be one of them and e ∈ D(k). The maps λ ↦ λ + e and t ↦ φ(t) − φ(e) are inverse isomorphisms between D and Pic⁰. In particular φ is an isomorphism D ≃ φ(e) + Pic⁰, and (i), (ii) follow.

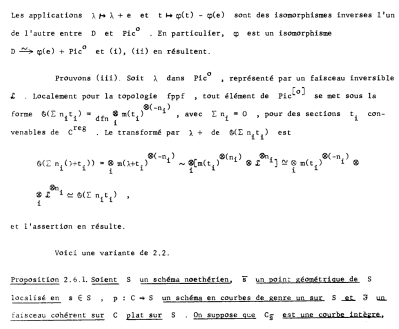
Let us prove (iii). Let λ ∈ Pic⁰, represented by an invertible sheaf $\mathcal{L}$. Locally for the fppf topology, every element of Pic^[0] has the form $\mathcal{O}(\Sigma\ n_i\ t_i) =_{dfn} \otimes_i m(t_i)^{\otimes(-n_i)}$, with Σ n_i = 0, for suitable sections t_i of C^reg. Transforming by λ + gives

$\mathcal{O}(\Sigma\ n_i(\lambda+t_i)) = \otimes_i\ m(\lambda+t_i)^{\otimes(-n_i)} \simeq \otimes_i\ [m(t_i)^{\otimes(n_i)} \otimes \mathcal{L}^{\otimes\ n_i}] \simeq \otimes_i\ m(t_i)^{\otimes(-n_i)} \otimes \otimes_i\ \mathcal{L}^{\otimes\ n_i} \simeq \mathcal{O}(\Sigma\ n_i\ t_i)$,

with Σ n_i = 0, so it remains equivalent to itself.

Here is a variant of 2.2.

Proposition 2.6.1. Let S be noetherian, s̄ a geometric point of S localized at s ∈ S, p : C → S a genus-one curve scheme over S, and $\mathcal{J}$ a coherent sheaf on C flat over S. Suppose that C_{s̄} is an integral curve,



Les applications $\lambda \mapsto \lambda + e$ et $t \mapsto \varphi(t) - \varphi(e)$ sont des isomorphismes inverses l'un de l'autre entre D et Pic^0. En particulier, φ est un isomorphisme $D \xrightarrow{\sim} \varphi(e) + \mathrm{Pic}^0$ et (i), (ii) en résultent.

Prouvons (iii). Soit λ dans Pic^0, représenté par un faisceau inversible $\mathcal{L}$. Localement pour la topologie fppf, tout élément de $\mathrm{Pic}^{[o]}$ se met sous la forme $\mathcal{O}(\sum n_i t_i) \otimes \bigotimes_i m(t_i)^{\otimes(-n_i)}$, avec $\sum n_i = 0$, pour des sections t_i convenables de C^{reg}. Le transformé par $\lambda +$ de $\mathcal{O}(\sum n_i t_i)$ est

$$\mathcal{O}(\textstyle\sum n_i(\cdot + t_i)) \otimes \bigotimes_i m(i+t_i)^{\otimes(-n_i)} \sim \bigotimes_i [m(t_i)^{\otimes(n_i)} \otimes \mathcal{L}^{\otimes n_i}] \simeq \bigotimes_i m(t_i)^{\otimes(-n_i)} \otimes \bigotimes_i \mathcal{L}^{\otimes n_i} \simeq \mathcal{O}(\textstyle\sum n_i t_i),$$

et l'assertion en résulte.

Voici une variante de 2.2.

Proposition 2.6.1. Soient S un schéma noethérien, $\bar{s}$ un point géométrique de S localisé en $s \in S$, $p : C \to S$ un schéma en courbes de genre un sur S et $\mathcal{J}$ un faisceau cohérent sur C plat sur S. On suppose que $C_{\bar{s}}$ est une courbe intègre,

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that $\mathcal{J}_{\bar{s}}$ is torsion-free and generically of rank one, and that $\chi(C_{\bar{s}},\mathcal{J}_{\bar{s}}) = -1$. Then over a neighborhood U of s there exists a unique section $t \in C(U)$ such that $\mathcal{J}|_{p^{-1}(U)}$ is isomorphic to m(t).

As observed in 2.1, C/S satisfies (2.1.1) near s; by 2.2 it is therefore enough to treat the case where S is the spectrum of an algebraically closed field, $S = \bar{s}$.

Under this hypothesis, let us prove $H^0(C,\mathcal{J}) = 0$. If a nonzero $f \in H^0(C,\mathcal{J})$ existed, the sequence

$$0 \to \mathcal{O}_C \to^{f} \mathcal{J} \to \mathcal{J}/\mathcal{O}_C\cdot f \to 0$$

would be exact, since $\mathcal{J}$ is torsion-free on integral C. We would obtain $\chi(C,\mathcal{J}) = \chi(C,\mathcal{O}_C) + \chi(C,\mathcal{J}/\mathcal{O}_C\cdot f) = 0 + \chi(C,\mathcal{J}/f\cdot\mathcal{J}) \geq 0$, contradicting the hypothesis.

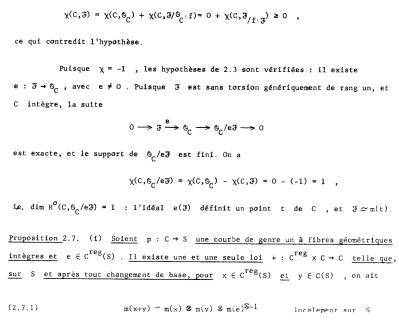
Since $\chi = -1$, the hypotheses of 2.3 hold: there is a nonzero $e : \mathcal{J} \to \mathcal{O}_C$. Since $\mathcal{J}$ is torsion-free of generic rank one and C is integral, the sequence

$$0 \to \mathcal{J} \to^{e} \mathcal{O}_C \to \mathcal{O}_C/e\mathcal{J} \to 0$$

is exact and $\mathcal{O}_C/e\mathcal{J}$ has finite support. We have $\chi(C,\mathcal{O}_C/e\mathcal{J}) = \chi(C,\mathcal{O}_C) - \chi(C,\mathcal{J}) = 0 - (-1) = 1$, i.e. $\dim H^0(C,\mathcal{O}_C/e\mathcal{J}) = 1$: the ideal $e(\mathcal{J})$ defines a point t of C, and $\mathcal{J} \simeq m(t)$.

Proposition 2.7. (i) Let $p : C \to S$ be a genus-one curve with geometrically integral fibers and $e \in C^{reg}(S)$. There is a unique law $+ : C^{reg} \times C \to C$ such that, over S and after every base change, for $x \in C^{reg}(S)$ and $y \in C(S)$,

(2.7.1) $m(x+y) \simeq m(x) \otimes m(y) \otimes m(e)^{\otimes -1}$ locally on S



$\chi(C,\mathcal{J}) = \chi(C,\mathcal{O}_C) + \chi(C,\mathcal{J}/\mathcal{O}_C\cdot f) = 0 + \chi(C,\mathcal{J}/f\cdot\mathcal{J}) \geq 0$,

ce qui contredit l'hypothèse.

Puisque $\chi = -1$, les hypothèses de 2.3 sont vérifiées : il existe $e : \mathcal{J} \to \mathcal{O}_C$, avec $e \neq 0$. Puisque $\mathcal{J}$ est sans torsion génériquement de rang un, et C intègre, la suite

$$0 \longrightarrow \mathcal{J} \xrightarrow{e} \mathcal{O}_C \longrightarrow \mathcal{O}_C/e\mathcal{J} \longrightarrow 0$$

est exacte, et le support de $\mathcal{O}_C/e\mathcal{J}$ est fini. On a

$$\chi(C,\mathcal{O}_C/e\mathcal{J}) = \chi(C,\mathcal{O}_C) - \chi(C,\mathcal{J}) = 0 - (-1) = 1,$$

i.e. $\dim H^0(C,\mathcal{O}_C/e\mathcal{J}) = 1$: l'idéal $e(\mathcal{J})$ définit un point t de C, et $\mathcal{J} \simeq m(t)$.

Proposition 2.7. (i) Soient $p : C \to S$ une courbe de genre un à fibres géométriques intègres et $e \in C^{reg}(S)$. Il existe une et une seule loi $+ : C^{reg} \times C \to C$ telle que, sur S et après tout changement de base, pour $x \in C^{reg}(S)$ et $y \in C(S)$, on ait

(2.7.1) $m(x+y) \simeq m(x) \otimes m(y) \otimes m(e)^{\otimes -1}$ localement sur S

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(ii) This law makes C^reg a commutative group scheme with identity e acting on C; for this action, C^reg acts trivially on Pic⁰_{C/S}.

(iii) The symmetry x ↦ −x of C^reg extends to C and, for t ∈ C(S), locally on S,

(2.7.2) m(−t) ≃ Hom(m(t),$\mathcal{O}$(−2(e))).

(iv) If C/S is stable, + is the unique generalized elliptic-curve structure on C with identity e.

Proof. We may suppose S noetherian. By 2.1, condition (2.1.1) holds. Assertion (i) follows from 2.4. By 2.6, the morphism

φ_e : C^reg → Pic⁰_{C/S} : x ↦ φ(x) − φ(e)

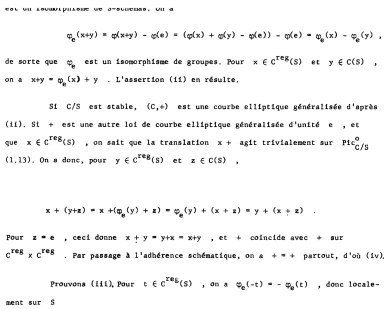
is an isomorphism of S-schemes. We have φ_e(x+y) = φ(x+y) − φ(e) = (φ(x) + φ(y) − φ(e)) − φ(e) = φ_e(x) − φ_e(y), so φ_e is a group isomorphism. For x ∈ C^reg(S) and y ∈ C(S), x+y = φ_e(x) + y, which proves (ii).

If C/S is stable, (C,+) is a generalized elliptic curve by (ii). If +′ is another generalized elliptic-curve law with identity e and x ∈ C^reg(S), translation by x +′ acts trivially on Pic⁰_{C/S} (1.13). Thus, for y ∈ C^reg(S) and z ∈ C(S),

x +′ (y+z) = x +′ (φ_e(y) + z) = φ_e(y) + (x +′ z) = y + (x +′ z).

Putting z = e gives x +′ y = y+x = x+y, and +′ agrees with + on C^reg × C^reg. By schematic closure, +′ = + everywhere, proving (iv).

Let us prove (iii). For t ∈ C^reg(S), φ_e(−t) = −φ_e(t), and therefore local- ly on S



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$\mathcal{O}$((-t)) ⊗ $\mathcal{O}$(-(e)) ≃ $\mathcal{O}$(-(t)) ⊗ $\mathcal{O}$((e)), i.e.

m(-t) ≃ m(t)^{⊗−1} ⊗ $\mathcal{O}$(-2(e)).

It remains to prove that, for arbitrary t, the second member of (2.7.2) is locally on S of the form m(x). The exact sequence

(2.7.3) 0 → m(t) → $\mathcal{O}$_C → t_*$\mathcal{O}$_S → 0

gives the exact sequence

(2.7.4) 0 → Hom(t_*$\mathcal{O}$_S,$\mathcal{O}$(-2(e))) → Hom($\mathcal{O}$_C,$\mathcal{O}$(-2(e))) →

→ Hom(m(t),$\mathcal{O}$(-2(e))) → Ext¹(t_*$\mathcal{O}$_S,$\mathcal{O}$(-2(e))) → 0.

Since ω_{C/S} is an invertible sheaf on C (cf. 1.6.1), $\mathcal{O}$(-2(e)) is locally on C isomorphic to ω_{C/S}. Hence, locally on C,

$\mathrm{Ext}^1(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) \simeq \mathrm{Ext}^1(t_*\mathcal{O}_S,\omega_{C/S})$.

We now apply the coherent-sheaf duality formula for the morphism $t : S \to C$:

(2.7.5) $R\,t_*\,R\,\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{O}_S,t^!\omega_{C/S}) \tilde{\to} R\,\mathrm{Hom}_{\mathcal{O}_C}(R\,t_*\mathcal{O}_S,\omega_{C/S})$.

Observe that

$t^!\omega_{C/S} = t^!p^!\mathcal{O}_S[1] \simeq (p\circ t)^!\mathcal{O}_S[1] = \mathcal{O}_S[1]$.

Moreover, $R^1t_*\mathcal{O}_S = 0$. Taking H^0 of the two sides of (2.7.5), we obtain

$\mathrm{Ext}^1(t_*\mathcal{O}_S,\omega_{C/S}) \simeq t_*\mathcal{O}_S$.

Furthermore, it is easy to see that $\mathrm{Hom}(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) = 0$. The resulting exact sequence

(2.7.6) $0 \to \mathcal{O}(-2(e)) \to \mathrm{Hom}(m(t),\mathcal{O}(-2(e))) \to t_*\mathcal{O}_S \to 0$

shows that $\mathrm{Hom}(m(t),\mathcal{O}(-2(e)))$ is flat over S, that its formation commutes with every base change $S' \to S$, and that its fibers are generically of rank 1 and torsion-free. To see that this sheaf is locally on S' of the form m(t'), it is enough to see that, for every geometric point $\bar{s}$ of S',

$\chi(C_{\bar{s}},\mathrm{Hom}(m(t),\mathcal{O}(-2(e)))) = -1$ (cf. 2.6).

But

$\deg_{C_{\bar{s}}}(\mathcal{O}(-2(e))) = -2$,

so

$\chi(C_{\bar{s}},\mathcal{O}(-2(e))) = -2$, and $\chi(C_{\bar{s}},t_*\mathcal{O}_{\bar{s}}) = 1$.

Thus indeed

$\mathcal{O}((-t))\mathcal{O}(-(e)) \simeq \mathcal{O}(-(t))\mathcal{O}((e))$, i.e.

$m(-t) \simeq m(t)^{\otimes -1} \otimes \mathcal{O}(-2(e))$.

Il reste à prouver que pour t quelconque, le second membre de (2.7.2) est, localement sur S , de la forme m(x) . La suite exacte

(2.7.3) $0 \longrightarrow m(t) \longrightarrow \mathcal{O}_C \longrightarrow t_*\mathcal{O}_S \longrightarrow 0$

donne la suite exacte

(2.7.4) $0 \longrightarrow \mathrm{Hom}(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) \longrightarrow \mathrm{Hom}(\mathcal{O}_C,\mathcal{O}(-2(e))) \longrightarrow$

$\longrightarrow \mathrm{Hom}(m(t),\ \mathcal{O}(-2(e))) \longrightarrow \mathrm{Ext}^1(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) \longrightarrow 0$.

Comme $\omega_{C/S}$ est un faisceau inversible sur C (cf. 1.6.1) , $\mathcal{O}(-2(e))$ est, localement sur C, isomorphe à $\omega_{C/S}$. Donc localement sur C $\mathrm{Ext}^1(t_*\mathcal{O}_S,\mathcal{O}(-2(e))) \simeq \mathrm{Ext}^1(t_*\mathcal{O}_S,\omega_{C/S})$. On applique maintenant la formule de dualité des faisceaux cohérents pour le morphisme t : S → C :

(2.7.5) $Rt_*R\,\mathrm{Hom}_{\mathcal{O}_S}(\mathcal{O}_S,t^!\omega_{C/S}) \xrightarrow{\sim} R\,\mathrm{Hom}_{\mathcal{O}_C}(Rt_*\mathcal{O}_S,\omega_{C/S})$.

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$\chi(C_{\bar{s}},\mathrm{Hom}(m(t),\mathcal{O}(-2(e)))) = -1$,

and consequently $\mathrm{Hom}(m(t),\mathcal{O}(-2(e))) = m(t')$ for a section t', which completes the proof.

Corollary 2.8. Let $p : C \to S$ be a generalized elliptic curve over S. The map

$x \mapsto -x : C^{\mathrm{reg}} \to C^{\mathrm{reg}}$

extends to an automorphism of C/S.

The required extension is unique because C^reg is schematically dense in C. The problem is therefore local for the fpqc topology. It is trivial over the open subset where C is smooth. Let s̄ be a geometric point of S, localized at s ∈ S, such that C_{s̄} is an n-gon. Near this point C_n is finite and flat and, locally for the fppf topology, admits a section t whose multiples meet the various components of C_{s̄}. Near s, t generates a subgroup H of C_n isomorphic to ℤ/n. The group H_{s̄} acts freely on C_{s̄}, and C_{s̄}/H is a 1-gon. On a neighborhood of s we then have:

a) H acts freely on C, and C/H is a generalized elliptic curve over S.

b) C/H has geometrically irreducible fibers (apply 1.15 to C/H).

c) C is a finite étale covering of C/H.

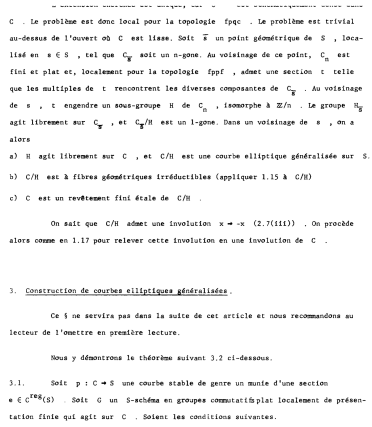
We know that C/H admits an involution x ↦ -x (2.7(iii)). We then proceed as in 1.17 to lift this involution to an involution of C.

3. Construction of generalized elliptic curves.

This section will not be used later in this article, and we recommend that the reader omit it on a first reading.

Here we prove Theorem 3.2 below.

3.1. Let p : C → S be a stable genus-one curve equipped with a section e ∈ C^reg(S). Let G be a commutative S-group scheme, flat and locally of finite presentation, acting on C. Consider the following conditions.



C . Le problème est donc local pour la topologie fpqc . Le problème est trivial au-dessus de l'ouvert où C est lisse. Soit $\bar{s}$ un point géométrique de S , localisé en s ∈ S , tel que $C_{\bar{s}}$ soit un n-gone. Au voisinage de ce point, C_n est fini et plat et, localement pour la topologie fppf , admet une section t telle que les multiples de t rencontrent les diverses composantes de $C_{\bar{s}}$. Au voisinage de s , t engendre un sous-groupe H de C_n , isomorphe à ℤ/n . Le groupe $H_{\bar{s}}$ agit librement sur $C_{\bar{s}}$, et $C_{\bar{s}}/H$ est un 1-gone. Dans un voisinage de s , on a alors

a) H agit librement sur C , et C/H est une courbe elliptique généralisée sur S.

b) C/H est à fibres géométriques irréductibles (appliquer 1.15 à C/H)

c) C est un revêtement fini étale de C/H .

On sait que C/H admet une involution x ↦ -x (2.7(iii)) . On procède alors comme en 1.17 pour relever cette involution en une involution de C .

3. Construction de courbes elliptiques généralisées.

Ce § ne servira pas dans la suite de cet article et nous recommandons au lecteur de l'omettre en première lecture.

Nous y démontrons le théorème suivant 3.2 ci-dessous.

3.1. Soit p : C → S une courbe stable de genre un munie d'une section e ∈ $C^{reg}(S)$. Soit G un S-schéma en groupes commutatifs plat localement de présentation finie qui agit sur C . Soient les conditions suivantes.

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(3.1.1) G acts trivially on Pic^o_{C/S}.

(3.1.2) For every geometric point s̄ of S, G(s̄) acts transitively on the set of irreducible components of C_{s̄}.

In the important special case where G is the constant group scheme ℤ, the action ζ of G is defined by the automorphism g = ζ(1) of C, and (3.1.1) and (3.1.2) become

(3.1.3) g acts trivially on Pic^o_{C/S}

(3.1.4) For every geometric point s̄ of S, g transitively permutes the irreducible components of S_{s̄}.

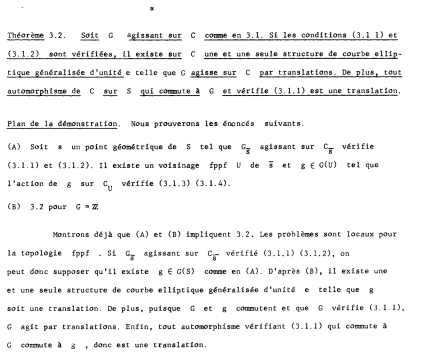
Theorem 3.2. Let G act on C as in 3.1. If conditions (3.1.1) and (3.1.2) hold, there is on C a unique generalized-elliptic-curve structure with identity e for which G acts on C by translations. Moreover, every automorphism of C over S that commutes with G and satisfies (3.1.1) is a translation.

Plan of the proof. We shall prove the following assertions.

(A) Let s be a geometric point of S such that G_{s̄} acting on C_{s̄} satisfies (3.1.1) and (3.1.2). There are an fppf neighborhood U of s̄ and an element g ∈ G(U) such that the action of g on C_U satisfies (3.1.3) and (3.1.4).

(B) Theorem 3.2 for G = $\mathbb{Z}$.

Let us first show that (A) and (B) imply 3.2. The questions are local for the fppf topology. If G_{s̄} acting on C_{s̄} satisfies (3.1.1) and (3.1.2), we may therefore suppose that there is a g ∈ G(S) as in (A). By (B), there is a unique generalized-elliptic-curve structure with identity e for which g is a translation. Moreover, since G and g commute and G satisfies (3.1.1), G acts by translations. Finally, every automorphism satisfying (3.1.1) that commutes with G also commutes with g, and hence is a translation.



Théorème 3.2. Soit G agissant sur C comme en 3.1. Si les conditions (3.1.1) et (3.1.2) sont vérifiées, il existe sur C une et une seule structure de courbe elliptique généralisée d'unité e telle que G agisse sur C par translations. De plus, tout automorphisme de C sur S qui commute à G et vérifie (3.1.1) est une translation.

Plan de la démonstration. Nous prouverons les énoncés suivants.

(A) Soit s un point géométrique de S tel que $G_{\bar{s}}$ agissant sur $C_{\bar{s}}$ vérifie (3.1.1) et (3.1.2). Il existe un voisinage fppf U de $\bar{s}$ et $g \in G(U)$ tel que l'action de g sur C_U vérifie (3.1.3) (3.1.4).

(B) 3.2 pour $G = \mathbb{Z}$.

Montrons déjà que (A) et (B) impliquent 3.2. Les problèmes sont locaux pour la topologie fppf. Si $G_{\bar{s}}$ agissant sur $C_{\bar{s}}$ vérifié (3.1.1) (3.1.2), on peut donc supposer qu'il existe $g \in G(S)$ comme en (A). D'après (B), il existe une et une seule structure de courbe elliptique généralisée d'unité e telle que g soit une translation. De plus, puisque G et g commutent et que G vérifie (3.1.1), G agit par translations. Enfin, tout automorphisme vérifiant (3.1.1) qui commute à G commute à g, donc est une translation.

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Proof of (A). If C_{s̄} is smooth, take for U the open subset of S over which C is smooth, and take g = e. Otherwise C_{s̄} is an n-gon, and the image of G(s̄) in the group of automorphisms of the component diagram Γ(C_{s̄}) has order at least n. By (3.1.1), G(s̄) acts by rotations on Γ(C_{s̄}). Since the rotation group $\mathrm{Aut}^+(\Gamma(C_{\bar{s}}))$ is cyclic of order n, there is a g_{s̄} ∈ G(s) whose image generates this group, and g_{s̄} transitively permutes the irreducible components of C_{s̄}. Locally for the fppf topology there is a section g of G inducing g_{s̄}. We may apply 1.16 to g to verify (3.1.4) near s̄. That g satisfies 3.1.3 near s̄ follows from 1.13.

Proof of 3.2 for G = $\mathbb{Z}$.

Let ζ be the action of G on C and let g = ζ(1). Applying 1.16 to g, we reduce to the case in which the geometric fibers of C are all smooth or n-gons (with n fixed). In this case $g^n e$ lies fiber by fiber in the same connected component of C as e, and g^n is covered by the following lemma.

Lemma 3.3. Let h be an automorphism of C that commutes with g and satisfies (3.1.3). If, fiber by fiber, he lies in the same irreducible component of C as e, then h is of the form

$x \mapsto \lambda + x$

for a section λ of Pic^o_{C/S}.

Proof. We must have λ equal to the class of $\mathcal{O}$(he - e). Hypothesis 3.1.3 ensures that x ↦ λ + x commutes with g. Replacing h by h(x - λ), we reduce to the case he = e. Since h satisfies (3.1.3), for every section μ of Pic^o_{C/S} we have

$h(g^k(\mu+e)) = g^k(\mu+e)$.

Every section of C^reg is locally for the fppf topology of the form $g^k(\mu+e)$. Thus h acts trivially on C^reg, and hence on C.

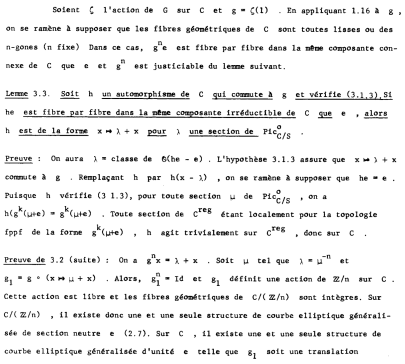
Continuation of the proof of 3.2. We have

$g^n x = \lambda + x$.

Choose μ such that λ = μ^{-n}, and put

$g_1 = g \circ (x \mapsto \mu + x)$.

Then g_1^n = Id and g_1 defines an action of $\mathbb{Z}/n$ on C. This action is free, and the geometric fibers of C/($\mathbb{Z}/n$) are integral. On C/($\mathbb{Z}/n$) there is therefore a unique generalized-elliptic-curve structure with identity section e (2.7). On C there is a unique generalized-elliptic-curve structure with identity e for which g_1 is a translation



Soient ζ l'action de G sur C et g = ζ(1) . En appliquant 1.16 à g , on se ramène à supposer que les fibres géométriques de C sont toutes lisses ou des n-gones (n fixe) Dans ce cas, $g^n e$ est fibre par fibre dans la même composante connexe de C que e et g^n est justiciable du lemme suivant.

Lemme 3.3. Soit h un automorphisme de C qui commute à g et vérifie (3.1.3). Si he est fibre par fibre dans la même composante irréductible de C que e , alors h est de la forme x ↦ λ + x pour λ une section de $Pic^o_{C/S}$.

Preuve : On aura λ = classe de δ(he - e) . L'hypothèse 3.1.3 assure que x ↦ λ + x commute à g . Remplaçant h par h(x - λ) , on se ramène à supposer que he = e . Puisque h vérifie (3 1.3), pour toute section μ de $Pic^o_{C/S}$, on a $h(g^k(\mu+e)) = g^k(\mu+e)$. Toute section de C^{reg} étant localement pour la topologie fppf de la forme $g^k(\mu+e)$, h agit trivialement sur C^{reg} , donc sur C .

Preuve de 3.2 (suite) : On a $g^n x = \lambda + x$. Soit μ tel que $\lambda = \mu^{-n}$ et $g_1 = g \circ (x \mapsto \mu + x)$. Alors, g_1^n = Id et g_1 définit une action de $\mathbb{Z}/n$ sur C . Cette action est libre et les fibres géométriques de C/($\mathbb{Z}/n$) sont intègres. Sur C/($\mathbb{Z}/n$) , il existe donc une et une seule structure de courbe elliptique généralisée de section neutre e (2.7). Sur C , il existe une et une seule structure de courbe elliptique généralisée d'unité e telle que g_1 soit une translation

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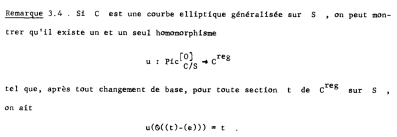
(cf. 1.17). By 3.3, for a section λ of Pic^o and for every generalized-elliptic-curve structure, (λ + e) + x equals λ + x. Thus the law obtained is also the unique one for which g is a translation. Finally, if h commutes with g and satisfies (3.1.3), locally there is a k such that hg^k is covered by 3.3, and h is a translation.

Remark 3.4. If C is a generalized elliptic curve over S, one can show that there is a unique homomorphism

u : Pic^[0]_{C/S} → C^reg

such that, after every base change, for every section t of C^reg over S,

u($\mathcal{O}$((t)-(e))) = t.



Remarque 3.4 . Si C est une courbe elliptique généralisée sur S , on peut montrer qu'il existe un et un seul homomorphisme

$$u : \mathrm{Pic}^{[0]}_{C/S} \to C^{reg}$$

tel que, après tout changement de base, pour toute section t de C^{reg} sur S , on ait

$$u(\delta((t)-(e))) = t \ .$$

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III. Representability theorems.

0. Introduction and notation.

In this chapter and those that follow, we define and study various algebraic stacks classifying generalized elliptic curves equipped with various additional structures.

0.1. Write $\mathfrak{M}$ for the category fibered in groupoids over the category (Sch) of schemes defined as follows: $\mathfrak{M}$(S) is the category whose objects are generalized elliptic curves over S and whose morphisms are S-isomorphisms; the functor ”inverse image by u : S → T” is

E ↦ E ×_T S.

The category $\mathfrak{M}$ is a stack over (Sch) for the fpqc topology (2.1), but it is not an algebraic stack.

0.2. Let $\mathfrak{M}$_* be the following substack of $\mathfrak{M}$: $\mathfrak{M}$_* classifies generalized elliptic curves C/S such that, for every geometric point s̄ of S, the characteristic of k(s̄) does not divide the number of irreducible components of the geometric fiber C_{s̄}. The principal result 2.5 of this chapter is that $\mathfrak{M}$_* is an algebraic stack. The representability theorems of the following chapters will follow from this technical theorem. We shall deduce it from M. Artin’s general criteria. To apply them, the essential point will be to prove the effective prorepresentability of certain functors; this will be done in Section 1.

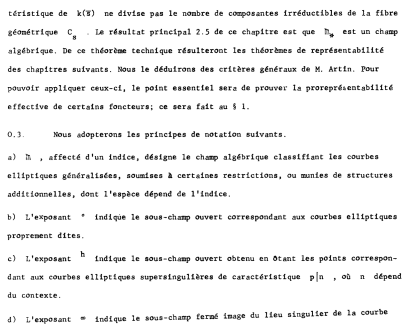
0.3. We adopt the following notational conventions.

a) $\mathfrak{M}$ with a subscript denotes the algebraic stack classifying generalized elliptic curves subject to certain restrictions or equipped with additional structures of the type indicated by the subscript.

b) The superscript o denotes the open substack corresponding to elliptic curves in the strict sense.

c) The superscript h denotes the open substack obtained by removing the points corresponding to supersingular elliptic curves in characteristic p dividing n, where n depends on the context.

d) The superscript ∞ denotes the closed substack that is the image of the singular locus of the curve



téristique de k(S) ne divise pas le nombre de composantes irréductibles de la fibre géométrique $C_{\bar{s}}$. Le résultat principal 2.5 de ce chapitre est que $\mathfrak{M}_g$ est un champ algébrique. De ce théorème technique résulteront les théorèmes de représentabilité des chapitres suivants. Nous le déduirons des critères généraux de M. Artin. Pour pouvoir appliquer ceux-ci, le point essentiel sera de prouver la proreprésentabilité effective de certains foncteurs; ce sera fait au § 1.

0.3. Nous adopterons les principes de notation suivants.

a) $\mathfrak{M}$, affecté d'un indice, désigne le champ algébrique classifiant les courbes elliptiques généralisées, soumises à certaines restrictions, ou munies de structures additionnelles, dont l'espèce dépend de l'indice.

b) L'exposant * indique le sous-champ ouvert correspondant aux courbes elliptiques proprement dites.

c) L'exposant $^{\flat}$ indique le sous-champ ouvert obtenu en ôtant les points correspondant aux courbes elliptiques supersingulières de caractéristique $p|n$, où n dépend du contexte.

d) L'exposant $^{=}$ indique le sous-champ fermé image du lieu singulier de la courbe

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universal.

e) A notation $\mathfrak{m}[1/n]$ indicates either that a stack over $\mathbb{Z}[1/n]$ is being considered or, if $\mathfrak{m}$ has already been defined, denotes $\mathfrak{m} \otimes \mathbb{Z}[1/n]$.

f) The coarse moduli scheme defined by an algebraic stack (I.8) is denoted by the same symbol, with $\mathfrak{m}$ replaced by M.

We shall extend these notations to certain stacks over $\mathfrak{m}$, not necessarily defined by a "moduli problem" stricto sensu.

1. A prorepresentability theorem.

1.1 Let Λ be a complete local ring with residue field k and let C_o be a generalized elliptic curve over k. Assume that C_o is smooth or is the base change to k of the standard generalized elliptic curve with n sides (II.1.9). Let D be the deformation functor of C_o: for a complete local Noetherian Λ-algebra A with residue field k, D(A) is the set of isomorphism classes of generalized elliptic curves C/Spec(A), equipped with an isomorphism $C \otimes_A k \simeq C_o$. This section is devoted to the proof of the following theorem.

Theorem 1.2. Suppose that C_o is smooth or that n is prime to the characteristic exponent p of k. Then:

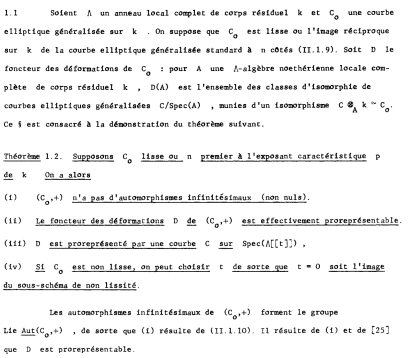
(i) (C_o,+) has no nonzero infinitesimal automorphisms.

(ii) The deformation functor D of (C_o,+) is effectively prorepresentable.

(iii) D is prorepresented by a curve C over $\mathrm{Spec}(\Lambda[[t]])$.

(iv) If C_o is not smooth, t may be chosen so that $t = 0$ is the image of the non-smoothness subscheme.

The infinitesimal automorphisms of (C_o,+) form the group Lie Aut(C_o,+), so (i) follows from (II.1.10). It follows from (i) and [25] that D is prorepresentable.



1.1 Soient Λ un anneau local complet de corps résiduel k et C_o une courbe elliptique généralisée sur k. On suppose que C_o est lisse ou l'image réciproque sur k de la courbe elliptique généralisée standard à n côtés (II.1.9). Soit D le foncteur des déformations de C_o : pour A une Λ-algèbre noethérienne locale complète de corps résiduel k, $D(A)$ est l'ensemble des classes d'isomorphie de courbes elliptiques généralisées $C/\mathrm{Spec}(A)$, munies d'un isomorphisme $C \otimes_A k \simeq C_o$. Ce § est consacré à la démonstration du théorème suivant.

Théorème 1.2. Supposons C_o lisse ou n premier à l'exposant caractéristique p de k. On a alors

(i) $(C_o,+)$ n'a pas d'automorphismes infinitésimaux (non nuls).

(ii) Le foncteur des déformations D de $(C_o,+)$ est effectivement prorepresentable.

(iii) D est prorepresenté par une courbe C sur $\mathrm{Spec}(\Lambda[[t]])$,

(iv) Si C_o est non lisse, on peut choisir t de sorte que $t = 0$ soit l'image du sous-schéma de non lissité.

Les automorphismes infinitésimaux de $(C_o,+)$ forment le groupe Lie $\mathrm{Aut}(C_o,+)$, de sorte que (i) résulte de (II.1.10). Il résulte de (i) et de [25] que D est proreprésentable.

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1.3. The case in which C_o is irreducible.

In this case the deformation functor of (C_o,+) is identified by II.2.7 with the functor D' of deformations of (C_o,e), the pair consisting of the curve C_o and the marked section e. This latter functor is effectively prorepresentable, since a formal scheme in complete curves is always projective and hence algebraizable. That D' satisfies (iii) follows from the two formulas

(1.3.1) $\dim \mathrm{Ext}^1(\Omega^1_{C_o}(e), \mathcal{O}_{C_o}) = 1$

(1.3.2) $\dim \mathrm{Ext}^2(\Omega^1_{C_o}(e), \mathcal{O}_{C_o}) = 0.$

Indeed, Ext^2 is the group in which the obstructions to deformations live. If it vanishes, D' is prorepresented by a formal power-series algebra over Λ. Ext^1 is identified with $D'(k[\varepsilon]/(\varepsilon^2))$ and determines the relative dimension.

Let us prove (1.3.1) and (1.3.2). Since ω_{C_o} is isomorphic to $\mathcal{O}_{C_o}$, duality (I.2.2) gives

$\dim \mathrm{Ext}^i(\Omega^1_{C_o}(e), \mathcal{O}_{C_o}) = \dim H^{1-i}(\Omega^1_{C_o}(e))$.

Formula (1.3.2) follows trivially. For (1.3.1), we may suppose that k is algebraically closed.

If C_o is smooth, $\Omega_{C_o}(e) \simeq \omega_{C_o}(e) \simeq \mathcal{O}_{C_o}(e)$ is an invertible sheaf of degree 1, whence $\dim H^0(C_o, \Omega_{C_o}(e)) = 1$ in this case.

Otherwise, the normalization $\pi : \tilde{C}_o \to C_o$ of C_o is isomorphic to $\mathbb{P}^1$, and a local calculation shows that the canonical morphism

$\Omega_{C_o}(e) \to \pi_* \Omega_{\tilde{C}_o}(e)$

is surjective with a kernel of length one, concentrated at the unique singular point of C_o. In this case the assertion follows from the long exact cohomology sequence: we use that

Ext s'identifie à $D'(k[\varepsilon]/(\varepsilon^2))$, et détermine la dimension relative.

Prouvons (1.3.1) et (1.3.2). Puisque ω_{C_o} est isomorphe à $\mathcal{O}_{C_o}$, on a par dualité (I.2.2)

$$\dim \mathrm{Ext}^1(\Omega^1_{C_o}(e), \mathcal{O}_{C_o}) = \dim H^{1-1}(\Omega^1_{C_o}(e)) .$$

La formule (1.3.2) en résulte trivialement. Pour (1.3.1), on peut supposer k algébriquement clos.

Si C_o est lisse, $\Omega_{C_o}(e) \simeq \omega_{C_o}(e) \simeq \mathcal{O}_{C_o}(e)$ est un faisceau inversible de degré 1, d'où $\dim H^0(C_o, \Omega_{C_o}(e)) = 1$ dans ce cas.

Sinon, le normalisé $\pi : \mathbb{P}^1_k \to C_o$ de C_o est isomorphe à $\mathbb{P}^1$ et un calcul local montre que le morphisme canonique

$$\Omega_{C_o}(e) \longrightarrow \pi_* \Omega_{\mathbb{P}^1}(e)$$

est surjectif à noyau de longueur un, concentré au seul point singulier de C_o. L'assertion dans ce cas résulte de la suite exacte de cohomologie : on utilise que

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$H^0(C, \pi_* \Omega_C(e)) \simeq H^0(\mathbb{P}^1, \Omega_{\mathbb{P}^1}(e)) = 0$ (since $\deg_{\mathbb{P}^1}(e) = -1$).

This proves (iii).

Let us prove (iv). Here C_o is the Néron 1-gon. The deformation theory of ordinary double points shows that there is a $u \in \Lambda[[t]]$ such that $u = 0$ is the image of the non-smooth locus of C. By (II.1.15), over $\mathrm{Spec}(\Lambda[[t]]/(u))$, C is isomorphic to the base change of the standard Néron 1-gon (1.1). The universal property of $\Lambda[[t]]$ then implies that $\mathrm{Spec}(\Lambda[[t]]/(u))$ is unramified over $\mathrm{Spec}(\Lambda)$, and hence that $\Lambda[[t]] \simeq \Lambda[[u]]$. This completes the proof for irreducible C_o.

1.4 The general case.

We may suppose that C_o is the standard Néron n-gon over k, obtained from $\mathbb{P}^1 \times \mathbb{Z}/n$ by gluing. Let H_o be the cyclic subgroup of order n in C_o, the image of $\{1\} \times \mathbb{Z}/n$.

Let D'' be the deformation functor of $(C_o,+,H_o)$: $D''(A)$ is the set of isomorphism classes of systems $(C,+,H)$, where $(C,+)$ is a generalized elliptic curve over A and H is a subgroup of C^{reg} isomorphic to $\mathbb{Z}/n$, equipped with an isomorphism of their special fiber with $(C_o,+,H_o)$.

Lemma 1.4.1. We have $D'' \simeq D$.

This follows because C_n^{reg} is finite étale over A (since n is prime to p).

1.4.2 For the rest of the proof we no longer suppose that n is prime to p (this hypothesis will be used only through 1.4.1). Let $(C,+,H)$ be as above (over a complete local A). The group H acts freely on C (because H_o acts freely on C_o), so C/H (which exists because C is projective) is flat over A. The law $+$ descends to the quotient, and $(C/H,+)$ is a deformation of the Néron 1-gon $(C_o/H_o,+)$. Writing D_1 for the deformation functor of $(C_o/H_o,+)$, we obtain

$\Lambda[[t]] \simeq \Lambda[[u]]$. Ceci achève la démonstration pour C_o irréductible.

1.4 Le cas général.

On peut supposer que C_o est le polygone de Néron standard à n côtés sur k, déduit de $P^1_k \times \mathbb{Z}/n$ par recollement. Soit H_o le sous-groupe cyclique d'ordre n de C_o, image de $\{1\} \times \mathbb{Z}/n$.

Soit D'' le foncteur des déformations de $(C_o, +, H_o)$: $D''(A)$ est l'ensemble des classes d'isomorphie de systèmes $(C,+,H)$ ($(C,+)$ courbe elliptique généralisée sur A, H sous-groupe de C^{reg} isomorphe à $\mathbb{Z}/n$), munis d'un isomorphisme de leur fibre spéciale avec $(C_o,+,H_o)$.

Lemme 1.4.1. On a $D'' \xrightarrow{\sim} D$.

Cela résulte de ce que C_n^{reg} est fini étale sur A (car n est premier à p).

1.4.2 Dans la fin de la démonstration, on ne suppose plus que n est premier à p (cette hypothèse ne servira que via 1.4.1). Soit $(C,+,H)$ comme plus haut (sur A local complet). Le groupe H agit librement sur C (car H_o agit librement sur C_o), de sorte que C/H (qui existe car C est projectif) est plat sur A. La loi $+$ passe au quotient, et $(C/H,+)$ est une déformation du polygone de Néron à n côté $(C_o/H_o,+)$. Notant D_1 le foncteur des déformations de $(C_o/H_o,+)$, on obtient

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in this way a morphism from D” to D_1.

Lemma 1.4.3. The morphism 1.4.2 from D” to D_1 is an isomorphism (here n is not assumed prime to p).

By EGA IV.18.1.2 when A is Artinian, together with the existence theorem in formal geometry, EGA III.5, when A is complete local, for every deformation (C,+) of (C_o/H_o,+) over A there exists one and only one deformation C of the curve C_o over A, equipped with π_o : C_o → C_o/H_o. We then apply II.1.17.

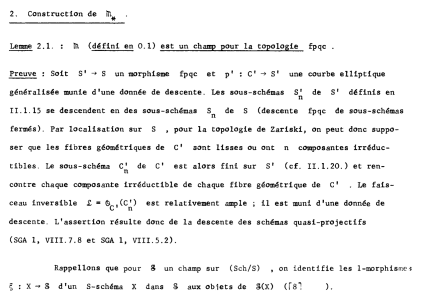
Let us complete the proof of 1.2. By 1.4.1 and 1.4.3, D is isomorphic to D_1, studied in 1.3, and therefore satisfies 1.2(ii) and (iii). Assertion (iv) follows from the analogous assertion for D_1, because the image of the non-smoothness subscheme is the same for C and C/H (notation of 1.4.2).

2. Construction of $\mathfrak{m}$_*.

Lemma 2.1. The stack $\mathfrak{m}$ (defined in 0.1) is a stack for the fpqc topology.

Proof. Let S’ → S be an fpqc morphism and let p’ : C’ → S’ be a generalized elliptic curve equipped with descent data. The subschemes S’_n of S’ defined in II.1.15 descend to subschemes S_n of S (fpqc descent of closed subschemes). By localizing on S for the Zariski topology, we may therefore suppose that the geometric fibers of C’ are smooth or have n irreducible components. The subscheme C’_n of C’ is then finite over S’ (cf. II.1.20) and meets every irreducible component of every geometric fiber of C’. The invertible sheaf $\mathcal{L} = \mathcal{O}$_{C’}(C’_n) is relatively ample and is equipped with descent data. The assertion thus follows from descent for quasi-projective schemes (SGA 1, VIII.7.8 and SGA 1, VIII.5.2).

Recall that for a stack $\mathcal{G}$ over (Sch/S), the 1-morphisms ξ : X → $\mathcal{G}$ from an S-scheme X to $\mathcal{G}$ are identified with the objects of $\mathcal{G}$(X) ([8]).



2. Construction de $\mathfrak{M}_*$.

Lemme 2.1. : $\mathfrak{M}$ (défini en 0.1) est un champ pour la topologie fpqc .

Preuve : Soit S' → S un morphisme fpqc et p' : C' → S' une courbe elliptique généralisée munie d'une donnée de descente. Les sous-schémas S'_n de S' définis en II.1.15 se descendent en des sous-schémas S_n de S (descente fpqc de sous-schémas fermés). Par localisation sur S , pour la topologie de Zariski, on peut donc supposer que les fibres géométriques de C' sont lisses ou ont n composantes irréductibles. Le sous-schéma C'_n de C' est alors fini sur S' (cf. II.1.20.) et rencontre chaque composante irréductible de chaque fibre géométrique de C' . Le faisceau inversible $\mathcal{L} = \mathcal{O}_{C'}(C'_n)$ est relativement ample ; il est muni d'une donnée de descente. L'assertion résulte donc de la descente des schémas quasi-projectifs (SGA 1, VIII.7.8 et SGA 1, VIII.5.2).

Rappellons que pour $\mathfrak{F}$ un champ sur (Sch/S) , on identifie les 1-morphismes $\xi : X \to \mathfrak{F}$ d'un S-schéma X dans $\mathfrak{F}$ aux objets de $\mathfrak{F}(X)$ ([8]).

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2.2. M. Artin proved a manageable criterion for showing that a functor is representable by an algebraic space ([2], [3]). The same proof gives the following criterion for verifying that a stack is algebraic.

Theorem 2.3 (M. Artin). Let S be a scheme of finite type over a field or over an excellent Dedekind ring. Let $\mathcal{G}$ be a category fibered in groupoids over (Sch/S).

$\mathcal{G}$ is an algebraic stack locally of finite type over S if (and only if):

(0) $\mathcal{G}$ is a stack for the étale topology.

(1) $\mathcal{G}$ is locally of finite presentation; that is, for every filtered projective system of affine S-schemes Spec(A_i), the canonical functor

lim→_i $\mathcal{G}$(Spec(A_i)) → $\mathcal{G}$(lim←_i Spec(A_i))

defines an equivalence of categories.

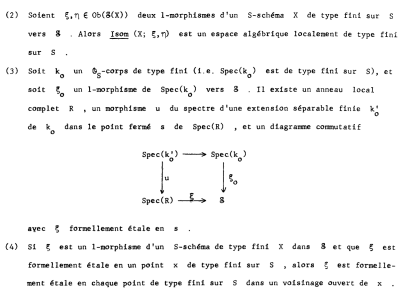
(2) Let ξ,η ∈ Ob($\mathcal{G}$(X)) be two 1-morphisms from an S-scheme X of finite type over S to $\mathcal{G}$. Then Isom(X; ξ,η) is an algebraic space locally of finite type over S.

(3) Let k_o be an $\mathcal{O}$_S-field of finite type (that is, Spec(k_o) is of finite type over S), and let ξ_o be a 1-morphism from Spec(k_o) to $\mathcal{G}$. There exist a complete local ring R, a morphism u from the spectrum of a finite separable extension k'_o of k_o into the closed point s of Spec(R), and a commutative diagram

Spec(k'_o) → Spec(k_o) | | u | | ξ_o ↓ ↓ Spec(R) —ξ—→ $\mathcal{G}$

with ξ formally étale at s.

(4) If ξ is a 1-morphism from an S-scheme X of finite type into $\mathcal{G}$ and ξ is formally étale at a point x of finite type over S, then ξ is formally étale at every point of finite type over S in an open neighborhood of x.


(2) Soient $\xi,\eta \in \mathrm{Ob}(\mathcal{G}(X))$ deux 1-morphismes d'un S-schéma X de type fini sur S vers $\mathcal{G}$. Alors $\underline{\mathrm{Isom}}(X;\xi,\eta)$ est un espace algébrique localement de type fini sur S.

(3) Soit k_o un $\mathcal{O}_S$-corps de type fini (i.e. $\mathrm{Spec}(k_o)$ est de type fini sur S), et soit ξ_o un 1-morphisme de $\mathrm{Spec}(k_o)$ vers $\mathcal{G}$. Il existe un anneau local complet R, un morphisme u du spectre d'une extension séparable finie k'_o de k_o dans le point fermé s de Spec(R), et un diagramme commutatif

$$\begin{array}{ccc} \mathrm{Spec}(k'_o) & \longrightarrow & \mathrm{Spec}(k_o) \\ \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle \xi_o} \\ \mathrm{Spec}(R) & \xrightarrow{\xi} & \mathcal{G} \end{array}$$

avec ξ formellement étale en s.

(4) Si ξ est un 1-morphisme d'un S-schéma de type fini X dans $\mathcal{G}$ et que ξ est formellement étale en un point x de type fini sur S, alors ξ est formellement étale en chaque point de type fini sur S dans un voisinage ouvert de x.

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Remarks 2.4.

(i) If the residue fields of finite type of S are perfect, then, to establish that $\mathcal{G}$ is an algebraic stack, one may replace (0)-(4) by (0), (1), (2), (3'), and (4), where (3') is the following condition.

(3') Let s ∈ S and let k_o be a finite extension of k(s). Suppose that u : Spec(k_o) → S is of finite type. By hypothesis k(s) is then perfect and k_o/k(s) is separable. There is therefore a unique complete local ring Λ(k_o) with residue field k_o, equipped with a formally étale morphism ū : Spec(Λ(k_o)) → S extending u.

For ξ_o ∈ Ob $\mathcal{G}$(k_o), write Def(ξ_o) for the following category over the dual of $\hat{C}$_{Λ(k_o)}: for A ∈ Ob $\hat{C}$_{Λ(k_o)}, an object of Def(ξ_o)(A) is an object ξ of $\mathcal{G}$(A), equipped with an isomorphism ξ_o $\tilde{\to}$ (the image of ξ in $\mathcal{G}$(k_o)). Condition (3') is the conjunction of (3'a) and (3'b):

(3'a) The objects of $\mathcal{G}$ have no infinitesimal automorphisms; more precisely, for k_o and ξ_o as above, and for ξ'_o the pullback of ξ_o to k_o[ε] (ε² = 0), one has Aut(ξ'_o) $\tilde{\to}$

Aut(ξ_o). This condition, together with (2), implies that the objects of Def(ξ_o)(A) have no nontrivial automorphisms.

(3'b) For k_o and ξ_o as above, the functor

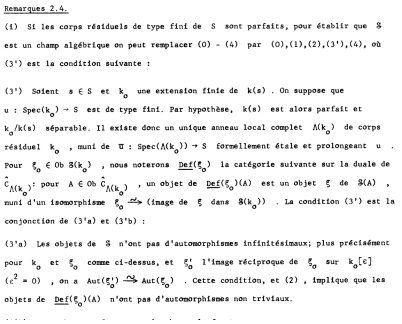
A → the set of isomorphism classes in Def(ξ_o)(A)

is effectively prorepresentable (by a complete local ring R and ξ ∈ $\mathcal{G}$(R) reducing to ξ_o).

It even suffices to verify (3'b) after extension of scalars to a finite extension of k_o.

(ii) If S is of finite type over a field or over an excellent Dedekind ring having infinitely many points, and if the complete local rings R that occur in condition (3'b) are all normal and of the same Krull dimension, condition (4) may be omitted.

(For a reader familiar with [2], (i) is easy to prove. For (ii), see [2], Thm. 3.9).



Remarques 2.4.

(1) Si les corps résiduels de type fini de S sont parfaits, pour établir que S est un champ algébrique on peut remplacer (0) - (4) par (0),(1),(2),(3'),(4), où (3') est la condition suivante :

(3') Soient $s \in S$ et k_o une extension finie de k(s). On suppose que $u : \mathrm{Spec}(k_o) \to S$ est de type fini. Par hypothèse, k(s) est alors parfait et $k_o/k(s)$ séparable. Il existe donc un unique anneau local complet $\Lambda(k_o)$ de corps résiduel k_o, muni de $\pi : \mathrm{Spec}(\Lambda(k_o)) \to S$ formellement étale et prolongeant u. Pour $\xi_o \in \mathrm{Ob}\, \mathcal{S}(k_o)$, nous noterons $\underline{\mathrm{Def}}(\xi_o)$ la catégorie suivante sur la duale de $\hat{C}_{\Lambda(k_o)}$: pour $A \in \mathrm{Ob}\, \hat{C}_{\Lambda(k_o)}$, un objet de $\underline{\mathrm{Def}}(\xi_o)(A)$ est un objet ξ de $\mathcal{S}(A)$, muni d'un isomorphisme $\xi_o \xrightarrow{\sim}$ (image de ξ dans $\mathcal{S}(k_o)$). La condition (3') est la conjonction de (3'a) et (3'b) :

(3'a) Les objets de $\mathcal{S}$ n'ont pas d'automorphismes infinitésimaux; plus précisément pour k_o et ξ_o comme ci-dessus, et ξ'_o l'image réciproque de ξ_o sur $k_o[\varepsilon]$ ($\varepsilon^2 = 0$), on a $\mathrm{Aut}(\xi'_o) \xrightarrow{\sim} \mathrm{Aut}(\xi_o)$. Cette condition, et (2), implique que les objets de $\underline{\mathrm{Def}}(\xi_o)(A)$ n'ont pas d'automorphismes non triviaux.

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The standard Néron n-gon defines a morphism

f_n : Spec($\mathbb{Z}$[1/n]) → $\mathfrak{m}$^*.

Theorem 2.5.

(i) $\mathfrak{m}$_* (defined in 0.2) is an algebraic stack smooth over Spec($\mathbb{Z}$).

(ii) The image $\mathfrak{m}$_*^∞ of the non-smoothness scheme of the universal generalized elliptic curve over $\mathfrak{m}$_* is the union of the images of f_n (n ≥ 1).

Proof. To prove that $\mathfrak{m}$_* is an algebraic stack, we use criterion 2.3 (with S = Spec($\mathbb{Z}$)). Let us review conditions (0) through (4).

(0) follows from 2.1.

(1) follows from standard arguments in EGA IV.8.8.

(2) Let X be an S-scheme of finite type and let (p_1 : C_1 → X,+), (p_2 : C_2 → X,+) be two generalized elliptic curves. Let us prove that Isom_X((C₁,+),(C₂,+)) is representable. This functor is an fpqc sheaf (2.1), so the problem is local. We may therefore suppose that the geometric fibers of C_1 are smooth or n-sided, for a suitable n, and that there is a φ : $\mathbb{Z}$/n ↪ C_1 whose image meets every irreducible component of every geometric fiber of C_1. The morphism f ↦ f∘φ(1) maps Isom into the open subscheme of C_n^reg consisting of those g for which the g^k meet all components of the geometric fibers. The following assertion implies that this morphism is relatively representable and hence that Isom is representable.

(*) For C_i with smooth or n-sided fibers, equipped with H_i ⊂ C_i^reg, with H_i isomorphic to $\mathbb{Z}/n$ and meeting all components of all geometric fibers (i=1,2), Isom((C₁,H₁,+),(C₂,H₂,+)) is representable.

By II 1.17, the morphism

Isom((C₁,H₁,+),(C₂,H₂,+)) → Isom((C₁/H₁,e),(C₂/H₂,e))

is representable by an étale morphism, while the theory of the Hilbert functor implies that Isom((C₁/H₁,e),(C₂/H₂,e)) is representable.

(avec S = Spec(ℤ)) . Passons en revue les conditions (0) à (4) .

(0) résulte de 2.1.

(1) résulte par des arguments standard de EGA IV.8.8.

(2) Soient X un S-schéma de type fini et $(p_1 : C_1 \to X,+)$, $(p_2 : C_2 \to X,+)$ deux courbes elliptiques généralisées. Prouvons que $\mathrm{Isom}_X((C_1,+),(C_2,+))$ est représentable. Ce foncteur est un faisceau fpqc (2.1) de sorte que le problème est local. On peut donc supposer que les fibres géométriques de C_1 sont lisses ou à n côtés (n convenable) et qu'il existe $\varphi : \underline{\mathbb{Z}/n} \hookrightarrow C_1$ dont l'image rencontre chaque composante irréductible de chaque fibre géométrique de C_1 . Le morphisme $f \to f\circ\varphi(1)$ envoye $\underline{\mathrm{Isom}}$ dans le sous-schéma ouvert de C_n^{reg} formé des g tels que les g^k rencontrent toutes les composantes des fibres géométriques. Il résulte de l'assertion suivante que ce morphisme est relativement représentable, donc $\underline{\mathrm{Isom}}$ représentable.

(*) Pour C_i à fibres lisses ou à n côté, et muni de $H_i \subset C_i^{reg}$, H_i isomorphe à $\underline{\mathbb{Z}/n}$, rencontrant toutes les composantes de toutes les fibres géométriques (i=1,2), $\mathrm{Isom}((C_1,H_1,+),(C_2,H_2,+))$ est représentable.

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(3’) Let k_o be a Spec(ℤ)-field of finite type; it is therefore a finite field.

(3’)a) follows from 1.2(i).

(3’)b) follows from 1.2(ii).

(4) By 1.2(iii), the deformation functors are prorepresented by regular rings of dimension 2. By 2.4(ii), condition (4) is verified automatically.

The smoothness of $\mathfrak{m}_*$ follows from its formal smoothness in 1.2(iii). The second assertion of 2.5 follows from II 1.15 and 1.2(iv).

Remark 2.6. $\mathfrak{m}_*$ is horribly non-separated! It will follow from (IV.2.2) that the open substack $\mathfrak{m}_1$ of $\mathfrak{m}_*$ classifying generalized elliptic curves C/S with integral geometric fibers is proper and smooth over ℤ. The ”catch-all” $\mathfrak{m}_*$ will be useful to us for compactifying moduli of elliptic curves with level structure.

The open substack $\mathfrak{m}_(n)[1/n]$ of $\mathfrak{m}_*[1/n]$ classifying generalized elliptic curves C/S, with n invertible on S, whose geometric fibers are smooth or Néron n-gons is likewise proper and smooth over Spec(ℤ[1/n]). It is, however, less natural than $\mathfrak{m}_1[1/n]$.

Remarque 2.6. $\mathfrak{M}_*$ est horriblement non séparé! Il résultera de (IV.2.2) que le sous-champ ouvert $\mathfrak{M}_1$ de $\mathfrak{M}_*$ classifiant les courbes elliptiques généralisées C/S à fibres géométriques intègres est propre et lisse sur ℤ . Le "fourre-tout" $\mathfrak{M}_*$ nous sera utile pour compactifier des modules de courbes elliptiques avec niveau.

Le sous-champ ouvert $\mathfrak{M}_{(n)}[1/n]$ de $\mathfrak{M}_*[1/n]$ qui classifie les courbes elliptiques généralisées C/S (n inversible sur S) à fibres géométriques lisses ou des polygones de Néron à n côtés est lui aussi propre et lisse sur Spec(ℤ[1/n]). Il est toutefois moins naturel que $\mathfrak{M}_1[1/n]$.

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IV. Level structures.

In this chapter we define and study the moduli schemes of elliptic curves whose n-division points are equipped with additional structures. We study in detail the structure at infinity of these schemes. Their reduction modulo p, for p|n, will not be considered seriously until the following chapter.

1. Contractions.

1.1. Let p : C → S be a stable curve of genus one, and let H ⊂ C^reg be a finite locally free subscheme over S of the smooth locus of C. Fiber by fiber, one verifies that H is automatically a Cartier divisor on C. We suppose that the fibers of H are nonempty.

Proposition 1.2. There exists a unique S-scheme $\bar{C}$, equipped with u : C → $\bar{C}$, such that

(a) $\bar{C}$ is a stable curve of genus one over S;

(b) for every geometric point $\bar{s}$ of S, $\bar{C}_{\bar{s}}$ is obtained from $C_{\bar{s}}$ by contracting to a point each irreducible component of $C_{\bar{s}}$ that does not meet $H_{\bar{s}}$.

A. Existence. A limit argument reduces us to the case in which S is Noetherian. For every $\bar{s}$ and every n > 0,

$H^0(C_{\bar{s}}, \mathcal{O}(-nH_{\bar{s}})) = 0$.

By duality (I 2.2 and II 1.2), it follows that

$H^1(C_{\bar{s}}, \mathcal{O}(nH_{\bar{s}})) = 0$.

By EGA III 7.8, the $p_*\mathcal{O}(nH)$ for n > 0 are locally free and their formation is compatible with every base change. The same assertion holds for n = 0 (II.1.6). Let G be the homogeneous algebra

$G = \oplus_{n>0} p_* \mathcal{O}(nH)$.

When $C_{\bar{s}}$ is irreducible, $\mathcal{O}(H_{\bar{s}})$ is ample, and $G_{\bar{s}}$ is generated by

$\oplus_{n\le A} H^0(\mathcal{O}(nH_{\bar{s}}))$,

where A is independent of s.

Suppose that $C_{\bar{s}}$ is a Néron polygon, and let $\bar{C}_{\bar{s}}$ be the Néron polygon obtained from it by contracting to a point every irreducible component that

1.1. Soit p : C → S une courbe stable de genre un et H ⊂ C^{reg} un sous-schéma fini localement libre sur S de l'ouvert de lissité de C. On vérifie fibre par fibre que H est automatiquement un diviseur de Cartier sur C. On suppose que les fibres de H sont non vides.

Proposition 1.2. Il existe un et un seul S-schéma $\bar{C}$, muni de u : C → $\bar{C}$, tel que

(a) $\bar{C}$ est une courbe stable de genre un sur S ;

(b) pour tout point géométrique $\bar{s}$ de S, $\bar{C}_{\bar{s}}$ se déduit de $C_{\bar{s}}$ en contractant en un point chacune des composantes irréductibles de $C_{\bar{s}}$ qui ne rencontrent pas $H_{\bar{s}}$.

A. Existence. Un argument de passage à la limite nous ramène à supposer S noethérien. Pour tout $\bar{s}$ et tout n > 0, on a $H^0(C_{\bar{s}}, \mathcal{O}(-nH_{\bar{s}})) = 0$. Par dualité

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does not meet H. One verifies that

$H^0(\bar{C}_{\bar{s}}, \mathcal{O}(n\bar{H}_{\bar{s}})) \xrightarrow{\sim} H^0(C_{\bar{s}}, \mathcal{O}(nH_{\bar{s}}))$.

On $\bar{C}_{\bar{s}}$, $\mathcal{O}(n\bar{H}_{\bar{s}})$ is ample, and $\bar{G}_{\bar{s}}$ is generated by

$\oplus_{n\le B} H^0(\mathcal{O}(nH_{\bar{s}}))$,

where B is independent of s.

The homogeneous algebra G = ⊕_n p_*$\mathcal{O}$(nH) is therefore of finite type, and its homogeneous spectrum $\bar{C}$ is proper and flat over S, with formation compatible with every base change. The natural morphism u : C → $\bar{C}$ satisfies (a) and (b).

B. Uniqueness. Let

u C ————→ $\bar{C}$ ╲ p / $\bar{p}$ ↘ ↙ S

satisfy (i) and (ii). Fiber by fiber, one verifies that u induces an isomorphism

$u^{-1}(\bar{C}^{reg}) \xrightarrow{\sim} C^{reg}$ (EGA IV 17.8.2).

The arguments of A show that $\mathcal{O}_{\bar{C}}$(H) is ample and that $\bar{G} = \oplus\, \bar{p}_*\mathcal{O}_{\bar{C}}$(nH) is flat and commutes with every base change. Fiber by fiber one verifies that $\bar{G} \xrightarrow{\sim} G$ and that $\bar{C}$ = Sph($\bar{G}$).

Proposition 1.3. Let p : C → S be a generalized elliptic curve whose geometric fibers are smooth or are nm-gons. There exists a unique generalized elliptic curve $\bar{p} : \bar{C} \to S$, equipped with u : C → $\bar{C}$, such that

(i) The geometric fibers of $\bar{C}$ are smooth or n-gons. $\bar{C}_{\bar{s}}$ is obtained from $C_{\bar{s}}$ by contracting to a point the irreducible components of $C_{\bar{s}}$ that are the closures of those connected components of $C_{\bar{s}}^{reg}$ whose order in $\pi_0(C_{\bar{s}}^{reg})$ does not divide n.

(ii) The following diagram is commutative:

$u^{-1}(\bar{C}^{reg}) \times_S C$ —+—→ C
│ u │
u ↓ ↓ $\bar{C}^{reg} \times_S \bar{C}$ —+—→ $\bar{C}$

Proposition 1.3. Soit p : C → S une courbe elliptique généralisée dont les fibres géométriques sont lisses ou sont des nm-gones. Il existe une et une seule courbe elliptique généralisée $\bar{p} : \bar{C} \to S$, munie de u : C → $\bar{C}$, telle que

(i) Les fibres géométriques de $\bar{C}$ sont lisses ou des n-gones. $\bar{C}_{\bar{s}}$ se déduit de $C_{\bar{s}}$ en contractant en un point les composantes irréductibles de $C_{\bar{s}}$ adhérence de celles des composantes connexes de $C_{\bar{s}}^{reg}$ dont l'ordre dans $\pi_0(C_{\bar{s}}^{reg})$ ne divise pas n.

(ii) Le diagramme suivant est commutatif

$$\begin{array}{ccc} u^{-1}(\bar{C}^{reg}) \times_S C & \xrightarrow{+} & C \\ \downarrow u & & u \downarrow \\ \bar{C}^{reg} \times_S \bar{C} & \xrightarrow{+} & \bar{C} \end{array}$$

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Authority page 65; collected-paper page 558; original folio 207.

Locally for the fppf topology on S, there exists an H ⊂ C as in 1.1 that meets exactly those irreducible components of the $C_{\bar{s}}$ that we do not wish to contract; one may take H = C_n, which is finite and flat by (II.1.20). By 1.2, there exists a unique stable curve of genus one $\bar{C}$, equipped with u : C → $\bar{C}$, that satisfies (i). The subgroup $u^{-1}(\bar{C}^{reg})$ of C^{reg} acts on $\bar{C}$ by transport of structure; since $u^{-1}(\bar{C}^{reg}) \xrightarrow{\sim} C^{reg}$, this gives the desired generalized elliptic-curve structure on $\bar{C}$.

Example 1.4. Let C/S be a generalized elliptic curve and apply 1.3 with n = 1. Locally on S, the hypothesis of 1.3 is satisfied for a suitable m (II.1.15). Hence there exists a unique generalized elliptic curve c(C) over S, with irreducible geometric fibers, equipped with u : C → c(C), such that u induces an isomorphism

(1.4.1) $(C^{reg})^0 \xrightarrow{\sim} c(C)^{reg}$.

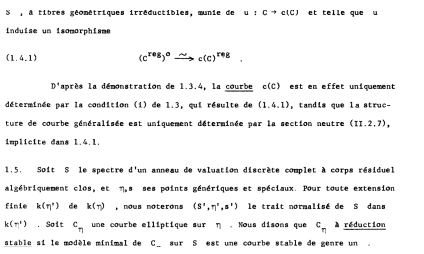
According to the proof of 1.3.4, the curve c(C) is in fact uniquely determined by condition (i) of 1.3, which follows from (1.4.1), while the generalized-curve structure is uniquely determined by the neutral section (II.2.7), implicit in 1.4.1.

1.5. Let S be the spectrum of a complete discrete valuation ring with algebraically closed residue field, and let η and s be its generic and special points. For every finite extension k(η') of k(η), write (S',η',s') for the normalization trait of S in k(η'). Let C_η be an elliptic curve over η. We say that C_η has stable reduction if the minimal model of C_η over S is a stable curve of genus one.

Proposition 1.6. (i) There exists a finite extension k(η') of k(η) such that the curve C_η' = C_η ⊗_η η' has stable reduction over η'.

(ii) If C_η has stable reduction, the minimal model C̃ of C_η over S has a unique generalized elliptic-curve structure extending that of C_η.

(iii) If, in addition, the points of order n of C_η are defined over k(η), then either the minimal model C̃ of C_η is smooth or C̃_s is an m-gon, with n|m, and there



S , à fibres géométriques irréductibles, munie de u : C → c(C) et telle que u induise un isomorphisme

(1.4.1) $(C^{reg}_\eta)^0 \xrightarrow{\sim} c(C)^{reg}_\eta$.

D'après la démonstration de 1.3.4, la courbe c(C) est en effet uniquement déterminée par la condition (i) de 1.3, qui résulte de (1.4.1), tandis que la structure de courbe généralisée est uniquement déterminée par la section neutre (II.2.7), implicite dans 1.4.1.

1.5. Soit S le spectre d'un anneau de valuation discrète complet à corps résiduel algébriquement clos, et η,s ses points génériques et spéciaux. Pour toute extension finie k(η') de k(η) , nous noterons (S',η',s') le trait normalisé de S dans k(η') . Soit C_η une courbe elliptique sur η . Nous disons que C_η à réduction stable si le modèle minimal de C_ sur S est une courbe stable de genre un .

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exists a generalized elliptic curve C over S whose special fiber is smooth or an n-gon, together with an isomorphism α from its generic fiber to C_η.

(iv) The pair (C,α) is unique up to a unique isomorphism.

Proof. (i) This is a theorem of Néron and Kodaira ([4] or [20] or [14]).

(ii) The universal property of the Néron model ensures that e ∈ C_η(η) extends to a section of C̃^reg over S. If u ∈ C̃^reg(S), the translation u+ on C_η extends by transport of structure to an automorphism of C̃; similarly, x → −x extends to C̃.

Let U be the largest open subset of C̃ such that the group law + of C_η extends to + : U × C → C. It is enough to prove that U = C̃^reg. Once this is established, the identities expressing that + is a generalized elliptic-curve structure follow from the schematic density of C_η in C̃. For v ∈ C̃^reg(S), U is stable under translation by v: extend + to v+U by (v+u) + x = v+(u+x). Since U is also stable under étale localization on S, as follows by descent and the compatibility of minimal-model formation with étale localization on S, it is enough to show that U_s ≠ ∅.

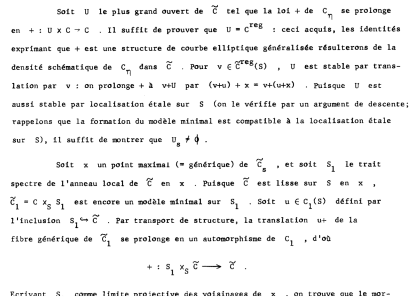
Let x be a maximal, hence generic, point of C̃_s, and let S_1 be the trait given by the spectrum of the local ring of C̃ at x. Since C̃ is smooth over S at x, $\tilde{C}_1 = \tilde{C} \times_S S_1$ is again a minimal model over S_1. Let u ∈ C_1(S) be defined by the inclusion $S_1 \hookrightarrow \tilde{C}$. By transport of structure, translation by u on the generic fiber of $\tilde{C}_1$ extends to an automorphism of C_1, whence

+ : $S_1 \times_S \tilde{C} \to \tilde{C}$.

Writing S_1 as the projective limit of the neighborhoods of x, one finds that the morphism + : C_η × C_η → C_η extends to + : U ×_S C̃ → C̃, with x ∈ U.

(iii) We know that C̃_n is quasi-finite and flat. By the hypothesis and the universal property of the Néron model, all geometric points of (C_η̄)_n come from sections of C̃_n

over S. This group is therefore finite over S, and (C̃_s)_n has rank n². Thus the special fiber C̃_s is smooth or an m-gon, with n|m (cf. II 1.13), and 1.3 is applied to obtain C by contraction.



Soit U le plus grand ouvert de $\bar{C}$ tel que la loi + de C_η se prolonge en $+ : U \times C \to C$. Il suffit de prouver que $U = C^{reg}$: ceci acquis, les identités exprimant que + est une structure de courbe elliptique généralisée résulterons de la densité schématique de C_η dans $\bar{C}$. Pour $v \in \bar{C}^{reg}(S)$, U est stable par translation par v : on prolonge $+ \grave{a}$ $v+U$ par $(v+u) + x = v+(u+x)$. Puisque U est aussi stable par localisation étale sur S (on le vérifie par un argument de descente; rappelons que la formation du modèle minimal est compatible à la localisation étale sur S), il suffit de montrer que $U_s \neq \emptyset$.

Soit x un point maximal (= générique) de $\bar{C}_s$, et soit S_1 le trait spectre de l'anneau local de $\bar{C}$ en x. Puisque $\bar{C}$ est lisse sur S en x, $\bar{C}_1 = C \times_S S_1$ est encore un modèle minimal sur S_1. Soit $u \in C_1(S)$ défini par l'inclusion $S_1 \hookrightarrow \bar{C}$. Par transport de structure, la translation $u+$ de la fibre générique de $\bar{C}_1$ se prolonge en un automorphisme de C_1, d'où

$$+ : S_1 \times_S \bar{C} \longrightarrow \bar{C} .$$

Ecrivant S comme limite projective des voisinages de x, on trouve que le mor-

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(iv) Let C be a generalized elliptic curve over S, with generic fiber C_η, and suppose that C_s is smooth or an n-gon. At a non-smooth point u of C, the formal completion of C is isomorphic to the completion at the origin of the affine curve in A_S² with equation xy = t^k, for a suitable k. By homogeneity, k does not depend on the point chosen. For k > 1, C is singular at u. These singularities are resolved by repeatedly blowing up the singular points, a process that replaces u by k−1 projective lines:

u ⋀/⋀ gives ⋀ ···· ···· ···· ⋀

(cf. [8]). Let C̃ be the resolved curve. There is no exceptional curve of the first kind contained in C̃_s, so C̃ is the minimal model of C_η. The curve C is smooth if and only if the minimal model C̃ of C_η is smooth, in which case C = C̃. Otherwise C̃_s is a Néron nk-gon, and C is obtained from it by contraction in the manner used in the existence proof.

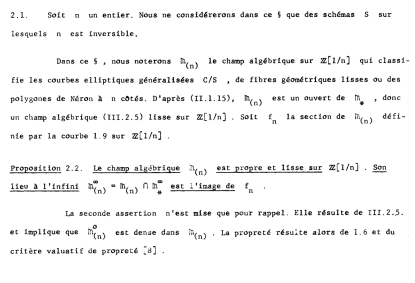
2. Level-n structures.

2.1. Let n be an integer. In this section we consider only schemes S on which n is invertible.

In this section, 𝔪_(n) denotes the algebraic stack over ℤ[1/n] classifying generalized elliptic curves C/S whose geometric fibers are smooth or Néron n-gons. By (II.1.15), 𝔪_(n) is an open substack of 𝔪_* and hence an algebraic stack, smooth over ℤ[1/n] by (III.2.5). Let f_n be the section of 𝔪_(n) defined by curve 1.9 over ℤ[1/n].

Proposition 2.2. The algebraic stack 𝔪_(n) is proper and smooth over ℤ[1/n]. Its locus at infinity 𝔪_(n)^∞ = 𝔪_(n) ∩ 𝔪_*^∞ is the image of f_n.

The second assertion is included only as a reminder. It follows from III.2.5 and implies that 𝔪_(n)^o is dense in 𝔪_(n). Properness then follows from 1.6 and the valuative criterion for properness [8].



2.1. Soit n un entier. Nous ne considérerons dans ce § que des schémas S sur lesquels n est inversible.

Dans ce §, nous noterons $\mathcal{M}_{(n)}$ le champ algébrique sur $\mathbb{Z}[1/n]$ qui classifie les courbes elliptiques généralisées C/S, de fibres géométriques lisses ou des polygones de Néron à n côtés. D'après (II.1.15), $\mathcal{M}_{(n)}$ est un ouvert de $\mathcal{M}_*$, donc un champ algébrique (III.2.5) lisse sur $\mathbb{Z}[1/n]$. Soit f_n la section de $\mathcal{M}_{(n)}$ définie par la courbe I.9 sur $\mathbb{Z}[1/n]$.

Proposition 2.2. Le champ algébrique $\mathcal{M}_{(n)}$ est propre et lisse sur $\mathbb{Z}[1/n]$. Son lieu à l'infini $\mathcal{M}_{(n)}^{\infty} = \mathcal{M}_{(n)} \cap \mathcal{M}_*^{\infty}$ est l'image de f_n.

La seconde assertion n'est mise que pour rappel. Elle résulte de III.2.5. et implique que $\mathcal{M}_{(n)}^{o}$ est dense dans $\mathcal{M}_{(n)}$. La propreté résulte alors de 1.6 et du critère valuatif de propreté [8].

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2.3. For C/S of the type under consideration, C_n^reg is finite étale of rank n^2 (II.1.20), hence locally isomorphic to $(\mathbb{Z}/n)^2$ in the étale topology. A level-n structure on C is an isomorphism

$\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$.

Definition 2.4. $\mathfrak{m}_n[1/n]$ is the following algebraic stack over $\mathbb{Z}[1/n]$: for a scheme S on which n is invertible, $\mathfrak{m}_n[1/n](S)$ is the category of generalized elliptic curves C/S whose geometric fibers are smooth or Néron n-gons, equipped with a level-n structure.

It is clear that $\mathfrak{m}_n[1/n]$ is finite étale and representable over $\mathfrak{m}_(n)$, and is in fact a principal homogeneous space under $GL(2,\mathbb{Z}/n)$. We write $\mathfrak{m}_n^\infty[1/n]$ for the inverse image of $\mathfrak{m}_(n)^\infty$, the image of the non-smoothness subscheme of the universal curve over $\mathfrak{m}_n[1/n]$. The following theorem follows immediately from 2.2.

Theorem 2.5. $\mathfrak{m}_n[1/n]$ is proper and smooth over $\mathbb{Z}[1/n]$, and $\mathfrak{m}_n^\infty[1/n]$ is finite étale over $\mathbb{Z}[1/n]$.

2.6. Recall that an algebraic stack $\mathcal{G}$ "is" an algebraic space when the objects it classifies have no automorphisms. More precisely, if for every algebraically closed field k the objects of $\mathcal{G}(k)$ have no nontrivial automorphisms, then

(a) for every scheme S, the objects of $\mathcal{G}(S)$ have no nontrivial automorphisms;

(b) the functor S ↦ (the set of isomorphism classes in $\mathcal{G}(S)$) is representable by an algebraic space.

Theorem 2.7. If $n \geq 3$, $\mathfrak{m}_n[1/n]$ is an algebraic space.

Let C be a generalized elliptic curve over an algebraically closed field k, either smooth or a Néron n-gon. One must prove that an automorphism σ of C that is trivial on C_n is the identity. When C is an elliptic curve, apply ([26], appendix to Exposé 17). When C is a Néron polygon, II.1.10 shows that Aut(C) acts faithfully on C_n.

2.6 Rappelons qu'un champ algébrique $\mathcal{G}$ "est" un espace algébrique si les objets qu'il classifie n'ont pas d'automorphismes. Plus précisément, si pour tout corps algébriquement clos, k , les objets de $\mathcal{G}(k)$ n'ont pas d'automorphisme non trivial, alors

a) pour tout schéma S , les objets de $\mathcal{G}(S)$ n'ont pas d'automorphisme non trivial;

b) le foncteur S ↦ (l'ensemble des classes d'isomorphie dans $\mathcal{G}(S)$) est représentable par un espace algébrique.

Théorème 2.7. Si $n \geq 3$, $\mathfrak{M}_n[1/n]$ est un espace algébrique.

Soient C une courbe elliptique généralisée sur un corps algébriquement clos k , lisse ou un polygone de Néron à n côtés. Il faut prouver qu'un automorphisme σ de C , trivial sur C_n , est l'identité. Pour C une courbe elliptique, on applique ([26], app. à Exp. 17). Pour C un polygone de Néron, on vérifie sur II.1.10 que Aut(C) agit fidèlement sur C_n .

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Variant 2.8. This argument, together with III.2.1, also immediately implies that the functor

F_n : S ↦ (the set of isomorphism classes of generalized elliptic curves over S equipped with a level-n structure)

is a sheaf for the fpqc topology. The arguments of III.2.5 then show that this functor satisfies M. Artin's representability criterion. To prove that F_n is representable by an algebraic space for n ≥ 3, it is therefore unnecessary first to generalize M. Artin's criterion to algebraic stacks.

Corollary 2.9. If n ≥ 3, $\mathfrak{m}$_n[1/n] is a projective smooth scheme over $\mathbb{Z}$[1/n].

Indeed, a regular algebraic space in curves over $\mathbb{Z}$ is always quasi-projective, and hence a scheme. Here is a more constructive proof: by transcendental methods (cf. §5), one verifies that after extension of scalars to $\mathbb{C}$, $\mathfrak{m}$_n^∞ meets every irreducible component of $\mathfrak{m}$_n. By specialization it follows that the divisor $\mathfrak{m}$_n^∞ meets every irreducible component of every geometric fiber of $\mathfrak{m}$_n; here one uses that $\mathfrak{m}$_n is proper and smooth. Consequently $\mathcal{O}$($\mathfrak{m}$_n^∞) is a relatively ample invertible sheaf.

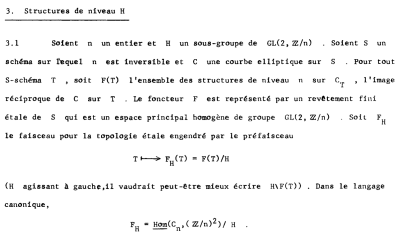
3. Level-H structures.

3.1. Let n be an integer and H a subgroup of GL(2,$\mathbb{Z}$/n). Let S be a scheme on which n is invertible and C an elliptic curve over S. For every S-scheme T, let F(T) be the set of level-n structures on C_T, the inverse image of C over T. The functor F is represented by a finite étale cover of S that is a principal homogeneous space under GL(2,$\mathbb{Z}$/n). Let F_H be the sheaf for the étale topology generated by the presheaf

T ↦ F_H(T) = F(T)/H.

(Since H acts on the left, it might be better to write H\F(T).) In canonical language,

F_H = Hom(C_n,($\mathbb{Z}$/n)²)/H.



3. Structures de niveau H

3.1 Soient n un entier et H un sous-groupe de GL(2, Z/n) . Soient S un schéma sur lequel n est inversible et C une courbe elliptique sur S . Pour tout S-schéma T , soit F(T) l'ensemble des structures de niveau n sur C_T , l'image réciproque de C sur T . Le foncteur F est représenté par un revêtement fini étale de S qui est un espace principal homogène de groupe GL(2, Z/n) . Soit F_H le faisceau pour la topologie étale engendré par le préfaisceau

$$T \longmapsto F_H(T) = F(T)/H$$

(H agissant à gauche,il vaudrait peut-être mieux écrire H\F(T)) . Dans le langage canonique,

$$F_H = \underline{\mathrm{Hom}}(C_n, (\mathbb{Z}/n)^2)/H .$$

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Authority page 70; collected-paper page 563; original folio 212.

This functor is represented by a finite étale cover of S.

A level-H structure on C is an element of F_H(S). An isomorphism α : C_n $\xrightarrow{\sim}$ ($\mathbb{Z}$/n)² defines a level-H structure, and two such isomorphisms define the same structure if and only if, locally on S, there is a g ∈ H such that gα = β. This g is uniquely determined and hence constant on each connected component of S. One should keep in mind that a level-H structure does not in general arise from an α : C_n $\xrightarrow{\sim}$ ($\mathbb{Z}$/n)². It arises from such an α only locally for the étale topology. For example, when H = GL(2,$\mathbb{Z}$/n), C has exactly one level-H structure; it comes from an α only if C_n is isomorphic to ($\mathbb{Z}$/n)² over S.

Definition 3.2. $\mathfrak{m}$_H^o[1/n] is the following algebraic stack: for a scheme S on which n is invertible, $\mathfrak{m}$_H^o[1/n](S) is the category of elliptic curves C/S equipped with a level-H structure, with isomorphisms as morphisms.

It is clear that $\mathfrak{m}$_H^o[1/n] is finite étale and locally representable over $\mathfrak{m}$_1^o[1/n].

Definition 3.3. The algebraic stack $\mathfrak{m}$_H is the normalization of $\mathfrak{m}$_1 in $\mathfrak{m}$_H^o[1/n].

Theorem 3.4. (i) $\mathfrak{m}$_H is proper over Spec($\mathbb{Z}$).

(ii) $\mathfrak{m}_H[1/n]$ is smooth over $\mathrm{Spec}(\mathbb{Z}[1/n])$, and $\mathfrak{m}_H^{o}[1/n]$ is the complement of a substack $\mathfrak{m}_H^{\infty}[1/n]$ that is finite étale over $\mathrm{Spec}(\mathbb{Z}[1/n])$.

Proof. (i) By definition, $\mathfrak{m}_H$ is finite and relatively representable over $\mathfrak{m}_1$. Its properness therefore follows from that of $\mathfrak{m}_1$.

(ii) The assertion follows from Abhyankar's lemma (SGA 1, XIII 5), which applies here because $\mathfrak{m}_1[1/n]$ is smooth over $\mathbb{Z}[1/n]$, because $\mathfrak{m}_1^{\infty}[1/n]$ is a divisor of $\mathfrak{m}_1[1/n]$, smooth over $\mathbb{Z}[1/n]$ and with generic point of characteristic zero, and because $\mathfrak{m}_H^{o}[1/n]$ is finite étale over $\mathfrak{m}_1[1/n] - \mathfrak{m}_1^{\infty}[1/n]$.

In 6.7 we shall give a "modular" proof of 3.4.

(ii) $\mathcal{M}_H[1/n]$ est lisse sur $\mathrm{Spec}\mathbb{Z}[1/n]$, et $\mathcal{M}_H^{o}[1/n]$ est le complément d'un sous-champ $\mathcal{M}_H^{\infty}[1/n]$ fini et étale sur $\mathrm{Spec}\mathbb{Z}[1/n]$).

Preuve (i) Par définition, $\mathcal{M}_H$ est fini et relativement représentable sur $\mathcal{M}_1$. Sa propreté résulte donc de celle de $\mathcal{M}_1$.

(ii) L'assertion résulte du lemme d'Abhyankar (SGA 1,XIII 5) appliquable ici parce que $\mathcal{M}_1[1/n]$, est lisse sur $\mathbb{Z}[1/n]$, que $\mathcal{M}_1^{\infty}[1/n]$ est un diviseur de $\mathcal{M}_1[1/n]$, lisse sur $\mathbb{Z}[1/n]$, et dont le point générique est de caractéristique 0, et que $\mathcal{M}_H^{o}[1/n]$ est fini étale sur $\mathcal{M}_1[1/n] - \mathcal{M}_1^{\infty}[1/n]$.

En 6.7, nous donnerons de 3.4 une démonstration "modulaire".

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For H = {e}, we set $\mathfrak{m}_n = \mathfrak{m}_H$. This notation is justified by the following proposition.

Proposition 3.5. For H = {e}, $\mathfrak{m}_H[1/n]$ is the stack $\mathfrak{m}_n[1/n]$ of 2.4.

Construction 1.4 defines a morphism of stacks

$c : \mathfrak{m}_n[1/n] \to \mathfrak{m}_1[1/n] : (C,\alpha) \mapsto c(C)$.

Since $\mathfrak{m}_n[1/n]$ is normal, indeed smooth over $\mathbb{Z}[1/n]$, and proper over $\mathbb{Z}[1/n]$, and since plainly $\mathfrak{m}_n^{o}[1/n] \simeq \mathfrak{m}_H^{o}[1/n]$, it is enough to prove that c is finite and locally representable. Its fibers are plainly finite. To test local representability, it therefore suffices to verify the following (cf. 2.6).

Lemma 3.5.1. Let k be an algebraically closed field, let C be an object of $\mathfrak{m}_n[1/n](k)$, and let c(C) be its image in $\mathfrak{m}_1(k)$. The natural map $i : \mathrm{Aut}(C) \to \mathrm{Aut}(c(C))$ is injective.

For $n \geq 3$, Aut(C) is trivial (cf. 2.7). For n = 2, $\mathrm{Aut}(C) = \{\pm 1\}$. Finally, for n = 1, i is the identity.

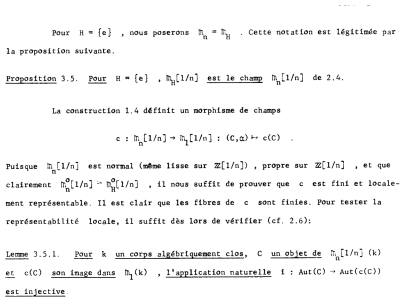
3.6. Let n and m be integers with n|m. Let $H \subset GL(2,\mathbb{Z}/n)$, and let H' be its inverse image in $GL(2,\mathbb{Z}/nm)$. For an elliptic curve E over S, a level-nm structure

$$\begin{array}{ccc} E_{nm} & \xrightarrow{\alpha'}\tilde{\to} & (\mathbb{Z}/nm)^2 \\ \big\downarrow x^m & & \big\downarrow \text{reduction} \\ E_n & \xrightarrow{\alpha}\tilde{\to} & (\mathbb{Z}/n)^2 \end{array}$$

defines a level-n structure α and makes the displayed square commute.

When nm is invertible on S, this construction defines an isomorphism from the sheaf of H'-structures on E to the sheaf of H-structures. Thus

$\mathfrak{m}_{H'}^{o}[1/nm] \simeq \mathfrak{m}_H^{o}[1/n][1/nm]$.



Pour $H = \{e\}$, nous poserons $\mathfrak{M}_n = \mathfrak{M}_H$. Cette notation est légitimée par la proposition suivante.

Proposition 3.5. Pour $H = \{e\}$, $\mathfrak{M}_H[1/n]$ est le champ $\mathfrak{M}_n[1/n]$ de 2.4.

La construction 1.4 définit un morphisme de champs

$$c : \mathfrak{M}_n[1/n] \to \mathfrak{M}_1[1/n] : (C,\alpha) \mapsto c(C) \quad .$$

Puisque $\mathfrak{M}_n[1/n]$ est normal (même lisse sur $\mathbb{Z}[1/n]$) , propre sur $\mathbb{Z}[1/n]$, et que clairement $\mathfrak{M}_n^0[1/n] \simeq \mathfrak{M}_H^0[1/n]$, il nous suffit de prouver que c est fini et localement représentable. Il est clair que les fibres de c sont finies. Pour tester la représentabilité locale, il suffit dès lors de vérifier (cf. 2.6):

Lemme 3.5.1. Pour k un corps algébriquement clos, C un objet de $\mathfrak{M}_n[1/n](k)$ et $c(C)$ son image dans $\mathfrak{M}_1(k)$, l'application naturelle $i : \mathrm{Aut}(C) \to \mathrm{Aut}(c(C))$ est injective

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Since $\mathfrak{m}_H^{o}[1/n]$ is finite étale over $\mathfrak{m}_1^{o}[1/n]$, it is the normalization of $\mathfrak{m}_1^{o}[1/n]$ in $\mathfrak{m}_{H'}^{o}[1/nm]$. It follows that

$$\mathfrak{m}_{H'} \simeq \mathfrak{m}_H.$$

Thus the stack $\mathfrak{m}_H$ depends only on the open subgroup K of $GL(2,\hat{\mathbb{Z}})$ that is the inverse image of H. We shall sometimes denote it by $\mathfrak{m}_K$.

3.7. Let n and m be relatively prime integers, let $H \subset GL(2,\mathbb{Z}/n)$, and let $I \subset GL(2,\mathbb{Z}/m)$. Let K and L be the inverse images of H and I in $GL(2,\hat{\mathbb{Z}})$. Reduction modulo n and m induces an isomorphism

$$GL(2,\mathbb{Z}/nm) \xrightarrow{\sim} GL(2,\mathbb{Z}/n) \times GL(2,\mathbb{Z}/m).$$

Via this isomorphism, $K \cap L$ is the inverse image of $H \times I \subset GL(2,\mathbb{Z}/n) \times GL(2,\mathbb{Z}/m)$.

Construction 3.8. With the preceding notation,

$$\mathfrak{m}_K^{o}[1/n] \times_{\mathfrak{m}_1^{o}} \mathfrak{m}_L^{o}[1/m] \xrightarrow{\sim} \mathfrak{m}_{K\cap L}^{o}[1/nm].$$

Let C/S be an elliptic curve. The maps $x \to x^m$ and $x \to x^n$ define an isomorphism

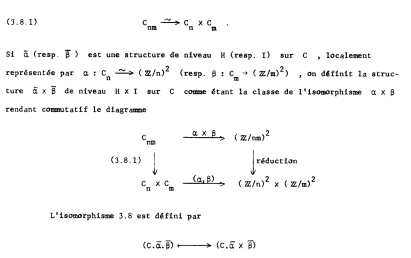
$$C_{nm} \xrightarrow{\sim} C_n \times C_m. \tag{3.8.1}$$

If $\bar{\alpha}$ (respectively $\bar{\beta}$) is a level-H (respectively level-I) structure on C, locally represented by $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ (respectively $\beta : C_m \xrightarrow{\sim} (\mathbb{Z}/m)^2$), define the level-$H \times I$ structure $\bar{\alpha} \times \bar{\beta}$ on C to be the class of the isomorphism $\alpha \times \beta$ that makes the diagram commute:

$$\begin{array}{ccc} C_{nm} & \xrightarrow{\alpha\times\beta} & (\mathbb{Z}/nm)^2 \\ \big\downarrow (3.8.1) & & \big\downarrow \text{réduction} \\ C_n \times C_m & \xrightarrow{(\alpha,\beta)} & (\mathbb{Z}/n)^2 \times (\mathbb{Z}/m)^2 \end{array}$$

The isomorphism in 3.8 is defined by

$$(C,\bar{\alpha},\bar{\beta}) \mapsto (C,\bar{\alpha} \times \bar{\beta}).$$



(3.8.1) $\quad C_{nm} \xrightarrow{\sim} C_n \times C_m$.

Si $\bar{\alpha}$ (resp. $\bar{\beta}$) est une structure de niveau H (resp. I) sur C , localement représentée par $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ (resp. $\beta : C_m \to (\mathbb{Z}/m)^2$) , on définit la structure $\bar{\alpha} \times \bar{\beta}$ de niveau $H \times I$ sur C comme étant la classe de l'isomorphisme $\alpha \times \beta$ rendant commutatif le diagramme

$$\begin{array}{ccc} C_{nm} & \xrightarrow{\alpha\times\beta} & (\mathbb{Z}/nm)^2 \\ \big\downarrow (3.8.1) & & \big\downarrow \text{réduction} \\ C_n \times C_m & \xrightarrow{(\alpha,\beta)} & (\mathbb{Z}/n)^2 \times (\mathbb{Z}/m)^2 \end{array}$$

L'isomorphisme 3.8 est défini par

$$(C,\bar{\alpha},\bar{\beta}) \longmapsto (C,\bar{\alpha} \times \bar{\beta})$$

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Proposition 3.9. With the notation of 3.7, the isomorphism of 3.8 extends to

(3.9.1) $\mathfrak{m}_K \times_{\mathfrak{m}_1} \mathfrak{m}_L \tilde{\to} \mathfrak{m}_{K\cap L}$.

Proof. It is enough to prove that $\mathfrak{m}_K \times_{\mathfrak{m}_1} \mathfrak{m}_L$ is normal. We prove this over $\mathbb{Z}[1/m]$; over $\mathbb{Z}[1/n]$ the argument is symmetric. The stack $\mathfrak{m}_L[1/m]$ is smooth over $\mathbb{Z}$ and hence flat over $\mathfrak{m}_1$ by (I.7.1). Thus $\mathfrak{m}_K \times_{\mathfrak{m}_1} \mathfrak{m}_L[1/m]$ is flat over $\mathfrak{m}_K$ and Cohen-Macaulay by (I.7.2 and I.7.1). Moreover $\mathfrak{m}_L^o[1/m]$ is étale over $\mathfrak{m}_1$, so $\mathfrak{m}_K \times_{\mathfrak{m}_1} \mathfrak{m}_L^o[1/m]$, being étale over $\mathfrak{m}_K$, is normal.

By Abhyankar's lemma, the ramification of $\mathfrak{m}_K[1/n]$ over $\mathfrak{m}_1[1/n]$ (respectively of $\mathfrak{m}_L[1/m]$ over $\mathfrak{m}_1[1/m]$) is tame. The ramification indices for $\mathfrak{m}_K$ and $\mathfrak{m}_L$ are therefore relatively prime; in fact they divide n and m. Hence $\mathfrak{m}_K[1/mn] \times_{\mathfrak{m}_1} \mathfrak{m}_L[1/mn]$ is smooth, and Serre's criterion (I.7.2) completes the proof.

Corollary 3.9.2. If n is the product of two relatively prime integers and $n \geq 3$, then $\mathfrak{m}_n$ is a scheme.

Write $n=n'n''$. The stack $\mathfrak{m}_n[1/n']$ is a scheme because it is representable over $\mathfrak{m}_{n'}[1/n]$. Likewise $\mathfrak{m}_n[1/n'']$ is a scheme; therefore $\mathfrak{m}_n$ is a scheme.

Proposition 3.10. (i) The coarse space M_H defined by $\mathfrak{m}_H$ is proper and flat over $\mathrm{Spec}(\mathbb{Z})$.

(ii) It is the normalization of M_1, which is isomorphic to $\mathbb{P}^1$ by VI.1.1, in $\mathfrak{m}_H^o[1/n]$.

(iii) One has $M_H = M_n/H$.

(iv) The geometric fibers of $M_H[1/n] \to \mathrm{Spec}(\mathbb{Z}[1/n])$ are geometrically unibranch.

Proof. Properness of M_H follows from properness of $\mathfrak{m}_H$. The description in I.8.2 of the strict henselization of a local ring of M_H, together with normality of $\mathfrak{m}_H$, shows that M_H is normal. Similarly M_H is flat over $\mathbb{Z}$. The finite map $\mathfrak{m}_H \to \mathfrak{m}_1$ induces a finite map $M_H \to M_1$, and (ii) follows from (i) and normality.

Soit $n = n'.n''$. Le champ $\mathfrak{M}_H[1/n']$ est un schéma, car représentable sur $\mathfrak{M}_{n'}[1/n]$. De même, $\mathfrak{M}_H[1/n'']$, donc $\mathfrak{M}_n$, est un schéma.

Proposition 3.10. (i) L'espace grossier M_H défini par $\mathfrak{M}_H$ est propre et plat sur $\mathrm{Spec}(\mathbb{Z})$

(ii) C'est le normalisé dans $M_H^0[1/n]$ de M_1 (isomorphe à $\mathbb{P}^1$: voir VI 1.1).

(iii) On a $M_H = M_n/H$.

(iv) Les fibres géométriques de $M_H[1/n] \to \mathrm{Spec}(\mathbb{Z}[1/n])$ sont géométriquement unibranches.

Preuve. La propreté de M_H résulte de celle de $\mathfrak{M}_H$. De la description de l'hensélisé strict d'un anneau local de M_H donnée en I.8.2, et de la normalité de $\mathfrak{M}_H$, il résulte que M_H est normal. De même, M_H est plat sur $\mathbb{Z}$. Puisque $\mathfrak{M}_H \to \mathfrak{M}_1$, donc $M_H \to M_1$ est fini, (ii) résulte de (i) et de la normalité.

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It remains to prove (iii) and (iv) when $n \geq 3$, by 3.6, in which case $M_n^o[1/n] = \mathfrak{m}_n^o[1/n]$. With the notation of [8], §4, one has

$\mathfrak{m}_H^o[1/n] = [M_n^o[1/n]/H]$,

and consequently $M_H^o[1/n] = M_n^o[1/n]/H$. Assertion (iii) now follows from (ii).

The map

$(M_n \otimes \bar{F}_p)/H \to (M_n/H) \otimes \bar{F}_p = M_H \otimes \bar{F}_p$

is radicial. If p does not divide n, then $M_n \otimes \bar{F}_p$ is smooth, so $(M_n \otimes \bar{F}_p)/H$ is smooth,

proving (iv).

3.11. The category of elliptic curves up to isogeny over a scheme S is obtained from the category of elliptic curves by inverting isogenies. Writing E⊗ℚ for the elliptic curve up to isogeny over S underlying E/S, one has

Hom_S(E⊗ℚ,F⊗ℚ)=Hom_S(E,F)⊗ℚ.

3.12. Let k be an algebraically closed field of characteristic zero. For every elliptic curve E over k, put

$\hat{T}(E)=\lim\leftarrow E_n$ (isomorphic to $\hat{\mathbb{Z}}^2$),

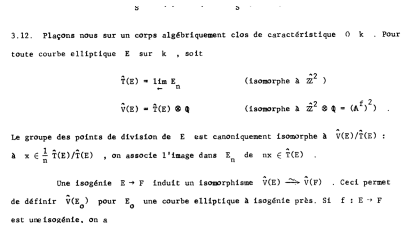
$\hat{V}(E)=\hat{T}(E)\otimes\mathbb{Q}$ (isomorphic to $\hat{\mathbb{Z}}^2\otimes\mathbb{Q}=(\mathbb{A}^f)^2$).

The group of division points of E is canonically isomorphic to $\hat{V}(E)/\hat{T}(E)$: to $x\in(1/n)\hat{T}(E)/\hat{T}(E)$ one associates the image in E_n of $nx\in\hat{T}(E)$.

An isogeny E→F induces an isomorphism $\hat{V}(E)\tilde{\to}\hat{V}(F)$. This permits one to define $\hat{V}(E_o)$ for an elliptic curve E_o up to isogeny. If f:E→F is an isogeny, then

$\mathrm{Ker}(f)=\mathrm{Ker}(\hat{V}(E)/\hat{T}(E)\to\hat{V}(F)/\hat{T}(F))=\mathrm{coker}(\hat{T}(E)\to\hat{T}(F))$.

The assignment $f\mapsto\hat{T}(F)\subset\hat{V}(E)$ identifies isogenies with source E with the lattices $\hat{T}'$ containing $\hat{T}(E)$. It follows that the functor



3.12. Plaçons nous sur un corps algébriquement clos de caractéristique 0 k . Pour toute courbe elliptique E sur k , soit

$\hat{T}(E) = \varprojlim E_n$ (isomorphe à $\hat{\mathbb{Z}}^2$)

$\hat{V}(E) = \hat{T}(E)\otimes\mathbb{Q}$ (isomorphe à $\hat{\mathbb{Z}}^2\otimes\mathbb{Q} = (\mathbb{A}^f)^2$).

Le groupe des points de division de E est canoniquement isomorphe à $\hat{V}(E)/\hat{T}(E)$: à $x \in \frac{1}{n}\hat{T}(E)/\hat{T}(E)$, on associe l'image dans E_n de $nx \in \hat{T}(E)$.

Une isogénie $E \to F$ induit un isomorphisme $\hat{V}(E) \xrightarrow{\sim} \hat{V}(F)$. Ceci permet de définir $\hat{V}(E_o)$ pour E_o une courbe elliptique à isogénie près. Si $f : E \to F$ est une isogénie, on a

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E ↦ (E⊗ℚ, $\hat{T}(E)\subset\hat{V}(E\otimes\mathbb{Q})$)

from elliptic curves to elliptic curves E_o up to isogeny equipped with a lattice $\hat{T}\subset\hat{V}(E)$ is an equivalence of categories.

3.13. Let E_o be an elliptic curve up to isogeny over k. A lattice $\hat{T}\subset\hat{V}(E_o)$ can be interpreted as a class, modulo $GL(2,\hat{\mathbb{Z}})$, of isomorphisms

$\beta:\hat{V}(E_o)\to(\mathbb{A}^f)^2$;

to β one associates $\beta^{-1}(\hat{\mathbb{Z}}^2)$.

Let $K\subset GL(2,\hat{\mathbb{Z}})$ be the inverse image of $H\subset GL(2,\mathbb{Z}/n)$. If $(E_o,\hat{T})$ corresponds to an elliptic curve E, then a level-H structure on E gives the set of isomorphisms $\beta:\hat{V}(E_o)\tilde{\to}(\mathbb{A}^f)^2$ such that $\beta^{-1}(\hat{\mathbb{Z}}^2)=\hat{T}$ and

$\beta_n:E_n=\hat{T}/n\hat{T}\tilde{\to}\hat{\mathbb{Z}}^2/n\hat{\mathbb{Z}}^2=(\mathbb{Z}/n\mathbb{Z})^2$

defines that level-H structure. The maps β form a left coset under K, and the set of elliptic curves with level-H structure $(E,\bar{\alpha})$ and $E\otimes\mathbb{Q}=E_o$ identifies with

$K\backslash\mathrm{Isom}(\hat{V}(E_o),(\mathbb{A}^f)^2)$.

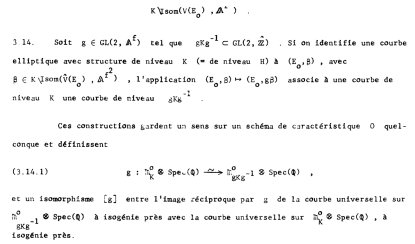
3.14. Let $g\in GL(2,\mathbb{A}^f)$ satisfy $gKg^{-1}\subset GL(2,\hat{\mathbb{Z}})$. Identifying an elliptic curve with level-K

structure, equivalently level-H structure, with (E_o,β), where β∈K\Isom(V̂(E_o),(𝔸^f)²), the assignment (E_o,β)↦(E_o,gβ) associates to a curve of level K a curve of level gKg⁻¹.

These constructions remain meaningful over any scheme of characteristic zero and define

(3.14.1) g:𝔪_K^o⊗Spec(ℚ)→̃𝔪_{gKg⁻¹}^o⊗Spec(ℚ),

together with an isomorphism [g] between the pullback by g of the universal curve over 𝔪_{gKg⁻¹}^o⊗Spec(ℚ), considered up to isogeny, and the universal curve over 𝔪_K^o⊗Spec(ℚ), considered up to isogeny.



K\Isom(V(E$_o$) , $\mathbb{A}^{\infty}$) .

3.14. Soit $g \in GL(2, \mathbb{A}^f)$ tel que $gKg^{-1} \subset GL(2, \hat{\mathbb{Z}})$. Si on identifie une courbe elliptique avec structure de niveau K (= de niveau H) à (E_o, β), avec $\beta \in K\backslash Isom(\hat{V}(E_o), \mathbb{A}^{f2})$, l'application $(E_o, \beta) \mapsto (E_o, g\beta)$ associe à une courbe de niveau K une courbe de niveau gKg^{-1}.

Ces constructions gardent un sens sur un schéma de caractéristique 0 quelconque et définissent

(3.14.1) $g : \mathfrak{M}^0_K \otimes Spec(\mathbb{Q}) \xrightarrow{\sim} \mathfrak{M}^0_{gKg^{-1}} \otimes Spec(\mathbb{Q})$,

et un isomorphisme [g] entre l'image réciproque par g de la courbe universelle sur $\mathfrak{M}^0_{gKg^{-1}} \otimes Spec(\mathbb{Q})$ à isogénie près avec la courbe universelle sur $\mathfrak{M}^0_K \otimes Spec(\mathbb{Q})$, à isogénie près.

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Authority page 76; collected-paper page 569; original folio 218.

Passing to the inverse limit over K gives an action of GL(2,𝔸^f) on the scheme lim←_K 𝔪_K^o⊗Spec(ℚ) and on the universal elliptic curve up to isogeny.

3.15. We now make the composite map explicit:

(3.15.1) 𝔪_K^o⊗ℚ →̃ 𝔪_{gKg⁻¹}^o⊗ℚ ⟶c 𝔪_1^o⊗ℚ.

Choose n and m such that

$\hat{\mathbb{Z}}^2 \subset (1/m)g^{-1}(\hat{\mathbb{Z}}^2) \subset (1/n)\hat{\mathbb{Z}}^2$.

Let B be the subgroup of $(\mathbb{Z}/n\mathbb{Z})^2$ that is the image of $(1/m)g^{-1}(\hat{\mathbb{Z}}^2)/\hat{\mathbb{Z}}^2$ under multiplication by n on

$(1/n)\hat{\mathbb{Z}}^2/\hat{\mathbb{Z}}^2 \xrightarrow{\sim} (\mathbb{Z}/n\mathbb{Z})^2$.

To an elliptic curve with level-H structure (E,ᾱ), associate the elliptic curve $E/\alpha^{-1}(B)$, where α:E_n→̃(ℤ/n)² is any representative of ᾱ. The isogeny [g] is 1/m times the projection $E \to E/\alpha^{-1}(B)$.

Proposition 3.16. The isomorphism of 3.14.1 extends to an isomorphism

g:𝔪_K^o→̃𝔪_{gKg⁻¹}^o.

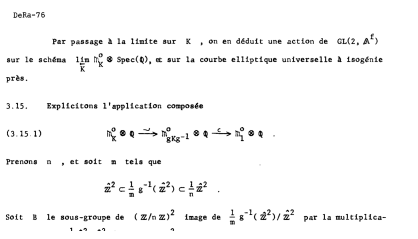
The isomorphism [g] extends to an isomorphism

[g]:g^*(the elliptic curve over 𝔪_{gKg⁻¹}^o)⊗ℚ ≃ (the elliptic curve over 𝔪_K^o)⊗ℚ.

Proof. We first show that (3.15.1) extends to

cg:𝔪_K^o→𝔪_1^o.

Let b be the order of B as in 3.15, and let I be the stack, relatively representable over 𝔪_1^o, that represents the functor of finite locally free subgroup schemes of rank b of the universal elliptic curve E. The theory of Hilbert schemes shows that the projection I→𝔪_1^o is proper and representable. By the next lemma it is finite.



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Par passage à la limite sur K, on en déduit une action de $GL(2, \mathbb{A}^f)$ sur le schéma $\varprojlim_K h^0_K \otimes \mathbb{Q}$, $\mathfrak{m}$ sur la courbe elliptique universelle à isogénie près.

3.15. Explicitons l'application composée

(3.15.1) $h^0_K \otimes \mathbb{Q} \xrightarrow{\sim} h^0_{gKg^{-1}} \otimes \mathbb{Q} \xrightarrow{c} h^0_1 \otimes \mathbb{Q}$.

Prenons n, et soit m tels que

$$\hat{\mathbb{Z}}^2 \subset \frac{1}{m} g^{-1}(\hat{\mathbb{Z}}^2) \subset \frac{1}{n} \hat{\mathbb{Z}}^2 .$$

Soit B le sous-groupe de $(\mathbb{Z}/n\mathbb{Z})^2$ image de $\frac{1}{m} g^{-1}(\hat{\mathbb{Z}}^2)/\hat{\mathbb{Z}}^2$ par la multiplica-

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Authority page 77; collected-paper page 570; original folio 219.

Lemma 3.17. Let E be an elliptic curve over an algebraically closed field k and let b be an integer. There are only finitely many subgroup schemes of E of rank b.

Proof of 3.17. Proceed by induction on b.

(a) There are only finitely many étale subgroup schemes of E of rank b: they identify with the subgroups of order b in E_b(k).

(b) If H⊂E is not étale, then k has characteristic p>0 and H contains Ker(F). Thus the non-étale subgroup schemes of E of rank b correspond to the subgroup schemes of rank b/p of E^(p)=E/Ker(F), and the induction hypothesis applies.

Proof of 3.16, continued. Construction 3.14 defines a section s over $\mathfrak{m}_K^o \otimes \mathrm{Spec}(\mathbb{Q})$ of $I \times_{\{\mathfrak{m}_1^o\}} \mathfrak{m}_K^o$.

By Zariski's Main Theorem and the normality of $\mathfrak{m}_K^o$, this section extends over $\mathfrak{m}_K^o$, defining a subgroup B of the pullback E, along c, of the universal elliptic curve over $\mathfrak{m}_1^o$. The required extension cg is defined by E/B, and the isomorphism [g] is defined as in 3.15.

The resulting morphism cg is quasi-finite and representable. It therefore identifies $\mathfrak{m}_K^o$ with an open substack of the normalization $\mathfrak{m}_{\{gKg^{-1}\}}^o$ of $\mathfrak{m}_1^o$ in

$\mathfrak{m}_K^o \otimes \mathrm{Spec}(\mathbb{Q}) \xrightarrow{g} \mathfrak{m}_{\{gKg^{-1}\}}^o \otimes \mathrm{Spec}(\mathbb{Q})$,

and hence gives

$g: \mathfrak{m}_K^o \hookrightarrow \mathfrak{m}_{\{gKg^{-1}\}}^o$.

This morphism is an isomorphism, with inverse g^{-1}.

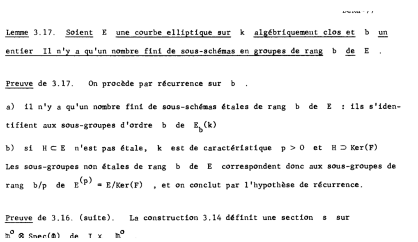
3.18. One can show that g extends to an isomorphism

$\mathfrak{m}_K \xrightarrow{\sim} \mathfrak{m}_{\{gKg^{-1}\}}$.

We shall only prove the analogous result for the coarse schemes.

Proposition 3.19. The morphism of coarse schemes induced by 3.16 extends to

$g: M_K \xrightarrow{\sim} M_{\{gKg^{-1}\}}$.



Lemme 3.17. Soient E une courbe elliptique sur k algébriquement clos et b un entier. Il n'y a qu'un nombre fini de sous-schémas en groupes de rang b de E.

Preuve de 3.17. On procède par récurrence sur b.

a) il n'y a qu'un nombre fini de sous-schémas étales de rang b de E : ils s'identifient aux sous-groupes d'ordre b de $E_b(k)$

b) si $H \subset E$ n'est pas étale, k est de caractéristique $p > 0$ et $H \supset \mathrm{Ker}(F)$ Les sous-groupes non étales de rang b de E correspondent donc aux sous-groupes de rang b/p de $E^{(p)} = E/\mathrm{Ker}(F)$, et on conclut par l'hypothèse de récurrence.

Preuve de 3.16. (suite). La construction 3.14 définit une section s sur $h^0 \otimes \mathrm{Spec}(\mathbb{Q})$ de $I \times h^0$

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Let Γ⊂M_K×M_{gKg⁻¹} be the closure of the graph G of g:M_K^o→̃M_{gKg⁻¹}^o. Set-theoretically,

Γ=G∪(Γ∩(M_K^∞×_ℤM_{gKg⁻¹}^∞)).

It follows that Γ is finite over both M_K and M_{gKg⁻¹}. By the Main Theorem and 3.10, the projections Γ→M_K and Γ→M_{gKg⁻¹} are isomorphisms. This proves 3.19.

3.20. Let C be an elliptic curve over S, with n invertible on S. The e_n pairing is well known to define an isomorphism

(3.20.1) e_n:Λ²C_n→̃μ_n.

A level-n structure α:C_n→̃(ℤ/n)² therefore defines an isomorphism

det(α)⁻¹:ℤ/n→̃μ_n,

that is, a primitive n-th root of unity

d(α)=det(α)⁻¹(1)

on S.

Let ℤ[ζ_n] be the ring of integers in the field of n-th roots of unity. The construction α↦d(α) defines a morphism

d:𝔪_n^o[1/n]→Spec(ℤ[ζ_n]).

This morphism extends to the normalization and defines

(3.20.2) d:𝔪_n→Spec(ℤ[ζ_n]).

Similarly, a level-H structure α defines

d(α)∈Isom(ℤ/n,μ_n)/det(H).

The group (ℤ/n)* acts on ℤ[ζ_n] by ζ_n↦ζ_n^i. If ℤ[ζ_n]^{det H} is the subring of invariants under the action of det H⊂(ℤ/n)*, the same construction gives

(3.20.3) d:𝔪_H→Spec(ℤ[ζ_n]^{det H}),

which defines

3.20. Soit C une courbe elliptique sur S, avec n inversible sur S. Il est bien connu que le "e_n-pairing" définit un isomorphisme,

(3.20.1) $$e_n : \overset{2}{\Lambda} C_n \xrightarrow{\sim} \mu_n .$$

Une structure de niveau n $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ définit donc un isomorphisme $\det(\alpha)^{-1} : \mathbb{Z}/n \xrightarrow{\sim} \mu_n$, i.e. une racine primitive $n^{\text{ième}}$ de l'unité $d(\alpha) = \det(\alpha)^{-1}(1)$ sur S

Notons $\mathbb{Z}[\zeta_n]$ l'anneau des entiers du corps des racines $n^{\text{ièmes}}$ de l'unité. La construction $\alpha \mapsto d(\alpha)$ définit un morphisme

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(3.20.4) d:M_H→Spec(ℤ[ζ_n]^{det H}).

3.21. Work over 𝔪_{(m)}[1/m], with n dividing m. If C is the universal generalized elliptic curve, the e_n pairing extends to an isomorphism

Λ²C_n≃μ_n.

In particular, for the standard Néron m-gon one obtains an isomorphism

(3.21.1) e_n:Λ²(𝔾_m×ℤ/m)_n≃μ_n.

Identify ℤ/n with the n-torsion subgroup (ℤ/m)_n by i↦(m/n)i. Then

(3.21.2) Λ²(𝔾_m×ℤ/m)_n=Λ²(μ_n×ℤ/n)→̃μ_n.

According to the sign conventions used to define e_n, the isomorphisms (3.21.1) and (3.21.2) are equal or opposite.

4. Examples.

4.1. Let Γ_o(n), Γ_{oo}(n), Γ'_{oo}(n), and Γ(n) be the subgroups of GL(2,$\hat{\mathbb{Z}}$) formed by the matrices with entries a,b,c,d satisfying, respectively,

Γ_o(n): c≡0 (mod n); Γ_{oo}(n): c≡0 and a≡1 (mod n); Γ'_{oo}(n): c≡0 and a≡d≡1 (mod n); Γ(n): the matrix is congruent to the identity (mod n).

Each of these is the inverse image of a subgroup H of GL(2,ℤ/n).

4.2. For Γ_o(n), H is the subgroup of GL(2,ℤ/n) that stabilizes ℤ/n×{0} in (ℤ/n)². If C is an elliptic curve over S and

Selon les conventions de signe utilisées pour définir e_n, (3.21.1) et (3.21.2) sont égaux ou opposés.

4. Exemples.

4.1. Nous noterons $\Gamma_o(n)$, $\Gamma_{oo}(n)$, $\Gamma'_{oo}(n)$ et $\Gamma(n)$ les sous-groupes de $GL(2,\hat{\mathbb{Z}})$ formé des matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ telles que :

$\Gamma_o(n)$: $c \equiv 0 \pmod n$;

$\Gamma_{oo}(n)$: $c \equiv 0 \pmod n$, $a \equiv 1 \pmod n$;

$\Gamma'_{oo}(n)$: $c \equiv 0 \pmod n$, $a \equiv d \equiv 1 \pmod n$;

$\Gamma(n)$: $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod n$.

Ces sous-groupes sont chacun image réciproque de sous-groupes H de

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Authority page 80; collected-paper page 573; original folio 222.

ᾱ is a level-H structure on C represented by an isomorphism α:C_n→̃(ℤ/n)², then the subgroup scheme α⁻¹(ℤ/n×{0}) of C_n depends only on ᾱ. When n is invertible on S, this gives an isomorphism between the sheaf of level-H structures on C and the sheaf of subgroup schemes of C locally isomorphic to ℤ/n. In other words:

Construction 4.3. The algebraic stack 𝔪^o_{Γ_o(n)}[1/n] classifies elliptic curves C/S, with n invertible on S, equipped with a subgroup scheme A locally isomorphic to ℤ/n.

4.4. For C/S and A as in 4.3, the subgroup C_n/A of the quotient curve C/A is locally isomorphic to ℤ/n. Hence there is a morphism

w:𝔪^o_{Γ_o(n)}[1/n]→𝔪^o_{Γ_o(n)}[1/n], (C,A)↦w(C,A)=(C/A,C_n/A).

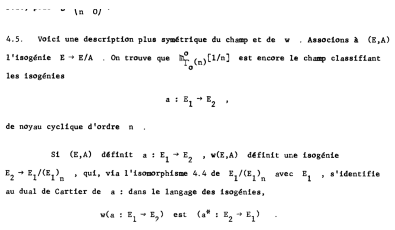
Multiplication by n factors through an isomorphism C/C_n→̃C. It yields an isomorphism

ww(C,A)=((C/A)/(C_n/A),(C/A)_n/(C_n/A))=(C/C_n,...)→̃(C,A),

and therefore w²≃Id: w is an involution. It is of the type described in 3.16 for the matrix with rows (0,1) and (n,0).

4.5. A more symmetric description is obtained by associating to (E,A) the isogeny E→E/A. The stack 𝔪^o_{Γ_o(n)}[1/n] also classifies isogenies a:E_1→E_2 with cyclic kernel of order n. If (E,A) defines a, then w(E,A) defines E_2→E_1/(E_1)_n; through the isomorphism of 4.4 between E_1/(E_1)_n and E_1, this is the Cartier dual of a. Thus

w(a:E_1→E_2)=(a^*:E_2→E_1).



4.5. Voici une description plus symétrique du champ et de w. Associons à (E,A) l'isogénie $E \to E/A$. On trouve que $\mathcal{M}^0_{\Gamma_0(n)}[1/n]$ est encore le champ classifiant les isogénies

$$a : E_1 \to E_2 ,$$

de noyau cyclique d'ordre n.

Si (E,A) définit $a : E_1 \to E_2$, $w(E,A)$ définit une isogénie $E_2 \to E_1/(E_1)_n$, qui, via l'isomorphisme 4.4 de $E_1/(E_1)_n$ avec E_1, s'identifie au dual de Cartier de a : dans le langage des isogénies,

$$w(a : E_1 \to E_2) \text{ est } (a^* : E_2 \to E_1) .$$

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Authority page 81; collected-paper page 574; original folio 223.

4.6. Let C/S and A be as in 4.3. For each prime divisor p of n, let A_p be the p-primary component of A, so that A=∏_{p|n}A_p, and let p^{n(p)} be the order of A_p, so n=∏_p p^{n(p)}. Put

w_p(E,A)=(E/A_p,(A+E_{p^{n(p)}})/A_p).

The w_p are pairwise commuting involutions of $\mathfrak{m}$^o_{Γ_o(n)}[1/n], and their composite is w.

4.7. The subgroup Γ_{oo}(n) of GL(2,$\hat{\mathbb{Z}}$) is the inverse image of the subgroup H of GL(2,$\mathbb{Z}$/n) that fixes (1,0) in ($\mathbb{Z}$/n)², or equivalently the homomorphism β:$\mathbb{Z}$/n→($\mathbb{Z}$/n)², x↦(x,0). Reasoning as in 4.2 gives:

Construction 4.8. The algebraic stack $\mathfrak{m}$^o_{Γ_{oo}(n)}[1/n] classifies elliptic curves C/S, with n invertible on S, equipped with an embedding β:$\mathbb{Z}$/n↪C_n, equivalently a section β(1) having exact order n everywhere.

4.9. The group Γ'_{oo}(n) is the inverse image of the subgroup H that fixes (1,0)∈($\mathbb{Z}$/n)² and the class of (0,1) in ($\mathbb{Z}$/n)²/($\mathbb{Z}$/n×{0}). Thus $\mathfrak{m}$^o_{Γ'_{oo}(n)}[1/n] classifies elliptic curves C/S with n invertible, equipped with β:$\mathbb{Z}$/n↪C_n and an isomorphism γ:$\mathbb{Z}$/n→̃C_n/Im(β). The exterior product gives the chain of isomorphisms

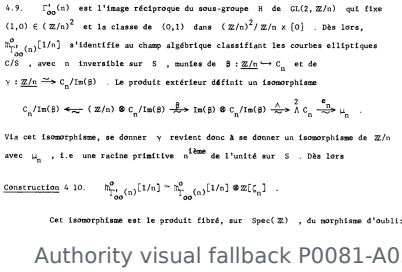
C_n/Im(β)←̃($\mathbb{Z}$/n)⊗C_n/Im(β)→̃^{β}Im(β)⊗C_n/Im(β)→̃^{Λ}Λ²C_n→̃^{e_n}μ_n.

Through it, choosing γ is equivalent to choosing an isomorphism $\mathbb{Z}$/n→̃μ_n, that is, a primitive n-th root of unity on S. Hence

Construction 4.10. $\mathfrak{m}$^o_{Γ'_{oo}(n)}[1/n]≃$\mathfrak{m}$^o_{Γ_{oo}(n)}[1/n]⊗$\mathbb{Z}$[ζ_n].

This is the fiber product over Spec($\mathbb{Z}$) of the forgetful morphism, defined by Γ'_{oo}(n)⊂Γ_{oo}(n), and the morphism d of (3.20.3).

Definition 4.11. The stack ´$\mathfrak{m}$_{Γ_o(n)}[1/n] classifies generalized elliptic curves C/S, with n invertible on S, equipped with a subgroup scheme A locally isomorphic to $\mathbb{Z}$/n and meeting every irreducible component of every geometric fiber of C.



4.9. $\Gamma'_{oo}(n)$ est l'image réciproque du sous-groupe H de $GL(2, \mathbb{Z}/n)$ qui fixe $(1,0) \in (\mathbb{Z}/n)^2$ et la classe de $(0,1)$ dans $(\mathbb{Z}/n)^2/\mathbb{Z}/n \times [0]$. Dès lors, $\mathcal{M}^0_{\Gamma'_{oo}(n)}[1/n]$ s'identifie au champ algébrique classifiant les courbes elliptiques C/S, avec n inversible sur S, munies de $\beta : \underline{\mathbb{Z}/n} \hookrightarrow C_n$ et de $\gamma : \underline{\mathbb{Z}/n} \xrightarrow{\sim} C_n/\mathrm{Im}(\beta)$. Le produit extérieur définit un isomorphisme

$$C_n/\mathrm{Im}(\beta) \xleftarrow{\sim} (\mathbb{Z}/n) \otimes C_n/\mathrm{Im}(\beta) \xrightarrow{\beta} \mathrm{Im}(\beta) \otimes C_n/\mathrm{Im}(\beta) \xrightarrow{\wedge} \wedge^2 C_n \xrightarrow{e_n} \mu_n .$$

Via cet isomorphisme, se donner γ revient donc à se donner un isomorphisme de $\mathbb{Z}/n$ avec μ_n, i.e une racine primitive $n^{\text{ième}}$ de l'unité sur S. Dès lors

Construction 4.10. $\mathcal{M}^0_{\Gamma'_{oo}(n)}[1/n] \simeq \mathcal{M}^0_{\Gamma_{oo}(n)}[1/n] \otimes \mathbb{Z}[\zeta_n]$.

Cet isomorphisme est le produit fibré, sur $\mathrm{Spec}(\mathbb{Z})$, du morphisme d'oubli:

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Authority page 82; collected-paper page 575; original folio 224.

(We shall see in a moment that ′𝔪_{Γ_o(n)}[1/n]≃𝔪_{Γ_o(n)}[1/n].)

If an m-gon is classified by ′𝔪_{Γ_o(n)}, then m divides n. The map that forgets A therefore sends ′𝔪_{Γ_o(n)}[1/n] to 𝔪_*. It is étale, and hence the source is a smooth algebraic stack.

4.12. As in 3.5, define the morphism that forgets A and contracts as in (1.3):

c:′𝔪_{Γ_o(n)}[1/n]→𝔪_1[1/n].

Let k be algebraically closed and let C be an object of the source over k. The map Aut(C)→Aut(c(C)) is injective. This is trivial for smooth C. For an m-gon equipped with A⊂C, II.1.10 shows that x↦-x is the only nontrivial automorphism of (C,A). As in 3.4, ′𝔪_{Γ_o(n)}[1/n] is proper over ℤ[1/n]; the argument of 3.5 yields:

Construction 4.13. 𝔪_{Γ_o(n)}[1/n]≃′𝔪_{Γ_o(n)}[1/n].

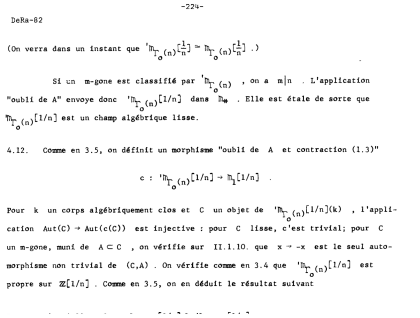
Analogous arguments give:

Construction 4.14. The algebraic stack 𝔪_{Γ_{oo}(n)}[1/n] classifies generalized elliptic curves C/S, with n invertible on S, equipped with an embedding β:ℤ/n↪C whose image meets every irreducible component of every geometric fiber.

Because ℤ[ζ_n][1/n] is étale over ℤ[1/n], normalization in 4.10 also gives:

Construction 4.15. 𝔪_{Γ'_{oo}(n)}[1/n]≃𝔪_{Γ_{oo}(n)}[1/n]⊗ℤ[ζ_n].

4.16. Let A be the subgroup of GL(2,ℤ/n) consisting of scalar matrices. Level-A structures will be called projective level-n structures. They play an important role in Ihara's work. More generally: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(On verra dans un instant que $'\mathfrak{M}_{\Gamma_o(n)}[\frac{1}{n}] \simeq \mathfrak{M}_{\Gamma_o(n)}[\frac{1}{n}]$.)

Si un m-gone est classifié par $'\mathfrak{M}_{\Gamma_o(n)}$, on a $m|n$. L'application "oubli de A" envoye donc $'\mathfrak{M}_{\Gamma_o(n)}[1/n]$ dans $\mathfrak{M}_{\infty}$. Elle est étale de sorte que $\mathfrak{M}_{\Gamma_o(n)}[1/n]$ est un champ algébrique lisse.

4.12. Comme en 3.5, on définit un morphisme "oubli de A et contraction (1.3)"

$$c : '\mathfrak{M}_{\Gamma_o(n)}[1/n] \to \mathfrak{M}_1[1/n] \quad .$$

Pour k un corps algébriquement clos et C un objet de $'\mathfrak{M}_{\Gamma_o(n)}[1/n](k)$, l'application Aut(C) → Aut(c(C)) est injective : pour C lisse, c'est trivial; pour C un m-gone, muni de A ⊂ C , on vérifie sur II.1.10. que x → -x est le seul automorphisme non trivial de (C,A) . On vérifie comme en 3.4 que $'\mathfrak{M}_{\Gamma_o(n)}[1/n]$ est propre sur $\mathbb{Z}[1/n]$. Comme en 3.5, on en déduit le résultat suivant

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Authority page 83; collected-paper page 576; original folio 225.

Definition 4.17. Let C/S be a generalized elliptic curve over S, with n invertible on S, and suppose that the singular geometric fibers of C are n-gons. A projective level-n structure on C is a section of

Isom(C_n,(ℤ/n)²)/A.

Lemma 4.18. Let (C,α) be a generalized elliptic curve over an algebraically closed field, equipped with a projective level-n structure. If n≥2, the only nontrivial automorphism of C preserving α is x↦-x.

For an n-gon the lemma follows easily from II.1.10 and is true even for n=1. For smooth C it is a special case of the following lemma.

Lemma 4.19. Let n≥2 and let g be a finite-order automorphism of an abelian variety X over an algebraically closed field k. If g acts as a scalar on X_n, then g²=1.

By hypothesis there is r∈ℤ/n such that g-r vanishes on X_n. Hence g-r=n·h for some endomorphism h of X. If ζ is a root of the characteristic polynomial of g, then ζ is a root of unity and the algebraic integer ζ-r is divisible by n. Let R be the ring of integers of ℚ(ζ). The image of ζ in R/nR is therefore the image of r. Since R=ℤ[ζ], this is impossible unless ζ=±1. Since g is semisimple, the assertion follows.

Combining Lemma 4.18 for n-gons with the now familiar arguments gives:

Construction 4.20. For A as in 4.16, 𝔪_A[1/n] is the algebraic stack classifying generalized elliptic curves equipped with a projective level-n structure.

For (C,α) classified by 𝔪_A[1/n] and n≥2, Lemma 4.18 shows that Aut(C,α) is constant. Hence, by (1.8):

Proposition 4.21. For A as in 4.16 and n≥2, the map

si n = 1 . Si C est lisse, c'est un cas particulier du lemme suivant.

Lemme 4.19. Soient n ≥ 2 et g un automorphisme d'ordre fini d'une variété abélienne X sur un corps algébriquement clos k . Si g induit une homothétie sur X_n, alors $g^2 = 1$.

Par hypothèse, il existe r ∈ℤ/n tel que g - r soit nul sur X_n . Il existe donc un endomorphisme h de X tel que g-r = n.h . Si ζ est une racine du polynôme caractéristique de g , ζ est une racine de l'unité et l'entier algébrique ζ-r est divisible par n . Si R est l'anneau des entiers du corps ℚ(ζ) , l'image de ζ dans R/nR est donc égale à celle de r . Puisque R = ℤ[ζ] , c'est absurde si ζ ≠ ± 1 . Puisque g est semi-simple, 4.19 en résulte.

Du lemme 4.18 pour les n-gones, et d'arguments maintenant familiers, on déduit le résultat suivant.

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𝔪_A[1/n]→M_A[1/n]

is étale.

Remark 4.22. When n=2, 𝔪_A is simply ′𝔪_2, so the map ′𝔪_2[1/2]→M_2[1/2] is étale.

5. Transcendental theory (review).

5.1. Let E be an elliptic curve over ℂ. The exponential map defines an exact sequence of analytic groups

0→H_1(E,ℤ)→^{ι}Lie(E)→^{exp}E(ℂ)→0.

If β:ℤ²→̃H_1(E,ℤ) is a basis of E, put

z(β)=ι(β(1,0))/ι(β(0,1))∈ℂ-ℝ.

For ι_1,ι_2∈Lie(E), with ι_2≠0, the quotient ι_1/ι_2 denotes the complex number λ satisfying ι_1=λι_2. If γ is the matrix with rows (a,c) and (b,d) in GL(2,ℤ), then

z(βγ)=(a z(β)+b)/(c z(β)+d).

5.2. The map x↦exp((1/n)ι(x)) induces an isomorphism

(5.2.1) H_1(E,ℤ)⊗ℤ/n→̃E_n.

Put X^±=ℂ-ℝ, and let X^+ be the upper half-plane. If α:E_n→̃(ℤ/n)² is a level-n structure, the class of

(z(β),α∘β), with z(β)∈X^± and α∘β∈GL(2,ℤ/n),

in X^±×GL(2,ℤ/n)/GL(2,ℤ) depends only on (E,α).

$\mathcal{H}_2[1/2] \to M_2[1/2]$ est donc étale.

5. Théorie transcendante (rappels).

5.1 Soit E une courbe elliptique sur $\mathbb{C}$. L'application exponentielle définit une suite exacte de groupes analytiques

$$0 \to H_1(E,\mathbb{Z}) \xrightarrow{\iota} \mathrm{Lie}(E) \xrightarrow{\exp} E(\mathbb{C}) \to 0 .$$

Si $\beta : \mathbb{Z}^2 \xrightarrow{\sim} H_1(E,\mathbb{Z})$ est une base de E, nous poserons

$$s(\beta) = s(\beta(1,0))/s(\beta(0,1)) \in \mathbb{C} - \mathbb{R}$$

(pour ι_1 et ι_2 deux éléments de $\mathrm{Lie}(E)$, avec $\iota_2 \neq 0$, on note ι_1/ι_2 le nombre complexe λ tel que $\iota_1 = \lambda\iota_2$).

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Authority page 85; collected-paper page 578; original folio 227.

5.3. Let H and K be as in 3.6. Construction 5.2 defines an isomorphism

$M_H^o(\mathbb{C}) \tilde{\to} X^{\pm} \times (H\backslash GL(2,\mathbb{Z}/n)/GL(2,\mathbb{Z}))$.

There are also isomorphisms

$X \times (H\backslash GL(2,\mathbb{Z}/n))/SL(2,\mathbb{Z}) \tilde{\to} X^{\pm} \times (H\backslash GL(2,\mathbb{Z}/n))/GL(2,\mathbb{Z}) \tilde{\to} K\backslash X^{\pm} \times GL(2,\mathbb{A}_f)/GL(2,\mathbb{Q})$.

5.4. The reduction morphism $SL(2,\mathbb{Z}) \to SL(2,\mathbb{Z}/n)$ is surjective. Hence the connected components of $M_H^o(\mathbb{C})$ identify with the fibers of

(5.4.1) $X \times (H\backslash GL(2,\mathbb{Z}/n))/SL(2,\mathbb{Z}) \to^{\det} \det H\backslash(\mathbb{Z}/n)^*$.

Let $\mu_n^*(\mathbb{C})$ be the set of primitive n-th roots of unity in $\mathbb{C}$. After extension of scalars from $\mathbb{Z}$ to $\mathbb{C}$, (3.20.4) defines

(5.4.2) $M_H(\mathbb{C}) \to^{d} \det H\backslash \mu_n^*(\mathbb{C})$.

Through the bijection $x \mapsto \exp(2\pi i x/n) : \mathbb{Z}/n \tilde{\to} \mu_n(\mathbb{C})$, (5.4.1) identifies with the restriction of this map to $M_H^o(\mathbb{C})$. The geometric characteristic-zero fibers of (3.20.3) and (3.20.4) are therefore irreducible. The next corollary follows from Zariski's connectedness theorem.

Corollary 5.5. In every characteristic, the geometric fibers of $M_H \to \mathrm{Spec}(\mathbb{Z}[\zeta_n]^{\det H})$ in (3.20.4) are connected.

Corollary 5.6. In characteristic p prime to n, the geometric fibers of (3.20.4) are irreducible. They are connected and geometrically unibranch by 3.10(iv).

Soit $\mu_n^*(\mathbb{C})$ l'ensemble des racines primitives $n^{\text{èmes}}$ de l'unité dans $\mathbb{C}$. Par extension des scalaires de $\mathbb{Z}$ à $\mathbb{C}$, (3.20.4) définit une application.

(5.4.2) $M_H(\mathbb{C}) \xrightarrow{d} \det H\backslash \mu_n^*(\mathbb{C})$.

Via la bijection $x \mapsto \exp(\frac{2\pi i x}{n}) : \mathbb{Z}/n \xrightarrow{\sim} \mu_n(\mathbb{C})$, (5.4.1) s'identifie à la restriction de cette application à $M_H^0(\mathbb{C})$. Les fibres géométriques de caractéristiques 0 de (3.20.3) (3.20.4) sont donc irréductibles. Le corollaire suivant résulte dès lors du théorème de connexion de Zariski.

Corollaire 5.5. En toute caractéristique, les fibres géométriques de (3.20.4) : $M_H \to \mathrm{Spec}(\mathbb{Z}[\zeta_n]^{\det H})$ sont connexes.

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Authority page 86; collected-paper page 579; original folio 228.

6. Level-H structures: study at infinity.

This section will not be used later in the article. We recommend omitting it on a first reading.

Let H be a subgroup of $GL(2,\mathbb{Z}/n)$. We seek a modular interpretation of $\mathfrak{m}_H[1/n]$ generalizing 3.5, 4.13, 4.14, 4.15, and 4.20. We shall first define modularly a stack $'\mathfrak{m}_H[1/n]$,

proper and smooth over $\mathbb{Z}[1/n]$, with $\mathfrak{m}_H^o[1/n]$ as a dense open substack. As in 3.5 we define a finite morphism $c:'\mathfrak{m}_H[1/n]\to\mathfrak{m}_1[1/n]$. We then prove the analogue of (3.5.1); the proof of 3.5 gives $'\mathfrak{m}_H[1/n]=\mathfrak{m}_H[1/n]$, the desired modular interpretation.

6.1. Let A be a group isomorphic to $(\mathbb{Z}/n)^2$, let $A'\subset A$ be a subgroup isomorphic to $\mathbb{Z}/n$, and let B contain A'. For a suitable divisor m of n, (A,A',B) is isomorphic to

$((\mathbb{Z}/n)^2,\mathbb{Z}/n\times\{0\},\mathbb{Z}/n\times m\mathbb{Z}/n\mathbb{Z})$.

Let μ also be isomorphic to $\mathbb{Z}/n$, let $\beta_0:\Lambda^2A\xrightarrow{\sim}\mu$, and denote by β_1 its restriction to Λ^2B.

The essential example is a generalized elliptic curve C over a complete trait S with points η and s. Let $\bar\eta$ be the spectrum of an algebraic closure of $k(\eta)$, and $\bar s$ the spectrum of the corresponding algebraic closure of k(s). If n is invertible on S, $C_{\bar\eta}$ is smooth, and $C_{\bar s}$ is an m-sided Néron polygon with m|n, take

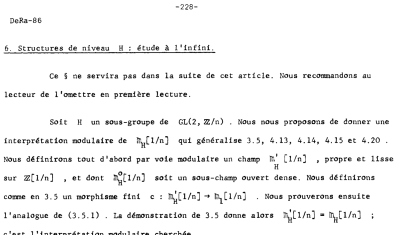
$\mu=\mu_n(\bar\eta)\simeq\mu_n(\bar s)$, $A=C_n(\bar\eta)$, $A'=C_n^o(\bar s)$, $B=(C_s^{reg})_n(\bar s)$, and $\beta_0=e_n$.

The restriction β_1 of β_0 to Λ^2B depends only on the special fiber $C_{\bar s}$; see 3.21.

6.2. Let I be the set of isomorphisms $\alpha:A\to(\mathbb{Z}/n)^2$, and let $\bar I$ be the set of pairs $(\bar\alpha,\beta)$, with $\bar\alpha:B\hookrightarrow(\mathbb{Z}/n)^2$ and $\beta:\mu\xrightarrow{\sim}\mathbb{Z}/n$, for which

$$\begin{array}{ccc}\Lambda^2\bar\alpha\ \Lambda^2B & \longrightarrow & \Lambda^2(\mathbb{Z}/n)^2\\ \beta_1\ \downarrow & & \downarrow\ \mu\\ s & \xrightarrow{\beta} & \mathbb{Z}/n\end{array}$$

commutes. This is diagram (6.2.1).



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6. Structures de niveau H : étude à l'infini.

Ce § ne servira pas dans la suite de cet article. Nous recommandons au lecteur de l'omettre en première lecture.

Soit H un sous-groupe de GL(2, $\mathbb{Z}/n$) . Nous nous proposons de donner une interprétation modulaire de $\mathfrak{M}_H[1/n]$ qui généralise 3.5, 4.13, 4.14, 4.15 et 4.20 . Nous définirons tout d'abord par voie modulaire un champ $\mathfrak{M}'_H[1/n]$, propre et lisse sur $\mathbb{Z}[1/n]$, et dont $\mathfrak{M}^0_H[1/n]$ soit un sous-champ ouvert dense. Nous définirons comme en 3.5 un morphisme fini $c : \mathfrak{M}'_H[1/n] \to \mathfrak{M}_1[1/n]$. Nous prouverons ensuite l'analogue de (3.5.1) . La démonstration de 3.5 donne alors $\mathfrak{M}'_H[1/n] = \mathfrak{M}_H[1/n]$; c'est l'interprétation modulaire cherchée.

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Authority page 87; collected-paper page 580; original folio 229.

Define $c:I\to\bar I$ by $\alpha\mapsto(\alpha|_B,\Lambda^2\alpha\circ\beta_0^{-1})$. The group $GL(2,\mathbb{Z}/n)$ acts on I and $\bar I$ on the left by $g.\alpha=g\circ\alpha$ and $g.(\bar\alpha,\beta)=(g\bar\alpha,\det(g).\beta)$. Let U be the subgroup of Aut(A) acting trivially on A' and A/A'. It acts on the right by $\alpha.u=\alpha\circ u$ and $(\bar\alpha,\beta)u=(\bar\alpha\circ(u|B),\beta)$. These actions are compatible with c.

Lemma 6.2.2. The map c is surjective.

The verification is left to the reader.

Subgroups U' of U correspond bijectively to subgroups B' of A containing A': U' maps to the invariant subgroup $A^{U'}$, while B' maps to the subgroup U(B') of U acting trivially on B'. Moreover $\bar I=I/U(B)$.

Let $\bar a\in H\backslash\bar I$ be the image of $\alpha\in I$. The set of $\alpha'\in I$ mapping to $\bar a$ is $H\alpha U(B)$. We call B adequate when U(B) is the subgroup of U stabilizing the image a of α in $H\backslash I$. Equivalently, in the square

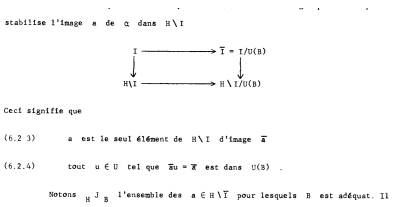
$$\begin{array}{ccc}I & \longrightarrow & \bar I=I/U(B)\\ \downarrow & & \downarrow\\ H\backslash I & \longrightarrow & H\backslash I/U(B),\end{array}$$

(6.2.3) a is the unique element of $H\backslash I$ mapping to $\bar a$, and (6.2.4) every $u\in U$ satisfying $\bar a u=\bar a$ lies in U(B).

Let {}_H J_B be the set of a∈H\Ī for which B is adequate. The preceding discussion gives:

Lemma 6.2.5. H\I→̃∐_{A'⊂B'⊂A}{}_H J_{B'}, the disjoint sum over B'.

6.3. Suppose only a group B and a subgroup A' are given, with (A',B)


stabilise l'image a de α dans H\I

Ī = I/U(B)

H\I → H\I/U(B)

Ceci signifie que

(6.2.3) a est le seul élément de H\I d'image ā

(6.2.4) tout u ∈ U tel que āu = ā est dans U(B) .

Notons $_H J_B$ l'ensemble des a ∈ H\Ī pour lesquels B est adéquat. Il

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isomorphic to (ℤ/n×{0},ℤ/nℤ×mℤ/nℤ) for a suitable divisor m of n. Let μ be isomorphic to ℤ/n and let β_1:Λ²B↪μ. Again let Ī be the set of pairs (ᾱ,β), where ᾱ:B↪(ℤ/nℤ)^2 and β:μ→̃ℤ/n make diagram (6.2.1) commute.

The group GL(2,ℤ/n) acts on the left by g.(ᾱ,β)=(g∘ᾱ,det(g).β), and the group Ū of automorphisms of B inducing the identity on A' and B/A' acts on the right by (ᾱ,β)u=(ᾱ∘u,β). Let {}_H J_B be the set of a∈H\Ī such that:

(a) if ã∈Ī maps to ā and gã=ã, then g∈H;

(b) every u∈Ū satisfying āu=ā is the identity.

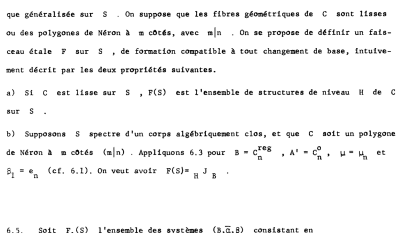
This notation agrees with that of 3.11 whenever the latter makes sense.

6.4. Let S be a scheme on which n is invertible, and let C be a generalized elliptic curve over S whose geometric fibers are smooth or m-sided Néron polygons with m|n. We define an étale sheaf F on S, compatible with arbitrary base change, by the following intended values.

(a) If C is smooth over S, F(S) is the set of level-H structures on C over S.

(b) If S is the spectrum of an algebraically closed field and C is an m-sided Néron polygon with m|n, apply 6.3 with B=C_n^{reg}, A'=C_n^o, μ=μ_n, and β_1=e_n. We require F(S)={}_H J_B.

6.5. Let F_1(S) be the set of systems (B,ᾱ,β) consisting of (a) a finite étale subgroup scheme B⊂C_n and (b) maps ᾱ:B↪(ℤ/n)^2 and β:μ_n→̃ℤ/n such that (c), locally on S, B contains a subgroup isomorphic to ℤ/n.


que généralisée sur S . On suppose que les fibres géométriques de C sont lisses ou des polygones de Néron à m côtés, avec m|n . On se propose de définir un faisceau étale F sur S , de formation compatible à tout changement de base, intuitivement décrit par les deux propriétés suivantes.

a) Si C est lisse sur S , F(S) est l'ensemble de structures de niveau H de C sur S .

b) Supposons S spectre d'un corps algébriquement clos, et que C soit un polygone de Néron à m côtés (m|n) . Appliquons 6.3 pour $B = C_n^{reg}$, $A' = C_n^o$, $\mu = \mu_n$ et $\beta_1 = e_n$ (cf. 6.1). On veut avoir $F(S) = {}_H J_\beta$.

6.5. Soit F.(S) l'ensemble des systèmes (B,ᾱ,β) consistant en

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(d) The diagram

$$\begin{array}{ccc} \bar{\alpha}\ \Lambda^2 B & \longrightarrow & \Lambda^2\mathbb{Z}/n \\ \big\downarrow e_n & & \big\downarrow s \\ \mu_n & \xrightarrow{\ \beta\ } & \mathbb{Z}/n \end{array}$$

commutes.

(e) For every geometric point $\bar{s}$ of S, if $C_{\bar{s}}$ is singular, $(\bar{\alpha},\beta)$ modulo H is of the type described in 6.4. In every case we require that if $g\in GL(2,\mathbb{Z}/n)$ satisfies $(g\bar{\alpha}_{\bar{s}},\det(g).\beta_{\bar{s}})=(\bar{\alpha}_{\bar{s}},$ then $g\in H$.

If C/S is smooth, a system $(B,\bar{\alpha},\beta)$ determines a unique level-H structure on C, locally represented by isomorphisms $\alpha: C_n \tilde{\to} (\mathbb{Z}/n)^2$ extending $\bar{\alpha}$ and having determinant β. Local existence follows from (6.2.1), and uniqueness modulo H from condition (e).

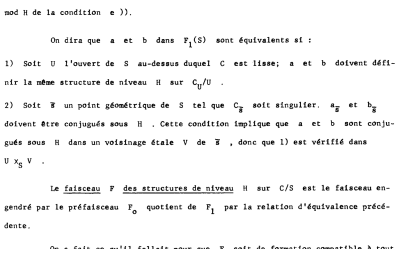
Elements $a,b\in F_1(S)$ are equivalent when:

(1) over the open $U\subset S$ where C is smooth, they define the same level-H structure on C_U/U;

(2) for every geometric point $\bar{s}$ with $C_{\bar{s}}$ singular, $a_{\bar{s}}$ and $b_{\bar{s}}$ are H-conjugate. This implies étale-locally near $\bar{s}$ that a and b are H-conjugate, so (1) holds over $U\times_S V$ for an étale neighborhood V of $\bar{s}$.

The sheaf F of level-H structures on C/S is generated by the presheaf F_0 obtained by quotienting F_1 by this equivalence relation. It is compatible with arbitrary base change, has the values prescribed in 6.4, and is represented by an étale S-scheme S'.

Definition 6.6. (i) A level-H structure on a generalized elliptic curve C/S as in 6.4 is an element of F(S).



mod H de la condition e)).

On dira que a et b dans $F_1(S)$ sont équivalents si :

1) Soit U l'ouvert de S au-dessus duquel C est lisse; a et b doivent définir la même structure de niveau H sur C_U/U .

2) Soit $\bar{s}$ un point géométrique de S tel que $C_{\bar{s}}$ soit singulier. $a_{\bar{s}}$ et $b_{\bar{s}}$ doivent être conjugués sous H . Cette condition implique que a et b sont conjugués sous H dans un voisinage étale V de $\bar{s}$, donc que 1) est vérifié dans $U \times_S V$.

Le faisceau F des structures de niveau H sur C/S est le faisceau engendré par le préfaisceau F_0 quotient de F_1 par la relation d'équivalence précédente.

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(ii) The algebraic stack $\acute{\mathfrak{m}}_H[1/n]$ over $\mathbb{Z}[1/n]$ classifies generalized elliptic curves C/S, with n invertible on S, whose geometric fibers are smooth or m-sided Néron polygons with m|n, equipped with a level-H structure.

Forgetting the level-H structure defines an étale morphism $\acute{\mathfrak{m}}_H[1/n]\to\mathfrak{m}_*$. Thus $\acute{\mathfrak{m}}_H[1/n]$ is smooth over $\mathbb{Z}[1/n]$, and $\acute{\mathfrak{m}}_H^\infty[1/n]$ is étale over $\mathbb{Z}[1/n]$. Contracting as in 1.3 for n=1 and forgetting the level structure defines $c:\acute{\mathfrak{m}}_H[1/n]\to\mathfrak{m}_1$.

Theorem 6.7. (i) The stack $\acute{\mathfrak{m}}_H[1/n]$ is proper and smooth over $\mathbb{Z}[1/n]$, and the image $\acute{\mathfrak{m}}_H^\infty[1/n]$ of the non-smoothness subscheme of the universal curve is finite étale over $\mathbb{Z}[1/n]$.

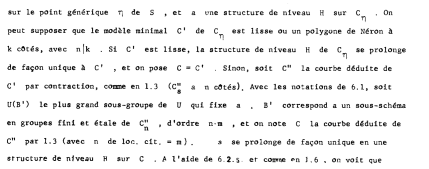
(ii) The morphism c is finite and representable. Consequently $\acute{\mathfrak{m}}_H[1/n]$ agrees with $\mathfrak{m}_H[1/n]$; see 3.5.

It remains to prove that $\acute{\mathfrak{m}}_H[1/n]$ is proper over $\mathbb{Z}[1/n]$ and that c is representable, that is, for every object C of $\acute{\mathfrak{m}}_H[1/n]$ over an algebraically closed field k, the map

(6.7) Aut(C)→Aut(c(C))

is injective.

Properness follows from the valuative criterion and 6.2.5. Let S be a complete trait with algebraically closed residue field, let C_η be an elliptic curve over the generic point η of S, and let a be a level-H structure on C_η. We may suppose that the minimal model C’ of C_η is smooth or a k-sided Néron polygon, with n|k. If C’ is smooth, the level-H structure extends uniquely to C’, and set C=C’. Otherwise let C” be obtained from C’ by contraction as in 1.3, so C”_s has n sides. With the notation of 6.1, let U(B’) be the largest subgroup of U fixing a. Then B’ corresponds to a finite étale subgroup scheme of C”_n of order n·m. Let C be the curve obtained from C” by 1.3, with the n of that reference equal to m. The structure a extends uniquely to a level-H structure on C. Using 6.2.5 and the argument of 1.6, one sees that


sur le point générique η de S, et a une structure de niveau H sur C_η. On peut supposer que le modèle minimal C' de C_η est lisse ou un polygone de Néron à k côtés, avec $n|k$. Si C' est lisse, la structure de niveau H de C_η se prolonge de façon unique à C', et on pose $C = C'$. Sinon, soit C'' la courbe déduite de C' par contraction, comme en 1.3 (C''_s a n côtés). Avec les notations de 6.1, soit $U(B')$ le plus grand sous-groupe de U qui fixe a. B' correspond à un sous-schéma en groupes fini et étale de C''_n, d'ordre $n\cdot m$, et on note C la courbe déduite de C'' par 1.3 (avec n de loc. cit. $= m$). a se prolonge de façon unique en une structure de niveau H sur C. A l'aide de 6.2.5, et comme en 1.6, on voit que

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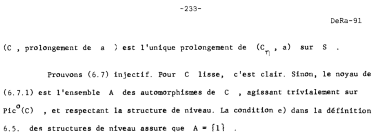
Authority page 91; collected-paper page 584; original folio 233.

(C, an extension of a) is the unique extension of (C_η,a) over S.

Let us prove that (6.7) is injective. For C smooth, this is clear. Otherwise, the kernel of (6.7.1) is the set A of automorphisms of C that act trivially on $\mathrm{Pic}^0(C)$ and respect the level structure. Condition e) in Definition 6.5 of level structures ensures that A = {1}.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$(C$, prolongement de $a)$ est l'unique prolongement de (C_η, a) sur S.

Prouvons (6.7) injectif. Pour C lisse, c'est clair. Sinon, le noyau de (6.7.1) est l'ensemble A des automorphismes de C, agissant trivialement sur $\mathrm{Pic}^0(C)$, et respectant la structure de niveau. La condition e) dans la définition 6.5. des structures de niveau assure que $A = \{1\}$.

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V. Reduction modulo p.

In this chapter we study the reduction modulo p of the stacks $\mathfrak{m}_K$, for K = Γ_0(p), Γ_∞(p), and Γ(p^k). Comparison of the results obtained for Γ_0(p) and Γ_∞(p) allows us to show that, in accordance with a conjecture of Shimura, certain abelian schemes studied by Shimura and Casselman [5] have good reduction over the ring of integers of suitable cyclotomic extensions of $\mathbb{Q}$.

1. Study of $\mathfrak{m}_{Γ_0(p)}$.

In this section we propose to give a modular interpretation of the stack $\mathfrak{m}_{Γ_0(p)}$ when p is prime. Some preliminary results will be used again in the following sections.

1.1. Let $\mathcal{G}$_n be the stack classifying generalized elliptic curves C/S with geometrically irreducible fibers, equipped with a subgroup scheme A locally isomorphic to μ_n (for the étale topology), that is, finite flat locally free of rank n with étale Cartier dual.

Proposition 1.2. The morphism "forget A" : $\mathcal{G}$_n → $\mathfrak{m}$_1 is étale; in particular, $\mathcal{G}$_n is an algebraic stack smooth over $\mathbb{Z}$.

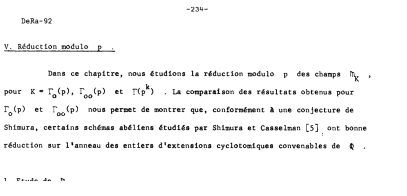
Let C/S be classified by $\mathfrak{m}$_1. It must be checked that the functor

F : (Sch/S) → (Sets),

which assigns to T/S the set of subgroups A of (C_T)_n locally isomorphic to μ_n, is represented by a scheme M étale over S. Representability follows from the theory of the Hilbert scheme; M is étale because it is formally étale (SGA 3 IX 3.6).

1.3. Let $\mathcal{B}$_n be the stack classifying generalized elliptic curves C/S whose geometric fibers are smooth or n-gons, equipped with a subgroup scheme B locally isomorphic to $\mathbb{Z}$/n (for the étale topology) that meets every irreducible component of every geometric fiber.

Let (C,B) be as above. The group scheme B acts freely on C



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V. Réduction modulo p .

Dans ce chapitre, nous étudions la réduction modulo p des champs $\mathcal{M}_K$, pour $K = \Gamma_0(p)$, $\Gamma_{00}(p)$ et $\Gamma(p^N)$. La comparaison des résultats obtenus pour $\Gamma_0(p)$ et $\Gamma_{00}(p)$ nous permet de montrer que, conformément à une conjecture de Shimura, certains schémas abéliens étudiés par Shimura et Casselman [5] ont bonne réduction sur l'anneau des entiers d'extensions cyclotomiques convenables de $\mathbb{Q}$.

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and C/B has geometrically irreducible fibers. In the exact sequence

0 → B → C_n → A → 0,

the extreme terms are Cartier dual. The pair (C/B,C_n/B) is therefore classified by $\mathcal{G}$_n, and this defines

(1.3.1) w : $\mathcal{B}$_n → $\mathcal{G}$_n.

Proposition 1.4. The morphism 1.3.1 is an isomorphism. In particular, $\mathcal{B}$_n is an algebraic stack smooth over $\mathbb{Z}$.

Proof: Again denote by w the morphism from $\mathcal{G}$_n^o to $\mathcal{B}$_n^o

w : (C,A) ↦ (C/A,C_n/A).

A construction identical to that of IV.4.4 shows that this morphism is an inverse of the restriction w : $\mathcal{B}$_n^o → $\mathcal{G}$_n^o of w to $\mathcal{B}$_n^o.

Let (C,A) over S be classified by $\mathcal{G}$_n. It must be proved that there exist ($\tilde{C}$,$\tilde{A}$) over S, classified by $\mathcal{B}$_n, and an isomorphism β : w($\tilde{C}$,$\tilde{A}$) $\tilde{\to}$ (C,A), and that ($\tilde{C}$,$\tilde{A}$,β) is unique up to unique isomorphism. By abuse of notation, we again denote by β the composite morphism

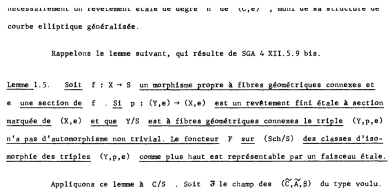
$\tilde{C}$ → $\tilde{C}$/$\tilde{A}$ $\tilde{\to}$^β C.

The curve with marked section ($\tilde{C}$,$\tilde{e}$) will necessarily be an étale covering of degree n of (C,e), endowed with its generalized elliptic-curve structure.

Recall the following lemma, which follows from SGA 4 XII.5.9 bis.

Lemma 1.5. Let $f : X \to S$ be a proper morphism with geometrically connected fibers, and let e be a section of f. If $p : (Y,e) \to (X,e)$ is a finite étale covering with marked section of (X,e), and Y/S has geometrically connected fibers, the triple (Y,p,e) has no nontrivial automorphism. The functor F on (Sch/S) of isomorphism classes of triples (Y,p,e) as above is represented by an étale sheaf.

Apply this lemma to C/S. Let $\mathcal{J}$ be the stack of triples ($\tilde{C}$,$\tilde{A}$,β) of the required type. The lemma shows that the objects of $\mathcal{J}$ have no nontrivial automorphism, and that the



nécessairement un revêtement étale de degré n de (C,e), muni de sa structure de courbe elliptique généralisée.

Rappelons le lemme suivant, qui résulte de SGA 4 XII.5.9 bis.

Lemme 1.5. Soit $f : X \to S$ un morphisme propre à fibres géométriques connexes et e une section de f. Si $p : (Y,e) \to (X,e)$ est un revêtement fini étale à section marquée de (X,e) et que Y/S est à fibres géométriques connexes le triple (Y,p,e) n'a pas d'automorphisme non trivial. Le foncteur F sur (Sch/S) des classes d'isomorphie des triples (Y,p,e) comme plus haut est représentable par un faisceau étale.

Appliquons ce lemme à C/S. Soit $\mathcal{F}$ le champ des $(\tilde{C},\tilde{A},\beta)$ du type voulu.

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functor $[\mathcal{J}]$ of isomorphism classes of objects of $\mathcal{J}$ is the subfunctor of F formed by the triples ($\tilde{C}$,p;e) such that $p^{-1}(e)$ is locally isomorphic to $\mathbb{Z}/n$ and A is the image of $\tilde{C}_n$. These conditions are open (because A is of multiplicative type), so $[\mathcal{J}]$ is represented by T étale over S.

Where C is smooth, one knows that $T \tilde{\to} S$. Moreover, a 1-gon over an algebraically closed field k admits a unique étale covering by an n-gon. All geometric fibers of T/S are therefore reduced to one point, and $T \tilde{\to} S$. This proves 1.4.

Let p be a prime number.

Theorem 1.6. $\mathfrak{m}_{\Gamma_0(p)}$ is the algebraic stack classifying generalized elliptic curves C/S equipped with a subgroup scheme A locally free of rank p that meets every irreducible component of every geometric fiber of C/S.

Proof: Recall [22] that a (commutative) group scheme locally free of rank p is killed by p. For A as above, one therefore has $A \subset C_p$.

Provisionally denote by $'\mathfrak{m}_{\Gamma_0(p)}$ the stack which assigns to each scheme S the category whose objects are the pairs (C,A) as in 1.6 and whose morphisms are the isomorphisms. In IV.4.3 we constructed an isomorphism between $\mathfrak{m}_{\Gamma_0(p)}{}^{o}[1/p]$ and $'\mathfrak{m}_{\Gamma_0(p)}{}^{o}[1/p]$. The isomorphism 1.6 will extend this isomorphism; reasoning as in IV.3.5, one finds that 1.6 follows from the conjunction of the following assertions:

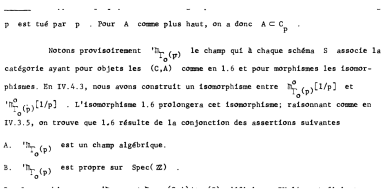
A. $'\mathfrak{m}_{\Gamma_0(p)}$ is an algebraic stack.

B. $'\mathfrak{m}_{\Gamma_0(p)}$ is proper over $\mathrm{Spec}(\mathbb{Z})$.

C. The morphism $c : '\mathfrak{m}_{\Gamma_0(p)} \to \mathfrak{m}_1 : (C,A) \mapsto c(C)$, defined by IV.1.4, is finite and representable.

D. $'\mathfrak{m}_{\Gamma_0(p)}$ is normal.

E. $'\mathfrak{m}_{\Gamma_0(p)}{}^{o}[1/p]$ is dense in $'\mathfrak{m}_{\Gamma_0(p)}$.



p est tué par p . Pour A comme plus haut, on a donc A ⊂ C_p .

Notons provisoirement 'h_{Γ_0(p)} le champ qui à chaque schéma S associe la catégorie ayant pour objets les (C,A) comme en 1.6 et pour morphismes les isomorphismes. En IV.4.3, nous avons construit un isomorphisme entre h^0_{Γ_0(p)}[1/p] et 'h^0_{Γ_0(p)}[1/p] . L'isomorphisme 1.6 prolongera cet isomorphisme; raisonnant comme en IV.3.5, on trouve que 1.6 résulte de la conjonction des assertions suivantes

A. 'h_{Γ_0(p)} est un champ algébrique.

B. 'h_{Γ_0(p)} est propre sur Spec(ℤ) .

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(1) Points at infinity of ´m_{Γ_0(p)}.

Let C be the one-sided Néron polygon over Spec(ℤ). One has C^{reg} ≃ G_m, C_p ≃ μ_p, and (C,C_p) is a section e_1 of ´m_{Γ_0(p)} over ℤ.

Let C be the p-gon over Spec(ℤ). One has C^{reg} ≃ G_m × ℤ/p, C_p ≃ μ_p × ℤ/p, and (C,e × ℤ/p) is a section e_2 of ´m_{Γ_0(p)} over ℤ.

Let k be an algebraically closed field. One checks that the only objects (C,A) of ´m_{Γ_0(p)}(k), with C singular, are e_1(k) and e_2(k).

(2) Proof of A (representability of ´m_{Γ_0(p)}).

Let $\mathcal{U}$_1 and $\mathcal{U}$_2 be the following two open substacks of ´m_{Γ_0(p)}: $\mathcal{U}$_1 classifies the curves (C,A) as in 1.6 for which the geometric fibers of C are irreducible, and $\mathcal{U}$_2 classifies the curves (C,A) for which the geometric fibers of C are smooth or p-gons.

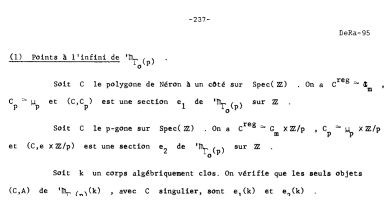
Lemma 1.7. The ”forgetful” morphism φ : $\mathcal{U}$_1 → m_1 is representable.

Proof. It must be seen that, for C/S a generalized elliptic curve with geometrically irreducible fibers over S, the following functor F : (Sch/S) → (Sets) is representable. To T/S, F assigns the set of finite locally free subgroup schemes of rank p of C_T^{reg}. The representability of F follows from the theory of Hilbert schemes.

By 1.7 and IV.2, $\mathcal{U}$_1 is an algebraic stack. We shall see below that $\mathcal{U}$_1 and $\mathcal{U}$_2 are isomorphic. The stack $\mathcal{U}$_2 is therefore algebraic, as is ´m_{Γ_0(p)}, the union of $\mathcal{U}$_1 and $\mathcal{U}$_2 by (1).

1.8. Let (C,A) be classified by $\mathcal{U}$_2. By II.1.20, C_p is locally free of rank p². One checks fiber by fiber that A acts freely on C, and this makes it possible to define

(1.8.1) w : $\mathcal{U}$_2 → $\mathcal{U}$_1 : (C,A) ↦ (C/A,C_p/A_p).



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(1) Points à l'infini de 'h_{Γ_0(p)} .

Soit C le polygone de Néron à un côté sur Spec(ℤ) . On a C^{reg} ≃ G_m , C_p ≃ μ_p et (C,C_p) est une section e_1 de 'h_{Γ_0(p)} sur ℤ .

Soit C le p-gone sur Spec(ℤ) . On a C^{reg} ≃ G_m × ℤ/p , C_p ≃ μ_p × ℤ/p et (C,e × ℤ/p) est une section e_2 de 'h_{Γ_0(p)} sur ℤ .

Soit k un corps algébriquement clos. On vérifie que les seules objets (C,A) de 'h_{Γ_0(p)}(k) , avec C singulier, sont e_1(k) et e_2(k) .

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Lemma 1.8.2. The morphism (1.8.1) is an isomorphism.

With the notation of 1.1 and 1.3, one has

$\mathcal{U}$_1 = $\mathcal{G}$_p ∪ m_{Γ_0(p)}° ↗ ↘ ´m_{Γ_0(p)}° = $\mathcal{U}$_1 ∩ $\mathcal{U}$_2 ´m_{Γ_0(p)} ↘ ↗ $\mathcal{U}$_2 = $\mathcal{B}$_p ∪ m_{Γ_0(p)}°.

Construction IV.4.4 again shows that w induces an involution of ′𝔪_{Γ_0(p)}°. In 1.4 we saw that w induces an isomorphism $\mathcal{G}$_p → $\mathcal{B}$_p, and 1.8.2 follows.

This completes the proof of (A). The proof has supplied us with the following construction.

Construction 1.9. By gluing w : $\mathcal{U}$_2 → $\mathcal{U}$_1 and w^{-1} : $\mathcal{U}$_1 → $\mathcal{U}$_2, one obtains an involution w : ′𝔪_{Γ_0(p)} → ′𝔪_{Γ_0(p)} that extends the involution (IV.4.4).

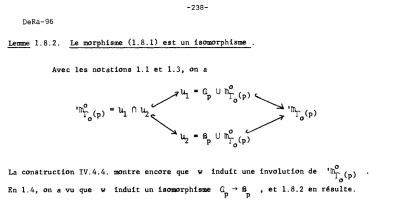
The union of $\mathcal{G}$_p and $\mathcal{B}$_p is the open substack of ′𝔪_{Γ_0(p)} corresponding to non-supersingular curves of characteristic p. Propositions 1.2 and 1.4 therefore give:

Proposition 1.10. ′𝔪_{Γ_0(p)} is smooth over Spec(ℤ) away from the supersingular points of characteristic p.

(3) Proof of (B) (properness of ′𝔪_{Γ_0(p)}).

By (1.10), ′𝔪_{Γ_0(p)}° is dense in ′𝔪_{Γ_0(p)}. The valuative criterion of properness therefore reduces us to proving the following lemma.

Lemma 1.11. Let (S,η,s) be a complete trait with algebraically closed residue field. Let (C,A) be an object of ′𝔪_{Γ_0(p)}°(η). There exists a finite extension k(η′) of k(η) such that, writing S′ for the normalization of S in k(η′), the image (C′,A′) of (C,A) in ′𝔪_{Γ_0(p)}°(η′) comes by base change from an object ($\bar{C}$,$\bar{A}$) of ′𝔪_{Γ_0(p)}(S′), unique up to unique isomorphism.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Lemme 1.8.2. Le morphisme (1.8.1) est un isomorphisme .

Avec les notations 1.1 et 1.3, on a

$'\mathfrak{h}^{o}_{\Gamma_0(p)} = U_1 \cap U_2$; $U_1 = G_p \cup \mathfrak{h}^{o}_{\Gamma_0(p)}$; $U_2 = \mathcal{B}_p \cup \mathfrak{h}^{o}_{\Gamma_0(p)}$; $'\mathfrak{h}^{o}_{\Gamma_0(p)}$

La construction IV.4.4. montre encore que w induit une involution de $'\mathfrak{h}^{o}_{\Gamma_0(p)}$.
En 1.4, on a vu que w induit un isomorphisme $G_p \to \mathcal{B}_p$, et 1.8.2 en résulte.

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Proof: After first making an extension of k(η), one may suppose that C has semistable reduction and that the special fiber of its minimal model $\tilde{C}$ is smooth or an m-gon, with p|m (IV.1.6). The group $\tilde{C}$_p is then finite over S (II.1.20), and the schematic closure of A is a finite locally free subgroup scheme of rank p of $\tilde{C}$. One obtains ($\bar{C}$,$\bar{A}$) by contracting the components of $\tilde{C}$_s that do not meet A (IV.1.3).

Uniqueness is proved as in (IV.1.6 (iv)): if ($\bar{C}$_1,$\bar{A}$_1) has generic fiber C, one obtains the minimal model $\tilde{C}$ of C by successively blowing up the singularities, as in (IV.1.6 (iv)). The inverse image of $\bar{A}$_1 is the schematic closure $\tilde{A}$ of A, and it is clear that $\bar{C}$ is obtained by contracting the irreducible components of $\tilde{C}$_s whose intersection with $\tilde{A}$ is empty; hence uniqueness.

(4) Proof of C (c finite and representable).

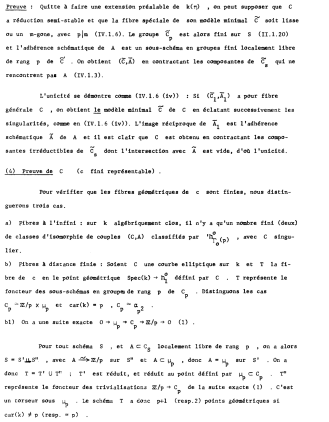
To verify that the geometric fibers of c are finite, we distinguish three cases.

a) Fibers at infinity: over an algebraically closed field k there are only finitely many (two) isomorphism classes of pairs (C,A), classified by ′𝔪_{Γ_0(p)}°, with C singular.

b) Fibers at finite distance: let C be an elliptic curve over k, and let T be the fiber of c over the geometric point Spec(k) → 𝔪_1° defined by C. The scheme T represents the functor of rank-p subgroup schemes of C_p. Distinguish the cases C_p ≃ ℤ/p × μ_p and, when char(k) = p, C_p ≃ α_{p^2}.

b1) There is an exact sequence $0 \to \mu_p \to C_p \to \mathbb{Z}/p \to 0$ (1).

For every scheme S and every $A \subset C_S$ locally free of rank p, one then has $S = S' \amalg S''$, with $A \xrightarrow{\sim} \mathbb{Z}/p$ over S'' and $A \subset \mu_p$, hence $A = \mu_p$ over S'. Thus $T = T' \cup T''$; T' is reduced and consists of the point defined by $\mu_p \subset C_p$. The scheme T'' represents the functor of trivializations $\mathbb{Z}/p \to C_p$ of the exact sequence (1). It is a torsor under μ_p. Consequently T has p+1 (respectively 2) geometric points if $\mathrm{char}(k) \neq p$ (respectively $= p$).



Preuve : Quitte à faire une extension préalable de k(η) , on peut supposer que C a réduction semi-stable et que la fibre spéciale de son modèle minimal $\overline{C}$ soit lisse ou un n-gone, avec p|n (IV.1.6). Le groupe G_p est alors fini sur S (II.1.20) et l'adhérence schématique de A est un sous-schéma en groupes fini localement libre de rang p de $\overline{C}$. On obtient $(\overline{C},\overline{A})$ en contractant les composantes de $\overline{C}_s$ qui ne rencontrent pas A (IV.1.3).

L'unicité se démontre comme (IV.1.6 (iv)) : Si $(\overline{C}_1,\overline{A}_1)$ a pour fibre générale C , on obtient le modèle minimal $\overline{C}$ de C en éclatant successivement les singularités, comme en (IV.1.6 (iv)). L'image réciproque de $\overline{A}_1$ est l'adhérence schématique $\overline{A}$ de A et il est clair que C est obtenu en contractant les composantes irréductibles de $\overline{C}_s$ dont l'intersection avec $\overline{A}$ est vide, d'où l'unicité.

(4) Preuve de C (c fini représentable) .

Pour vérifier que les fibres géométriques de c sont finies, nous distinguerons trois cas.

a) Fibres à l'infini : sur k algébriquement clos, il n'y a qu'un nombre fini (deux) de classes d'isomorphie de couples (C,A) classifiés par $\mathcal{M}^{ns}_{1,p}(p)$, avec C singulier.

b) Fibres à distance finie : Soient C une courbe elliptique sur k et T la fibre de c en le point géométrique $\mathrm{Spec}(k) \to \mathcal{M}^{ns}_1$ défini par C . T représente le foncteur des sous-schémas en groupes de rang p de C_p . Distinguons les cas $C_p = \mathbb{Z}/p \times \mu_p$ et $\mathrm{car}(k) = p$, $C_p = \alpha_{p^2}$.

b1) On a une suite exacte $0 \to \mu_p \to C_p \to \mathbb{Z}/p \to 0$ (1) .

Pour tout schéma S , et $A \subset C_S$ localement libre de rang p , on a alors $S = S' \amalg S''$, avec $A \xrightarrow{\sim} \mathbb{Z}/p$ sur S'' et $A \subset \mu_p$, donc $A = \mu_p$ sur S' . On a donc $T = T' \cup T''$; T' est réduit, et réduit au point défini par $\mu_p \subset C_p$. T'' représente le foncteur des trivialisations $\mathbb{Z}/p \to C_p$ de la suite exacte (1) . C'est un torseur sous μ_p . Le schéma T a donc p+1 (resp.2) points géométriques si $\mathrm{car}(k) \neq p$ (resp. $= p$) .

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b2) $C_p \simeq \alpha_{p^2}$. Every subgroup of α_{p^2} is infinitesimal. If it has order p, it equals the Frobenius kernel α_p, and T has only one geometric point.

Since $\overline{\mathfrak{m}}_{\Gamma_0(p)}$ is proper, c, being proper with finite fibers, is finite. To prove that c is representable, one proceeds as in IV.3.5: one checks over ⟦a blank in the authority⟧ that, for a singular pair (C,A), $\mathrm{Aut}(C,A) = \{\pm 1\}$, so that

$$\mathrm{Aut}(C,A) \hookrightarrow \mathrm{Aut}(c(C)).$$

(5) The key point of the proof is the following lemma.

Lemma 1.12. The morphism $c : \overline{\mathfrak{m}}_{\Gamma_0(p)}{}^{\circ} \to \mathfrak{m}_1{}^{\circ}$ is finite and flat.

Proof: We shall prove 1.12 using the following criterion, which applies because $\mathfrak{m}_1{}^{\circ}$ is reduced.

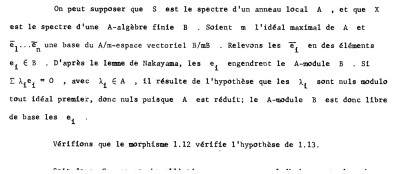
Lemma 1.13. Let $f : X \to S$ be a finite morphism of Noetherian schemes, with S reduced. If the rank of the geometric fibers of f is constant, then f is flat.

One may suppose that S is the spectrum of a local ring A and that X is the spectrum of a finite A-algebra B. Let $\mathfrak{m}$ be the maximal ideal of A, and let $\bar{e}_1,\ldots,\bar{e}_n$ be a basis of the $A/\mathfrak{m}$-vector space $B/\mathfrak{m}B$. Lift the $\bar{e}_i$ to elements $e_i \in B$. By Nakayama's lemma, the e_i generate the A-module B. If $\sum \lambda_i e_i = 0$, with $\lambda_i \in A$, the hypothesis implies that the λ_i vanish modulo every prime ideal, and hence vanish because A is reduced; thus the A-module B is free with basis the e_i.

Let us verify that the morphism 1.12 satisfies the hypothesis of 1.13.

Let C be an elliptic curve over an algebraically closed field k. The fiber T of c over the geometric point $\mathrm{Spec}(k) \to \mathfrak{m}_1{}^{\circ}$ defined by C represents the functor of rank-p subgroup schemes of C_p. We prove that it has rank p+1.

If C_p ≃ μ_p × ℤ/p, we saw in (4) that T is isomorphic to Spec(k) ⨿ μ_p,



On peut supposer que S est le spectre d'un anneau local A , et que X est le spectre d'une A-algèbre finie B . Soient m l'idéal maximal de A et $\bar{e}_1 \ldots \bar{e}_n$ une base du A/m-espace vectoriel B/mB . Relevons les $\bar{e}_i$ en des éléments $e_i \in B$. D'après le lemme de Nakayama, les e_i engendrent le A-module B . Si $\Sigma \lambda_i e_i = 0$, avec $\lambda_i \in A$, il résulte de l'hypothèse que les λ_i sont nuls modulo tout idéal premier, donc nuls puisque A est réduit; le A-module B est donc libre de base les e_i .

Vérifions que le morphisme 1.12 vérifie l'hypothèse de 1.13.

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Authority page 99; collected-paper page 592; original folio 241.

and hence has rank p+1.

It remains to treat the case in which k has characteristic p and C_p ≃ α_{p²}. In this case T represents the functor of locally free rank-p subschemes A of 𝔾_a contained in α_{p²}.

Over an arbitrary base, a locally free rank-p subscheme A of 𝔾_a is defined by an equation

(*) X^p + ∑_{0}^{p-1} a_i X^i = 0.

For A to be stable under the homothety X ↦ nX (n ∈ (ℤ/p)*), it is necessary and sufficient that (*) be identical to the equation

X^p + ∑_{0}^{p-1} n^{p-i} a_i X^i.

In particular, if A is a subgroup scheme of 𝔾_a, its equation reduces to

X^p - aX = 0.

For A to be a subscheme of α_{p²}, it is necessary and sufficient that the left-hand side of this equation divide X^{p²}. Since

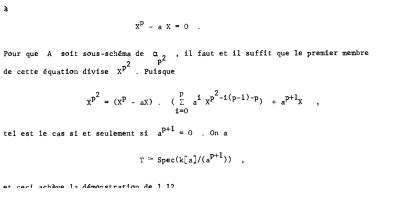
X^{p²} = (X^p - aX) · (∑_{i=0}^{p} a^i X^{p²-i(p-1)-p}) + a^{p+1}X,

this holds if and only if a^{p+1} = 0. Thus

T ≃ Spec(k[a]/(a^{p+1})),

which completes the proof of 1.12.

Since c is finite and flat and 𝔪_1° is regular, ′𝔪_{Γ_0(p)}° is Cohen–Macaulay (EGA IV.15 4.2), purely of relative dimension one over Spec(ℤ). At infinity, 𝔪_{Γ_0(p)} is even smooth (1.10); consequently the whole of ′𝔪_{Γ_0(p)} is Cohen–Macaulay and, again by 1.10, smooth over Spec(ℤ) outside a finite set (of codimension 2). Assertion (D) therefore follows from Serre's criterion for normality (EGA IV.5.8.6).



à

$$X^p - aX = 0 \ .$$

Pour que A soit sous-schéma de α_{p^2} , il faut et il suffit que le premier membre de cette équation divise X^{p^2} . Puisque

$$X^{p^2} = (X^p - aX) \ . \ \left(\sum_{i=0}^{p} a^i X^{p^2-i(p-1)-p}\right) + a^{p+1}X \ ,$$

tel est le cas si et seulement si $a^{p+1} = 0$. On a

$$T \simeq \mathrm{Spec}(k[a]/(a^{p+1})) \ ,$$

et ceci achève la démonstration de 1.12

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1.14. Variants: Let n be an integer prime to p, let H ⊂ GL(2, $\mathbb{Z}$/n), and let K be the inverse image of H in GL(2, $\hat{\mathbb{Z}}$). Theorem 1.6 generalizes to $\mathfrak{m}_{K\cap\Gamma_0(p)}[1/n]$:

(1.14.1) $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}[1/n]$ classifies generalized elliptic curves C/S whose geometric fibers are smooth, n-gons, or np-gons, equipped with an isomorphism α : $C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ and a locally free rank-p subgroup H of C such that $C_n \cdot H$ meets every geometric component of every geometric fiber.

(.14.2) For square-free n, $\mathfrak{m}_{\Gamma_0(n)}$ classifies generalized elliptic curves C/S equipped with a locally free rank-n subgroup H of C that meets every geometric component of every geometric fiber.

1.15. Let S be a scheme of characteristic p and E an elliptic curve over S. To E one associates the following two objects of $\mathfrak{m}_{\Gamma_0(p)}{}^{\circ}(S)$:

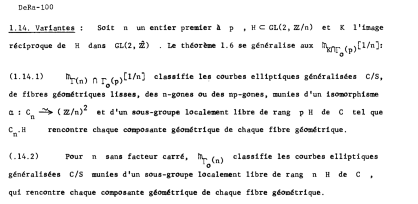
$\Phi_1(E) : (E, \mathrm{Ker}(F : E \to E^{(p)}))$

$\Phi_2(E) = w\Phi_1(E) : (E^{(p)}, \mathrm{Ker}(V : E^{(p)} \to E))$.

In these formulas, F denotes the Frobenius morphism and V its transpose, the Verschiebung: FV = VF = p. These constructions define a diagram

$$\begin{array}{ccccc} \mathfrak{m}_1{}^{\circ}\otimes\mathbb{F}_p & & \mathfrak{m}_1{}^{\circ}\otimes\mathbb{F}_p & & \\ & \searrow^{\Phi_1} \quad {}^{w\Phi_1}\swarrow & & & \\ & & \mathfrak{m}_{\Gamma_0(p)}{}^{\circ}\otimes\mathbb{F}_p & & \\ & & & \swarrow^{c} \quad {}_{cw}\searrow & \\ \mathfrak{m}_1{}^{\circ}\otimes\mathbb{F}_p & & & & \mathfrak{m}_1{}^{\circ}\otimes\mathbb{F}_p \end{array}$$

The composites $c\Phi_1 : E \mapsto (E, \mathrm{Ker}\, F) \mapsto E$ and $cw\Phi_1$ are the identity. The composites $cw\Phi_1 : E \mapsto (E^{(p)}, \mathrm{Ker}\, V) \mapsto E^{(p)}$ are the Frobenius endomorphism of $\mathfrak{m}_1{}^{\circ} \otimes \mathbb{F}_p$.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1.14. Variantes : Soit n un entier premier à p , H ⊂ GL(2, ℤ/n) et K l'image réciproque de H dans GL(2, $\hat{\mathbb{Z}}$) . Le théorème 1.6 se généralise aux $\mathfrak{M}_{K\cap\Gamma_0(p)}[1/n]$:

(1.14.1) $\mathfrak{M}_{\Gamma(n)\cap\Gamma_0(p)}[1/n]$ classifie les courbes elliptiques généralisées C/S, de fibres géométriques lisses, des n-gones ou des np-gones, munies d'un isomorphisme $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ et d'un sous-groupe localement libre de rang p H de C tel que $C_n \cdot H$ rencontre chaque composante géométrique de chaque fibre géométrique.

(.14.2) Pour n sans facteur carré, $\mathfrak{M}_{\Gamma_0(n)}$ classifie les courbes elliptiques généralisées C/S munies d'un sous-groupe localement libre de rang n H de C , qui rencontre chaque composante géométrique de chaque fibre géométrique.

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Theorem 1.16. (i) The maps Φ_1 and $\Phi_2 = w\Phi_1$ are closed immersions. Their images are the two irreducible components of $\mathfrak{m}_{\Gamma_0(p)}{}^{\circ} \otimes \mathbb{F}_p$.

(ii) These irreducible components meet transversely at the supersingular points. The stack $\mathfrak{m}_{\Gamma_0(p)} \otimes \mathbb{F}_p$, which is reduced, therefore has no singularities other than ordinary double points.

(iii) $\mathfrak{m}_{\Gamma_0(p)}$ is regular.

Use a superscript h to denote the stack obtained by removing the supersingular points of characteristic p. One checks that Φ_1 and $w\Phi_1$ define an isomorphism

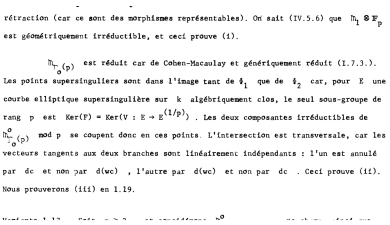
$\Phi_1 \amalg w\Phi_1 : \mathfrak{m}_1{}^{oh} \otimes \mathbb{F}_p \amalg \mathfrak{m}_1{}^{oh} \otimes \mathbb{F}_p \xrightarrow{\sim} \mathfrak{m}_{\Gamma_0(p)}{}^{oh} \otimes \mathbb{F}_p$.

The inverse isomorphism is defined by c and cw, restricted to the images of Φ_1 and $w\Phi_1$.

That Φ_1 (respectively $w\Phi_1$) is a closed immersion follows from the existence of a retraction (since these are representable morphisms). It is known (IV.5.6) that $\mathfrak{m}_1 \otimes \mathbb{F}_p$ is geometrically irreducible, and this proves (i).

The stack $\mathfrak{m}_{\Gamma_0(p)}$ is reduced because it is Cohen–Macaulay and generically reduced (I.7.3). The supersingular points lie in the images of both Φ_1 and Φ_2 because, for a supersingular elliptic curve E over an algebraically closed field k, the only subgroup of rank p is Ker(F) = Ker(V : E → E^{(1/p)}). Thus the two irreducible components of $\mathfrak{m}_{\Gamma_0(p)}{}^{\circ}$ modulo p meet at these points. The intersection is transverse because the tangent vectors to the two branches are linearly independent: one is killed by dc and not by d(wc), while the other is killed by d(wc) and not by dc. This proves (ii). We shall prove (iii) in 1.19.

Variant 1.17. Let n ≥ 3 and consider $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ}$. This stack, as well as $\mathfrak{m}_{\Gamma(n)}{}^{\circ}$, is a scheme: the classified objects have no automorphisms (cf. IV.2.7). By 1.14 these schemes represent the functors of isomorphism classes of objects of the following type


rétraction (car ce sont des morphismes représentables). On sait (IV.5.6) que $\mathfrak{h}_1 \otimes \mathbb{F}_p$ est géométriquement irréductible, et ceci prouve (i).

$\mathfrak{h}_{\Gamma_0(p)}$ est réduit car de Cohen-Macaulay et génériquement réduit (I.7.3.). Les points supersinguliers sont dans l'image tant de Φ_1 que de Φ_2 car, pour E une courbe elliptique supersingulière sur k algébriquement clos, le seul sous-groupe de rang p est Ker(F) = Ker(V : E → $E^{(1/p)}$) . Les deux composantes irréductibles de $\mathfrak{h}^{\circ}_{\Gamma_0(p)}$ mod p se coupent donc en ces points. L'intersection est transversale, car les vecteurs tangents aux deux branches sont linéairement indépendants : l'un est annulé par dc et non par d(wc) , l'autre par d(wc) et non par dc . Ceci prouve (ii). Nous prouverons (iii) en 1.19.

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$\mathfrak{m}_{\Gamma(n)}{}^{\circ}$: (E,α_n) : α_n : E_n ⥲ $(\mathbb{Z}/n\mathbb{Z})^2$;

$\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ}$: (E,α_n,H) : α_n : E_n ⥲ $(\mathbb{Z}/n\mathbb{Z})^2$; H ⊂ E of rank p,

or, more symmetrically (cf. IV.4.5), set F = E/H.

(E →^h F, α_n) : h a p-isogeny and α_n : E_n ⥲_{h̄} F_n ⥲ $(\mathbb{Z}/n\mathbb{Z})^2$.

After reduction modulo p, there is a diagram analogous to (1.15.1). Its arrows (expressed as morphisms of functors) are

$\Phi_1 : \mathfrak{m}_{\Gamma(n)}{}^{\circ} \otimes \mathbb{F}_p \to \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} \otimes \mathbb{F}_p : (E,\alpha_n) \mapsto (E,\alpha_n, \mathrm{Ker}\, F)$

$w : \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} \xrightarrow{\sim} \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} : (E,\alpha_n,H) \mapsto (E/H,\alpha_n,E_p/H)$

$c : \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} \to \mathfrak{m}_{\Gamma(n)}{}^{\circ} : (E,\alpha_n,H) \mapsto (E,\alpha_n).$

Take care that the isomorphism $w^2(E,H) \simeq (E,H)$ defined in IV.4.4 induces an isomorphism $w^2(E,\alpha_n,H) \simeq (E,p\alpha_n,H)$. The composite $cw\Phi_1 : (E,\alpha_n) \mapsto (E^{(p)},\alpha_n)$ is Frobenius.

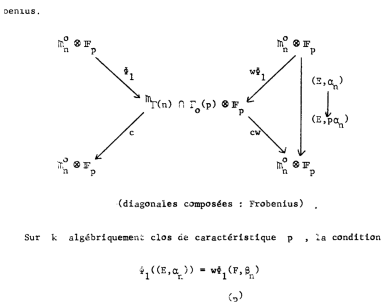
$\mathfrak{m}_n{}^{\circ} \otimes \mathbb{F}_p$ $\mathfrak{m}_n{}^{\circ} \otimes \mathbb{F}_p$ ╲ Φ_1 $w\Phi_1$ ╱ │ ↘ ↙ │ (E,α_n) $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}{}^{\circ} \otimes \mathbb{F}_p$ ↓ ↙ ↘ │ (E,pα_n) c ╱ ╲ cw $\mathfrak{m}_n{}^{\circ} \otimes \mathbb{F}_p$ $\mathfrak{m}_n{}^{\circ} \otimes \mathbb{F}_p$

(composite diagonals: Frobenius).

Over an algebraically closed field k of characteristic p, the condition

$\Phi_1((E,\alpha_r)) = w\Phi_1(F,\beta_n)$

means that F is supersingular, that E = F^{(p)}, and that the diagram



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$E_n = F_n^{(p)} \tilde{\leftarrow} F_n \mid \alpha_n \mid \beta_n \downarrow \downarrow (\mathbb{Z}/n\mathbb{Z})^2 = (\mathbb{Z}/n\mathbb{Z})^2$

is commutative, i.e. $(E,\alpha_n) = (F,\beta_n)^{(p)}$. Parts (i) and (ii) of 1.16 therefore admit the following variant.

Variant 1.18. The scheme $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}^{\circ} \otimes_{\mathbb{Z}} \mathbb{F}_p$ can be obtained by gluing two copies of $\mathfrak{m}_{\Gamma(n)}^{\circ} \otimes \mathbb{F}_p$ along the supersingular points: the supersingular point x of the second copy is identified with the point $x^{(p)}$ of the first copy.

1.19. Let us prove that $\mathfrak{m}_{\Gamma_0(p)}$ is regular. Away from the supersingular points, this follows from 1.10. Since at a supersingular point $\mathfrak{m}_{\Gamma_0(p)} \to \mathrm{Spec}(\mathbb{Z})$ has an ordinary quadratic singularity, (I.5.2) shows that there exists $k \in \mathbb{N}$ such that the completion of the strict local ring of $\mathfrak{m}_{\Gamma_0(p)}$ at such a point is isomorphic to

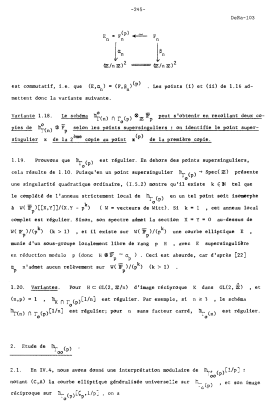
$W(\mathbb{F}_p)[[X,Y]]/(X.Y - p^k)$ (W = Witt vectors).

If $k = 1$, this complete local ring is regular. Otherwise, its spectrum has the section $X = Y = 0$ over $W(\mathbb{F}_p)/(p^k)$ $(k > 1)$, and over $W(\mathbb{F}_p)/(p^k)$ there exists an elliptic curve E equipped with a locally free subgroup H of rank p, with E supersingular after reduction modulo p (hence $H \otimes \mathbb{F}_p \simeq \alpha_p$). This is absurd, because by [22] α_p admits no lifting over $W(\mathbb{F}_p)/(p^k)$ when $k > 1$.

1.20. Variants. Let $H \subset GL(2, \mathbb{Z}/n)$, let K be its inverse image in $GL(2, \hat{\mathbb{Z}})$, and suppose $(n,p) = 1$. Then $\mathfrak{m}_{K\cap\Gamma_0(p)}[1/n]$ is regular. For example, if $n \geq 3$, the scheme $\mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)}[1/n]$ is regular; for square-free n, $\mathfrak{m}_{\Gamma_0(n)}$ is regular.

2. Study of $\mathfrak{m}_{\Gamma'_\infty(p)}$.

2.1. In IV.4 we gave a modular interpretation of $\mathfrak{m}_{\Gamma'_\infty(p)}[1/p]$: denoting by (C,A) the universal generalized elliptic curve over $\mathfrak{m}_{\Gamma_0(p)}$, and also its inverse image over $\mathfrak{m}_{\Gamma_0(p)}[\zeta_p,1/p]$, one has



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$\mathfrak{m}_{\Gamma'_\infty(p)}[1/p] = \mathrm{Isom}_{\mathfrak{m}_{\Gamma_0(p)}[\zeta_p,1/p]}(\mathbb{Z}/p,A)$.

By definition, $\mathfrak{m}_{\Gamma'_\infty(p)}$ is the normalization of $\mathfrak{m}_{\Gamma_0(p)}[\zeta_p]$ in this stack $\mathrm{Isom}_{\mathfrak{m}_{\Gamma_0(p)}[\zeta$
One readily deduces the description below of the stack $\mathfrak{m}_{\Gamma'_\infty(p)}^h$ obtained from $\mathfrak{m}_{\Gamma'_\infty(p)}$ by removing the supersingular points of characteristic p.

2.2. Let $\mathcal{G}'_p \subset \mathfrak{m}_{\Gamma_0(p)}$ be the open substack classifying the pairs (C,A) for which A is locally isomorphic to μ_p (for the étale topology), and let $\mathcal{B}'_p \subset \mathfrak{m}_{\Gamma_0(p)}$ be the open substack classifying the pairs (C,A) for which A is locally isomorphic to $\mathbb{Z}/p$. Set

$\mathcal{U} = \mathrm{Isom}_{\mathcal{G}'_p}(\mu_p,A)$

$\mathcal{V} = \mathrm{Isom}_{\mathcal{B}'_p}(\mathbb{Z}/p,A)$;

the schemes $\mathcal{U}$ and $\mathcal{V}$ are finite étale over $\mathcal{G}'_p$ and $\mathcal{B}'_p$, respectively. Over $\mathbb{Z}[\zeta_p,1/p]$, $\mathbb{Z}/p$ and μ_p are canonically isomorphic, hence there is an isomorphism

$\mathcal{U}[\zeta_p,1/p] \simeq \mathcal{V}[\zeta_p,1/p]$.

Proposition 2.3. The stack $\mathfrak{m}_{\Gamma'_\infty(p)}^h$ is obtained by gluing $\mathcal{U}[\zeta_p]$ and $\mathcal{V}[\zeta_p]$ along the common open substack $\mathcal{U}[\zeta_p,1/p]$. In particular, $\mathfrak{m}_{\Gamma'_\infty(p)}^h$ is smooth over $\mathbb{Z}[\zeta_p]$.

To study $\mathfrak{m}_{\Gamma'_\infty(p)}$ near the supersingular points, we shall use the results of [22] recalled below.

2.4. The functor F that assigns to a scheme S the set of isomorphisms

$\chi : \mathbb{F}_p^* \tilde{\to} \mu_{p-1}$

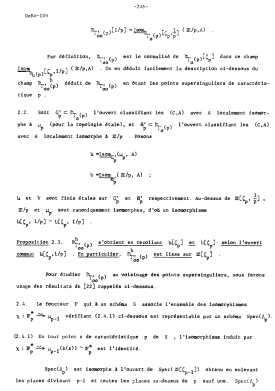
satisfying (2.4.1) below is represented by a scheme $\mathrm{Spec}(\Lambda_p)$.

(2.4.1) At every point s of S of characteristic p, the induced isomorphism

$\chi : \mathbb{F}_p^* \tilde{\to} \mu_{p-1}(k(s)) \simeq \mathbb{F}_p^*$

is the identity.

The scheme $\mathrm{Spec}(\Lambda_p)$ is isomorphic to the open subscheme of $\mathrm{Spec}(\mathbb{Z}[\zeta_{p-1}])$ obtained by removing the places dividing p-1 and all but one of the places above p. $\mathrm{Spec}(\Lambda_p)$



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is étale over Spec(ℤ), and $\mathbb{F}_p \tilde{\to} \Lambda_p \otimes \mathbb{F}_p$.

Let S be equipped with χ satisfying (2.4.1) (so S is a Λ_p-scheme), and let A be a locally free group scheme of rank p over S. Let L be the submodule of the affine algebra $\mathcal{O}$(A) of A (a sheaf of $\mathcal{O}$_S-algebras) formed by those f such that

f(ix) = χ(i) f(x) (i ∈ $\mathbb{F}_p$^*).

By [22], L is an invertible module over S, and there exists a unique section a of L^{⊗(1-p)} such that, for f as above,

f^p = ⟨a,f^{p-1}⟩.f in L ⊂ $\mathcal{O}$(A).

Let A* be the Cartier dual of A, and let L* be defined by A*. The duality e between $\mathcal{O}$(A) and $\mathcal{O}$(A*) puts L and L* in duality. Let b ∈ (L*)^{⊗ 1-p} be the analogue for A* of the section a above. Then

a.b = w(χ)

for a certain number w(χ) ∈ Λ_p, with (w(χ)) = (p) in Λ_p.

The functor (on isomorphisms)

(A : a locally free group scheme of rank p) ↦ ((L,a,b) : L invertible, a ∈ L^{⊗(1-p)}, b ∈ L^{⊗(p-1)}, a.b = w(χ))

is an equivalence of categories.

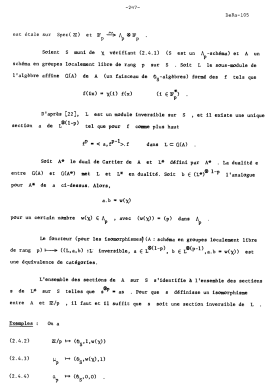
The set of sections of A over S is identified with the set of sections s of L* over S such that s^{⊗p} = as. For s to define an isomorphism between A and ℤ/p, it is necessary and sufficient that s be an invertible section of L*.

Examples: One has

(2.4.2) ℤ/p ↦ ($\mathcal{O}$_S,1,w(χ))

(2.4.3) μ_p ↦ ($\mathcal{O}$_S,w(χ),1)

(2.4.4) α_p ↦ ($\mathcal{O}$_S,0,0).



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Authority page 106; collected-paper page 599; original folio 248.

2.5. Let an isomorphism ζ : $\mathbb{Z}/p \xrightarrow{\sim} \mu_p$ be given over S (where S is a Λ_p[ζ_p,1/p]-scheme). The isomorphism ζ defines ζ* : $\mathcal{O}(\mu_p) \to \mathcal{O}(\mathbb{Z}/p)$ and, through (2.4.2) and (2.4.3), induces

ζ* : $\mathcal{O}$_S = L(μ_p) → L($\mathbb{Z}$/p) = $\mathcal{O}$_S,

namely multiplication by a number w(χ,ζ) ∈ Λ_p[ζ_p]. One has

w(χ,ζ)^{p-1} = w(χ).

An isomorphism s : $\mathbb{Z}/p \xrightarrow{\sim} A$ defines by duality

s* : A* $\xrightarrow{\sim}$ μ_p $\xleftarrow{\sim}$ $\mathbb{Z}$/p,

whose inverse is t. The sections s(1) and t(1) are identified with sections x and y of L* and L satisfying

x^{p-1} = a y^{p-1} = b

xy = w(χ,ζ).

2.6. Let ′𝔪_{Γ′_∞(p)} ⊗ Λ_p be the following algebraic Λ_p[ζ_p]-stack over 𝔪_{Γ_0(p)} ⊗ Λ_p[ζ_p]: an object (C,A,x,y) of ′𝔪_{Γ′_∞(p)} ⊗ Λ_p over a Λ_p[ζ_p]-scheme S consists of

a) an object (C,A) of 𝔪_{Γ_0(p)}. Let (L,a,b) be deduced from A.

b) sections x of L* and y of L such that x^{p-1} = a, y^{p-1} = b, and xy = w(χ,ζ).

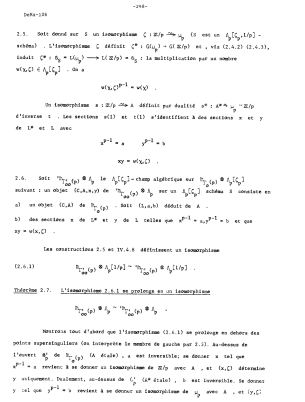
The constructions of 2.5 and IV.4.8 define an isomorphism

(2.6.1) 𝔪_{Γ′_∞(p)} ⊗ Λ_p[1/p] ≃ ′𝔪_{Γ′_∞(p)} ⊗ Λ_p[1/p].

Theorem 2.7. The isomorphism 2.6.1 extends to an isomorphism

𝔪_{Γ′_∞(p)} ⊗ Λ_p ≃ ′𝔪_{Γ′_∞(p)} ⊗ Λ_p.

Let us first show that the isomorphism (2.6.1) extends away from the supersingular points (the left-hand side is interpreted using 2.3). Over the open substack $\mathcal{B}'$_p of 𝔪_{Γ_0(p)} (where A is étale), a is invertible; giving x such that x^{p-1} = a amounts to giving an isomorphism from $\mathbb{Z}$/p to A, and (x,ζ) uniquely determines y. Dually, over $\mathcal{G}'$_p (where A* is étale), b is invertible. Giving y such that y^{p-1} = b amounts to giving an isomorphism from μ_p to A, and (y,ζ)



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uniquely determines x. Where p is invertible, these two constructions are compatible through the isomorphism ζ : ℤ/p ⥲ μ_p.

The stack ´𝔪_{Γ´_∞(p)} ⊗ Λ_p is finite and representable over 𝔪_{Γ_0(p)} ⊗ Λ_p. It therefore suffices to prove that ´𝔪_{Γ´_∞(p)} ⊗ Λ_p is normal at the supersingular points. The following lemma shows that ´𝔪_{Γ´_∞(p)} ⊗ Λ_p is in fact regular at these points.

Lemma 2.8. Let s̄ be a supersingular geometric point of characteristic p of 𝔪_{Γ_0(p)}. Let Ŝ be the completion of the strict henselization of 𝔪_{Γ_0(p)} at s̄, equipped with its unique Λ_p-algebra structure. Let (C,A) over Ŝ be defined by Ŝ → 𝔪_{Γ_0(p)}, and let (L,a,b) be defined by A.

(i) Ŝ ≃ W(𝔽_p)[[u,v]]/(uv-w(χ)) with (L,a,b) ≃ (𝒪,u,v).

(ii) The normalization ´𝔪_{Γ´_∞(p)} ⊗_{𝔪_{Γ_0(p)}} Ŝ of Ŝ in Isom_{Ŝ[ζ_p,1/p]}(ℤ/p,A) is Ŝ-isomorphic to

W(𝔽_p)[[ζ_p,x,y]]/(xy-w(χ,ζ)) ≃ W(𝔽_p)[[x,y]]/(x^{p-1} y^{p-1} - w(χ)),

with x^{p-1} = u and y^{p-1} = v.

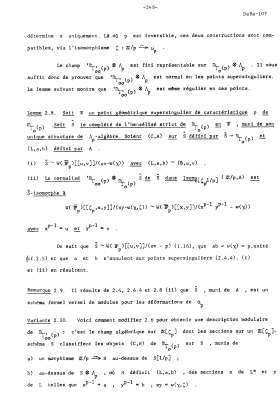
It is known that Ŝ ≃ W(𝔽_p)[[u,v]]/(uv - p) (1.16), that ab = w(χ) = p times a unit (cf. 2.5), and that a and b vanish at the supersingular points (2.4.4). Assertions (i) and (ii) follow.

Remark 2.9. It follows from 2.4, 2.4.4, and 2.8(ii) that Ŝ, equipped with A, is a versal formal moduli scheme for deformations of α_p.

Variant 2.10. Here is how to modify 2.6 to obtain a modular description of 𝔪_{Γ´_∞(p)}: it is the algebraic stack over ℤ[ζ_p] whose sections over a ℤ[ζ_p]-scheme S classify the objects (C,H) of 𝔪_{Γ_0(p)} over S, equipped with

a) a morphism ℤ/p ⥲ H over S[1/p];

b) over S ⊗ Λ_p, where H defines (L,a,b), sections x of L* and y of L such that x^{p-1} = a, y^{p-1} = b, and xy = w(χ,ζ).



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These two data are required to be compatible, through the dictionary of 2.4, over S ⊗ Λ_p[1/p].

2.11. With the notation of 2.2 and of 1.1 and 1.3, $\mathcal{U}$ ⊗ $\mathbb{F}$_p and $\mathcal{V}$ ⊗ $\mathbb{F}$_p are étale covers of $\mathcal{G}$_p ⊗ $\mathbb{F}$_p and $\mathcal{B}$_p ⊗ $\mathbb{F}$_p, respectively. The morphisms Φ_1 and wΦ_1 (cf. 1.15) identify $\mathcal{G}$_p ⊗ $\mathbb{F}$_p and $\mathcal{B}$_p ⊗ $\mathbb{F}$_p with open substacks of $\mathfrak{m}$_1 ⊗ $\mathbb{F}$_p; through this identification, the local structure theorem 2.8 shows that the covers $\mathcal{U}$ ⊗ $\mathbb{F}$_p and $\mathcal{V}$ ⊗ $\mathbb{F}$_p of $\mathfrak{m}$_1^h ⊗ $\mathbb{F}$_p are totally ramified at the supersingular points of $\mathfrak{m}$_1 ⊗ $\mathbb{F}$_p. In particular, $\mathcal{U}$ ⊗ $\mathbb{F}$_p and $\mathcal{V}$ ⊗ $\mathbb{F}$_p are geometrically irreducible, and $\mathfrak{m}$_{Γ′_∞(p)} ⊗ $\mathbb{F}$_p has two irreducible components, the inverse images of the irreducible components of $\mathfrak{m}$_{Γ_0(p)} ⊗ $\mathbb{F}$_p. The proof of 2.7 gives the following result.

Theorem 2.12. (i) The two irreducible components of $\mathfrak{m}$_{Γ′_∞(p)} ⊗ $\mathbb{F}$_p are smooth and meet transversely at the supersingular points.

(ii) The scheme $\mathfrak{m}$_{Γ′_∞(p)} is regular; the morphism $\mathfrak{m}$_{Γ′_∞(p)} → Spec($\mathbb{Z}$[ζ_p]) has only quadratic singularities.

(iii) The morphism $\mathfrak{m}$_{Γ′_∞(p)} → $\mathfrak{m}$_{Γ_0(p)} induces a bijection (an equivalence of categories) on the supersingular geometric points of characteristic p.

2.13. Let n be an integer ≥ 3 and prime to p. As in 1.17, one can make 2.12 more concrete by passing to the scheme $\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)}. Let $\mathcal{J}$_n^h be the étale cover Isom(μ_p,(C_p)^o) of $\mathfrak{m}$_{Γ(n)}^h ⊗ $\mathbb{F}$_p. It extends to a cover $\mathcal{J}$_n of $\mathfrak{m}$_{Γ(n)} ⊗ $\mathbb{F}$_p, totally ramified at the supersingular points, and

$\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)} ⊗_{$\mathbb{Z}$[ζ_p]} $\mathbb{F}$_p

is obtained by gluing two copies of $\mathcal{J}$_{n,p} ⊗_{$\mathbb{F}$_p} $\mathbb{F}$_p along the supersingular points, identifying x in the second copy with x^{(p)} in the first (as in 1.17). In particular, the morphism

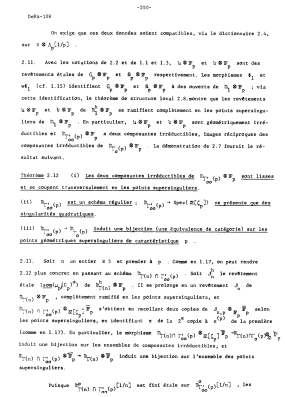
$\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)} ⊗_{$\mathbb{Z}$[ζ_p]} $\mathbb{F}$_p → $\mathfrak{m}$_{Γ(n)∩Γ_0(p)} ⊗_$\mathbb{Z}$ $\mathbb{F}$_p

induces a bijection on the sets of irreducible components, and

$\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)} ⊗ $\mathbb{F}$_p → $\mathfrak{m}$_{Γ(n)} ⊗ $\mathbb{F}$_p

induces a bijection on the set of supersingular points.

Since $\mathfrak{m}$_{Γ(n)∩Γ′_∞(p)}°[1/n] is finite étale over $\mathfrak{m}$_{Γ′_∞(p)}°[1/n], the



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Authority page 109; collected-paper page 602; original folio 251.

the local properties are the same.

2.14. Let H be a subgroup of (ℤ/p)^*, and let d be its index. Write Γ_∞(H) and Γ′_∞(H) for the subgroups of GL(2, $\hat{\mathbb{Z}}$) that are the inverse images of the following subgroups of GL(2, ℤ/p):

Γ_∞(H) : (H ℤ/p) Γ′_∞(H) : (H ℤ/p) (0 (ℤ/p)^*) (0 H).

For H = {e} (respectively, H = (ℤ/p)^*), one has Γ_∞(H) = Γ_∞(p) and Γ′_∞(H) = Γ′_∞(p) (respectively, Γ_∞(H) = Γ′_∞(H) = Γ_0(p)).

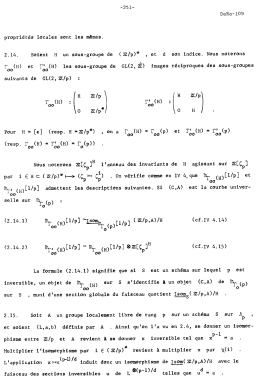
We denote by ℤ[ζ_p]^H the ring of invariants for H acting on ℤ[ζ_p] by i ∈ H ⊂ (ℤ/p)^* ↦ (ζ_p ↦ ζ_p^i). One verifies, as in IV.4, that 𝔪_{Γ_∞(H)}[1/p] and 𝔪_{Γ′_∞(H)}[1/p] admit the following descriptions. If (C,A) is the universal curve over 𝔪_{Γ_0(p)},

(2.14.1) 𝔪_{Γ_∞(H)}[1/p] ≃ Isom_{𝔪_{Γ_0(p)}[1/p]}(ℤ/p,A)/H (cf. IV.4.14)

(2.14.2) 𝔪_{Γ′_∞(H)}[1/p] ≃ 𝔪_{Γ_∞(H)}[1/p] ⊗ ℤ[ζ_p]^H (cf. IV.4.15)

Formula (2.14.1) means that if S is a scheme on which p is invertible, an object of 𝔪_{Γ_∞(H)} over S is identified with an object (C,A) of 𝔪_{Γ_0(p)} over S, equipped with a global section of the quotient sheaf Isom_S(ℤ/p,A)/H.

2.15. Let A be a locally free group scheme of rank p over a Λ_p-scheme S, and let (L,a,b) be defined by A. As seen in 2.4, giving an isomorphism between ℤ/p and A amounts to giving an invertible x such that x^{p-1} = a. Multiplying the isomorphism by i ∈ (ℤ/p)^* amounts to multiplying x by χ(i). The map x ↦ x^{(p-1)/d} therefore induces an isomorphism from Isom(ℤ/p,A)/H to the sheaf of invertible sections u of L^{⊗(1-p)/d} such that u^d = a.



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One verifies that

$w(\chi,\zeta)^{(p-1)/d} \in \Lambda_p[\zeta_p]^H$.

2.16. Let $'\mathfrak{m}_{\Gamma'_\infty(H)} \otimes \Lambda_p$ be the following algebraic stack over $\Lambda_p[\zeta_p]^H$: an object of $'\mathfrak{m}_{\Gamma'_\infty(H)} \otimes \Lambda_p$ over a $\Lambda_p[\zeta_p]^H$-scheme S is an object (C,A) of $\mathfrak{m}_{\Gamma_0(p)}$ over S, with A defining (L,a,b), together with sections u of $L^{\otimes(1-p)/d}$ and v of $L^{\otimes(p-1)/d}$ such that

$u^d = a$, $v^d = b$, $uv = w(\chi,\zeta)^{(p-1)/d}$.

The constructions of 2.14 and 1.15 define an isomorphism

(2.16.1) $'\mathfrak{m}_{\Gamma'_\infty(H)} \otimes \Lambda_p[1/p] \simeq \mathfrak{m}_{\Gamma'_\infty(H)} \otimes \Lambda_p[1/p]$.

2.17. As in 2.10, this definition can be modified to obtain a stack $'\mathfrak{m}_{\Gamma'_\infty(H)}$ over $\mathbb{Z}[\zeta_p]^H$: an object of $'\mathfrak{m}_{\Gamma'_\infty(H)}$ over a $\mathbb{Z}[\zeta_p]^H$-scheme S is an object (C,A) of $\mathfrak{m}_{\Gamma_0(H)}$ over S, equipped with

a) a global section u_1 of $\mathrm{Isom}_{S[1/p]}(\mathbb{Z}/p,A)/H$;

b) over $S \otimes \Lambda_p$, where A defines (L,a,b), sections u and v of $L^{\otimes(1-p)/d}$ and $L^{\otimes(p-1)/d}$, with $u^d = a$, $v^d = b$, and $uv = w(\chi,\zeta)^{(p-1)/d}$.

These data are required to be compatible, through (2.16.1), over $S \otimes \Lambda_p[1/p]$.

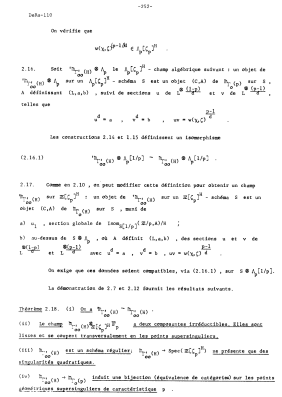
The proof of 2.7 and 2.12 gives the following results.

Theorem 2.18. (i) One has $\mathfrak{m}_{\Gamma'_\infty(H)} \simeq '\mathfrak{m}_{\Gamma'_\infty(H)}$.

(ii) The stack $\mathfrak{m}_{\Gamma'_\infty(H)} \otimes_{\mathbb{Z}[\zeta_p]^H} \mathbb{F}_p$ has two irreducible components. They are smooth and meet transversely at the supersingular points.

(iii) The stack $\mathfrak{m}_{\Gamma'_\infty(H)}$ is a regular scheme; the morphism $\mathfrak{m}_{\Gamma'_\infty(H)} \to \mathrm{Spec}(\mathbb{Z}[\zeta_p]^H)$ has only quadratic singularities.

(iv) The morphism $\mathfrak{m}_{\Gamma'_\infty(H)} \to \mathfrak{m}_{\Gamma_0(p)}$ induces a bijection (an equivalence of categories) on the supersingular geometric points of characteristic p.



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2.19. Let n be an integer prime to p, with n ≥ 3. Theorem 2.18 generalizes to the schemes $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)}[1/n]$.

(i) $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)}[1/n]$ is regular.

(ii) The morphism $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)}[1/n] \to \mathrm{Spec}(\mathbb{Z}[\zeta_p,1/p]^H)$ is smooth away from the supersingular points of characteristic p, where it has ordinary quadratic singularities.

(iii) The morphism $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes_{\mathbb{Z}[\zeta_p]^H} \mathbb{F}_p \to \mathfrak{m}_{\Gamma(n)} \otimes_{\mathbb{Z}} \mathbb{F}_p$ induces a bijection on supersingular points.

(iv) The morphism $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes_{\mathbb{Z}[\zeta_p]^H} \mathbb{F}_p \to \mathfrak{m}_{\Gamma(n)\cap\Gamma_0(p)} \otimes_{\mathbb{Z}} \mathbb{F}_p$ induces a bijection between the sets of irreducible components.

(v) The scheme $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes_{\mathbb{Z}[\zeta_p]^H} \mathbb{F}_p$ is obtained by gluing two copies of $\mathcal{J}_n/H$ along the supersingular points (cf. 2.13).

3. A good-reduction theorem.

3.1. Let p be a prime, let $H \subset (\mathbb{Z}/p)^*$, let $\Gamma'_\infty(H)$ be the group of 2.14, let $\mathbb{Z}[\zeta_p]^H$ be as in 2.14, and let n be an integer prime to p. Assume n ≥ 3, so that $\mathfrak{m}_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes \mathbb{Q} = M_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes \mathbb{Q}$ is a scheme. It is a scheme over $\mathbb{Q}[\zeta_n,\zeta_p]^H$.

In this section, J(n,H) denotes the abelian variety over $\mathbb{Q}[\zeta_n,\zeta_p]^H$ given by the identity component of the Picard scheme of $M_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes \mathbb{Q}$:

$J(n,H) = \mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma'_\infty(H)} \otimes \mathbb{Q} \,/\, \mathbb{Q}[\zeta_n,\zeta_p]^H)$.

For $H = (\mathbb{Z}/p)^*$, one has $\Gamma'_\infty(H) = \Gamma_0(p)$, and the map

$\mathfrak{m}_{\Gamma'_\infty(H)} \to \mathfrak{m}_{\Gamma_0(p)}$

induces a morphism

(3.1.1) $J(n,(\mathbb{Z}/p)^*) \otimes_{\mathbb{Q}[\zeta_n]} \mathbb{Q}[\zeta_n,\zeta_p]^H \to J(n,H)$

with finite kernel.

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Theorem 3.2. The abelian variety I(n,H) over ℚ[ζ_n,ζ_p]^H, defined as the cokernel of (3.1.1), has good reduction at the places dividing p.

Let v be a characteristic-p point of ℤ[ζ_n,ζ_p]^H, and let \overline{k(v)} be an algebraic closure of its residue field. Let ν be the number of supersingular points of M_{Γ(n)} ⊗_{ℤ[ζ]} \overline{k(v)}. The scheme M_{Γ(n)} ⊗_{ℤ[ζ]} \overline{k(v)} is known to be irreducible. It follows that the connected scheme M_{Γ(n)∩Γ(H)} ⊗_{ℤ[ζ_n,ζ_p]^H} \overline{k(v)} has two irreducible components meeting transversely at ν points (cf. 1.18), and that the Picard group variety over \overline{k(v)},

Pic^o(M_{Γ(n)∩Γ(H)} ⊗_{ℤ[ζ_n,ζ_p]^H} \overline{k(v)}/\overline{k(v)}),

is an extension of an abelian scheme by a torus of dimension ν - 1 (cf. I. 3.7). In particular:

Proposition 3.3. The abelian variety J(n,H) over ℚ[ζ_n,ζ_p]^H has semistable reduction at the places dividing p.

Suppose 0 → A → B → C → 0 is an exact sequence up to isogeny of abelian varieties over the fraction field of a discrete valuation ring. Let α_ab, α_m, and α_add denote the abelian, multiplicative, and unipotent ranks of the special fiber of the Néron model. Then

α_ab(A) - α_ab(B) + α_ab(C) = 0,

and likewise for α_m and α_add. Moreover, if α_add = 0, then α_ab and α_m are invariant under scalar extension (the semistable case). The condition α_add = α_m = 0 is equivalent to good reduction. Here α_add(J(n,H)) = 0 and α_m(J(n,H)) = ν-1, independently of H; the theorem follows.

Variant 3.4. It follows from 2.14.2 that

J(n,H) = Pic^o(M_{Γ(n)∩Γ_∞(H)} ⊗ ℚ / ℚ(ζ_n)) ⊗_{ℚ(ζ_n)} ℚ(ζ_n,ζ_p)^H.

Let I_1(n,H) be the abelian variety over ℚ(ζ_n) that is the cokernel of the morphism from

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a finite kernel defined by $\mathfrak{m}_{\Gamma_\infty(H)} \to \mathfrak{m}_{\Gamma_0(p)}$:

$$\mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma_0(p)} \otimes \mathbb{Q}/\mathbb{Q}(\zeta_n)) \to \mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma_\infty(H)} \otimes \mathbb{Q}/\mathbb{Q}(\zeta_n)) \to I_1(n,H) \to 0.$$

One has $I(n,H) = I_1(n,H) \otimes_{\mathbb{Q}(\zeta_n)} \mathbb{Q}(\zeta_n,\zeta_p)^H$, so 3.2 can still be formulated by saying that, after extension of scalars from $\mathbb{Q}(\zeta_n)$ to $\mathbb{Q}(\zeta_n,\zeta_p)^H$, $I_1(n,H)$ acquires good reduction at the places dividing p.

Variant 3.5. The abelian variety over $\mathbb{Q}(\zeta_p)^H$

$$RJ(n,H) =_{\mathrm{dfn}} \mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma'_\infty(H)}/\mathbb{Q}(\zeta_p)^H)$$

is obtained from $J(n,H)$ by Weil restriction of scalars from $\mathbb{Q}(\zeta_n,\zeta_p)^H$ to $\mathbb{Q}(\zeta_p)^H$. The cokernel $RI(n,H)$ of

$$\mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma_0(p)}/\mathbb{Q}) \otimes \mathbb{Q}(\zeta_p)^H \to \mathrm{Pic}^o(M_{\Gamma(n)\cap\Gamma'_\infty(p)}/\mathbb{Q}(\zeta_p)^H)$$

is obtained similarly from $I(n,H)$ by Weil restriction of scalars. Since $\mathbb{Q}(\zeta_n)$ is unramified at p, $RI(n,H)$ still has good reduction over $\mathbb{Q}(\zeta_p)^H$ at the places dividing p.

Variant 3.6. Let K be a subgroup of $GL(2,\hat{\mathbb{Z}})$ containing $\Gamma(n)$, and let $RI_1(K,H)$ be the abelian variety over $\mathbb{Q}$ that is the cokernel of

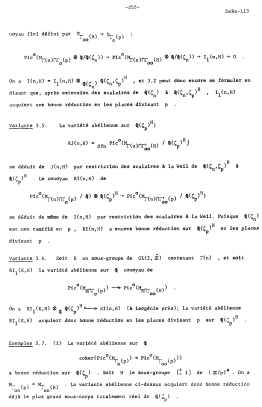
$$\mathrm{Pic}^o(M_{K\cap\Gamma_0(p)}) \to \mathrm{Pic}^o(M_{K\cap\Gamma_\infty(H)}).$$

One has $RI_1(K,H) \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_p)^H \leftrightarrow RI(n,H)$ (up to isogeny); hence the abelian variety $RI_1(K,H)$ acquires good reduction over $\mathbb{Q}(\zeta_p)^H$ at the places dividing p.

Examples 3.7. (i) The abelian variety over $\mathbb{Q}$

$$\mathrm{coker}(\mathrm{Pic}^o(M_{\Gamma_0(p)}) \to \mathrm{Pic}^o(M_{\Gamma_\infty(p)}))$$

has good reduction over $\mathbb{Q}(\zeta_p)$. Let H be the subgroup $\{\pm 1\}$ of $(\mathbb{Z}/p)^*$. One has $M_{\Gamma_\infty(p)} = M_{\Gamma_\infty(H)}$. The abelian variety in the preceding variant therefore acquires good reduction already over the maximal totally real subfield of $\mathbb{Q}(\zeta_p)$.



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(ii) Let p be a prime of the form 4n + 1, and let H ⊂ (ℤ/p)^* be the subgroup of squares. The abelian variety over ℚ

coker(Pic^o(M_{Γ_0(p)}) → Pic^o(M_{Γ_∞(H)}))

acquires good reduction everywhere over the real quadratic field ℚ(√p).

(iii) Let n be a square-free integer, let H be a subgroup of (ℤ/n)^*, and let Γ_∞(H) be the subgroup of GL(2,$\hat{\mathbb{Z}}$) that is the inverse image of

(H ℤ/n) (0 (ℤ/n)^*) ⊂ GL(2,ℤ/n).

If n = pm, then ℤ/n^* ≃ (ℤ/p)^* × (ℤ/m)^*. We set

H_p = (ℤ/p)^* ∩ H

and write H.(ℤ/p)^* for the subgroup of (ℤ/n)^* generated by H and (ℤ/p)^*. The abelian variety over ℚ

coker(Σ_{p|n} Pic^o(M_{Γ_∞(H.(ℤ/p)^*)}) → Pic^o(M_{Γ_∞(H)}))

acquires good reduction over ℚ(ζ_n)^H.

Proof. Let p be a prime divisor of n: n = mp with (m,p) = 1. Let us prove that the abelian variety A under consideration has good reduction at the places of ℚ(ζ_n)^H dividing p. Since ℚ(ζ_n)^{H_p} is unramified over ℚ(ζ_n)^H at the places dividing p, it is enough to prove that A has good reduction at the places of ℚ(ζ_n)^{H_p} dividing p. By 3.6,

B = coker(Pic^o(M_{Γ_0(p)∩Γ_∞(m)}) → Pic^o(M_{Γ_∞(H_p)}))

has good reduction at these places. The abelian variety

C = coker(Pic^o(M_{Γ_∞(H.(ℤ/p)^*)}) → Pic^o(M_{Γ_∞(H)}))

is isogenous to the one obtained from B by passing to the invariants under H (note that Γ_∞(H).(Γ_0(p) ∩ Γ_∞(m)) = Γ_∞(H.(ℤ/p)^*)). The abelian variety A is a quotient of

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C therefore also has good reduction.

(iv) Since M_{Γ_∞(H)} = M_{±Γ_∞(H)}, no generality is lost by assuming that -1 ∈ H. In particular, in (iii), ℚ(ζ_p)^H may be replaced by its totally real subfield.

Remark 3.8. The abelian varieties of 3.7(ii) are those studied by Casselman in [5].

4. Study of $\mathfrak{m}_n$

4.1. Let C be a generalized elliptic curve over a scheme S, and let n be an integer.

Assume that

a) the geometric fibers of C are smooth or m-gons, with n|m;

b) they are never supersingular of characteristic p|n.

Conditions a) and b) ensure that, locally for the étale topology, C_n is an extension of $\mathbb{Z}$/n by μ_n. The e_n pairing defines an isomorphism of fppf sheaves

e_n : ∧^2 C_n ≃ μ_n.

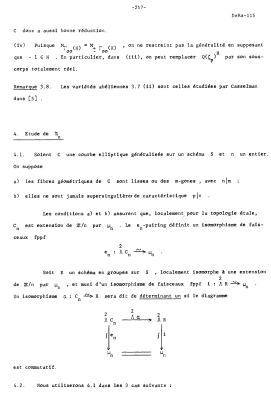
Let R be a group scheme over S, locally isomorphic to an extension of $\mathbb{Z}$/n by μ_n, and equipped with an isomorphism of fppf sheaves i : ∧^2 R ≃ μ_n.

An isomorphism α : C_n ≃ R is said to have determinant one if the diagram

∧^2 C_n –∧^2 α–> ∧^2 R | e_n | i v v μ_n = μ_n

commutes.

4.2. We shall use 4.1 in the following three cases:



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a) S is a scheme over $\mathbb{Z}[\zeta_n, 1/n]$, and $R = (\mathbb{Z}/n)^2$.

In this case one has over S the map $\zeta : \mathbb{Z}/n \simeq \mu_n$, and i is the composite

$\wedge^2(\mathbb{Z}/n)^2 = \mathbb{Z}/n \simeq \mu_n$.

b) $R = \mu_n \times \mathbb{Z}/n$ and i is the evident isomorphism: if $a \in \mu_n$ and $b \in \mathbb{Z}/n$, then $i(a \wedge b) = b \cdot a$ (= a^b when the group law of μ_n is written multiplicatively).

c) S is a scheme over $\mathbb{Z}[\zeta_n]$ and $A \subset (\mathbb{Z}/n)^2$ is a cyclic subgroup of order n.

In this case, write R(A) for the S-group scheme obtained by the pushout

$A \to (\mathbb{Z}/n)^2 \to (\mathbb{Z}/n)^2/A \downarrow \zeta \downarrow \| A \otimes \mu_n \to R(A) \to (\mathbb{Z}/n)^2/A.$

One also has

$\wedge^2 R(A) \simeq (A \otimes \mu_n) \otimes (\mathbb{Z}/n)^2/A \simeq (A \otimes (\mathbb{Z}/n)^2/A) \otimes \mu_n \simeq \wedge^2(\mathbb{Z}/n)^2 \otimes \mu_n \simeq \mathbb{Z}/n \otimes \mu_n \simeq \mu_n$.

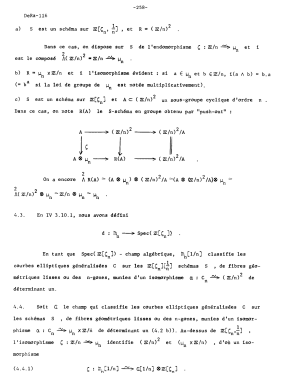
4.3. In IV.3.10.1 we defined

$d : \mathfrak{m}_n \to \mathrm{Spec}(\mathbb{Z}[\zeta_n])$.

As an algebraic stack over $\mathrm{Spec}(\mathbb{Z}[\zeta_n])$, $\mathfrak{m}_n[1/n]$ classifies generalized elliptic curves C over $\mathbb{Z}[\zeta_n][1/n]$-schemes S whose geometric fibers are smooth or n-gons, equipped with a determinant-one isomorphism $\alpha : C_n \simeq (\mathbb{Z}/n)^2$.

4.4. Let $\mathfrak{a}$ be the stack classifying generalized elliptic curves C over schemes S whose geometric fibers are smooth or n-gons, equipped with a determinant-one isomorphism $\alpha : C_n \simeq \mu_n \times \mathbb{Z}/n$ (4.2(b)). Over $\mathbb{Z}[\zeta_n, 1/n]$, the isomorphism $\zeta : \mathbb{Z}/n \simeq \mu_n$ identifies $(\mathbb{Z}/n)^2$ with $\mu_n \times \mathbb{Z}/n$, hence gives an isomorphism

(4.4.1) $\zeta : \mathfrak{m}_n[1/n] \simeq \mathfrak{a}[1/n] \otimes \mathbb{Z}[\zeta_n]$.



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We denote by $\mathcal{G}'$ the open substack of $\mathcal{G}$ classifying the (C,α) satisfying

(4.4.2) For every geometric point \overline{s} of S, if C_{\overline{s}} is an n-gon,

α((C^o_{\overline{s}})_n) = μ_n.

If (C,α) is an object of $\mathcal{G}'$ over S, then (C,α^{-1}(e × $\mathbb{Z}$/n)) is an object of the stack $\mathcal{B}$_n (1.3), hence a morphism

(4.4.3) γ : $\mathcal{G}'$ → $\mathcal{B}$_n.

Proposition 4.5. The morphism 4.3.3 is étale and separated. In particular, $\mathcal{G}'$ is a smooth algebraic stack over Spec($\mathbb{Z}$).

Let (C,β) be an object of $\mathcal{B}$_n over S. Then $\mathcal{G}'$ ×_{$\mathcal{B}$_n} S represents the following functor F : (Sch/S) → (Ens): F(T) is the set of pairs (A,u) consisting of a subgroup scheme A of C_n, locally isomorphic to μ_n, such that

A × B ⥲ C_n, together with u : B ⥲ $\mathbb{Z}$/n.

For such a pair (A,u), there is indeed a unique isomorphism

v : A ⥲ μ_n such that

α : C_n ↞ A × B –(v,u)⥲ μ_n × $\mathbb{Z}$/n$\mathbb{Z}$

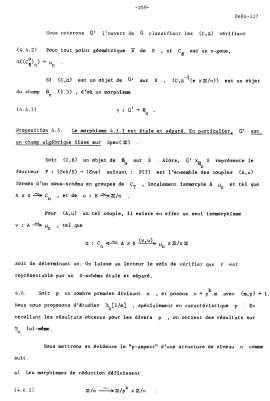
has determinant one. We leave it to the reader to verify that F is representable by an étale and separated S-scheme.

4.6. Let p be a prime divisor of n, and put n = p^k·m with (m,p) = 1. We propose to study $\mathfrak{m}$_n[1/m], especially in characteristic p. Gluing together the results obtained for the various p gives results on $\mathfrak{m}$_n itself.

We will exhibit the ”p-aspect” of a level-n structure as follows.

a) The reduction morphisms define

(4.6.1) $\mathbb{Z}$/n ⥲ $\mathbb{Z}$/p^k × $\mathbb{Z}$/m.



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b) The maps $x \mapsto x^m$ and $x \mapsto x^{p^k}$ define

(4.6.2) $\mu_n \xrightarrow{\sim} \mu_{p^k} \times \mu_m$.

c) compatibility: if $d \mid n$, identify $\mathbb{Z}[\zeta_d]$ with the subring $\mathbb{Z}[\zeta_{n^{n/d}}]$ of $\mathbb{Z}[\zeta_n]$. With this convention, a $\mathbb{Z}[\zeta_n]$-scheme S is also a $\mathbb{Z}[\zeta_{p^k}]$-scheme and a $\mathbb{Z}[\zeta_m]$-scheme; the morphisms $\zeta : \mathbb{Z}/p^k \to \mu_{p^k}$ and $\zeta : \mathbb{Z}/m \to \mu_m$ are the components, via (4.6.1) and (4.6.2), of $\zeta : \mathbb{Z}/n \to \mu_n$.

d) The maps $x \mapsto x^m$ and $x \mapsto x^{p^k}$ define

(4.6.3) $C_n \xrightarrow{\sim} C_{p^k} \times C_m$.

A morphism $\alpha : C_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$ has determinant one if and only if its components, via (4.6.2) (4.6.4), namely $\alpha' : C_{p^k} \xrightarrow{\sim} (\mathbb{Z}/p^k)^2$ and $\alpha'' : C_m \xrightarrow{\sim} (\mathbb{Z}/m)^2$, do.

e) We identify $\mathfrak{m}_n[1/n]$ with the stack classifying curves C over a $\mathbb{Z}[\zeta_n,1/n]$-scheme S as in 4.3, equipped with $\alpha' : C_{p^k} \xrightarrow{\sim} (\mathbb{Z}/p^k)^2$ and $\alpha'' : C_m \xrightarrow{\sim} (\mathbb{Z}/m)^2$ of determinant one.

f) Likewise, $\mathcal{G}$ classifies the (C,α',α'') such that $\alpha' : C_{p^k} \xrightarrow{\sim} \mu_{p^k} \times \mathbb{Z}/p^k$, $\alpha'' : C_m \xrightarrow{\sim} \mu_m \times \mathbb{Z}/m$, and $\det(\alpha') = \det(\alpha'') = 1$.

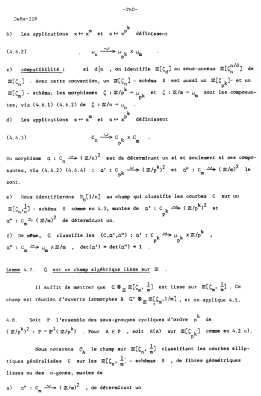
Lemma 4.7. $\mathcal{G}$ is a smooth algebraic stack over $\mathbb{Z}$.

It is enough to show that $\mathcal{G} \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_m,1/m]$ is smooth over $\mathbb{Z}[\zeta_m,1/m]$. This stack is a union of open substacks isomorphic to $\mathcal{G}' \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_m,1/m]$, and 4.5 applies.

4.8. Let P be the set of cyclic subgroups of order p^k in $(\mathbb{Z}/p^k)^2$: $P \simeq \mathbb{P}^1(\mathbb{Z}/p^k)$. For $A \in P$, let R(A) over $\mathbb{Z}[\zeta_{p^k}]$ be as in 4.2(c).

We denote by $\mathcal{C}_A$ the stack over $\mathbb{Z}[\zeta_n,1/m]$ classifying generalized elliptic curves C over $\mathbb{Z}[\zeta_n,1/m]$-schemes S whose geometric fibers are smooth or n-gons, equipped with

a) $\alpha'' : C_m \xrightarrow{\sim} (\mathbb{Z}/m)^2$, of determinant one



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b) α′ : C_{p^k} ⥲ R(A), with determinant one.

Since R(A) is isomorphic to μ_{p^k} × ℤ/p^k, the stack C_A is isomorphic to C ⊗_ℤ ℤ[ζ_n, 1/m], and hence is a smooth algebraic stack over ℤ[ζ_n, 1/m].

For p invertible, ζ : (ℤ/p^k)^2 ⥲ R(A) is an isomorphism, hence an isomorphism of stacks over ℤ[ζ_n, 1/n]

(4.8.1) 𝔪_n[1/n] ≃ C_A[1/p].

4.9. Let 𝔪_1^h be the complement in 𝔪_1 of the finite set of points corresponding to supersingular elliptic curves of characteristic dividing n. Define

c : C_A → 𝔪_1^h

by associating with (C,α′,α″) over S the curve c(C) (IV 1.4).

Lemma 4.10. The morphism c : C_A → 𝔪_1^h is representable and quasi-finite.

The verification is left to the reader.

4.11. Let C[1/m] be the stack obtained by gluing the stacks C_A (A ∈ P) along their common open C_A[1/p] (4.8.1). Let C be obtained by gluing the C[1/m] along their common open C[1/m,1/p] ≃ 𝔪_n[1/n]. The stack C is smooth over ℤ[ζ_n]; one has

(4.11.1) C[1/n] ≃ 𝔪_n[1/n]

and the morphisms 4.9 glue to a morphism

(4.11.2) c : C → 𝔪_1^h.

Lemma 4.11.3. The representable morphism (4.11.2) is finite and separated.

Proof. It suffices to show that, for every p|n, c induces a proper morphism (in particular, a separated one)

c : C[1/m] → 𝔪_1^h[1/m].

We use the valuative criterion ([8]). Thus let (S,η,s) be a complete trait with

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algebraically closed residue field, u : S → 𝔪_1^h[1/m], and v_η : η → C such that

c v_η = u_η.

We must show that, after replacing S by its normalization in a finite extension of k(η), u admits a lift v : S → C extending v_η, and that v is unique up to unique isomorphism. Since 𝔪_n^o[1/n] is dense in C, we may suppose that v_η sends η into 𝔪_n^o[1/n]. Since C[1/p] ≃ 𝔪_n[1/n] is finite over 𝔪_1[1/n] (cf. IV.2.5.), it is enough to consider the case in which s has characteristic p.

Let (C_η,α′_η,α″_η) be the elliptic curve with level-n structure over η that defines v_η. A preliminary finite extension of k(η) allows us to suppose that C_η extends to a generalized elliptic curve C over S, with special fiber either smooth or an n-gon. This curve is unique (IV 1.6). The group scheme C_m is finite étale over S, so α″_η extends uniquely to

α″ : C_m ⃗→ (ℤ/m)^2.

By hypothesis, the special fiber C_s is not supersingular. The multiplicative subgroup (C_s)^o_{p^k} of (C_s)_{p^k} lifts uniquely to a subgroup of C_{p^k}, which becomes an extension

0 → A → C_{p^k} → B → 0

with A ≃ μ_n and B ≃ ℤ/n. Let A = α′(A_η) ∈ P. The isomorphism α′_η extends to

α′ : C_{p^k} ⃗→ R(A),

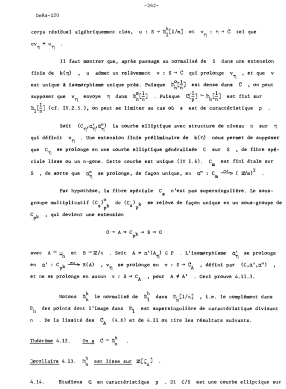
v_η extends to v : S → C_A, defined by (C,α′,α″), and does not extend to any v : S → C_{A′} for A ≠ A′. This proves 4.11.3.

Let 𝔪_n^h denote the normalization of 𝔪_1^h in 𝔪_n[1/n], i.e. the complement in 𝔪_n of the points whose image in 𝔪_1 is supersingular of characteristic dividing n. From the smoothness of the C_A (4.8) and from 4.11 we obtain the following results.

Theorem 4.12. One has C ≃ 𝔪_n^h.

Corollary 4.13. 𝔪_n^h is smooth over ℤ[ζ_n].

4.14. Let us study C in characteristic p. If C/S is an elliptic curve over



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Authority page 121; collected-paper page 614; original folio 263.

S, with p nilpotent on S, whose geometric fibers are smooth and nonsupersingular or are n-gons, then (C_{p^k})^o is locally isomorphic to μ_{p^k} (for the étale topology), and giving an isomorphism α′ : C_{p^k} ⥲ μ_{p^k} × ℤ/p^k amounts to giving an isomorphism of extensions

(4.14.1) 0 → (C_{p^k})^o → C_{p^k} → C_{p^k}/(C_{p^k})^o → 0 ↓ α′_1 ↓ α′ ↓ α′_2 0 → μ_{p^k} → μ_{p^k} × ℤ/p^k → ℤ/p^k → 0.

This amounts to giving

a) a trivialization τ of the extension in the first row of 4.14.1;

b) isomorphisms α′_1 : C^o_{p^k} ⥲ μ_{p^k} and α′_2 : C_{p^k}/C^o_{p^k} ⥲ ℤ/p^k.

Giving τ amounts to giving the étale subgroup α′^{-1}(ℤ/p^k) of C_{p^k}, so that (C,α′) ↦ (C,τ) corresponds to the morphism $\mathcal{G} \to \mathcal{B}_{p^k}$ already considered. Moreover, if det(α′) = 1, α′_2 is uniquely determined by α′_1.

4.15. Let $\mathcal{J}_k^h \otimes \mathbb{Z}/p^N$ be the algebraic stack over ℤ/p^N classifying generalized elliptic curves with irreducible fibers C/S, with p^N = 0 on S, equipped with an isomorphism α′_1 : C^o_{p^k} ⥲ μ_{p^k}. The discussion in 4.14 shows that, after reduction modulo p^N, the stack $\mathcal{G} = \mathcal{G}(p^k)$ is a fiber product

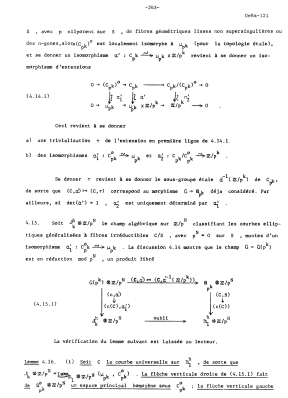
(4.15.1) $\mathcal{G}(p^k) \otimes \mathbb{Z}/p^N$ —(C,α′) ↦ (C,α′^{-1}(ℤ/p^k))⟶ $\mathcal{B}_{p^k} \otimes \mathbb{Z}/p^N$ │ (C,α′) │ (C,B) ↓ ↓ │ (c(C),α′_1) │ (c(C)) $\mathcal{J}_k^h \otimes \mathbb{Z}/p^N$ ——————— forgetful ———————→ $\mathfrak{m}_1^h \otimes \mathbb{Z}/p^N$.

Verification of the following lemma is left to the reader.

Lemma 4.16. (i) Let C be the universal curve over $\mathfrak{m}_1^h$, so that

$\mathcal{J}_k^h \otimes \mathbb{Z}/p^N = \mathrm{Isom}_{\mathfrak{m}_1^h \otimes \mathbb{Z}/p^N}(\mu_{p^k}, C^o_{p^k})$.

The right vertical arrow in (4.15.1) makes $\mathcal{B}_{p^k}^o \otimes \mathbb{Z}/p^N$ a principal homogeneous space under C^o_{p^k}; the left vertical arrow



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makes $\mathcal{G}$^o ⊗ $\mathbb{Z}$/p^N a principal homogeneous space under μ_{p^k}.

(ii) In particular, for N = 1, these arrows identify $\mathcal{J}$_k^h ⊗ $\mathbb{F}$_p with ($\mathcal{G}$ ⊗ $\mathbb{F}$_p)^{(p^k)} and $\mathfrak{m}$_1^h ⊗ $\mathbb{F}$_p with ($\mathcal{B}$_n ⊗ $\mathbb{F}$_p)^{(p^k)} (the arrow becoming an iterated Frobenius morphism) (a bijective radicial morphism between smooth schemes of dimension one is always an iterated Frobenius).

(iii) The horizontal arrows in (4.15.1) are étale, and make ($\mathcal{J}$_n^h ⊗ $\mathbb{Z}$/p^N) (respectively $\mathcal{G}$ ⊗ $\mathbb{Z}$/p^N) a principal homogeneous space under the group ($\mathbb{Z}$/p^k)^* over ($\mathfrak{m}$_1^h ⊗ $\mathbb{Z}$/p^N) (respectively $\mathcal{B}$_n ⊗ $\mathbb{Z}$/p^N), with the group ($\mathbb{Z}$/p^k)^* acting by (i,(C,α′_1)) ↦ (C,iα′_1).

Let $\mathcal{J}$_k ⊗ $\mathbb{Z}$/p be the normalization of $\mathfrak{m}$_1 ⊗ $\mathbb{Z}$/p in $\mathcal{J}$_k^h. We shall use the following theorem of Igusa [35] (see Katz [13] 4.3 (bis)).

Theorem 4.17 (Igusa). The covering $\mathcal{J}$_k ⊗ $\mathbb{Z}$/p → $\mathfrak{m}$_1 ⊗ $\mathbb{Z}$/p is totally ramified at the supersingular points.

4.18. For general n, $\mathcal{G}$^o = $\mathcal{G}$^o(n) is a fiber product

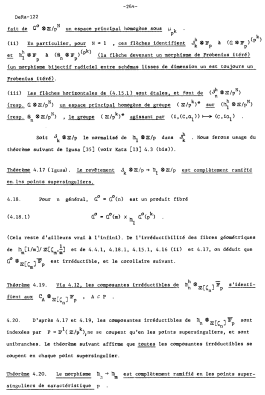
(4.18.1) $\mathcal{G}$^o = $\mathcal{G}$^o(m) ×_{$\mathfrak{m}$_1} $\mathcal{G}$^o(p^k).

(This moreover remains true at infinity.) From the irreducibility of the geometric fibers of $\mathfrak{m}$_m[1/m]/$\mathbb{Z}$[ζ_m,1/m] and from 4.4.1, 4.18.1, 4.15.1, 4.16(ii), and 4.17, it follows that $\mathcal{G}$^o ⊗_{$\mathbb{Z}$[ζ_m]} $\mathbb{F}$_p is irreducible, and the following corollary.

Theorem 4.19. Via 4.12, the irreducible components of $\mathfrak{m}$_n^h ⊗_{$\mathbb{Z}$[ζ_n]} $\mathbb{F}$_p identify with C_A ⊗_{$\mathbb{Z}$[ζ_n]} $\mathbb{F}$_p, A ∈ P.

4.20. By 4.17 and 4.19, the irreducible components of $\mathfrak{m}$_n ⊗_{$\mathbb{Z}$[ζ_n]} $\mathbb{F}$_p are indexed by P = $\mathbb{P}$^1($\mathbb{Z}$/p^k), meet only at the supersingular points, and are unibranch. The following theorem asserts that all the irreducible components meet at every supersingular point.

Theorem 4.20. The morphism $\mathfrak{m}$_n → $\mathfrak{m}$_m is totally ramified at the supersingular points of characteristic p.



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Proof. By 4.6, the validity of 4.20 depends only on p^k, not on m; we may therefore suppose that m ≥ 3, so that we have to deal only with schemes.

Let E_0 be a supersingular elliptic curve over $\bar{F}_p$, and let $E/\hat{S}$ be the formal moduli scheme of E_0 over $W(\bar{F}_p)$. We have $\hat{S} \simeq W(\bar{F}_p)[[t]]$; it is the completion of the strict henselization of $\mathfrak{m}_1$ at the given geometric point.

Let $\hat{T}$ be the normalization of $\hat{S}$ in the $\hat{S}[1/p]$-scheme Isom_$\hat{S}$(E_{p^k},$(\mathbb{Z}/p^k)^2$). Since completions and normalizations commute,

$\hat{T} = \mathfrak{m}_{p^k} \times_{\mathfrak{m}_1} \hat{S}$.

By 4.17, the irreducible components of $\hat{T} \otimes \bar{F}_p$ are naturally indexed by $P = \mathbb{P}^1(\mathbb{Z}/p^k)$. The connected components of $\hat{T}$ define a partition $\mathcal{P}$ of P. Assertion 4.20 means that $\hat{T}$ is always connected, i.e. that $\mathcal{P}$ is trivial.

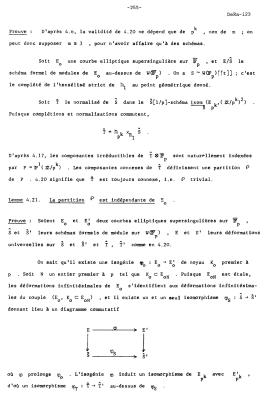
Lemma 4.21. The partition $\mathcal{P}$ is independent of E_0.

Proof. Let E_0 and E'_0 be two supersingular elliptic curves over $\bar{F}_p$, let $\hat{S}$ and $\hat{S}'$ be their formal moduli schemes over $W(\bar{F}_p)$, let E and E′ be their universal deformations over $\hat{S}$ and $\hat{S}'$, and let $\hat{T}$ and $\hat{T}'$ be as in 4.20.

It is known that there exists an isogeny $\varphi_0 : E_0 \to E'_0$ whose kernel K_0 is prime to p. Let N be an integer prime to p such that $K_0 \subset E_{oN}$. Since E_{oN} is étale, the infinitesimal deformations of E_0 identify with the infinitesimal deformations of the pair $(E_0, K_0 \subset E_{oN})$, and there exists a unique isomorphism $\varphi_S : \hat{S} \to \hat{S}'$ giving rise to a commutative diagram

E ⟶φ⟶ E′ | | ↓ ↓ $\hat{S}$ ⟶φ_S⟶ $\hat{S}'$

in which φ extends φ_0. The isogeny φ induces an isomorphism from E_{p^k} to E′_{p^k}, hence an isomorphism $\varphi_T : \hat{T} \to \hat{T}'$ above φ_S.



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The bijection between the set I(T̂ ⊗ F̄_p) of irreducible components of T̂ ⊗ F̄_p and P can be described as follows: on T̂ there is a morphism

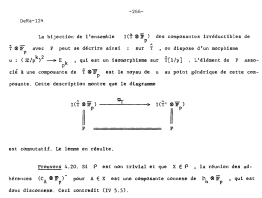
u : (ℤ/p^k)^2 → E_{p^k},

which is an isomorphism on T̂[1/p]. The element of P associated with a component of T̂ ⊗ F̄_p is the kernel of u at the generic point of that component. This description shows that the diagram

I(T̂ ⊗ F̄_p) ⟶φ_T⟶ I(T̂′ ⊗ F̄_p) ‖ ‖ P ═══════════ P

is commutative. The lemma follows.

Let us prove 4.20. If $\mathcal{P}$ is nontrivial and X ∈ $\mathcal{P}$, the union of the closures ($\mathcal{C}$_A ⊗ F̄_p)⁻ for A ∈ X is a connected component of 𝔪_n ⊗ F̄_p, which is therefore disconnected. This contradicts (IV 5.5).



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VI. Coarse moduli schemes.

In this chapter we study coarse moduli schemes, in particular the relation between 𝔪_H and M_H. Some classical results are proved by ”pure thought”. The result of § 5 is made more precise in the following chapter.

1. The modular invariant.

Theorem 1.1. The coarse space M_1 and $\mathfrak{m}_1$ is isomorphic to $\mathbb{P}^1_{\mathbb{Z}}$.

Characteristics 2 and 3 make the direct verification of 1.1 somewhat laborious. We shall deduce 1.1 from its corollary

(1.1.1) $M_{1\mathbb{C}} \simeq \mathbb{P}^1_{\mathbb{C}}$.

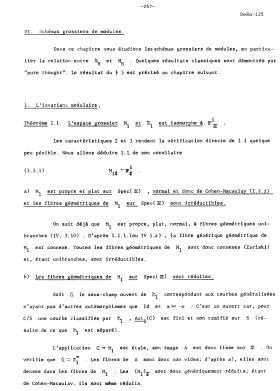
a) M_1 is proper and flat over Spec($\mathbb{Z}$), normal and hence Cohen–Macaulay (I.7.2), and the geometric fibers of M_1 over Spec($\mathbb{Z}$) are irreducible.

We already know that M_1 is proper, flat, normal, with geometrically unibranch fibers (IV. 3.10). By 1.1.1 (or IV 5.4), the geometric generic fiber of M_1 is connected. All geometric fibers of M_1 are therefore connected (Zariski) and, being unibranch, are irreducible.

b) The geometric fibers of M_1 over Spec($\mathbb{Z}$) are reduced.

Let G be the open substack of $\mathfrak{m}_1$ corresponding to generalized curves having no automorphisms other than Id and x ↦ -x. It is open because, for C/S a curve classified by $\mathfrak{m}_1$, Aut_S(C) is finite and unramified over S (a consequence of the fact that $\mathfrak{m}_1$ is separated).

The map G → M_1 is étale, so its image A is smooth over $\mathbb{Z}$. One verifies that G ⊃ $\mathfrak{m}_1^\infty$. The fibers of A are therefore nonempty; by a), they are dense in the fibers of M_1. The $(M_1)_{\bar{s}}$ are thus generically reduced; being Cohen–Macaulay, they are in fact reduced.



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VI Schémas grossiers de modules.

Dans ce chapitre nous étudions les schémas grossiers de modules, en particulier la relation entre D_N et M_1. Quelques résultats classiques sont démontrés par "pure thought". Le résultat du § 5 est précédé au chapitre suivant.

1. L'invariant modulaire.

Théorème 1.1. L'espace grossier M_1 et $\mathbb{P}^1_{\mathbb{Z}}$ sont isomorphe à $\mathbb{P}^1_{\mathbb{Z}}$.

Les caractéristiques 2 et 3 rendent la vérification directe de 1.1 quelque peu pénible. Nous allons déduire 1.1 de son corollaire

(1.1.1) $M_{1\mathbb{C}} \simeq \mathbb{P}^1_{\mathbb{C}}$

a) M_1 est propre et plat sur Spec($\mathbb{Z}$), normal et donc de Cohen-Macaulay (I.7.2), et les fibres géométriques de M_1 sur Spec($\mathbb{Z}$) sont irréductibles.

On sait déjà que M_1 est propre, plat, normal, à fibres géométriques unibranches (IV. 3.10). D'après 1.1.1 (ou IV 5.4), la fibre générique géométrique de M_1 est connexe. Toutes les fibres géométriques de M_1 sont donc connexes (Zariski) et, étant unibranches, sont irréductibles.

b) Les fibres géométriques de M_1 sur Spec($\mathbb{Z}$) sont réduites.

Soit G le sous-champ ouvert de D_1 correspondant aux courbes généralisées n'ayant pas d'autres automorphismes que Id et x↦-x . C'est un ouvert car, pour C/S une courbe classifiée par D_1, $\mathrm{Aut}_S(C)$ est fini et non ramifié sur S (résulte de ce que D_1 est séparé).

L'application G → M_1 est étale, son image A est donc lisse sur $\mathbb{Z}$. On vérifie que G ⊃ M_1^∞. Les fibres de A sont donc un vidon; d'après a), elles sont denses dans les fibres de M_1. Les $(M_1)_{\bar{k}}$ sont donc géométriquement réduits; étant de Cohen-Macaulay, ils sont même réduits.

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c) End of the proof.

By invariance of the arithmetic genus under specialization, and by (1.1.1), one finds that the geometric fibers of M_1 are integral and have arithmetic genus 0. They are therefore isomorphic to the projective line. In particular, M_1 is proper and smooth over $\mathbb{Z}$.

The Néron polygon with one side over Spec($\mathbb{Z}$) defines a section of M_1, with image M_1^∞. Fiber by geometric fiber one verifies that $H^1(M_1,\mathcal{O}(M_1^\infty)) = 0$, and then that $H^0(M_1,\mathcal{O}(M_1^\infty))$ is locally free of rank 2 over $\mathbb{Z}$ and its formation is compatible with passage to geometric fibers. Since $\mathbb{Z}$ is principal, $H^0(M_1,\mathcal{O}(M_1^\infty))$ is free and

has a basis {1,j}. Fiber by geometric fiber one verifies that $\mathcal{O}(M_1^{\wedge}\infty)$ is very ample and that j defines an isomorphism

$M_1 \tilde{\to} \mathbb{P}^1_\mathbb{Z}$.

Proof of (1.1.1) Here are two classical proofs of (1.1.1).

(1) One verifies that $M_1^{\wedge}o(\mathbb{C})$ is the quotient of the Poincaré half-plane by $SL(2,\mathbb{Z})$ (IV 5.4). Drawing a fundamental domain, one finds that $M_1^{\wedge}o(\mathbb{C})$ is homeomorphic to $\mathbb{R}^2$. The space $M_1(\mathbb{C})$, obtained from it by adjoining one point, is therefore a sphere.

(2) Over $\mathbb{C}$, the moduli scheme classifying integral generalized elliptic curves equipped with an invariant differential is

$\mathrm{Spec}(\mathbb{C}[g_2,g_3])$ - (the point $g_2 = g_3 = 0$),

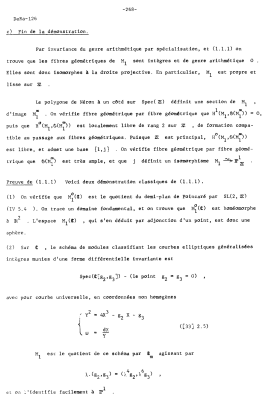
with universal curve, in nonhomogeneous coordinates,

$\lceil\ Y^2 = 4X^3 - g_2\ X - g_3\ \rfloor\ \omega = dX/Y$ ([33] 2.5)

M_1 is the quotient of this scheme by $\mathbb{G}_m$ acting by

$\lambda.(g_2,g_3) = (\lambda^4\ g_2,\lambda^6\ g_3)$,

and is easily identified with $\mathbb{P}^1$.



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The irreducibility of the fibers of M_1 over $\mathbb{Z}$ has the following corollary.

Corollary 1.2. Let k be an algebraically closed field of characteristic p. There are only finitely many isomorphism classes of supersingular elliptic curves over k.

The set of points of $(M_1)_k$ corresponding to generalized elliptic curves E for which the Frobenius kernel is of multiplicative type is open and contains the point at infinity. Its complement, which corresponds to supersingular curves, is therefore finite.

1.3. The function j introduced in 1.1 c) is not uniquely determined; the ambiguity is $j \mapsto \pm j + n$ ($n \in \mathbb{Z}$). A choice of j, recalled below, is classical.

Let C/S be a generalized elliptic curve with irreducible fibers over S. Tate's formulary [33] explains how to associate with C/S a section j of $\mathbb{P}^1$ over S, with $j = \infty$ wherever C is singular. By the universal property of M_1, this morphism $j : \mathfrak{m}_1 \to \mathbb{P}^1$ factors through $j : M_1 \to \mathbb{P}^1$. One readily checks that $j : M_1(\mathbb{C}) \to \mathbb{P}^1(\mathbb{C})$ is bijective [33].

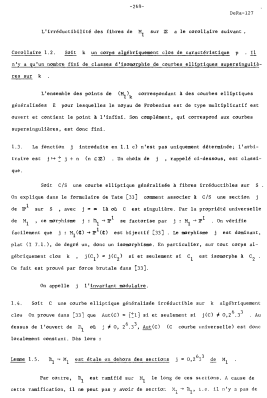
The morphism j is dominant, flat (I 7.1.), and of degree one, hence an isomorphism. In particular, over every algebraically closed field k, j(C_1) = j(C_2) if and only if C_1 is isomorphic to C_2. This is proved by brute force in [33].

The function j is called the modular invariant.

1.4. Let C be an irreducible generalized elliptic curve over an algebraically closed field k. It is proved in [33] that Aut(C) = {±1} if and only if j(C) ≠ 0,2^6.3^3. Over the open subset of $\mathfrak{m}$_1 where j ≠ 0, 2^6.3^3, Aut(C) (with C the universal curve) is therefore locally constant. Hence:

Lemma 1.5. $\mathfrak{m}$_1 → M_1 is étale away from the sections j = 0,2^6 3^3 of M_1.

By contrast, $\mathfrak{m}$_1 is ramified over M_1 along these sections. Because of this ramification, there can be no section M_1 → $\mathfrak{m}$_1, i.e. there is no



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L'irréductibilité des fibres de $\overline{M}_1$ sur $\mathbb{Z}$ a le corollaire suivant :

Corollaire 1.8. Soit k un corps algébriquement clos de caractéristique p . Il n'y a qu'un nombre fini de classes d'isomorphie de courbes elliptiques supersingulières sur k .

L'ensemble des points de $\mathcal{O}_{\mathbb{Q}}$ correspondant à des courbes elliptiques généralisées E pour lesquelles le sous de Frobenius est de type multiplicatif est ouvert et contient le point à l'infini. Son complément, qui correspond aux courbes supersingulières, est donc fini.

1.3. La fonction j introduite en 1.1 c) n'est pas uniquement déterminée; l'arbitraire est $j \mapsto \frac{a}{c} j + n$ ($n \in \mathbb{Z}$) . Un choix de j , rappelé ci-dessous, est classique.

Soit C/S une courbe elliptique généralisée à fibres irréductibles sur S . On explique dans le formulaire de Tate [31] comment associer à C/S une section j de $\mathbb{P}^1$ sur S , avec $j = \infty$ là où C est singulière. Par la propriété universelle de $\overline{M}_1$, ce morphisme $j : S \to \mathbb{P}^1$ se factorise par $j : S \to \overline{M}_1 - \mathbb{P}^1$. On vérifie facilement que $j : \overline{M}_1(K) \to \mathbb{P}^1(K)$ est injectif [32] . Le morphisme j est donc fini, plat (I 7.1.), de degré 12, donc un isomorphisme. En particulier, sur tout corps algébriquement clos k , $j(C_1) = j(C_2)$ si et seulement si C_1 est isomorphe à C_2 . Ce fait est prouvé par force brutale dans [33].

On appelle j l'invariant modulaire.

1.4. Soit C une courbe elliptique généralisée irréductible sur k algébriquement clos. On prouve dans [33] que $\mathrm{Aut}(C) = \{\pm 1\}$ si et seulement si $j(C) \neq 0, 2^6.3^3$. Au dessus de l'ouvert de $\overline{M}_1$ où $j \neq 0, 2^6.3^3$, $\mathrm{Aut}(C)$ (C courbe universelle) est donc localement constant. Dès lors :

Lemme 1.5. $\overline{M}_1 - M_1$ est étale au dehors des sections $j = 0, 2^6 3^3$ de M_1 .

Par contre, $\overline{M}_1$ est ramifié sur M_1 le long de ces sections. A cause de cette ramification, il ne peut pas y avoir de section: $M_1 - \overline{M}_1$, c.-à. il n'y a pas de

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generalized elliptic curve over $\mathbb{P}^1$ with modular invariant j. Tate has nevertheless shown that over $\mathbb{P}^1_{\mathbb{Z}}$ minus the two sections 0 and 2^6 3^3, i.e. over

S = Spec $\mathbb{Z}$[1/j] (2^6 3^3 1/j - 1)^{-1},

there are two curves with invariant j. Put k = j-2^6 3^3, so that 1/j and 1/k are sections of $\mathcal{O}$_S and j/k is an invertible section.

Proposition 1.6 (Tate). The curves

(1.6.1) y^2 + xy = x^3 - (2^2 3^2/k) x - 1/k (1.6.2) y^2 + xy = x^3 + (2^2 3^2/j) x - (j-2^7 3^2)/j^2

have modular invariant j.

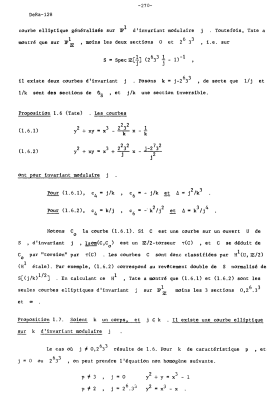
For (1.6.1), c_4 = j/k, c_6 = - j/k, and Δ = j^2/k^3. For (1.6.2), c_4 = k/j, c_6 = - k^2/j^2, and Δ = k^3/j^4.

Write C_o for the curve (1.6.1). If C is a curve over an open subset U of S with invariant j, Isom(C,C_o) is a $\mathbb{Z}$/2-torsor τ(C), and C is obtained from C_o by ”twisting” by τ(C). The curves C are therefore classified by H^1(U,$\mathbb{Z}$/2) (étale H^1). For example, (1.6.2) corresponds to the double cover of S with normalization S[(j/k)^{1/2}]. By calculating this H^1, Tate showed that (1.6.1) and (1.6.2) are the only elliptic curves with invariant j over $\mathbb{P}^1_{\mathbb{Z}}$ minus the three sections 0,2^6.3^3 and ∞.

Proposition 1.7. Let k be a field and let j ∈ k. There exists an elliptic curve over k with modular invariant j.

The case j ≠ 0,2^6 3^3 follows from 1.6. If k has characteristic p and j = 0 or 2^6 3^3, one may use the following affine equation.

p ≠ 3, j = 0 y^2 + y = x^3 - 1 p ≠ 2, j = 2^6.3^3 y^2 = x^3 - x.



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1.8. Let C be the universal elliptic curve over $\mathfrak{m}_1^o$, and let γ be the morphism

Aut(C) → Aut(Lie(C)) ≃ $\mathbb{G}_m$. We extract from [33] the following result about Aut(C).

Lemma 1.9. Over the substack of $\mathfrak{m}_1^o[1/6]$ defined by $c_4 = 0$ (resp. $c_6 = 0$) (i.e. $\sqrt[3]{j} = 0$, resp. $\sqrt[2]{(j-1728)} = 0$), γ is an isomorphism Aut(C) $\tilde{\to}$ μ_6 (resp. Aut(C) $\tilde{\to}$ μ_4).

2. Automorphisms of elliptic curves. Criterion for $\mathfrak{m}_H^o = M_H^o$.

2.1. For an algebraic stack to be an algebraic space, it is necessary and sufficient that its geometric points have no nontrivial automorphisms. Thus, if H ⊂ GL(2, $\mathbb{Z}/n$), in order that $\mathfrak{m}_H^o[1/n] = M_H^o[1/n]$ (resp. $\mathfrak{m}_H[1/n] = M_H[1/n]$), it is necessary and sufficient that an elliptic curve (resp. generalized elliptic curve) equipped with an H-level structure, over an algebraically closed field k of characteristic p ∤ n, have no nontrivial automorphism.

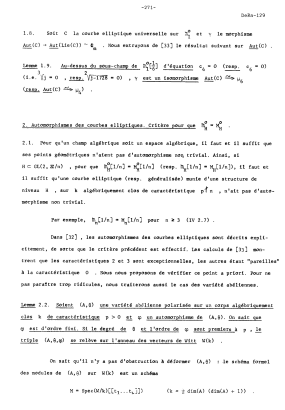
For example, $\mathfrak{m}_n[1/n] = M_n[1/n]$ for n ≥ 3 (IV 2.7).

In [32], the automorphisms of elliptic curves are described explicitly, so the preceding criterion is effective. The calculations of [33] show that characteristics 2 and 3 are exceptional, while the others are ”like” characteristic 0. We propose to verify this point a priori. So as not to seem too ridiculous, we shall also treat the case of abelian varieties.

Lemma 2.2. Let (A, θ) be a polarized abelian variety over an algebraically closed field k of characteristic p > 0, and let φ be an automorphism of (A, θ). It is known that φ has finite order. If the degree of θ and the order of φ are prime to p, the triple (A, θ, φ) lifts to the ring of Witt vectors W(k).

There is no obstruction to deforming (A, θ): the formal moduli scheme of (A, θ) over W(k) is a scheme

M = Spec(W/k)[[t_1...t_r]] (k = ½ dim(A) (dim(A) + 1)).



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Continuing the proof of Lemma 2.2, interpret φ as the action on (A, θ) of a cyclic group G whose order is prime to p. By transport of structure, G acts on the formal moduli scheme M: the automorphism φ of (A, θ) extends to an automorphism of M and of the universal deformation (A_M, θ_M) of (A, θ) over M. The formal moduli scheme of (A, θ, φ) is the G-fixed subscheme M^G of M. Since the order of G is prime to p, this subscheme is smooth over W(k), hence admits sections over W(k), and this proves 2.2.

Proposition 2.3. Let A be an abelian variety of dimension g over an algebraically closed field. Let φ be an automorphism of prime order q, and write

A^φ =_dfn Ker(φ - id_A).

There is an integer k such that

2.g = dim(A^φ) + k.(q-1).

Proof: Passing from A to A/A^φ, we reduce to the case in which A^φ is finite.

The characteristic polynomial of φ acting on A has integer coefficients, does not have 1 as a root, and has only qth roots of unity as roots. It is therefore a power of the cyclotomic polynomial, which has degree q-1. Comparing degrees proves the lemma.

Corollary 2.4. Let A be an abelian variety of dimension g over an algebraically closed field. Let φ be an automorphism of A of finite order. Every prime factor of the order of φ is smaller than 2g + 1.

Proof: This follows from 2.4.

Corollary 2.5. Let (A, θ) be a polarized abelian variety of dimension g over an algebraically closed field k of characteristic p, with polarization degree prime to p. Suppose that p > 2g + 1. If φ is an automorphism of (A, θ), then (A, θ, φ) lifts to characteristic 0 (over W(k)).

Proof: This follows from 2.2 and 2.4.

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2.6. An elliptic curve has a canonical principal polarization. By 2.6, if φ is an automorphism of an elliptic curve E over an algebraically closed field k of characteristic p > 3, (E,φ) lifts to characteristic 0 (over W(k)).

Theorem 2.7. For $\mathfrak{m}_H^o[1/6n]$ to be an algebraic space, it is necessary and sufficient that the inverse image Γ of H in $SL(2,\mathbb{Z})$ have no finite-order element other than 1.

Let $(E,\bar{\alpha})$ be an elliptic curve equipped with a level-H structure over an algebraically closed field k of characteristic p = 0 or p > 3, with $p \nmid n$. We must show that every automorphism φ of $(E,\bar{\alpha})$ is trivial. By 2.6, (E,φ), and hence $(E,\bar{\alpha},\varphi)$, lifts to characteristic 0, so it is enough to treat the case p = 0.

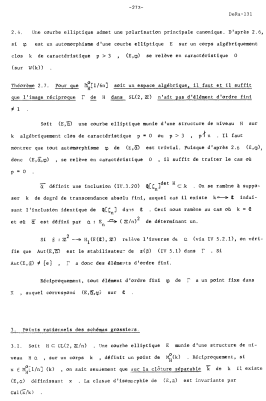
The structure $\bar{\alpha}$ defines an inclusion (IV.3.20) $\mathbb{Q}[\zeta_n]^{\det H} \subset k$. We may reduce to the case in which k has finite absolute transcendence degree; then there is an embedding $k \hookrightarrow \mathbb{C}$ inducing the identity inclusion of $\mathbb{Q}[\zeta_n]$ in $\mathbb{C}$. This reduces us to $k = \mathbb{C}$, with $\bar{\alpha}$ represented by a determinant-one isomorphism $\alpha : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$.

If $\beta : \mathbb{Z}^2 \to H_1(E(\mathbb{C}),\mathbb{Z})$ lifts the inverse of α (via IV 5.2.1), one verifies that $\mathrm{Aut}(E,\bar{\alpha})$ is the stabilizer of $z(\beta)$ (IV 5.1) in Γ. If $\mathrm{Aut}(E,\bar{\alpha}) \neq \{e\}$, Γ therefore has elements of finite order.

Conversely, every finite-order element φ of Γ has a fixed point in X, corresponding to $(E,\bar{\alpha},\varphi)$ over $\mathbb{C}$.

3. Rational points of coarse schemes.

3.1. Let $H \subset GL(2, \mathbb{Z}/n)$. An elliptic curve E equipped with a level-H structure α over a field k defines a point of $M_H^o(k)$. Conversely, if $x \in M_H^o[1/n](k)$, all one knows is that over the separable closure $\bar{k}$ of k there exists a pair (E,α) defining x. The isomorphism class of (E,α) is invariant under $\mathrm{Gal}(\bar{k}/k)$.



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If (E,α) comes from (E_0,α_0) over k, the set of k-isomorphism classes of curves (E_1,α_1) defining x is canonically H^1(Gal(k̄/k),Aut(E_0,α_0)(k̄)). The obstruction to the existence of an (E_0,α_0) lies in an H^2 (in Giraud's sense). In the special case where Aut(E,α) is commutative, Gal(k̄/k) acts on Aut(E,α), and the obstruction lies in H^2(Gal(k̄/k),Aut(E,α)

Proposition 3.2. Let k be a field of characteristic p and let H ⊂ GL(2, ℤ/n). Assume that p ∤ n. Every point x ∈ M_H^o(k) is defined by an elliptic curve E over k equipped with a level-H structure.

Proof (after J.P. Serre and J.S. Milne). Choose a separable closure k̄ of k and an object (E,α) over k̄ that defines the image of x in M_H^o(k̄). If H = GL(2, ℤ/n), which amounts to assuming n = 1, Proposition 3.2 reduces to 1.7. In the language of 3.1, a certain obstruction ε(E) is therefore zero. If Aut(E,α) is trivial, Proposition 3.2 is clear. If Aut(E) is reduced to ± 1 and Aut(E,α) is nontrivial, then Aut(E,α) $\xrightarrow{\sim}$ Aut(E); the obstructions ε(E,α) and ε(E) to defining (E,α) or E over k are therefore equal and zero. To complete the proof, it is enough to treat the following two cases.

Case 1: p ≠ 2,3. By 1.9, Aut(E) is then μ_2, μ_4, or μ_6. The automorphism group Aut(E,α) of (E,α) is a subgroup of Aut(E). Proposition 3.2 says that the obstruction class ε(E,α) ∈ H^2(Gal(k̄/k), Aut(E,α)) is zero. By 1.7 its image ε(E) in H^2(Gal(k̄/k), Aut(E)) is zero, and we apply the following lemma.

Lemma 3.3. Let n be prime to p.

(i) The subgroups of μ_n are the μ_m, for m|n.

(ii) The map H^2(Gal(k̄/k), μ_m) → Gal(k̄/k, μ_n) is injective.

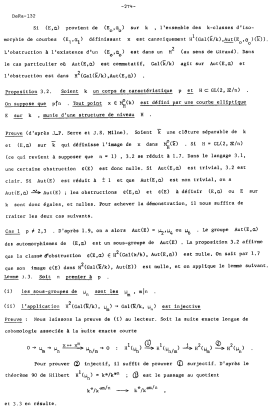
Proof. We leave the proof of (i) to the reader. Consider the long exact cohomology sequence associated with the short exact sequence

0 → μ_m → μ_n ⟶(x ↦ x^m)→ μ_{n/m} → 0 : H^1(μ_n) ⟶①→ H^1(μ_{n/m}) ⟶j→ H^2(μ_m) ⟶②→ H^2(μ_n).

To prove that ② is injective, it is enough to prove that ① is surjective. By Hilbert's Theorem 90, H^1(μ_n) = k^*/k^{*n}; ① is passage to the quotient

k^*/k^{*m/n} → k^*/k^{*m/n},

and 3.3 follows.



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Case 2: p ≠ 0 and Aut(E) is not reduced to ±1.

In this case, x is obtained by scalar extension from a point of M_H^o with values in a finite field (one has j = 0 or 2^6.3^3, and the field of definition of x is algebraic over the field generated by j(x)). We may therefore suppose that k is finite. Since Gal(k̄/k) = $\hat{\mathbb{Z}}$ and Aut(E,α) is finite, the Giraud H^2 in which the obstruction lies vanishes, and 3.2 follows. In other words, let x ∈ M_H^o($\mathbb{F}$_q); there is a finite extension $\mathbb{F}$_{q^n} of $\mathbb{F}$_q and a pair (E,α) over $\mathbb{F}$_{q^n} that defines x. The curve (E,α) is geometrically isomorphic to its conjugates under Gal($\mathbb{F}$_{q^n}/$\mathbb{F}$_q). After replacing $\mathbb{F}$_{q^n} by an extension, we may therefore suppose that there is an isomorphism

φ′ : (E,α)^(q) ⟶̃ (E,α).

This isomorphism defines descent data from $\mathbb{F}$_{q^n} to $\mathbb{F}$_q if and only if the nth power of φ : E —Fr→ E^(q) —φ′→ E is the Frobenius endomorphism F of E/$\mathbb{F}$_{q^n}. In general, φ^n = Fβ, where β has finite order m. Thus φ^{nm} = F^m, and after extending scalars to $\mathbb{F}$_{q^{nm}}, φ′ defines descent data from $\mathbb{F}$_{q^{nm}} to $\mathbb{F}$_q for (E,α). The descended curve solves the problem.

4. Numerical remarks

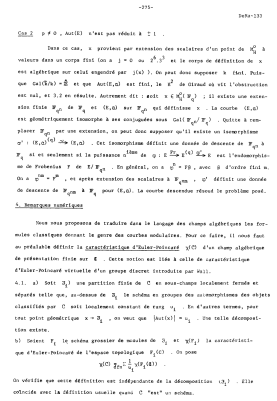
We propose to translate into the language of algebraic stacks the classical formulas giving the genera of modular curves. To do so, we must first define the Euler–Poincaré characteristic χ($\mathcal{C}$) of an algebraic stack of finite presentation over $\mathbb{C}$. This notion is related to Wall's virtual Euler–Poincaré characteristic of a discrete group.

4.1. a) Let ($\mathcal{Z}$_i) be a finite partition of $\mathcal{C}$ into separated locally closed substacks such that, over $\mathcal{Z}$_i, the group scheme of automorphisms of the objects classified by $\mathcal{C}$ is locally constant of rank u_i. In other words, for every geometric point x → $\mathcal{Z}$_i, we require |Aut(x)| = u_i. Such a decomposition exists.

b) Let F_i be the coarse moduli scheme of $\mathcal{Z}$_i, and let χ(F_i) be the Euler–Poincaré characteristic of the topological space F_i($\mathbb{C}$). Set

χ($\mathcal{C}$) =(dfn) Σ (1/u_i) χ(F_i($\mathbb{C}$)).

One checks that this definition is independent of the decomposition ($\mathcal{Z}$_i). It agrees with the usual definition when $\mathcal{C}$ "is" a scheme.



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Authority page 134; collected-paper page 627; original folio 276.

The Euler–Poincaré characteristic has the following properties:

α) additivity: if $\mathcal{Z}$ is a closed substack of $\mathcal{C}$ with open complement $\mathcal{U}$,

$\chi(\mathcal{C}) = \chi(\mathcal{U}) + \chi(\mathcal{Z})$;

if $\mathcal{U}$ and $\mathcal{V}$ are two open substacks,

$\chi(\mathcal{U} \cup \mathcal{V}) + \chi(\mathcal{U} \cap \mathcal{V}) = \chi(\mathcal{U}) + \chi(\mathcal{V})$.

β) multiplicativity: if $f : \mathcal{C} \to \mathcal{D}$ is a morphism and the geometric fibers $\mathcal{C}$_s of f have constant Euler–Poincaré characteristic, then

$\chi(\mathcal{C}) = \chi(\mathcal{D}).\chi(\mathcal{C}_s)$.

γ) algebraic character: the definition extends readily and makes it possible to define $\chi(\mathcal{C})$ for an algebraic stack $\mathcal{C}$ of finite presentation over an algebraically closed field k of characteristic zero.

δ) Let $f : \mathcal{C} \to \mathcal{D}$ be a finite flat morphism (i.e. flat, proper, with finite fibers), and let s be a geometric point of $\mathcal{D}$. For a geometric point x of $\mathcal{C}$_s, let Ã_x be the local ring of $\mathcal{C}$_s at x. It is an Artinian ring. Set

(4.1.1) deg_s(f) = Σ_x (1/|Aut(x)|) dim_{k(s)}(A_x).

This number is locally constant in s. If it is constant, denote it by deg(f). If f is finite étale, then $\chi(\mathcal{C}_s) = \deg_s(f)$, and β) gives

(4.1.2) $\chi(\mathcal{C}) = \deg(f)\, \chi(\mathcal{D})$.

More concretely, a closed geometric point s of $\mathcal{D}$ is simply an object of $\mathcal{D}$ over $\mathbb{C}$, and a geometric point of $\mathcal{C}$_s is an object x of $\mathcal{C}$ equipped with an isomorphism α : f(x) ↔ s. One has

(4.1.3) $\chi(\mathcal{C}_s) = \Sigma\, 1/\mathrm{Aut}(x,\alpha)$.

4.2. Apply Definition 4.1 to $\mathfrak{m}_1^{o} \otimes \mathbb{C}$.

We separate the closed substacks ”c_4 = 0” (6 automorphisms), ”c_6 = 0” (4 auto-

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Authority page 135; collected-paper page 628; original folio 277.

Continuing the preceding sentence, the complementary open (2 automorphisms) has as coarse scheme the affine line with the points j = 0 and j = 1728 removed. We find

(4.2.1) χ(𝔪_1^o ⊗ ℂ) = 1/6 + 1/4 + -1/2 = -1/12,

or, more pleasantly,

(4.2.2) χ(𝔪_1^o ⊗ ℂ) = ζ(-1).

For H ⊂ GL(2, ℤ/n), it follows from 4.1.5 that

(4.2.3) χ(𝔪_H^o ⊗ ℂ) = [GL(2, ℤ/n) : H] . -1/12.

To pass from this to the classical formula for χ(M_H ⊗ ℂ), the cusps and the automorphisms must be taken into account. Consider M_n ⊗ ℂ for n ≥ 3. Let ±U be the group of matrices ±(1 u; 0 1), with u ∈ ℤ/nℤ. There are [GL(2, ℤ/n) : ±U] cusps (5.1), and the objects of 𝔪_n ⊗ ℂ have no nontrivial automorphisms (IV 2.7).

Therefore

χ(M_n(ℂ)) = |GL(2, ℤ/n)| . -1/12 + [GL(2, ℤ/n) : ±U]

= -1/12 n^4 . ∏_{p|n} (1 - 1/p^2)(1 - 1/p) + n^3/2 ∏_{p|n} (1 - 1/p^2)(1 - 1/p).

The curve M_n(ℂ) has φ(n) connected components, all isomorphic. Their genus g is given by

2g - 2 = ∏_{p|n} (1 - 1/p^2) . (n^3/12 - n^2/2).

4.3. Let C be a proper Cohen-Macaulay algebraic stack, pure of dimension one, over an algebraically closed field k (for example, 𝔪_1 ⊗ ℂ). Let ω be an invertible sheaf on C. Its degree is defined as follows.

a) Choose a rational section f of ω.

b) Let x be a closed geometric point of C, and let Õ_x be the henselian local ring of C at x. If f is regular at x, set deg_x(f) = dim_k Õ_x/(f). If f^{-1} is regular there, set deg_x(f) = - dim_k Õ_x/(f^{-1}).

The definition continues on physical page 136.

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Continuing §4.3, set

deg(ω) = deg(C,ω) = Σ_x 1/|Aut(x)| deg_x(f).

When k is not algebraically closed, degree is defined after first extending scalars to $\bar{k}$. This degree is independent of the choice of f and agrees with the usual notion when C "is" a scheme.

It satisfies:

α) additivity in ω: deg(ω ⊗ ω') = deg(ω) + deg(ω').

β) coverings: Let f : C' → C be a finite flat morphism of constant degree. Then

deg(f*ω) = deg(f). deg(ω).

γ) specialization: Let S be a trait and C a proper flat Cohen-Macaulay algebraic stack, pure of relative dimension one over S. Let ω be an invertible sheaf on C. Then

deg(C_η,ω) = deg(C_s,ω).

4.4. Compute on $\mathfrak{m}_1$ the degree of ω, the dual of the Lie algebra of the universal elliptic curve. The discriminant Δ is a section of ω^(⊗12) vanishing simply at infinity. Since the point at infinity has two automorphisms, one finds

(4.4.1) deg(ω) = 1/24.

Starting from (4.4.1) and using (4.2.2) and 4.3(β), one obtains, for every H ⊂ GL(2, $\mathbb{Z}$/n),

(4.4.2) χ($\mathfrak{m}$_H^o ⊗ $\mathbb{C}$) = -2 deg($\mathfrak{m}$_H,ω).

4.5. Here is a direct proof of (4.4.2). At every point s of $\mathfrak{m}$_1^o over $\mathbb{C}$ corresponding to an elliptic curve E, deformation theory gives an isomorphism between the tangent space to $\mathfrak{m}$_1^o at s and H^1(E,T_E^1) ≃ H^1(E,O) ⊗ ω^(⊗-1) ≃ ω_s^(⊗-2). A study at infinity shows that the corresponding isomorphism Ω^1_($\mathfrak{m}$_1^o) ≃ ω^(⊗2) extends. The sentence continues on physical page 137.

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Authority page 137; collected-paper page 630; original folio 279.

At infinity, the identification is

(4.5.1) Ω^1_{𝔪_1}(𝔪_1^∞) ≃ ω^{⊗2}

In view of the multiplicativity properties of the two sides of (4.4.2), it is enough to consider 𝔪_n for an integer n ≥ 3. We then deal with genuine schemes, so that

χ(M_n ⊗ ℂ) = - deg(M_n, Ω^1_{M_n}).

Since the pullback of a differential form with logarithmic poles is again of this type, (4.5.1) gives us an isomorphism

(4.5.2) Ω^1_{𝔪_H}(𝔪_H^∞) ≃ ω^{⊗2} ;

if c is the number of points at infinity of 𝔪_n, we therefore have

χ(𝔪_n^o ⊗ ℂ) = χ(𝔪_n ⊗ ℂ) - c = - deg(Ω^1) - c = - deg(Ω^1(∞)) = - deg(ω^{⊗2}) = -2 deg(ω).

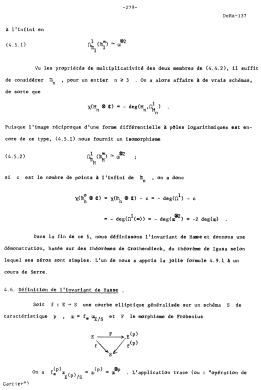
In the remainder of this section, we define the Hasse invariant and give a proof, based on theorems of Grothendieck, of Igusa's theorem that its zeros are simple. One of us learned the elegant formula 4.9.1 in a lecture by Serre.

4.6. Definition of the Hasse invariant.

Let f : E → S be a generalized elliptic curve over a scheme S of characteristic p, let ω = f_* ω_{E/S}, and let F be the Frobenius morphism

E ⟶F⟶ E^(p) f↘ ↙f^(p) S

We have f_*^(p) ω_{E^(p)/S} = ω^(p) = ω^{⊗p}. The trace map (also called the "Cartier operation")



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Tr_F : F_* ω_{E/S} → ω_{E^(p)/S}

defines a morphism

(4.6.1) Tr_F : ω → ω^{⊗p},

i.e. a section

(4.6.2) h ∈ Γ(S,ω^{⊗(p-1)}).

This is the Hasse invariant. Here is a dual description: coherent-sheaf duality pairs ω with R^1 f_* $\mathcal{O}$, and likewise pairs ω^(p) with R^1 f_*^(p) $\mathcal{O}$. The transpose of (4.6.1) is the pullback

(4.6.3) F^* : R^1 f_*^(p) $\mathcal{O}$ → R^1 f_* $\mathcal{O}$.

If R^1 f_* $\mathcal{O}$ and R^1 f_*^(p) $\mathcal{O}$ are identified with ω^{⊗(-1)} and ω^{⊗(-p)}, then (4.6.3) becomes multiplication by h from (4.6.2).

On $\mathfrak{m}_1 \otimes \mathbb{F}_p$, the Hasse invariant is known not to vanish at infinity, and its zeros correspond to supersingular elliptic curves (see Katz [13]).

The zeros of h on $\mathfrak{m}_1 \otimes \mathbb{F}_p$ are known to be simple. In 4.8 we give a direct proof based on Grothendieck's theory of deformations of abelian schemes; a special case of that theory is recalled below.

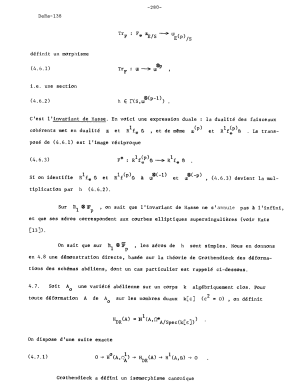
4.7. Let A_o be an abelian variety over an algebraically closed field k. For every deformation A of A_o over the dual numbers k[ε], where ε^2 = 0, define

H_DR(A) = H^1(A, Ω^*_{A/Spec(k[ε])}).

There is an exact sequence

(4.7.1) 0 → H^0(A, Ω^1_A) → H_DR(A) → H^1(A, $\mathcal{O}$) → 0.

Grothendieck defined a canonical isomorphism



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(4.7.2) H_DR(A) ≃ H_DR(A_o) ⊗_k k[ε],

and proved that

a) For every morphism f : A → B, the diagram

H_DR(B_o) ⊗ k[ε] —f_o ⊗ k[ε]→ H_DR(A_o) ⊗ k[ε] ‖ ‖ H_DR(B) —f*→ H_DR(A)

is commutative.

b) The map that assigns to each isomorphism class of deformations A the filtration (4.7.1) of H_DR(A_o) ⊗ k[ε], subject to lifting the filtration (4.7.1) of H_DR(A_o), is bijective.

4.8. Let E_o be a supersingular elliptic curve over k, let E be a nontrivial deformation of E_o over k[ε], and let us prove that h ≠ 0 (i.e. that h is a nonzero multiple of ε). Since E^(p) is the trivial deformation of E_o^(p), we have a commutative diagram

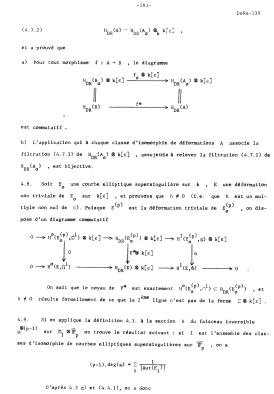
0 → H^0(E_o^(p),Ω^1) ⊗ k[ε] → H_DR(E_o^(p)) ⊗ k[ε] → H^1(E_o^(p),$\mathcal{O}$) ⊗ k[ε] ↓0 ↓F* ⊗ k[ε] ↓h 0 → H^0(E,Ω^1) → H_DR(E) ⊗ k[ε] → H^1(E,$\mathcal{O}$) → 0.

We know that the kernel of F* is exactly H^0(E_o^(p),Ω^1) ⊂ H_DR(E_o^(p)), and h ≠ 0 follows formally from the fact that the second row is not of the form Σ ⊗ k[ε].

4.9. If we apply Definition 4.3 to the section h of the invertible sheaf ω^{⊗(p-1)} on $\mathfrak{m}$_1 ⊗ $\mathbb{F}$_p, we obtain the following result: if I is the set of isomorphism classes of supersingular elliptic curves over $\mathbb{F}$_p, then

(p-1).deg(ω) = Σ_i 1/|Aut(E_i)|

By 4.3 α) and (4.4.1), we therefore have



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(4.9.1) Σ_{E supersingular} 1/|Aut(E)| = (p-1)/24.

For example, if p = 2 (respectively 3), only the elliptic curve with modular invariant 0 is supersingular. It has 24 (respectively 12) automorphisms. Likewise, if n ≥ 3 and (p,n) = 1, there are exactly (1-p)/2 χ(𝔪_n^o(ℂ)) supersingular points on 𝔪_n^o ⊗ 𝔽̄_p.

5. The Galois action on the cusps.

5.1. The set 𝔪_n^∞(ℂ) is the set of isomorphism classes, over ℂ, of singular generalized elliptic curves with n sides, equipped with a level-n structure.

Let (C,+) be the standard n-gon over ℂ, equipped with its generalized elliptic-curve structure II.1.9. Canonically,

C_n = μ_n × ℤ/n.

The automorphism group of (C,+) was computed in II 1.10. Its image in GL(μ_n × ℤ/n) is the group ± U of matrices

$$\pm \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \quad (u \in \mathrm{Hom}(\mathbb{Z}/n, \mu_n)).$$

The set of level-n structures on (C,+) is Isom(C_n,(ℤ/n)^2). Thus, canonically,

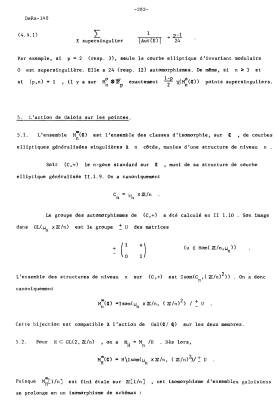
𝔪_n^∞(ℂ) = Isom(μ_n × ℤ/n, (ℤ/n)^2) / ± U.

This bijection is compatible with the action of Gal(ℂ/ℚ) on both sides.

5.2. For H ⊂ GL(2,ℤ/n), we have 𝔪_H = 𝔪_n/H. Consequently,

𝔪_H^∞(ℂ) = H\Isom(μ_n × ℤ/n, (ℤ/n)^2)/± U.

Since 𝔪_H^∞[1/n] is finite étale over ℤ[1/n], this isomorphism of Galois sets extends to an isomorphism of schemes:



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Construction 5.3. $\mathfrak{m}_H^\infty[1/n] = H\backslash\mathrm{Isom}_{\mathbb{Z}[1/n]}(\mu_n \times \mathbb{Z}/n, (\mathbb{Z}/n)^2) / \pm U$.

6. Study of $M_{\Gamma_0(n)}$.

6.1. In IV.4.5 we described $\mathfrak{m}^0_{\Gamma_0(n)}[1/n]$ as the stack over $\mathbb{Z}[1/n]$ classifying isogenies $E \to F$ with cyclic kernel of degree n. This exhibits two maps $\mathfrak{m}^0_{\Gamma_0(n)}[1/n] \to \mathfrak{m}_1^0[1/n]$, defined respectively by E and F and interchanged by an involution w (IV.4.4). By IV.3.19, the induced maps on the coarse schemes extend to

(6.1.1) c and cw : $M_{\Gamma_0(n)} \to M_1$.

6.2. The finite map (c,cw) : $M_{\Gamma_0(n)} \to M_1 \times M_1$ has as its image a divisor invariant under permutation of the coordinates. Since $M_{\Gamma_0(n)} \to M_1$ has degree equal to the number P(n) of points of the projective line over $\mathbb{Z}/n$ and is generically injective, this divisor has bidegree (P(n), P(n)). The maps c and cw carry points at infinity to points at infinity. Since $M_1^0 = \mathrm{Spec}(\mathbb{Z}[j])$ and $\mathbb{Z}[j,j']$ is factorial, the image divisor of

(6.2.1) (c,cw) : $M^0_{\Gamma_0(n)} \to M_1^0 \times M_1^0$

is defined by an equation $\Phi_n \in \mathbb{Z}[j,j']$, the modular equation. We have

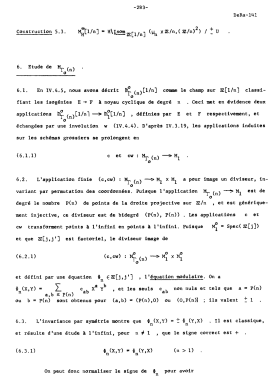
$\Phi_n(X,Y) = \Sigma_{a,b\le P(n)}\, c_{ab} X^a Y^b$,

and the only nonzero c_{ab} for which a = P(n) or b = P(n) occur at (a,b) = (P(n),0) or (0,P(n)); they are ± 1.

6.3. Symmetry shows that $\Phi_n(X,Y) = \pm\Phi_n(Y,X)$. It is classical, and follows from a study at infinity, that for $n \neq 1$ the correct sign is +.

(6.3.1) $\Phi_n(X,Y) = \Phi_n(Y,X)$ $(n > 1)$.

We may therefore normalize the sign of Φ_n so as to have



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(6.3.2) Φ_n(X,Y) = X^{P(n)} + Y^{P(n)} + Σ_{a,b<P(n)} c_ab X^a Y^b c_ab = c_ba ∈ ℤ.

6.4. One should note that the map (6.2.1) is not necessarily injective. Over an algebraically closed field k in which n is invertible, there may indeed exist two cyclic-kernel n-isogenies

(6.4.1) φ_1 : E_1 → F_1 φ_2 : E_2 → F_2

that are not isomorphic, even though there are isomorphisms e : E_1 ≃→ E_2 and f : F_1 → F_2. The endomorphism g = e^{-1} φ_2^* f φ_1 of E_1 then has degree n^2, but its kernel is not the kernel of multiplication by n. Thus E_1 has complex multiplication.

Conversely, let g be an endomorphism of degree n^2 of an elliptic curve E, suppose Ker(g) ≠ E_n, and let A ⊂ Ker(g) be cyclic of order n. A pair (6.4.1) is obtained by taking

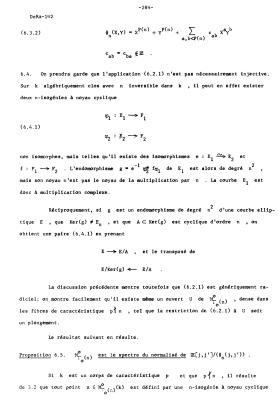
E → E/A, and the transpose of E/Ker(g) ← E/A.

The preceding discussion nevertheless shows that (6.2.1) is generically radicial; one easily shows that there is even an open subset U of M^o_{Γ_o(n)}, dense in the fibers of characteristic p ∤ n, such that the restriction of (6.2.1) to U is an immersion.

The following result follows.

Proposition 6.5. M^o_{Γ_o(n)} is the spectrum of the normalization of ℤ[j,j′]/(Φ_n(j,j′)).

If k is a field of characteristic p and p ∤ n, it follows from 3.2 that every point x ∈ M^o_{Γ_o(n)}(k) is defined by a cyclic-kernel n-isogeny



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E → F defined over k. It does not follow that if j,j’ belong to k and

Φ_n(j,j’) = 0,

there exists such an n-isogeny with j(E) = j and j(F) = j’.

If n is a prime p, the decomposition V.1.16 of $\mathfrak{m}_{\Gamma_o(p)}$ mod p translates as follows at the level of coarse schemes.

Theorem 6.6. (H. Weber-Lehrbuch der Algebra III, p.244 eq. (32)).

Φ_p(j,j’) ≡ (j-j’^p)(j^p-j’) mod p.

By studying infinity, one proves more generally that if (n,p) = 1, then

(6.6.1) $\Phi_{np}(j,j') \equiv \Phi_n(j^p,j')\,\Phi_n(j,j'^p) \pmod{p}$

(6.6.2) $\Phi_{np^2}(j,j') \equiv \Phi_n(j^{p^2},j')\,\Phi_n(j,j')^{p-1}\,\Phi_n(j,j'^{p^2}) \pmod{p}$

(6.6.3) $\Phi_{np^k}(j,j') \equiv \prod_{\ell+m=k,\ K=\inf(\ell,m)} \Phi_n(j^{p^{\ell-K}},j'^{p^{m-K}})^{\varphi(p^K)} \pmod{p}$

If p^2 | n, (6.6.3) gives

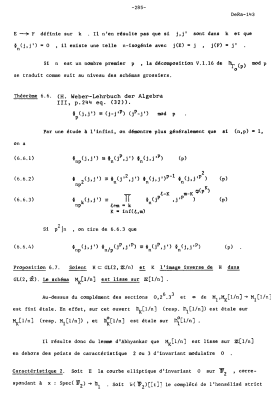
(6.6.4) $\Phi_{np}(j,j')\,\Phi_{n/p}(j^p,j'^p) \equiv \Phi_n(j^p,j')\,\Phi_n(j,j'^p) \pmod{p}$.

Proposition 6.7. Let H ⊂ GL(2,ℤ/n), and let K be the inverse image of H in $\mathrm{GL}(2,\hat{\mathbb{Z}})$. The scheme M_K[1/n] is smooth over ℤ[1/n].

Over the complement of the sections 0, 2^6·3^3, and ∞ of M_1, the map M_K[1/n] → M_1[1/n] is finite étale. Indeed, over this open set $\mathfrak{m}$_K[1/n] (respectively $\mathfrak{m}$_1[1/n]) is étale over M_K[1/n] (respectively M_1[1/n]), and $\mathfrak{m}$_K^o[1/n] is étale over $\mathfrak{m}$_1^o[1/n].

It therefore follows from Abhyankar’s lemma that M_K[1/n] is smooth over ℤ[1/n] away from the points of characteristic 2 or 3 with modular invariant 0.

Characteristic 2. Let E be the elliptic curve of invariant 0 over $\mathbb{F}$_2 corresponding to x : Spec($\mathbb{F}$_2) → $\mathfrak{m}$_1. Let W($\mathbb{F}$_2)[[t]] be the completion of the strict henselization



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of $\mathfrak{m}_1$ at x. The group A = Aut(E) acts on $W(\mathbb{F}_2)[[t]]$, and, in view of I.8, one must prove that for every subgroup B of A the algebra $W(\mathbb{F}_2)[[t]]^B$ is formally smooth over $W(\mathbb{F}_2)$. The group A embeds in $GL(2,\mathbb{F}_3)$ (IV.2.7). In fact,

$A = SL(2,\mathbb{F}_3)$.

It follows that if one of the algebras $W(\mathbb{F}_2)[[t]]^B$ were not formally smooth, then already one of the schemes M_H ($H \subset GL(2,\mathbb{Z}/3)$) would not be smooth over $\mathbb{Z}$.

The curve $M_3/\mathbb{Z}[\zeta_3]$ is known to have genus 0. Hence, for $H \subset GL(2,\mathbb{Z}/3)$, $M_H = M_3/H$ over $\mathbb{Z}[\zeta_3]^H$ has genus 0. The geometric fibers of $M_H[1/3]$ are also known to be generically smooth, irreducible, and reduced (being Cohen–Macaulay, with M_H normal). Since they have arithmetic genus 0, they are smooth.

Characteristic 3: The same argument applies, using the embedding $Aut(E) \subset GL(2,\mathbb{F}_4)$ and the fact that $M_4/\mathbb{Z}[\zeta_4]$ has genus 0.

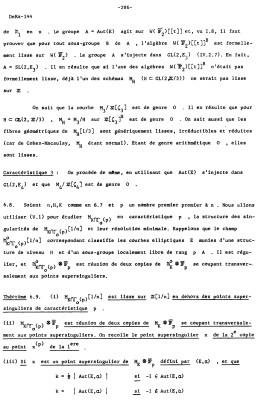
6.8. Let n,H,K be as in 6.7, and let p be a prime not dividing n. We shall use (V.1) to study $M_{K\cap\Gamma_o(p)}$ in characteristic p, the structure of the singularities of $M_{K\cap\Gamma_o(p)}[1/n]$, and their minimal resolution. Recall that the corresponding stack $\mathfrak{m}^o_{K\cap\Gamma_o(p)}[1/n]$ classifies elliptic curves E equipped with a level-H structure and a locally free subgroup A of rank p. It is regular, and $\mathfrak{m}^o_{K\cap\Gamma_o(p)} \otimes \mathbb{F}_p$ is the union of two copies of $\mathfrak{m}_K^o \otimes \mathbb{F}_p$ meeting transversely at the supersingular points.

Theorem 6.9. (i) $M_{K\cap\Gamma_o(p)}[1/n]$ is smooth over $\mathbb{Z}[1/n]$ away from the supersingular points of characteristic p.

(ii) $M_{K\cap\Gamma_o(p)} \otimes \mathbb{F}_p$ is the union of two copies of $M_K \otimes \mathbb{F}_p$ meeting transversely at the supersingular points. The supersingular point x of the second copy is glued to the point $x^{(p)}$ of the first.

(iii) If x is a supersingular point of $M_K \otimes \mathbb{F}_p$ defined by (E,α), and if

$k = \tfrac{1}{2}|\mathrm{Aut}(E,\alpha)|$ when $-1 \in \mathrm{Aut}(E,\alpha)$ $k = |\mathrm{Aut}(E,\alpha)|$ when $-1 \notin \mathrm{Aut}(E,\alpha)$



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Then, at the corresponding point of $M_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$, the scheme $M_{K\cap\Gamma_0(p)} \otimes W(\mathbb{F}_p)$ has a singularity of type A_{k-1}, formally isomorphic to

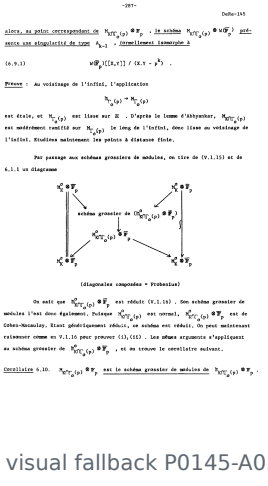
(6.9.1) $W(\mathbb{F}_p)[[X,Y]]/(X.Y-p^k)$.

Proof: In a neighborhood of infinity, the map $\mathcal{M}_{\Gamma_0(p)} \to M_{\Gamma_0(p)}$ is étale, and $M_{\Gamma_0(p)}$ is smooth over $\mathbb{Z}$. By Abhyankar's lemma, $M_{K\cap\Gamma_0(p)}$ is tamely ramified over $M_{\Gamma_0(p)}$ along infinity, and hence is smooth in a neighborhood of infinity. Let us now study the points at finite distance.

Passing to coarse moduli schemes, (V.1.15) and 6.1.1 yield the displayed diagram, with the central upper label "coarse moduli scheme of $\mathcal{M}^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$" and the caption "composed diagonals = Frobenius."

We know that $\mathcal{M}^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$ is reduced (V.1.16). Its coarse moduli scheme therefore is reduced as well. Since $M^o_{K\cap\Gamma_0(p)}$ is normal, $M^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$ is Cohen-Macaulay. Being generically reduced, this scheme is reduced. We may now argue as in V.1.16 to prove (i),(ii). The same arguments apply to the coarse moduli scheme of $\mathcal{M}^o_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$, and we obtain the following corollary.

Corollary 6.10. $M_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$ is the coarse moduli scheme of $\mathcal{M}_{K\cap\Gamma_0(p)} \otimes \mathbb{F}_p$.



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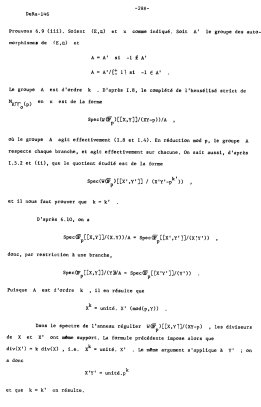
We prove 6.9 (iii). Let (E,α) and x be as indicated. Let A’ be the automorphism group of (E,α), and set

A=A’ if -1∉A’, A=A’/{±1} if -1∈A’.

The group A has order k. By I.8, the completion of the strict henselization of $M_{K\cap\Gamma_0(p)}$ at x is Spec(W($\mathbb{F}_p$)[[X,Y]]/(XY-p))/A, where A acts faithfully (I.8 and I.4). After reduction modulo p, A preserves each branch and acts faithfully on each one. We also know, by I.5.2 and (ii), that the quotient has the form Spec(W($\mathbb{F}_p$)[[X’,Y’]] /(X’Y’-p^{k’})), and we must prove k=k’.

By 6.10, Spec($\mathbb{F}_p$[[X,Y]]/(X.Y))/A = Spec($\mathbb{F}_p$[[X’,Y’]] /(X’.Y’)), and hence, upon restriction to one branch, Spec($\mathbb{F}_p$[[X,Y]]/(Y))/A = Spec($\mathbb{F}_p$[[X’Y’]] /(Y’)).

Since A has order k, it follows that X^k=unit·X’ (mod(p,Y)). In the spectrum of the regular ring W($\mathbb{F}_p$)[[X,Y]]/(XY-p), the divisors of X and X’ have the same support. The preceding formula therefore forces div(X’)=k div(X), i.e. X^k=unit·X’. The same argument applies to Y’; hence X’Y’=unit·p^k, and it follows that k=k’.



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Proposition 6.9 (iii). Soient (E,α) et x comme indiqué. Soit A' le groupe des automorphismes de (E,α) et

$$A = A' \text{ si } -1 \notin A'$$
$$A = A'/\{\pm 1\} \text{ si } -1 \in A' .$$

Le groupe A est d'ordre k. D'après I.8, le complété de l'hensélisé strict de $M_{K\cap\Gamma_0(p)}$ en x est de la forme

$$\mathrm{Spec}(W(\mathbb{F}_p)[[X,Y]]/(XY-p))/A ,$$

où le groupe A agit effectivement (I.8 et I.4). En réduction mod p, le groupe A respecte chaque branche, et agit effectivement sur chacune. On sait aussi, d'après I.5.2 et (ii), que le quotient étudié est de la forme

$$\mathrm{Spec}(W(\mathbb{F}_p)[[X',Y']] / (X'Y'-p^{k'})) ,$$

et il nous faut prouver que $k = k'$.

D'après 6.10, on a

$$\mathrm{Spec}\mathbb{F}_p[[X,Y]]/(X.Y))/A = \mathrm{Spec}\mathbb{F}_p[[X',Y']]/(X'.Y')) ,$$

donc, par restriction à une branche,

$$\mathrm{Spec}\mathbb{F}_p[[X,Y]]/(Y)/A = \mathrm{Spec}\mathbb{F}_p[[X'Y']]/(Y')) .$$

Puisque A est d'ordre k, il en résulte que

$$X^k = \text{unité} . X' \ (\mathrm{mod}(p,Y)) .$$

Dans le spectre de l'anneau régulier $W\mathbb{F}_p[[X,Y]]/(XY-p)$, les diviseurs de X et X' sont donc égaux. La formule précédente impose alors que $\mathrm{div}(X') = k\ \mathrm{div}(X)$, i.e. $X^k = \text{unité} . X'$. Le même argument s'applique à Y' ; on a donc

$$X'Y' = \text{unité}.p^k$$

et que $k = k'$ en résulte.

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Specialization invariance of the Euler-Poincaré characteristic with $\mathbb{Q}$-coefficients, together with 6.9(ii), gives the following topological Euler-Poincaré formula.

Corollary 6.11. Let s be the number of supersingular points on $M_K \otimes \mathbb{F}_p$. Then

(6.11.1) $\chi(M_{K\cap\Gamma_0(p)}) = 2\chi(M_K)-2s$.

If $M_{K\cap\Gamma_0(p)}$ is regarded as a geometrically irreducible curve over $R=\mathbb{Z}[\zeta_n]^H$, and if $s_0=s/[R:\mathbb{Z}]$ denotes the number of supersingular points of $M_K \otimes_R \mathbb{F}_p$, then 6.11 can be rewritten in terms of the genus as

(6.11.2) $g(M_{K\cap\Gamma_0(p)} \otimes_R \mathbb{C}) = 2g(M_K)+(s_0-1)$.

Corollary 6.12. $M_{K\cap\Gamma_0(p)} \otimes_R \mathbb{C}$ is an elliptic curve if and only if $M_K \otimes_R \mathbb{C}$ has genus 0 and $s_0=2$.

Reminder 6.13. Resolving the singularity (6.9.1) by blow-ups produces a chain of (k-1) projective lines of self-intersection -2. The locus p=0 in the resolution is reduced. This makes it possible to calculate the minimal resolution of $M_{K\cap\Gamma_0(p)}$.

The following lemma, taken from [20], will be proved in VII.2.7.

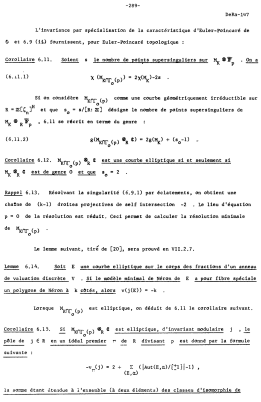
Lemma 6.14. Let E be an elliptic curve over the fraction field of a discrete valuation ring V. If the special fiber of the minimal Néron model of E is a Néron k-gon, then v(j(E))=-k.

When $M_{K\cap\Gamma_0(p)}$ is elliptic, Corollary 6.11 gives the following result.

Corollary 6.15. Suppose $M_{K\cap\Gamma_0(p)} \otimes_R \mathbb{C}$ is elliptic, with modular invariant j. The order of the pole of $j\in R$ at a prime ideal π of R dividing p is

$$-v_\pi(j) = 2 + \Sigma_{(E,\alpha)} (|\mathrm{Aut}(E,\alpha)/\{\pm 1\}|-1),$$

where the sum ranges over the two-element set of isomorphism classes of



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l'invariance par spécialisation de la caractéristique d'Euler-Poincaré de G et 6.9 (iii) fournissent, pour Euler-Poincaré cyclotomique :

Corollaire 6.11. Soient ℓ le nombre de points supersinguliers sur $X_0 \otimes \mathbb{F}_p$. On a

$$(6.11.1) \qquad \chi(X_{0,\mathrm{ét},\mathbb{Q}_p}) = 2\chi(X_0) - 2\ell \ .$$

Si on considère $X_{0,\mathrm{ét},\mathbb{Q}_p}$ comme une courbe géométriquement irréductible sur $k = \mathrm{ét}(\mathbb{Q}_p)^H$ et que $r_p = \mathrm{rg}(H,\mathbb{Z})$ désigne le nombre de points supersinguliers de $X_0 \otimes_{\mathbb{Z}} \mathbb{F}_p$, 6.11 se réécrit en terme du genre :

$$(6.11.2) \qquad g(X_{0,\mathrm{ét},\mathbb{Q}_p} \otimes_k \bar{k}) = 2g(X_0) + (r_p - 1) \ .$$

Corollaire 6.12. $X_{0,\mathrm{ét},\mathbb{Q}_p} \otimes_k \bar{k}$ est une courbe elliptique si et seulement si $X_0 \otimes_{\mathbb{Z}} \bar{k}$ est de genre 0 et que $r_p = 2$.

Exemple 6.13. Résolvant la singularité (4,9,1) par éclatements, on obtient une chaîne de (k-1) droites projectives de self intersection -2 , le lieu d'équation $y = 0$ de la résolution est réduit. Ceci permet de calculer la résolution minimale de $X_{0,\mathrm{ét},\mathbb{Q}_p}$.

Le lemme suivant, tiré de [IV], sera prouvé en VII.2.7.

Lemme 6.14. Soit E une courbe elliptique sur le corps des fractions d'un anneau de valuation discrète V ; si le modèle minimal de Néron de E a pour fibre spéciale un polygone de Néron à k côtés, alors $v(j(E)) = -k$.

Lorsque $X_{0,\mathrm{ét},\mathbb{Q}_p}$ est elliptique, on déduit de 6.11 le corollaire suivant.

Corollaire 6.15. Si $X_{0,\mathrm{ét},\mathbb{Q}_p} \otimes_k \bar{k}$ est elliptique, d'invariant modulaire j , le pôle de $j \in R$ en un idéal premier π de R divisant p est donné par la formule suivante :

$$-v_\pi(j) = 2 + \sum_{(E,\alpha)} (|\mathrm{Aut}(E,\alpha)/\{\pm 1\}|-1) \ ,$$

la somme étant étendue à l'ensemble (à deux éléments) des classes d'isomorphie de

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supersingular elliptic curves E, equipped with a level-H structure of determinant one, over an algebraic closure $\mathbb{F}_p$ of $R/(\pi)$.

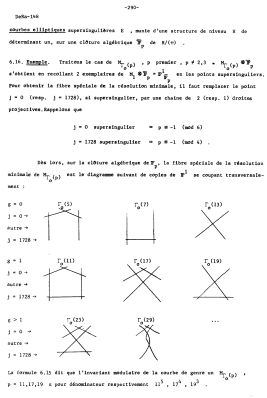
6.16. Example. Consider $M_{\Gamma_0(p)}$, where p is prime and $p\neq 2,3$. Its special fiber $M_{\Gamma_0(p)} \otimes \mathbb{F}_p$ is obtained by gluing two copies of $M_1 \otimes \mathbb{F}_p = \mathbb{P}^1_{\mathbb{F}_p}$ at the supersingular points.

To obtain the special fiber of the minimal resolution, a supersingular point j=0 must be replaced by a chain of two projective lines, while a supersingular point j=1728 must be replaced by a chain of one projective line. Recall that

j=0 is supersingular $\Leftrightarrow$ $p\equiv -1 \pmod 6$, j=1728 is supersingular $\Leftrightarrow$ $p\equiv -1 \pmod 4$.

Thus, over an algebraic closure of $\mathbb{F}_p$, the special fiber of the minimal resolution of $M_{\Gamma_0(p)}$ is represented by the displayed diagram of copies of $\mathbb{P}^1$ meeting transversely. Its left-hand labels are j=0, other, and j=1728.

Formula 6.15 says that the modular invariant of the genus-one curve $M_{\Gamma_0(p)}$, for p=11,17,19, has denominator $11^5,17^4,19^3$, respectively.



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VII. The Tate curve.

In §1 of this chapter, we use a method due to Raynaud to construct the Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ over $\mathbb{Z}[[q]]$. This enables us to give an explicit description of the formal completion of $\mathcal{M}_H$ along infinity (§2). In §§3 and 4, we deduce integrality theorems for the coefficients of the Fourier-series expansions of modular forms.

1. Construction of the Tate curve over $\mathbb{Z}[[q]]$.

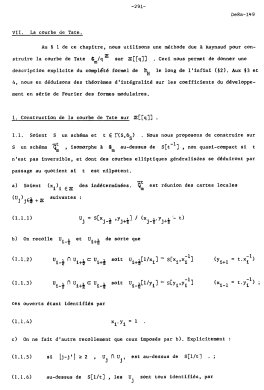
1.1. Let S be a scheme and $t \in \Gamma(S,\mathcal{O}_S)$. We propose to construct over S a scheme $\bar{\mathbb{G}}_m^t$, isomorphic to $\mathbb{G}_m$ over $S[t^{-1}]$, non-quasi-compact when t is not invertible, and from which generalized elliptic curves will be obtained by taking a quotient when t is nilpotent.

a) Let $(x_i)_{i\in\mathbb{Z}}$ be indeterminates. $\bar{\mathbb{G}}_m^t$ is the union of the local charts $(U_j)_{j\in 1/2+\mathbb{Z}}$:

(1.1.1) $U_j = S[x_{j-1/2}, y_{j+1/2}] / (x_{j-1/2} y_{j+1/2} - t)$.

b) Glue $U_{i-1/2}$ and $U_{i+1/2}$ so that their overlap is $U_{i+1/2}[1/x_i] \simeq S[x_i, x_i^{-1}]$, with $y_{i+1} = t\, x_i^{-1}$, and also $U_{i-1/2}[1/y_i] \simeq S[y_i, y_i^{-1}]$, with $x_{i-1} = t\, y_i^{-1}$; these opens are identified by $x_i\, y_i = 1$.

c) No gluings other than those imposed by b) are made. If $|j-j'| \geq 2$, $U_j \cap U_{j'}$ lies over $S[1/t]$; over $S[1/t]$, all the U_j are identified by



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x_i=t^{-i}x_0,

y_i=t^{-i}y_0.

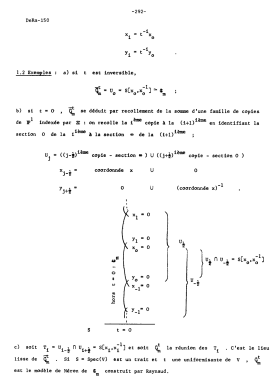
1.2 Examples: a) if t is invertible, $\bar{G}_m^t=U_0=S[x_0,x_0^{-1}]\simeq\mathbb{G}_m$.

b) if t=0, $\bar{G}_m^t$ is obtained by gluing the disjoint union of a family of copies of $\mathbb{P}^1$ indexed by $\mathbb{Z}$: the ith copy is glued to the (i+1)st by identifying the zero section of the ith copy with the infinity section of the (i+1)st.

For each j, U_j is the union of the (j-1/2)th copy minus its infinity section and the (j+1/2)th copy minus its zero section, with x_{j-1/2}=(coordinate x)∪0 and y_{j+1/2}=0∪(coordina x)^{-1}.

The schematic shows an infinite vertical chain with successive junctions x_1=0, y_1=0, x_0=0, y_0=0, x_{-1}=0, y_{-1}=0. Overlapping braces label U_{1/2}, U_{-1/2}, and U_{1/2}∩U_{-1/2}=S[x_0,x_0^{-1}]. The chain meets the base S at the special fiber t=0, and the generic part is labeled “away from t=0: $\mathbb{G}_m$.”

c) Let T_i=U_{i-1/2}∩U_{i+1/2}=S[x_i,x_i^{-1}], and let $\mathbb{G}_m^t$ be the union of the T_i. It is the smooth locus of $\bar{G}_m^t$. If S=Spec(V) is a trait and t is a uniformizer of V, then $\mathbb{G}_m^t$ is the Néron model of $\mathbb{G}_m$ constructed by Raynaud.



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The maps

$\bar{G}_m^t(S) \leftarrow \mathbb{G}_m^t(S) \rightarrow \mathbb{G}_m^t(K)=\mathbb{G}_m(K)=K^*$

are bijective, where K is the fraction field of V. To t^i u∈K^*, with u∈V^*, corresponds the section x_i=u of T_i. This example motivates the definitions that follow.

d) Abuse of notation: for a invertible on S and k∈$\mathbb{Z}$, we shall sometimes denote by at^k the section of T^k such that x_k(at^k)=a.

1.3. We define on $\bar{G}_m^t$ the structure of a commutative group scheme. Multiplication sends T_i×T_j to T_{i+j} and is given by x_{i+j}(a·b)=x_i(a)x_j(b). The intersection of distinct charts is the inverse image of S[1/t]; over S[1/t] the barred model equals $\mathbb{G}_m$ and this law is ordinary multiplication.

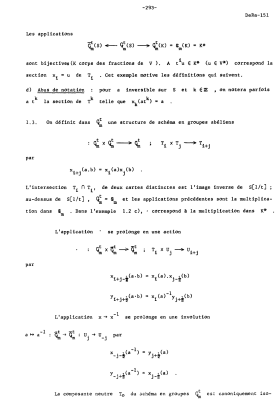
Multiplication extends to an action $\mathbb{G}_m^t \times \bar{G}_m^t \to \bar{G}_m^t$, carrying $T_i \times U_j$ to U_{i+j}, with

$x_{i+j-1/2}(a\cdot b)=x_i(a)x_{j-1/2}(b)$, $y_{i+j+1/2}(a\cdot b)=x_i(a)^{-1}y_{j+1/2}(b)$.

Inversion extends to $\bar{G}_m^t$ and carries U_j to U_{-j}, with

$x_{-j-1/2}(a^{-1})=y_{j+1/2}(a)$, $y_{-j+1/2}(a^{-1})=x_{j-1/2}(a)$.

The identity component T_0 of $\mathbb{G}_m^t$ is canonically iso-



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Les applications

$$\mathcal{G}_m^t(k) \longleftarrow \mathcal{G}_m^t(k) \longrightarrow \mathcal{G}_m^t(k) = \mathbb{G}_m(k) = k^*$$

avec bijectivité (k corps des fractions de V). A $t^k u \in k^*$ ($u \in V^*$) correspond la section $\pi_k = u \cdot$ de T_k. Cet exemple motive les définitions qui suivent.

d) Abus de notation : pour a inversible sur S et $k \in \mathbb{Z}$, on notera parfois $a.t^k$ la section de T^k telle que $\pi_k(a.t^k) = a$.

1.3. On définit dans $\mathcal{G}_m^t$ une structure de schéma en groupes abélien

$$\cdot : \mathcal{G}_m^t \times \mathcal{G}_m^t \longrightarrow \mathcal{G}_m^t \quad ; \quad T_i \times T_j \longrightarrow T_{i+j}$$

par

$$\pi_{i+j}(a.b) = \pi_i(a)\pi_j(b) .$$

L'intersection $T_i \cap T_{i'}$ de deux cartes disciuces est l'image inverse de $S[1/t]$; au-dessus de $S[1/t]$, $\mathcal{G}_m^t = \mathbb{G}_m$ et les applications précédentes sont la multiplication dans $\mathbb{G}_m$. Dans l'exemple 1.2 c), $\cdot$ correspond à la multiplication dans k^*.

L'application $\cdot$ se prolonge en une action

$$\cdot : \mathcal{G}_m^t \times \bar{\mathcal{G}}_m^t \longrightarrow \bar{\mathcal{G}}_m^t \quad ; \quad T_i \times U_j \longrightarrow U_{i+j}$$

par

$$\pi_{i+j-\frac{1}{2}}(a.b) = \pi_i(a).x_{j-\frac{1}{2}}(b)$$

$$\gamma_{i+j+\frac{1}{2}}(a.b) = \pi_i(a)^{-1}y_{j+\frac{1}{2}}(b)$$

L'application $x \mapsto x^{-1}$ se prolonge en une involution

$\sigma \mapsto \sigma^{-1} : \bar{\mathcal{G}}_m^t \to \bar{\mathcal{G}}_m^t$; $U_j \to U_{-j}$ par

$$x_{-j-\frac{1}{2}}(a^{-1}) = y_{j+\frac{1}{2}}(a)$$

$$y_{-j+\frac{1}{2}}(a^{-1}) = x_{j-\frac{1}{2}}(a) .$$

Le composante neutre T_0 du schéma en groupes $\bar{\mathcal{G}}_m^t$ est canoniquement iso-

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-morphic to $\mathbb{G}_m$, via x_0.

1.4. If $u \in \Gamma(S,\mathcal{O}_S)^*$ and $m \geq 1$, define $[u]:\bar{G}_m^t \to \bar{G}_m^{ut^m}$. On the indicated charts its coordinates are

$x_i^{(ut^m)}\circ[u]=x_{mi+k}^{(t)}t^k u^{-i}$, $y_i^{(ut^m)}\circ[u]=y_{mi-k}^{(t)}t^k u^i$, $0 \leq k < m$.

For $t=0$ this contracts each T_i with i not congruent to 0 modulo m to a point. If $m=1$ it is an isomorphism; in particular, the Néron model is independent of the chosen uniformizer.

1.5. Suppose t is nilpotent. Let g be a section of $\mathbb{G}_m^t$ lying in T_k, $k \neq 0$. Let $\mathbb{Z}$ act on $\bar{G}_m^t$ by $(i,x) \mapsto g^i \cdot x$.

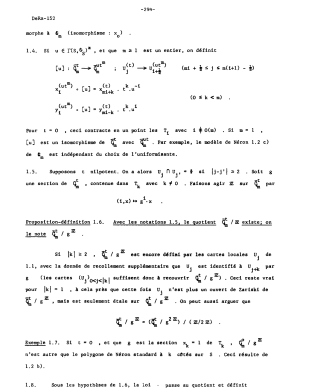
Proposition-definition 1.6. With the notation of 1.5, the quotient $\bar{G}_m^t/\mathbb{Z}$ exists and is denoted $\bar{G}_m^t/g^{\mathbb{Z}}$.

For $|k| \geq 2$ it is covered by the local charts U_j with the additional gluing that g identifies U_j with U_{j+k}. For $|k|=1$ the same description holds étale-locally rather than by Zariski opens. Equivalently,

$\bar{G}_m^t/g^{\mathbb{Z}}=(\bar{G}_m^t/g^{2\mathbb{Z}})/(\mathbb{Z}/2\mathbb{Z})$.

Example 1.7. For $t=0$ and g given by $x_k=1$ on T_k, the quotient is the standard Néron k-gon.

1.8. Under the hypotheses of 1.6, multiplication descends to the quotient and defines



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Authority page 153; collected-paper page 646; original folio 295.

a map

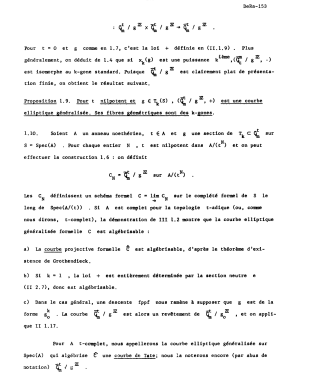
$\bar{G}_m\hat{}t/g\hat{}\mathbb{Z} \times \bar{G}_m\hat{}t/g\hat{}\mathbb{Z} \to \bar{G}_m\hat{}t/g\hat{}\mathbb{Z}$.

For t=0 and g as in 1.7, this is the addition law of (II.1.9). More generally, if x_k(g) is a kth power, the printed dot-marked quotient is isomorphic to the standard k-gon. Since the quotient is flat and of finite presentation, one obtains:

Proposition 1.9. For nilpotent t and g∈T_k(S), $(\bar{G}_m\hat{}t/g\hat{}\mathbb{Z},+)$ is a generalized elliptic curve whose geometric fibers are k-gons.

1.10. Let A be Noetherian, t∈A, and g a section of T_k over S=Spec(A). For every N define C_N=$\bar{G}_m\hat{}t/g\hat{}\mathbb{Z}$ over A/(t^N). The C_N define the direct-limit formal scheme C over the completion of S along Spec(A/(t)). If A is t-adically complete, the formal generalized elliptic curve is algebraizable: the formal projective curve $\hat{C}$ is algebraizable by Grothendieck existence; for k=1 the identity section determines the law; in general fppf descent reduces to g=g_0^k and II 1.17 applies to the resulting covering.

The algebraization over Spec(A) is called a Tate curve and is again denoted $\bar{G}_m\hat{}t/g\hat{}\mathbb{Z}$.



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Authority page 154; collected-paper page 647; original folio 296.

Proposition 1.11. The image of the nonsmoothness subscheme of the Tate curve $\bar{G}_m^t/g^{\mathbb{Z}}$ over Spec(A) is defined by t=0. It is the same image for the quotient, the formal scheme $\hat{C}$, and the barred model.

1.12. For nilpotent t, the identity components of $\bar{G}_m^t$ and $\bar{G}_m^t/g^{\mathbb{Z}}$ are canonically isomorphic to $\mathbb{G}_m$. Hence the formal completion of $\mathbb{G}_m$ along its identity is isomorphic to the completion of the quotient; their Lie algebras are identified with $\mathcal{O}$; and μ_n injects into the n-torsion.

For Noetherian t-complete A, formulas (1.12.1)–(1.12.3) give the corresponding formal-group isomorphism, Lie isomorphism, and inclusion $\mu_n \hookrightarrow (\bar{G}_m^t/g^{\mathbb{Z}})_n$. Tate curves therefore carry the canonical invariant differential dx/x.

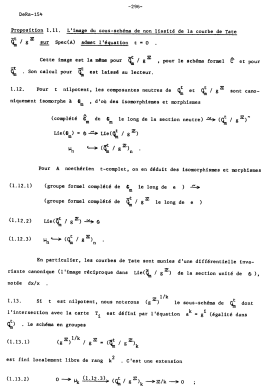
1.13. If t is nilpotent, let $(g^{\mathbb{Z}})^{1/k}$ be the subscheme whose intersection with T_i is defined by $a^k=g^i$. The group scheme

$$(g^{\mathbb{Z}})^{1/k}/g^{\mathbb{Z}}=(\bar{G}_m^t/g^{\mathbb{Z}})_k$$

is finite locally free of rank k^2 and is an extension

$$0\to\mu_k\to(\bar{G}_m^t/g^{\mathbb{Z}})_k\to\mathbb{Z}/k\to0,$$

whose explanation continues on page 155.



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Authority page 155; collected-paper page 648; original folio 297.

The projection to $\mathbb{Z}/k$ sends x to the residue class of the integers i for which x is the image of a section of T_i.

For Noetherian t-complete A, passage to the limit yields the exact sequence

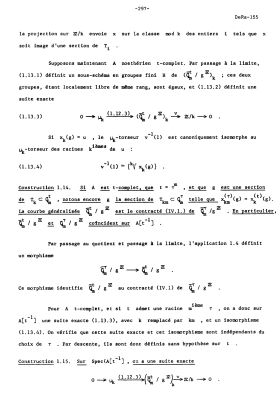
$$0\to\mu_k\to(\bar{G}_m^t/g^{\mathbb{Z}})_k\to{}^v\mathbb{Z}/k\to0. \quad (1.13.3)$$

If $x_k(g)=u$, the torsor $v^{-1}(1)$ is canonically the torsor of kth roots of u:

$$v^{-1}(1)\simeq\{\sqrt[k]{x_k(g)}\}. \quad (1.13.4)$$

Construction 1.14. If $t=\tau^m$ and g lies in T_k of $\bar{G}_m^t$, regard it as a section of T_{km} of $\bar{G}_m^\tau$ satisfying $x_{km}^{(\tau)}(g)=x_k^{(t)}(g)$. The t-curve is the contraction of the τ-curve, and the two coincide over $A[t^{-1}]$. The quotient map defines the contraction morphism. The exact sequence and root-torsor description are independent of τ and hence descend without a root hypothesis.

Construction 1.15. Over Spec(A[t^{-1}]) there is the same exact sequence with projection v.



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Authority page 156; collected-paper page 649; original folio 298.

The torsor v^{-1}(1) is canonically the torsor of kth roots of g∈𝔾_m^t(A[t^{-1}])≃A[t^{-1}]^*.

Let ℤ[[q^{1/k}]] be the power-series ring in q^{1/k}. Denote by q the section of T_k in Ḡ_m^{q^{1/k}} for which x_k(g)=1.

Definition 1.16. The k-sided Tate curve over ℤ[[q^{1/k}]] is Ḡ_m^{q^{1/k}}/q^ℤ. For k=1 it is the Tate curve over ℤ[[q]]. It carries: an isomorphism Ĉ≃Ĝ_m; the invariant differential dx/x; injections μ_n↪C_n; and an isomorphism C_k≃μ_k×ℤ/k.

The last isomorphism is defined by μ_k and the section q^{1/k}. For one natural pairing convention, despite the printed e_k/e_n mismatch,

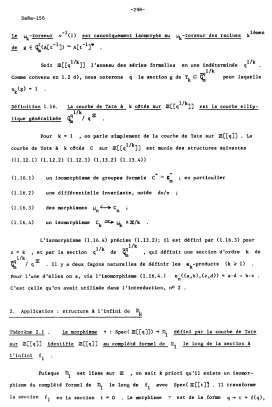
e_n((a,b),(c,d))=a·d-b·c.

2. Application: structure at infinity of $\mathcal{M}$_H

Theorem 2.1. The Tate-curve morphism τ:Spec(ℤ[[q]])→$\mathcal{M}$_1 identifies ℤ[[q]] with the formal completion of $\mathcal{M}$_1 along the section at infinity f_1. Smoothness gives a coordinate t with f_1 at t=0. The morphism has the form

q↦t=f(q),

and the proof continues on physical page 157.



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for a formal series f (that is, τ*(t) = f(q)). We know that the image, in $\mathcal{M}_1$, of the nonsmoothness subscheme of the universal curve is f_1 (IV 2.2). By 1.11, the image, in Spec($\mathbb{Z}$[[q]]), of the nonsmoothness subscheme of the Tate curve has equation q = 0. Since formation of this image commutes with base change, we have

(q) = (f(q)) in $\mathbb{Z}$[[q]],

and this implies that τ is an isomorphism.

Let H ⊂ GL(2, $\mathbb{Z}$/n).

Corollary 2.2. The fiber product

Spec($\mathbb{Z}$[[q]]) ×_{$\mathcal{M}_1$} $\mathcal{M}$_H

is the spectrum of the normalization of $\mathbb{Z}$[[q]] in the affine algebra of the finite étale scheme over $\mathbb{Z}$[[q]][1/n, 1/q]

Isom((𝔾_m^q/q^$\mathbb{Z}$)_n, ($\mathbb{Z}$/n)²)/H.

Proof. This follows from the definitions and from the fact that normalization and completion commute.

Since the group scheme (𝔾_m^q/q^$\mathbb{Z}$)_n over $\mathbb{Z}$[[q]][1/q] was computed in 1.15, 2.2 makes it possible to describe explicitly the structure of $\mathcal{M}$_H at infinity.

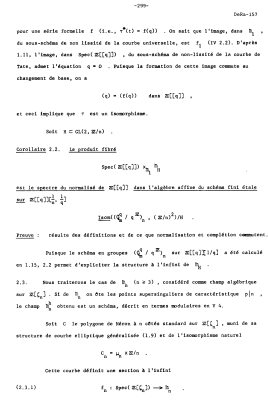
2.3. We shall treat the case of $\mathcal{M}$_n (n ≥ 3), considered as an algebraic stack over $\mathbb{Z}$[ζ_n]. If the supersingular points of characteristic p | n are removed from $\mathcal{M}$_n, the resulting stack $\mathcal{M}$_n^h is a scheme, described in modular terms in V 4.

Let C be the standard Néron n-gon over $\mathbb{Z}$[ζ_n], endowed with its structure of generalized elliptic curve (1.9) and with the natural isomorphism

C_n = μ_n × $\mathbb{Z}$/n.

This curve defines a section at infinity

(2.3.1) f_n : Spec($\mathbb{Z}$[ζ_n]) → $\mathcal{M}$_n.



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Over $\mathbb{Z}[\zeta_n][[q^{1/n}]]$, consider the generalized elliptic curve

$C' = \mathbb{G}_m^{q^{1/n}}/q^{\mathbb{Z}}$.

The map (1.16.3), $\mu_n \to C'_n$, and the section of order n "$q^{1/n}$" ($x_1 = 1$) define an isomorphism

$C'_n = \mu_n \times \mathbb{Z}/n$.

The corresponding morphism

(2.3.2) $\mathrm{Spec}(\mathbb{Z}[\zeta_n][[q^{1/n}]]) \to \mathcal{M}_n$

extends (2.3.1), and, by 1.14, the diagram

$$\begin{array}{ccc} \mathrm{Spec}(\mathbb{Z}[\zeta_n][[q^{1/n}]]) & \to & \mathcal{M}_n \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathbb{Z}[[q]]) & \to & \mathcal{M}_1 \end{array}$$

is commutative.

Corollary 2.4. The morphism (2.3.2) identifies $\mathbb{Z}[\zeta_n][[q^{1/n}]]$ with the formal completion of $\mathcal{M}_n$ along the section at infinity f_n.

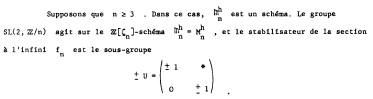
Proof. This can be deduced from 2.2. It is simpler to repeat the proof of 2.1, using the facts that

(a) $\mathcal{M}_n$ is smooth over $\mathbb{Z}[\zeta_n]$ in a neighborhood of infinity;

(b) the image on $\mathrm{Spec}(\mathbb{Z}[\zeta_n][[q^{1/n}]])$ of the nonsmooth locus of $\mathbb{G}_m^{q^{1/n}}$ is defined by the equation $q^{1/n} = 0$.

Suppose that $n \geq 3$. In this case, $\mathcal{M}_n^h$ is a scheme. The group $SL(2, \mathbb{Z}/n)$ acts on the $\mathbb{Z}[\zeta_n]$-scheme $\mathcal{M}_n^h = M_n^h$, and the stabilizer of the section at infinity f_n is the subgroup

$\pm U = \{ \pm \begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix} \}$.



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Corollary 2.5. If n ≥ 3, the completion of $\mathcal{M}$_n^h along infinity is a sum of copies of Spec(ℤ[ζ_n][[q^{1/n}]]), indexed by SL(2, ℤ/n)/±U.

Corollary 2.6. Let C be a generalized elliptic curve over a complete local ring A. Suppose that C is not smooth; let (t) be the ideal of A that defines the nonsmoothness subscheme. Suppose that the special fiber C_s of C has k sides and that Gal(s̄/s) acts trivially on the diagram Γ(C_s̄) of the irreducible components of C_s̄. Then there exists u ∈ A* such that

C ≅ 𝔾_m^t/(ut^k)^ℤ.

Proof. Let X_v be the group scheme over ℤ[v, v^{-1}], an extension of ℤ/k by μ_k,

0 → μ_k → X_v →^{φ} ℤ/k → 0,

such that φ^{-1}(1) is the μ_k-torsor of kth roots of v. Let $\mathcal{G}$_v be the algebraic stack over ℤ[v, v^{-1}] classifying generalized elliptic curves E over a ℤ[v, v^{-1}]-scheme, smooth or k-sided, endowed with an isomorphism E_k ≅ X_v. After scalar extension to ℤ[v^{1/k}], X_v becomes isomorphic to μ_k × ℤ/k, and $\mathcal{G}$_v to $\mathcal{G}$ of V 4.4. The map

Spec(ℤ[v, v^{-1}][[τ]]) → $\mathcal{G}$_v

defined by the curve 𝔾_m^τ/(vτ^k)^ℤ therefore identifies Spec(ℤ[v, v^{-1}][[τ]]) with the formal completion of $\mathcal{G}$_v along a section at infinity.

Under the hypotheses of 2.6, there exists v ∈ A* such that C_k is isomorphic to X_v and C_s to 𝔾_m^τ/(vτ^k)^ℤ over k(s). The curve C is therefore the pullback of the curve 𝔾_m^τ/(vτ^k)^ℤ over ℤ[v, v^{-1}][[τ]], and 2.6 follows.

2.7. Let us prove VI 6.14. With the notation of loc. cit., we reduce to the case in which V is complete with separably closed residue field. Let C be the Néron model of the elliptic curve E over the fraction field, and apply 2.6 to C. We find

C ≅ 𝔾_m^t/(ut^k)^ℤ

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and, because C is regular, t is a uniformizer. By 2.1 and VI 1.1, we have

j(E)^{-1} = ±(ut^k) + ∑_{n>0} d_n(ut^k)^n,

and j(E) therefore has valuation −k.

3. Application: expansion of a modular form as a Fourier series.

We shall show in § 4 that, after scalar extension to $\mathbb{C}$, the notions introduced in this section coincide with the classical notions.

Definition 3.1. The algebra of integral modular forms is the graded algebra

⊕_k $H^0(\mathcal{M}_1, \omega^{\otimes k})$.

In other words, an integral modular form of weight k is a rule, compatible with base change, which to C/S, an irreducible generalized elliptic curve over a scheme S, associates a section of $\omega^{\otimes k}$ over S.

Definition 3.2. The algebra of integral modular forms of level H ⊂ GL(2, $\mathbb{Z}$/n) is the graded algebra

⊕_k $H^0(\mathcal{M}_H, \omega^{\otimes k})$.

The following lemma makes it possible, in certain cases, to consider only the $\mathcal{M}$_n.

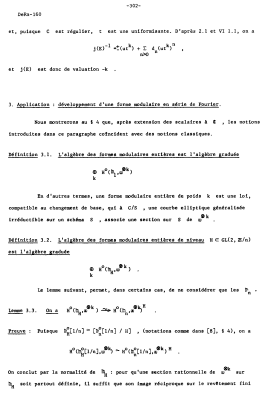
Lemma 3.3. We have

$H^0(\mathcal{M}_H, \omega^{\otimes k}) \simeq H^0(\mathcal{M}_n, \omega^{\otimes k})^H$.

Proof. Since $\mathcal{M}_H^\circ[1/n] \simeq [\mathcal{M}_n^\circ[1/n]/H]$ (notation as in [8], § 4), we have

$H^0(\mathcal{M}_H^\circ[1/n], \omega^{\otimes k}) \simeq H^0(\mathcal{M}_n^\circ[1/n], \omega^{\otimes k})^H$.

We conclude by the normality of $\mathcal{M}$_H: for a rational section of $\omega^{\otimes k}$ on $\mathcal{M}$_H to be defined everywhere, it is enough that its pullback to the finite covering



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Authority page 161; collected-paper page 654; original folio 303.

be so on $\mathcal{M}$_n.

Theorem 3.4. The algebra of integral modular forms of level H is a finitely generated $\mathbb{Z}$-algebra.

Proof. By 3.3, it is enough to treat the case of $\mathcal{M}$_n, for n large. In this case $\mathcal{M}$_n = M_n is a projective scheme over $\mathbb{Z}$, and ω is an ample invertible sheaf on M_n. Indeed, the divisor of the section Δ of $\omega^{\otimes 12}$ is a multiple of M_n^∞ (on $\mathcal{M}$_n, div(Δ) = $\mathcal{M}_n^\infty$); it meets every irreducible component of every fiber of M_n over $\mathbb{Z}$, and is therefore ample. The theorem consequently follows from the general theorem EGA III, 2.3.4.1.

3.5. One may also consider modular forms with coefficients in algebras over $\mathbb{Z}$. Since we are not sure of the correct definition in the general case, we shall for the most part restrict ourselves to the case of the $\mathcal{M}$_n, viewed as $\mathbb{Z}[\zeta_n]$-stacks.

Definition 3.6. Let A be an algebra over $\mathbb{Z}[\zeta_n]$. A modular form of level n and weight k over the $\mathbb{Z}[\zeta_n]$-algebra A is an element of

$H^0(\mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} A, \omega^{\otimes k})$.

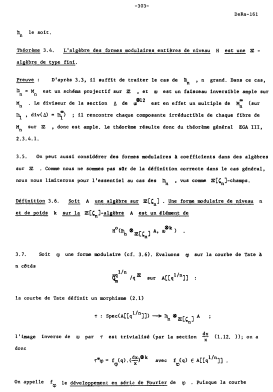
3.7. Let φ be a modular form (cf. 3.6). Evaluate φ on the n-sided Tate curve $\mathbb{G}_m^{q^{1/n}}/q^{\mathbb{Z}}$ over $A[[q^{1/n}]]$: the Tate curve defines a morphism (2.1)

$\tau : \mathrm{Spec}(A[[q^{1/n}]]) \to \mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} A$;

the pullback of ω by τ is trivialized (by the section dx/x (1.12)); hence

$\tau^*\varphi = f_\varphi(q)(dx/x)^{\otimes k}$, with $f_\varphi(q) \in A[[q^{1/n}]]$.

We call f_φ the Fourier-series expansion of φ. Since the curve



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$\mathcal{H}_n$ le suit.

Théorème 3.4. L'algèbre des formes modulaires entières de niveau N est une $\mathbb{Z}$-algèbre de type fini.

Preuve : D'après 3.3, il suffit de traiter le cas de $\mathcal{H}_n$, n grand. Dans ce cas, $\mathcal{H}_n = H_n$ est un schéma projectif sur $\mathbb{Z}$, et ω est un faisceau inversible ample sur H_n. Le diviseur de la section f de $\omega^{\otimes 12}$ est en effet un multiple de $\mathcal{H}_n^{\infty}$ (sur $\mathcal{H}_n$, $\mathrm{div}(f) = \mathcal{H}_n^{\infty}$; il rencontre chaque composante irréductible de chaque fibre de H_n sur $\mathbb{Z}$, donc est ample. Le théorème résulte donc du théorème général EGA III, 2.3.4.1.

3.5. On peut aussi considérer des formes modulaires à coefficients dans des algèbres sur $\mathbb{Z}$. Comme nous ne sommes pas sûr de la définition correcte dans le cas général, nous nous limiterons pour l'essentiel au cas des $\mathcal{H}_n$, vus comme $\mathbb{Z}[\zeta_n]$-champs.

Définition 3.6. Soit A une algèbre sur $\mathbb{Z}[\zeta_n]$. Une forme modulaire de niveau n et de poids k sur la $\mathbb{Z}[\zeta_n]$-algèbre A est un élément de

$$H^0(\mathcal{H}_n \otimes_{\mathbb{Z}[\zeta_n]} A, \omega^{\otimes k}) .$$

3.7. Soit φ une forme modulaire (cf. 3.6). Evaluons φ sur la courbe de Tate à n côtés

$$\mathbb{G}_m^{q^{1/n}}/q^{\mathbb{Z}} \text{ sur } A[[q^{1/n}]] :$$

la courbe de Tate définit un morphisme (2.1)

$$\tau : \mathrm{Spec}(A[[q^{1/n}]]) \longrightarrow \mathcal{H}_n \otimes_{\mathbb{Z}[\zeta_n]} A :$$

L'image inverse de ω par τ est trivialisé (par la section $\frac{dx}{x}$ (1.12.)); on a donc

$$\tau^*\omega = f_\varphi(q).(\frac{dx}{x})^{\otimes k} \quad \text{avec} \quad f_\varphi(q) \in A[[q^{1/n}]] .$$

On appelle f_φ le développement en série de Fourier de φ. Puisque la courbe

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Tate n-gon is already defined over $\mathbb{Z}[[q^{1/n}]] \otimes_{\mathbb{Z}} A$ (the subring of $A[[q^{1/n}]]$ formed by the formal series whose coefficients generate a finitely generated $\mathbb{Z}$-module), we have

(3.7.1) $f_\varphi(q) \in \mathbb{Z}[[q^{1/n}]] \otimes_{\mathbb{Z}} A \subset A[[q^{1/n}]]$.

3.8. Let φ be a modular form (cf. 3.6). For $g \in SL(2, \mathbb{Z}/n)$, we denote by gφ the modular form which to a curve with level-n structure (C, α) over S associates the section

$(g\varphi)(C, \alpha) = \varphi(C, g^{-1}\alpha)$

of $\omega^{\otimes k}$ over S.

The $f_{g\varphi}$ are the expansions of φ at the cusps. If

$u = \pm\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$

and $f_\varphi = \sum c_m q^m$ $(m \in (1/n)\mathbb{Z})$, then

(3.8.1) $f_{u\varphi} = (-1)^k \sum c_m \zeta_n^{amn} q^m$.

Theorem 3.9. Let φ be a modular form of level n and weight k with coefficients in the $\mathbb{Z}[\zeta_n]$-algebra A.

(i) If the expansions at the cusps f_{gφ} vanish, then φ vanishes.

(ii) If B is a subalgebra of A and the f_{gφ} lie in B[[q^{1/n}]], then φ has coefficients in B (it is obtained by scalar extension from a unique form over B).

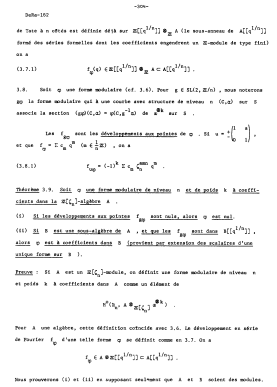
Proof. If A is a $\mathbb{Z}[\zeta_n]$-module, a modular form of level n and weight k with coefficients in A is defined to be an element of

$H^0(\mathcal{M}_n, A \otimes_{\mathbb{Z}[\zeta_n]} \omega^{\otimes k})$.

For A an algebra, this definition agrees with 3.6. The Fourier-series expansion f_φ of such a form φ is defined as in 3.7. We have

$f_\varphi \in A \otimes_{\mathbb{Z}} \mathbb{Z}[[q^{1/n}]] \subset A[[q^{1/n}]]$.

We shall prove (i) and (ii) under the sole assumption that A and B are modules.



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de Tate à n effets est définie déjà sur $\mathbb{Z}[\zeta_n^{1/n}]$; $\otimes_{\mathbb{Z}} A$ (le sous-anneau de $A[[q^{1/n}]]$ formé des séries formelles dont les coefficients engendrent un $\mathbb{Z}$-module de type fini) est à

(3.7.1) $\qquad f_\varphi(q) \in \mathbb{Z}[[q^{1/n}]] \otimes_{\mathbb{Z}} A \subset A[[q^{1/n}]]$.

3.8. Soit φ une forme modulaire (cf. 3.6). Pour $g \in SL(2, \mathbb{Z}/n)$, nous noterons $g\varphi$ la forme modulaire qui à une courbe avec structure de niveau n (C, ω) sur S associe la section $(g\varphi)(C, \omega) = \varphi(C, t^{-1}\omega)$ de $\omega^{\otimes k}$ sur S .

Les $f_{g\varphi}$ sont les développements aux pointes de φ . Si $u = \pm\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$, et que $f_\varphi = \sum c_n q^n$ $(q \in \frac{1}{n}\mathbb{Z})$, on a

(3.8.1) $\qquad f_{u\varphi} = (-1)^k \sum c_n \, e^{2\pi i n/n} q^n$.

Théorème 3.9. Soit φ une forme modulaire de niveau n et de poids k à coefficients dans le $\mathbb{Z}[\zeta_n]$-algèbre A .

(i) Si les développements aux pointes $f_{g\varphi}$ sont nuls, alors φ est nul.

(ii) Si B est une sous-algèbre de A , et que les $f_{g\varphi}$ sont dans $B[[q^{1/n}]]$, alors φ est à coefficients dans B (provient par extension des scalaires d'une unique forme sur B).

Preuve : Si A est un $\mathbb{Z}[\zeta_n]$-module, on définit une forme modulaire de niveau n et poids k à coefficients dans A comme un élément de

$$H^0(\mathcal{M}_n, A \otimes_{\mathbb{Z}[\zeta_n]} \omega^{\otimes k}) \ .$$

Pour A une algèbre, cette définition coïncide avec 3.6. Le développement en série de Fourier f_φ d'une telle forme φ se définit comme en 3.7. On a

$$f_\varphi \in A \otimes \mathbb{Z}[[q^{1/n}]] \subset A[[q^{1/n}]] \ .$$

Nous prouverons (i) et (ii) en supposant seulement que A et B soient des modules.

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Authority page 163; collected-paper page 656; original folio 305.

Let us prove (i). If A is a direct limit $\varinjlim A_i$, then

$H^0(\mathcal{M}_n, A \otimes \omega^k) = \varinjlim H^0(\mathcal{M}_n, A_i \otimes \omega^k)$.

It therefore suffices to treat the case in which A is a finitely generated $\mathbb{Z}[\zeta_n]$-module. If the f_{gφ} vanish, (2.4) shows that φ vanishes on the formal completion of $\mathcal{M}_n$ along infinity. The support of φ, disjoint from at least one point of every irreducible component of every geometric fibre of $\mathcal{M}_n$, is therefore finite over $\mathbb{Z}[\zeta_n]$. Since $\mathcal{M}_n$ is Cohen–Macaulay over $\mathbb{Z}[\zeta_n]$, and since $A \otimes \omega^k$ is locally the inverse image of a module over $\mathbb{Z}[\zeta_n]$, this is absurd if $\varphi \neq 0$.

Let us prove (ii). Let $\mathrm{Spec}(\hat{D})$ be the sum of the formal completions of $\mathcal{M}_n$ along the sections at infinity gf_n ($g \in SL(2, \mathbb{Z}/n)$). The expansions at the cusps define a map

$H^0(\mathcal{M}_n, A \otimes \omega^k) \to A \otimes_{\mathbb{Z}[\zeta_n]} \hat{D}$.

We prove (ii) by considering the following diagram, whose second row is exact because $\hat{D}$ is flat over $\mathbb{Z}[\zeta_n]$:

$0 \to H^0(\mathcal{M}_n, B \otimes \omega^k) \to H^0(\mathcal{M}_n, A \otimes \omega^k) \to H^0(\mathcal{M}_n, A/B \otimes \omega^k)$
$\downarrow \quad \downarrow \quad \downarrow$
$0 \to B \otimes \hat{D} \to A \otimes \hat{D} \to A/B \otimes \hat{D}$.

Theorem 3.10. (i) One has

$H^0(\mathcal{M}_n, \omega^k) \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{Q}(\zeta_n) \simeq H^0(\mathcal{M}_n \otimes \mathbb{Q}, \omega^k)$.

(ii) Let $\varphi \in H^0(\mathcal{M}_n \otimes \mathbb{Q}, \omega^k)$, let $f\varphi(q) = \sum a_i q^i$ ($i \in (1/n)\mathbb{Z}$, $a_i \in \mathbb{Q}(\zeta_n)$) be the Fourier-series expansion of φ at f_n, and let π be a place of $\mathbb{Q}(\zeta_n)$. Then

$$\nu\pi(f\varphi) = \deg(f_n)\ \inf(\nu\pi(a_i))$$

is the order of the zero (or the negative of the order of the pole) of φ at the generic point of the irreducible component of the fibre of $\mathcal{M}_n$ at π that meets the section at infinity f_n.

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From the point of view adopted here, this theorem is trivial; it follows from

Corollary 3.11. Let $\sum a_i q^i$ be the Fourier-series expansion of $\varphi \in H^0(\mathcal{M}_n \otimes \mathbb{Q}, \omega^k)$. The denominators of the a_i are bounded.

Corollary 3.12. Let π be a place of $\mathbb{Q}(\zeta_n)$ of characteristic p, let p^m be the largest power of p dividing n, and let g be an element of SL(2, $\mathbb{Z}$/n) whose image in SL(2, $\mathbb{Z}/p^m$) lies in the upper-triangular group. Then

$$\nu\pi(f\varphi) = \nu\pi(f_{g\varphi}).$$

Indeed, the cusps f_n and gf_n meet the same irreducible component in characteristic p.

Corollary 3.13. If the expansion at one cusp of $\varphi \in H^0(\mathcal{M}_n \otimes \mathbb{Q}, \omega^k)$ has coefficients in $\mathbb{Z}[\zeta_n][1/n]$, then the expansions at the other cusps have the same property.

3.14. Let $\varphi \in H^0(\mathcal{M}_n, \omega^k)$. If π is a place of $\mathbb{Q}(\zeta_n)$ of characteristic p and the first coefficients of the Fourier-series expansion of φ are divisible by π, the reduction of φ modulo π vanishes to some order at the point at infinity f_n. Write $\nu(\pi,g,\varphi)$ for the order of the zero of $g\varphi$:

if $f_{g\varphi}(q) = \sum a_i q^i$ ($i \in (1/n)\mathbb{Z}$), then $a_i \equiv 0$ (π) for $ni < \nu(\pi,g,\varphi)$.

The degree of ω^k on $\mathcal{M}_n$ can be calculated using φ. The result is already known (cf. VI 4.4). Carrying out the calculation gives:

Corollary 3.14. Let π be a place of $\mathbb{Q}(\zeta_n)$ of characteristic p, let p^m be the largest power of p dividing n, and let $\varphi \in H^0(\mathcal{M}_n, \omega^k)$. As g runs through the set BSL of elements of SL(2, $\mathbb{Z}$/n) whose image in SL(2, $\mathbb{Z}/p^m$) lies in the upper-triangular group, one has

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$|SL(2, \mathbb{Z}/n)|^{-1} \sum_{g \in BSL} \nu(\pi, g, \varphi) \leq |\mathbb{P}^1(\mathbb{Z}/p^m)|^{-1}\, k/12.$

For n = 1, this result may be restated as follows.

Corollary 3.15. Let p be a prime number and let $\varphi \in H^0(\mathcal{M}_1, \omega^k)$ be a modular form with series expansion $f\varphi(q) = \sum a_n q^n$. Then

$\nu_p(f\varphi) = \inf\{\nu_p(a_i) \mid i \leq k/12\}$.

In other words, to verify that the coefficients of a modular form of weight k are divisible by a power of p, it suffices to verify this for those whose index is ≤ k/12.

3.16. One can also relate what happens on the various irreducible components in characteristic p. As an example, we shall treat the case of modular forms of level $\Gamma_0(p)$. The same arguments apply to the study, at p, of modular forms of level $\Gamma_0(p) \cap \Gamma(n)$, with n prime to p. Passing to such a level would spare us the use of fractional intersection numbers, introduced on the model of VI 4.

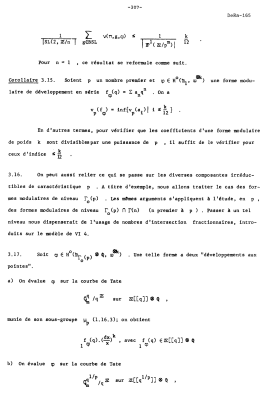
3.17. Let $\varphi \in H^0(\mathcal{M}_{\Gamma_0(p)} \otimes \mathbb{Q}, \omega^k)$. Such a form has two "expansions at the cusps."

a) Evaluate φ on the Tate curve $\overline{\mathbb{G}}_m\hat{}\,q/q^{\mathbb{Z}}$ over $\mathbb{Z}[[q]] \otimes \mathbb{Q}$, equipped with its subgroup μp (1.16.3); one obtains

${}_1f\varphi(q)\cdot(dx/x)^k$, with ${}_1f\varphi(q) \in \mathbb{Z}[[q]] \otimes \mathbb{Q}$.

b) Evaluate φ on the Tate curve

$\overline{\mathbb{G}}_m\hat{}\,\{q^{1/p}\}/q^{\mathbb{Z}}$ over $\mathbb{Z}[[q^{1/p}]] \otimes \mathbb{Q}$,



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equipped with its subgroup isomorphic to $\mathbb{Z}/p$ generated by q. One obtains

$_2f\varphi(q)\cdot(dx/x)^k$, with $_2f\varphi(q) \in \mathbb{Z}[[q^{1/p}]] \otimes \mathbb{Q}$.

For $\ell \neq p$, the ℓ-adic valuations of $_1f\varphi$ and $_2f\varphi$ (the lower bound of the ℓ-adic valuations of the coefficients) coincide. We shall write $\nu_1(\varphi)$ and $\nu_2(\varphi)$, or simply ν_1 and ν_2, for the p-adic valuations of $_1f\varphi$ and $_2f\varphi$.

3.18. We extend the involution w (V.1.3) to modular forms as follows. To an elliptic curve C/S equipped with a subgroup A ⊂ C[p], the form $w\varphi$ assigns the inverse image in ω^k of the section $\varphi(C/A, C[p]/A)$ of $(\omega_{(C/A)/S})^k$. One has

$ww\varphi = p^k\varphi$.

One checks that

$_1f_{w\varphi}(q) = p^k\ {}_2f\varphi(q^p)$,

$_2f_{w\varphi}(q) = {}_1f\varphi(q^{1/p})$.

Consequently

$(\nu_1(w\varphi), \nu_2(w\varphi)) = (k + \nu_2(\varphi), \nu_1(\varphi))$,

and $\nu_2 - \nu_1 + k/2$ changes sign when one passes from φ to $w\varphi$.

3.19. We propose to estimate $\nu_1 - \nu_2$. This number does not change when φ is multiplied by an integer; we may therefore suppose that

$\varphi \in H^0(\mathcal{M}_{\Gamma_0(p)}, \omega^k)$.

Assume that φ is not identically zero. If N_1 and N_2 are the two irreducible components of $\mathcal{M}_{\Gamma_0(p)} \otimes \mathbb{F}p$, corresponding respectively to curves (C,A) with A infinitesimal and A discrete, the divisor of φ may be written

$\mathrm{div}(\varphi) = D + \nu_1 N_1 + \nu_2 N_2$,

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where the divisor D meets N_1 and N_2 at only finitely many points. On the regular stack $\mathcal{M}_{\Gamma_0(p)}$, $\mathrm{div}(p) = N_1 + N_2$, whence the vanishing of the intersection numbers

$$(N_1 + N_2, N_1) = (N_1 + N_2, N_2) = 0.$$

On $N_1 \simeq \mathcal{M}_1 \otimes \mathbb{F}_p$, ω has degree $1/24$; therefore

$$(\mathrm{div}(\varphi), N_1) = k/24.$$

Similarly, since on $\mathcal{M}_{\Gamma_0(p)}$ the degree of ω is $(1/24)[\mathcal{M}_{\Gamma_0(p)}:\mathcal{M}_1] = (p+1)/24$, one has

$$\deg(D) = (D,N_1) + (D,N_2) = (p+1)k/24.$$

By VI.4.9, $(N_2,N_1) = (p-1)/24$.

One has

$$(D + \nu_1 N_1 + \nu_2 N_2, N_1) = (D,N_1) + \nu_1(N_1+N_2,N_1) + (\nu_2-\nu_1)(N_2,N_1)$$

$$= \tfrac{1}{2}((D,N_1)+(D,N_2)) + \tfrac{1}{2}((D,N_1)-(D,N_2)) + (\nu_2-\nu_1+k/2)(N_2,N_1) - (k/2)(N_2,N_1).$$

Substituting the expressions above gives

$$\tfrac{1}{2}\cdot k\cdot(p+1)/24 + \tfrac{1}{2}((D,N_1)-(D,N_2)) + (\nu_2-\nu_1+k/2)(p-1)/24 - (k/2)\cdot(p-1)/24 = k/24.$$

Thus

$$(\nu_2-\nu_1+k/2)(p-1)/24 = \tfrac{1}{2}((D,N_2)-(D,N_1)).$$

Proposition 3.20. One has

$$\nu_2-\nu_1+k/2 = \tfrac{1}{2}\cdot 24/(p-1)\ ((D,N_2)-(D,N_1)),$$

$$(D,N_1) + (D,N_2) = k\cdot(p+1)/24,$$

and hence

$$|\nu_2-\nu_1+k/2| \le \tfrac{1}{2}\cdot k\cdot(p+1)/(p-1).$$

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In particular, if $_1f\varphi(q)$ is p-integral, then the p-adic valuation of $_1f_{w\varphi}(q)$ is $\geq -k/(p-1)$.

4. Comparison with transcendental theory.

4.1. We denote by D the open unit disc contained in $\mathbb{C}$, by q the usual coordinate on D (the inclusion $D \hookrightarrow \mathbb{C}$), and by D* the punctured disc $D - \{0\}$.

We denote by $D[q^{1/k}]$ the Riemann surface of $q^{1/k}$: it is a copy of the unit disc, whose usual coordinate is called $q^{1/k}$, equipped with the map $u : D[q^{1/k}] \to D$ such that

$q \circ u = (q^{1/k})^k$.

4.2. A construction identical to Construction 1.1 gives an analytic space $\overline{\mathbb{G}}_m\hat{}q$ over D; translation by the section q ($x_1 = 1$) defines an action of $\mathbb{Z}$ on $\overline{\mathbb{G}}_m\hat{}q$, and this action is free (because $|q| < 1$). Passing to the quotient gives an analytic space $\overline{\mathbb{G}}_m\hat{}q/q\mathbb{Z}$ over D, which is a family of generalized elliptic curves parametrized by D (the Tate curve).

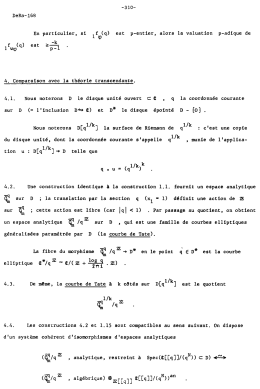
The fibre of the morphism $\overline{\mathbb{G}}_m\hat{}q/q\mathbb{Z} \to D^*$ at the point $q \in D^*$ is the elliptic curve $\mathbb{C}^*/q\mathbb{Z} \simeq \mathbb{C}/(\mathbb{Z} + (\log q)/(2\pi i)\cdot\mathbb{Z})$.

4.3. Similarly, the k-gon Tate curve over $D[q^{1/k}]$ is the quotient

$\overline{\mathbb{G}}_m\hat{}\{q^{1/k}\}/q\mathbb{Z}$.

4.4. Constructions 4.2 and 1.15 are compatible in the following sense. There is a coherent system of isomorphisms of analytic spaces

$(\overline{\mathbb{G}}_m\hat{}q/q\mathbb{Z}$, analytic, restricted to $\mathrm{Spec}(\mathbb{C}[[q]]/(q^N)) \subset D) \simeq ((\overline{\mathbb{G}}_m\hat{}q/q\mathbb{Z}$, algebraic) $\otimes_{\mathbb{Z}[[q]]} \mathbb{C}[[q]]/(q^N))\hat{}\{an\}$.



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Similarly, there are isomorphisms of analytic spaces over

$\mathbb{C}[[q^{1/k}]]/(q^N)$:

$(\overline{\mathbb{G}}_m\text{^}\{q^{1/k}\}/q^{\mathbb{Z}}$, analytic, restricted to $\mathrm{Spec}(\mathbb{C}[[q^{1/k}]]/(q^N)) \subset D[q^{1/k}]) \simeq ((\overline{\mathbb{G}}_m\text{^}\{q^{1/k}\}/q^{\mathbb{Z}}$, algebraic) $\otimes_{\{\mathbb{Z}[[q^{1/k}]]\}} \mathbb{C}[[q^{1/k}]]/(q^N))^{an}$;

these isomorphisms respect the structures 1.16 on the two sides.

4.5. Classically, a modular form of weight k and level n is a holomorphic function f on the Poincaré half-plane

$X = \{z \mid \mathrm{Im}(z) > 0\}$

satisfying the following conditions.

(A) If (a b; c d) $\in SL(2, \mathbb{Z})$ is congruent to (1 0; 0 1) mod n,

$f(z) = (cz+d)^{-k}\, f((az+b)/(cz+d))$.

This condition implies that, for $\gamma \in SL(2, \mathbb{Z}/n)$, the image of $\gamma_0 = (a\ b;\ c\ d) \in SL(2, \mathbb{Z})$, the function

$f|\gamma_0 : z \mapsto (cz+d)^{-k}\, f((az+b)/(cz+d))$

depends only on γ, not on the choice of γ_0. It is denoted $f|\gamma$.

(B) For every $\gamma \in SL(2, \mathbb{Z}/n)$,

$(f|\gamma)(z)$ remains bounded as $\mathrm{Im}(z) \to \infty$

(it is equivalent, and more honest, to require $f|\gamma$ to have polynomial growth in vertical strips as $\mathrm{Im}(z) \to \infty$).

Such a function is periodic with period n and therefore has a Fourier-series expansion

$f(z) = \sum a_m\, e\text{^}(2\pi i m z)$ $(m \in (1/n)\mathbb{Z},\ m \geq 0)$.

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Construction 4.6. Modular forms of weight k and level n are identified with the sections

$$H^0(\mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}, \omega^{\otimes k}) = H^0(\mathcal{M}_n, \omega^{\otimes k}) \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}.$$

Proof. Suppose first that $n \geq 3$. In that case, $\mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C} = M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ is a scheme. The analytic space underlying $M_n^\circ \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ is

$$X/\mathrm{Ker}(\mathrm{SL}(2, \mathbb{Z}) \to \mathrm{SL}(2, \mathbb{Z}/n)),$$

and the holomorphic sections of $\omega^{\otimes k}$ on this space are identified with the holomorphic functions on X satisfying 4.5 (A). The analytic space underlying $M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ is obtained from it by adding finitely many points. A holomorphic section of $\omega^{\otimes k}$ on $M_n^\circ \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ extends to $M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$ if and only if it satisfies (B). By GAGA, these sections are the algebraic sections of $\omega^{\otimes k}$ on $M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$.

The case of arbitrary n is reduced to the preceding one by passing to the invariants under

$$\mathrm{Ker}(\mathrm{SL}(2, \mathbb{Z}/3n) \to \mathrm{SL}(2, \mathbb{Z}/n))$$

in the space of modular forms of weight 3n. One could also invoke GAGA for the algebraic stack $\mathcal{M}_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}$.

4.7. For $n \geq 3$, $(M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C})^{an}$ is also the analytic space classifying generalized elliptic curves C/S over analytic spaces S, equipped with a level-n structure α such that $\det(\alpha) = 1$. From this point of view, a modular form f is a rule that assigns to every such curve a section of $\omega^{\otimes k}$ over S. The Fourier-series expansion is then obtained by evaluating the form on the Tate curve (4.3) (and multiplying the result by a power of $2\pi i$ depending on the normalizations chosen). For $\varphi \in H^0(M_n \otimes_{\mathbb{Z}[\zeta_n]} \mathbb{C}, \omega^{\otimes k})$, compatibility 4.4 shows that this expansion coincides with the expansion defined purely algebraically in § 3.

4.8. From this point of view, the results of § 3 give arithmetic properties of the coefficients of the Fourier-series expansions at the various cusps of the functions on X satisfying 2.5 (A) and (B). For example:

- The graded algebra of modular forms of level n (2.5) is the tensor product with $\mathbb{C}$ over $\mathbb{Z}[\zeta_n]$ of the finite-type $\mathbb{Z}[\zeta_n]$-subalgebra consisting of those

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for which the Fourier-series expansions of the f|γ have coefficients in $\mathbb{Z}[\zeta_n]$.

- If the expansion of f has coefficients in $\mathbb{Z}[\zeta_n][1/n]$, that of f|γ has the same property.

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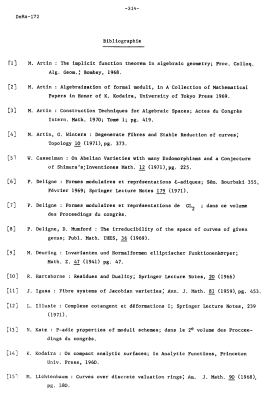

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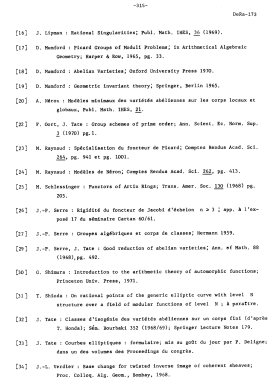

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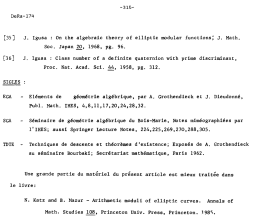

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